Microbiology

Workbook

Dr. Danish

MEDICAL INSTITUTE FOR SCREENING TEST
MIST

INSTRUCTIONS:

- Our initiative is to make a workbook in a precise manner to make you write the most important points, focus more on understanding the important concepts in class, save you precious time & energy, at the sametime keep you actively involved in the sessions.
- Workbook spaces are provided wherever necessary to help place your desired content.

PLEASE NOTE:

- This workbook has to be strictly used when the faculty is teaching as there are blank spaces which need to be filled, otherwise the inference may be misleading.
- The information contained in this book is strictly for educational purposes. Content is not intended as a substitute for professional medical advice, diagnosis or treatment.
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DISCLAIMER:

- The notes have been made taking care of all the relevant points required but special care has been taken to avoid exhaustive nature. These notes are only for the purpose of understanding in a crisp manner and help you in quick complete revision. Please do not use them for the treatment of your patients as each patient requirement differ at various times.

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MY TARGET → FMGE SCORE /300

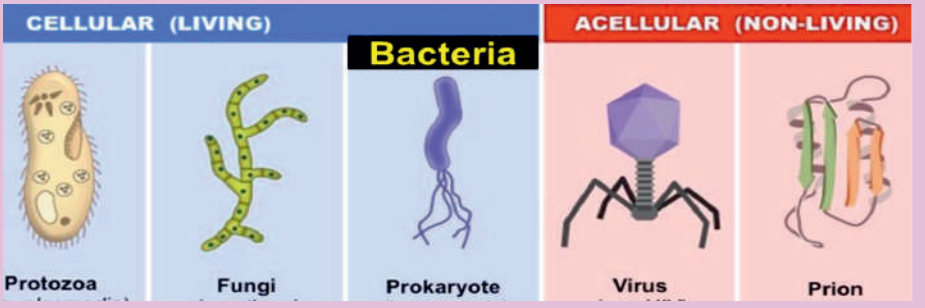
SECTION	TOPIC Class PYQs after every topic	PAGES (Page Numbers)	PYQS IN THIS WORKBOOK	HOW MANY MCQs IN FMGE EXAM
I	General Microbiology	18 (2-19)	62	2-5
II	Bacteriology	32 (20-51)	130 (78+52)	12-18
III	Virology	19 (52-70)	84	9-13
IV	Parasitology	9 (71-79)	56	3-5
V	Fungology = Mycology	3 (80-82)	28	2-6
VI	Immunology	6 (83-88)	56	2-3
Workbook 712 + Class >325 PYQ		38 (90-127)	351 Extra	1100 PYQs

Master Mind Map- 90 Logos	1 (89)
Anti- Confusion Antivenom	3 (128-130)
One liner NEET PG Buzzwords A→Z	46 (131-176)
FMGE - NEET PG MCQ Analysis	4 (177-180)
FMGE Master strategies for 200+	4 (181-184)

Disclaimer

1. Workbook for educational purpose
2. Workbook has blank spaces, to be used only with classes and guidance
3. Workbook is not a substitute for professional clinical diagnosis, treatment and patient consultation
4. Guidelines keep getting updated, author/publisher have no liability for errors/omissions

Main topic- Red, Subtopic- Blue, Keywords- Pink



AUG	Start Codon
UAA	Stop Codon-Ochre
UAG	Stop Codon-Amber
UGA	Stop Codon-Opal

Section I : General Microbiology

13 topics A-L, 19 pages, 62 PYQs

Microbes, Bioterrorism, Biomedical Waste, Infectious Disease, Transmission of Infection, Fathers, Types of Infection, Evolution and Spread of Infection, Infection Control (Standard precautions, Hand Hygiene, Isolation, Quarantine), Sterilization, Diagnosis of Infections

A. MICROBES

1. Bad = cause disease

2. Good/neutral = Normal flora = Commensals→

Lactobacillus (curd)→ Gut, vagina, Staph epidermidis→ Skin

Pathogens	Size	Cell type	Nucleic acid
B	0.5-5 µm		DNA (PCR) + RNA
F	~10 µm	Eukaryote	DNA (PCR) + RNA
P	~20 µm	Eukaryote	DNA (PCR) + RNA
V	0.02-3 µm		Either DNA or RNA
P 100% death	-F21/23 No cell- Only protein- No DNA/RNA Cruetzfeldt		

B. BIOTERRORISM -F1/23

3 categories- Cat A highest risk

Cat A- SBPAVT- Smallpox, Botox, Plague, Anthrax (MC), Viral hemorrhagic fev (Dengue, Ebola), Tularemia= Rabbit fever

Cat B- Moderate risk- Salmonella, Shigella, Vibrio, toxins

Cat C- Emerging Pathogens- Nipah, Hanta, Influenza, SARS

C. BIOMEDICAL = HEALTHCARE WASTE -F1/26,7/24/25,N22

1. BMW guidelines 2016/18- Non-chlorinated bags only -F7/25

2. BLACK-General waste- Food, paper/cover, construc waste

WHITE- Sharp waste- Blade, needle, lancet, scalpel, wire, sharp glass/metal, fix needle syringe (tuberculin, insulin)

4. BLUE- Glass/Metal- Blue+white- Secured landfill=Sharp pit Ampoule, vial, tube, beaker, metal scissor/implant/stent

5. RED- Plastic/rubber/silicone waste- Microwave/Autoclave

Red is cancerous- Dont burn- Make road/Fuel oil -F1/26

Syringe, Tube- Ryle, Line-iv, Catheter-Foley, Stent-DJ, Bag, TPN, Vaccutainer, plastic appron/goggle, glove, O_2 mask

6. YELLOW- Incinerable/Infected soiled waste- Incinerator

Yellow is fire- Burn in incinerator (chimney)- pollution -F1/26

Cotton, bandage, gauze, plaster, cloth/linen/gown/headcap Surg/N95 mask, organ, placenta, animal, culture, vaccine, toxin, chemicals, disinfectant, liquids, medicine, Cytotoxic drug in glass/plastic syringe/vial, bloodbag-India guideline

7. Blood spill- 1%Na hypochlorite NaClO x 20m→ Mop→

8. waste- Sulfur powder→ Syringe/Mop→

9. waste- Lead bin (absorbs radiation)→ AERB

10. Liquid waste (Drain/suction fluid)-Effluent treatment plant

11. waste- Xray film, pressurized cylinder/inhaler, battery, electronics (Glucometer/hearing aid/ thermometer/ infusion-insulin pump/ pacemaker/ defibrillator/ cardiac monitor/ pulse oximeter/ ECG-USG-CTG-Holter machine/ infant warmer/ portable O_2 concentrator-spirometer/ torch/ otoscope/ophthalmoscope/digital stethoscope/TENS/ESU)

D. INFECTIOUS DISEASE (INFECTION)

= Communicable = Transmissible = Contagious disease

1. Agent (Pathogen = Micro-organism causing infection)

↓

2. Reservoir (Source- Patient > animal > water > soil)

↓

3. Transmission (Resp, orofecal, sexual, vertical)

↓

4. Human Host (↑ infec in immunocompromised)

E. TRANSMISSION OF INFECTION

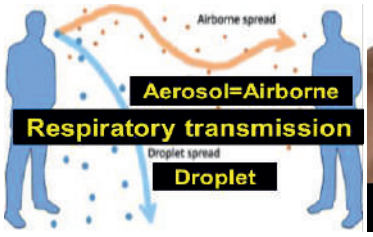
1. Resp (100µm)→ Surg=Med=3 ply/layer mask

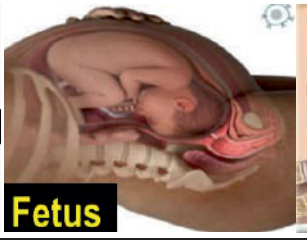
2. Resp (<5µm)→ Airborne spread→

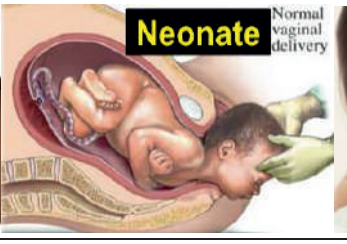
.

3. Direct touch/contact -N18 HSV, Ringworm,

4. (Hi-touch object) -N18/19 Door handle,







5. (Non-living) borne-F26-Food,water,air,soil,fomite

6. Orofecal- Typhoid, Cholera, Shigella, Campylobacter, E. coli, Polio, Rotavirus, Hepatitis A/E, Entamoeba, Giardia, Cryptosporidium, Helminths

7. Blood transfusion & Pregnancy screening -F1/22/23,N20

.

8. Sexual/ Venereal- HPV (MC),

9. Kissing (saliva) disease= Infectious mononucleosis

. Saliva- HSV, EBV, CMV, Mumps, Rabies, Nipah

10. Congenital = Vertical = Mother To Child Transmission (3)

a. =Transplacental=Fetal~Teratogenic infec

TORCHES/TORCH -F1/25 Toxoplasma, others (HIV, Hep B/C, Malaria, Chickenpox), Rubella-MC, CMV, HSV, EBV,

b. =Intrapartum =Intranatal=Natal=Neonatal

Ophthalmia neonatorum-N18/19- Gonorrhea, Chlamydia, HSV

c. Breastfeeding unsafe (Not necessarily contraindicated)

. Milk- HCL- HIV, chemo/radiation Rx,

. Breast infected- HCTB- Herpes, Chickenpox, TB, Abscess

. Baby-PMG-Phenylketonuria,Maple syp urine dis,

11. (Arthropods/Insects) borne-

transmits and causes Malaria

12. (70%)= Infection from animal reservoir→ man

. Bacteria- Leptospira (MC), Anthrax, Plague, Brucella, TB

. Virus- Rabies, Covid, Ebola, Nipah, bird flu, KFD

. Parasite- Toxoplasma, Hydatid cyst

. Fungus- Cryptococcus, Histoplasma, R

13. Anthroponosis- Man→ Animal (TB, Covid, influenza)

14. Amphixenoses- Both directions (TB, Covid, influenza)

F. FATHER of Microscopy- Leeuwenhoek- Didnt see bacteria

1. Vaccinology- except Anthrax, Rabies

2. Antibiotics- Discover Penicillin

3. Epidemiology- Cholera handpump

4. Antiseptic Surgery-

5. Microbiology- Germ theory of dis

6. Modern Bacterio- Kochs postulates

. No growth in agar= No Koch postulates- Virus,

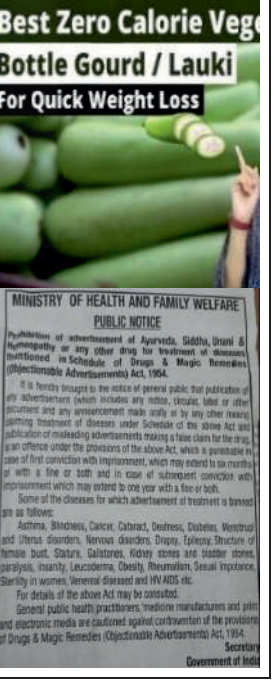
.

Top CSIR Scientist Dies After Drinking Lauki-Karela Juice

Lauki (bottle gourd) and Karela (bitter gourd) juice proved lethal for a 59-year-old diabetic Sushil Kumar Saxena, Deputy Secretary in Council for Scientific and Industrial Research, who died in Rockland Hospital after vomitting blood, reports

Consumption of bottle gourd juice unsafe,warns Delhi govt

Over a year after a doctor from Delhi succumbed to lauki juice poisoning ,the constituted by the Health Ministry



- Evidence Based Medicine→Drug/Vacc→C

. All medicines have side effects- Must be declared to patient

. Absence of evidence ≠ Evidence of absence -F7/23

G. TYPES OF INFECTIONS (INF)

1. Focal inf and Generalized/systemic/disseminated/invasive
2. Recurrent=Repetitive inf & Reactivated inf=Latent→Active
3. Resistant/Refractory inf- Difficult Rx & Relapse=Return
4. Superinfection/Coinfection- Two infec together
5. Primary inf= 1st inf and Secondary inf= Post primary inf
6. Acute bacterial inf- ↑ ↑
7. Acute viral inf- ↑ ↑
8. Acute parasitic inf- ↑ ↑
9. Chronic/ persistent- ↑ ↑
10. Cured= Treated inf- ↑ ↑
11. inf= Common =
12. Atypical inf= uncommon
13. Autoinfection -INI25 Pinworm, tapeworms
14. Inf- Endocarditis, hydatid cyst, Pneumonia
15. Active inf (TB, Chickenpox)- Symptomatic+
16. Subclinical/ state (Covid)- Asymp +
17. Latent inf (TB, Chickenpox)- Asymp +

18. inf = Suppurative inf- Pathogen→ pus (TB)

. Rx-

19. inf- Pathogens→ toxin (Tetanus, Diphtheria)

. Rx-

20. inf- Own immunity damages (RHD, PSGN)

. Rx-

21. Autoimmune dis-F26/24,N18/19 Rheumatoid arthritis(MC),SLE

22. D =Accidental inf (No human→human transm)

. HCLJT- Hydatid cyst, Cysticercosis, Leptospira,

. No herd immunity -F1/24/25

. Herd/Social/Population immunity immunized

23. O =Chance inf- Immunocompromised (HIV)

24. O inf in doctors- HIV, Hep B/C, TB, Covid

25. inf spreads from patient to patient -N20

. MC-

26. Community acquired inf- e.g.

27. Iatrogenic/ Nosocomial/Hosp/Healthcare assoc inf (≥48hr)

a. (MC)- Catheter(Foley) assoc urinary tract inf

b. - Ventilator associated pneumonia- MC P

c. - Central Line assoc Blood Stream Inf

d. - Surgical site inf -N20 MC delayed wound healing

. MC surgical prophylaxis -F6/19 1st gen-

e. Needle Stick Injury (Occupational infec) → Stat PEP asap

. Risk HIV 0.3%, Hepatitis C- 3%, Hepatitis B- 30-50%

. Wash → No squeeze/ suck. Dont recap/ burn/ cut needle

H. EVOLUTION AND SPREAD OF INFECTIOUS DISEASES

1. Sporadic inf- Infrequent inf→ Tetanus, Rabies -N22

2. Outbreak/ Cluster ~ Limited epidemic- Nipah

3. Epidemic - ↑↑ patients→ Dengue, Ebola, Zika, SARS

4. Pandemic = Worldwide/multicountry epidemics-

. Covid-19, Influenza 2009, Cholera, Plague (Black death)

5. Endemic - Infec occurs regularly-

6. Endemic+ Sporadic+ Epidemic+ Pandemic

7. Syndemic= Synergistic epidemic→ HIV + Drug abuse -N20

8. -N20 Public Health Emergency of International

H1N1 2009, Zika 2016, Covid 2020, Ebola 2014/18, Mpox 2022/24

I. INFECTION CONTROL

1. Hygiene- Spore, naked virus not killed

a. Handwash- Soap+water- Spore, naked virus washed away

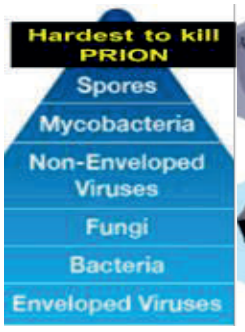
b. Handrub- 70% ethanol = ethyl alcohol- Spore on hands

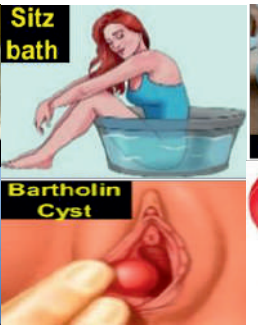
c. Moments → Steps → Steps before surgery

d. MC nosocomial transmission

e. Most important nosocomial prevention

1. precautions- For all healthcare scenarios . PPE-N19 6 items- Surg Mask/glove/gown, head/eye/foot wear . PPE 3 types- Standard, Full (N95- Covid, Ebola), enhanced 2. precaution- Old concept for HIV, Hep B/C	
1. -F19,20 For diagnosed (Covid RTPCR positive) . Isolation ward/home/self- 2. -N24 For exposed not yet diagnosed positive	
1. - Worldwide . India- Guineaworm 2000 Stepwell, World- Polio except . Incidence(New case)/Prevalence(Total)/Transmission- 2. -N18-Prevalence <1/10000 (Leprosy/Kalaazar) . India-Leprosy 2005, Mother-Child Tetanus 2015, Trachoma	
Antimicrobial stewardship- Right antibiotic/ patient/ time/ dose/ route/ duration (↑patient safety/outcome ↓resistance)	
J. MICROBIAL TOXINS -F7/19/24,8/20, N19 1. Neurotoxin- Cobra, Krait (Silent killer), scorpion a. T b. 2. Toxin ↑cAMP- Watery=secretory diarrhea- Rx CER 3. Toxin ↓protein synth- DPS- Diphtheria, Pseudomonas, Shi a. Diphtheria ↓60s ribosome – ADP ribosylation of	





K. STERILIZATION 100% kills all microbes + all spores

Most potent (γ ray, H_2O_2), Plasma Steriliz, Glutaral/ Formaldehyde

M common (Ethylene oxide gas), Hot air oven, Hypochlorite, Autoclave

- Disinfection- Spore no kill, Decontamination= Cleaning
- Antiseptic for skin/mucosa (H_2O_2, povidone iodine, alcohol)

1. Dry Heat (Max penetrating power) -F6/22 For dry item

. 160^0C x 2 hr- Operation theatre= OT

. Glass, cotton, gauze, plaster, surg eqpt, wax, oil, grease

. Quality Control- Browne tubes, Bacillus subtilis spores

2a. Moist Heat < 100^0C

. 85^0Cx 20mx 3day- LJ medium (TB)

. Flash (Milk,wine)- 72^0C for 15sec→ 13^0C

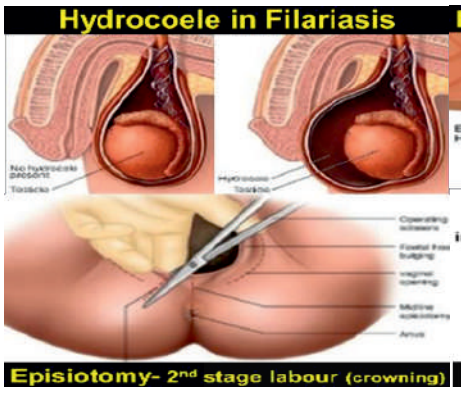
. Pateurization QC-

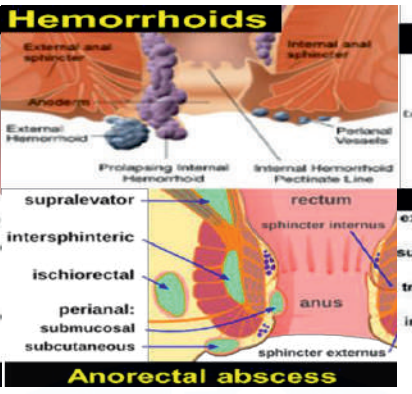
2b. Moist Heat = 100^0C- Tyndallization 100^0Cx20mx3d- Sugar

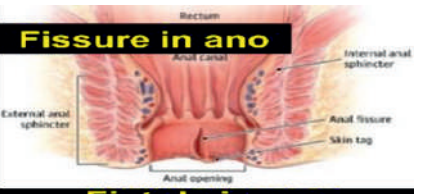
2c. Moist Heat > 100^0C -F18,22

. 121^0Cx 20 min @1atm/1bar/15psi/760 mmHg

. QC- Autoclave tape, Geobacillus sterothermophilus spore







3. Chlorination -N20 Water/swim pool/well/ tank

a. Water chlorinated by =$CaOCl_2$

b. Biolog chlorine demand- starch iodine

c. End user free residual chlorine test by

. Test/ Orthotoluidine arsenite test

d. Cl ppm after 1hr- Normal Disaster Swimpool

e. Most imp water contaminant- test

4. Povidone iodine=Betadine brown 5-10%- Wound, Sitz bath

5. Formaldehyde gas 40%-

6. 2%Glutaraldehyde=Cidex&Orthopthaldehyde-F1/21

7. 1% silver sulfadiazine/ nitrate-

. MC infec in burns-

8. Copper T- Life 10 yr- Spermicidal/ bactericidal -F1/20/21

. Best emergency contraception- >99% effective

. MC infec in Copper T-

9. Soap (↓surface tension)- Kill HIV, Hep B/C,Covid,influenza

10. HEPA= High efficiency particulate air filter- 0.3µm

11. 70% ethanol= ethyl alcohol- Hand rub, antiseptic, wound

12. Isopropanol- Thermometer, stethoscope

13. Phenol, cresol, lysol- Mopping (Phenol standard for test)

14. Dettol- Chloroxylenol, Savlon- Chlorhexidine + cetrimide

. MC bacteria growing in savlon-

15. Chlorhexidine- Mouthwash, contact

. MC contact lens keratitis= corneal ulcer-

16. Non-ionizing radiation- Ultraviolet C/ Infrared -F1/24 OT

17. Ionizing radiat (X/γ-ray)-Catgut suture, contraceptives -F21

(Most) .Radiosensitive- Bone marrow .Radioresistant- Brain

18. Ethylene oxide gas- Plastic, rubber, Heart lung machine, respirator QC- Magenta dye strip, Bacillus subtilis

19. H_2O_2 35-50% (White fluid- bubbles)-

20. Plasma sterilization- Arthroscope, urethroscope -F1/23,N18

21. Peracetic acid- Hemodialyzer and Ozone- Lab equipment

1. - Sterilization efficacy- Phenol

2. Award- PHC/Govt - Inf control/BMW/Quality

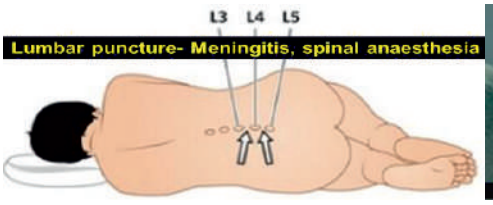
3. Strict Infec Control (Restricted entry)- OT, NICU, Cardiac Cath lab, labour room, burns/ dialysis/ transplant/ cancer/ isolation ward

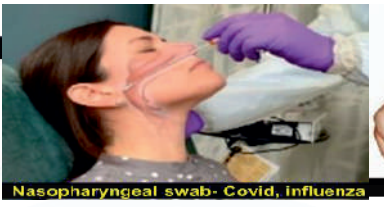
4. CSSD/TSSU- Central Sterile Stores Dept/ OT instrum sterilization

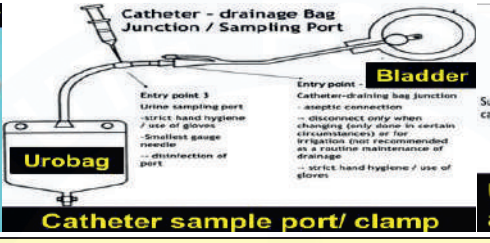
L. DIAGNOSIS OF INFECTION

Sample→ Test→ Report

Sample, Microscopy, Bacterial structure (Capsule, Cell Wall, stains, Flagella), Growth Curve, Culture Media (~100% specific), PCR, Genetics, Antimicrobial Resistance, Toxins







- **Sample**- Store CSF@ room temp, urine@ 2-8^{0}C/ boric acid
- **Microscope** **Light=Compound M** **Electron M**

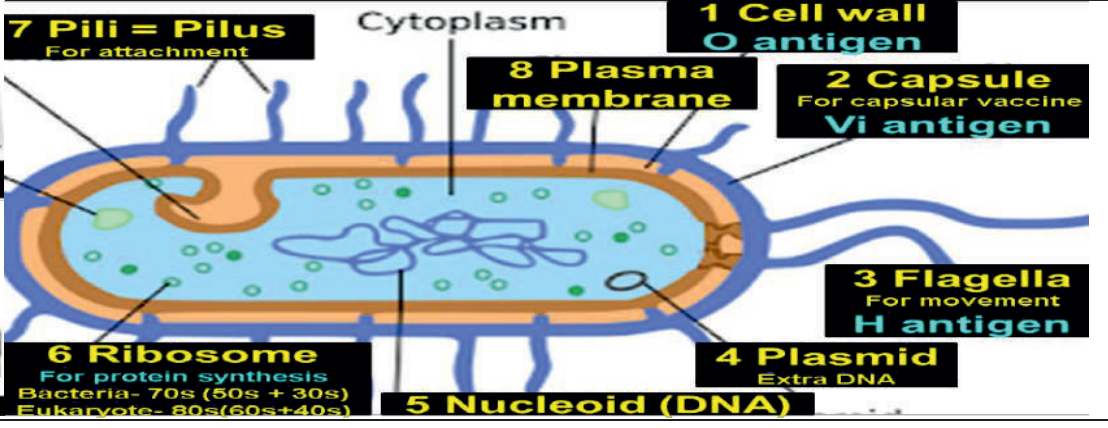
.Magnification/Resolution 1500times/0.2µm 100000x/0.5nm

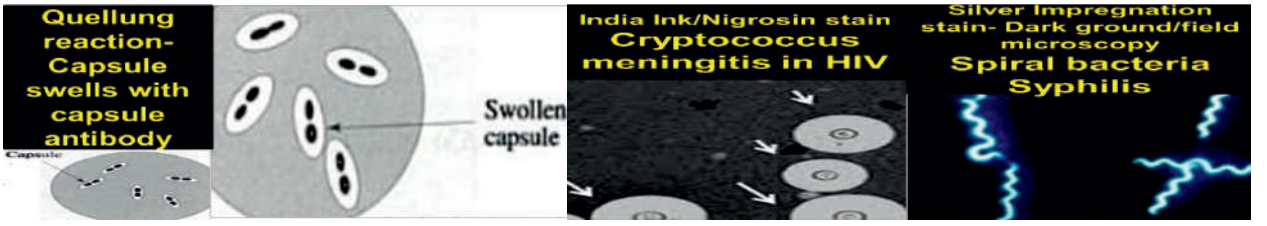
- **Bacteria-** Capsule, cell wall, memb, flagella, pili, plasmid, nucleoid
- **Capsule + capsule antibody**-F21/24,19,N21→

. → 3 capsulated bacteria- NHS

. Neisseria meningitides, Hemophilus influenzae, Strep pne

. Vacc made from capsule→ <2wk/ >1wk of



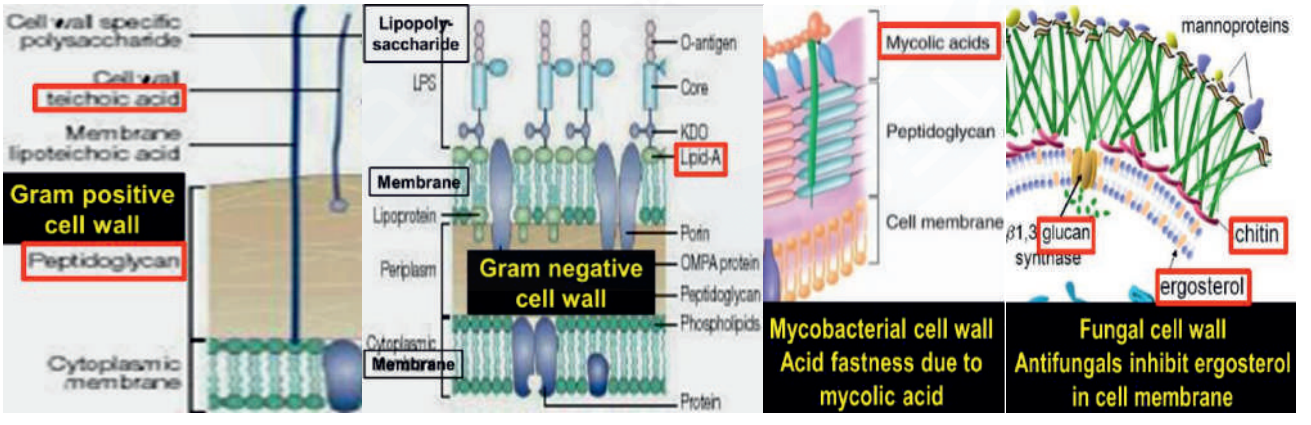


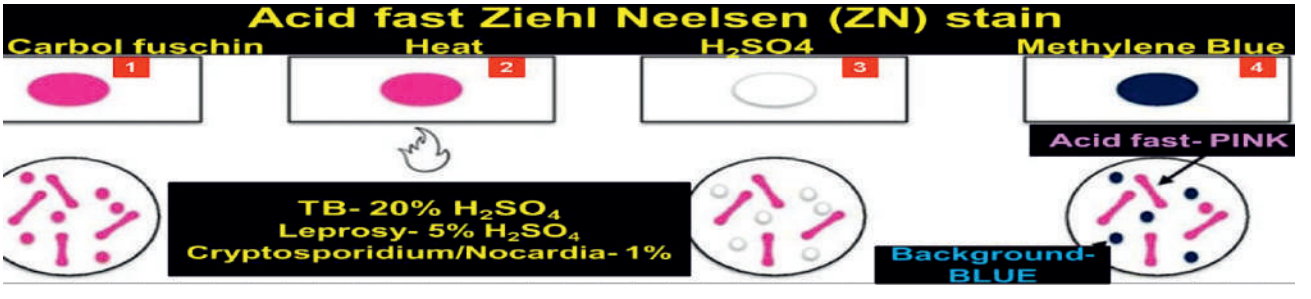
- **Capsule unstained** -F1/23/25 **Negative background stain**

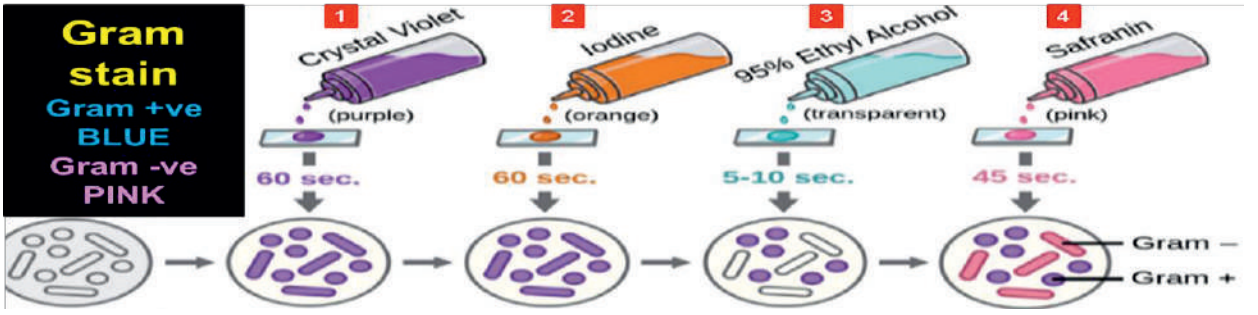
. Nigrosin=

- **Cell wall = cw (Seen in only bacteria, fungi)**

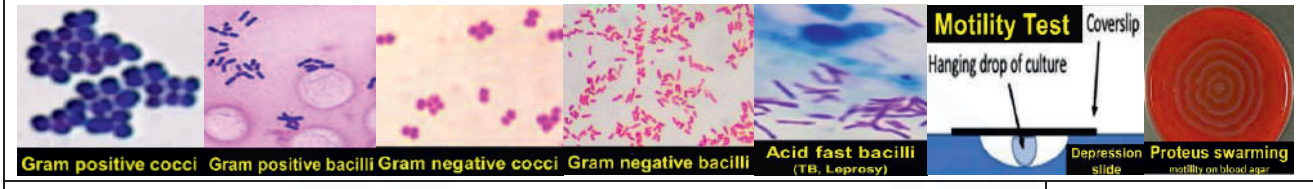
1. **Bacteria never having cw**
2. **Gram +ve cw-** **+ Teichoic acid** -F6/22
3. **Gram -ve cw (L)- Lipopolysaccharide+ Lipid A = Endotoxin**
4. **Mycobacteria cw-** **-Reason for acid fast**
5. **Fungal cw- Chitin+glucan**-F6/22 **Fung cell memb- Ergosterol**
6. **L= Lister form- Grow without cell wall in penicillin media**







1. Gram positive- Blue coccus 2. Gram negative- Pink ella



- Flagella→ Motile. Nonmotile→ Cocci, Shigella, Klebsiella

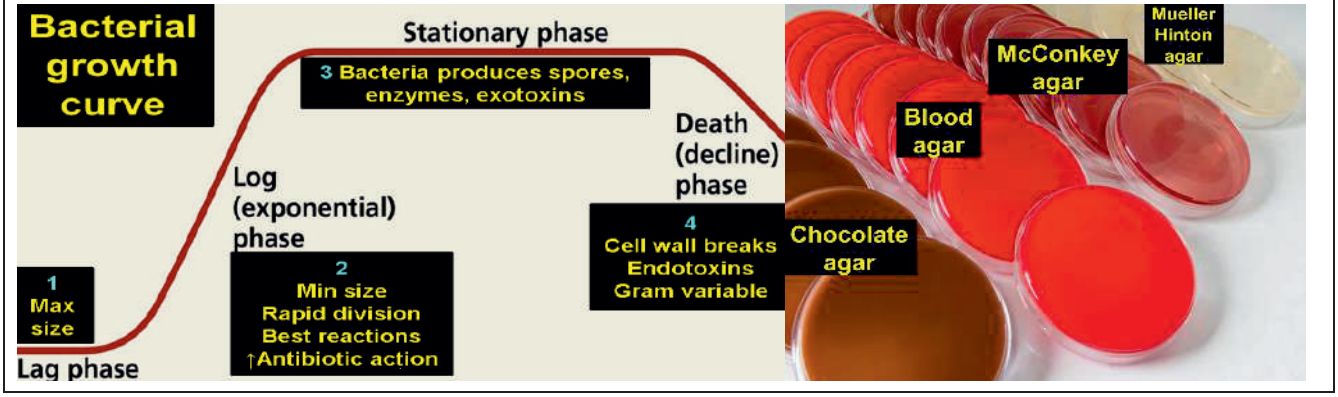
1. Darting= Shooting star-F21/22/23/24
2. Tumbling
4. Gliding
5. Twitching/ Jerky
6. Falling Leaf
7. Corkscrew
8. Swarming

→ Put chloral hydrate- Stops

- Tough life- Bacteria (Bacillus, Clostridium)

. Fungal Reproduction . Parasite Tough life

- E. coli 20min, TB 15hr, Leprosy 15days



- **Solid = Agar culture**- Bacteria grown on agar surface

. Agar→ Red algae (Gelidium)→ No food/color/taste/smell

1. Yellow — Test antibiotics
2. Pink-F1/22/23 — Differential agar
3. Red (RBC) 5% sheep — Enriched agar

Red -F1/23 Hematochezia (Fresh blood stool), Hematuria

4. Cola colour-

Brown- Malena (Dark tarry stool), Hemoglobinuria

- **Selective Media-** Allows growth of 1 bacteria -F22/24/25

1. Salt milk/ Mannitol salt/ Ludlams agar
2. Crystal violet blood agar
3. Potassium tellurite agar/ Loefflers serum slope
4. Lowenstein Jensen agar/ MGIT Midlebrook 7H9
5. Cetrimide agar — Savlon has cetrimide
6. Alkaline bile salt/ Monsurs GTTA/ TCBS
7. DCA, XLD — 8. Wilson Blair

- **Liquid=Broth culture→ Blood culture(Brain Heart Infusion)**

. -F24

- **Molecular Tests (PCR/RTPCR)** -F21/22/23/24, N18

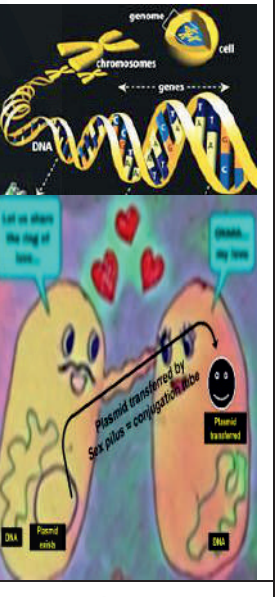
1. R DNA→DNA
2. T DNA→RNA
3. Reverse Transcription- RNA→DNA
4. T RNA→Proteins Multiple enzymes

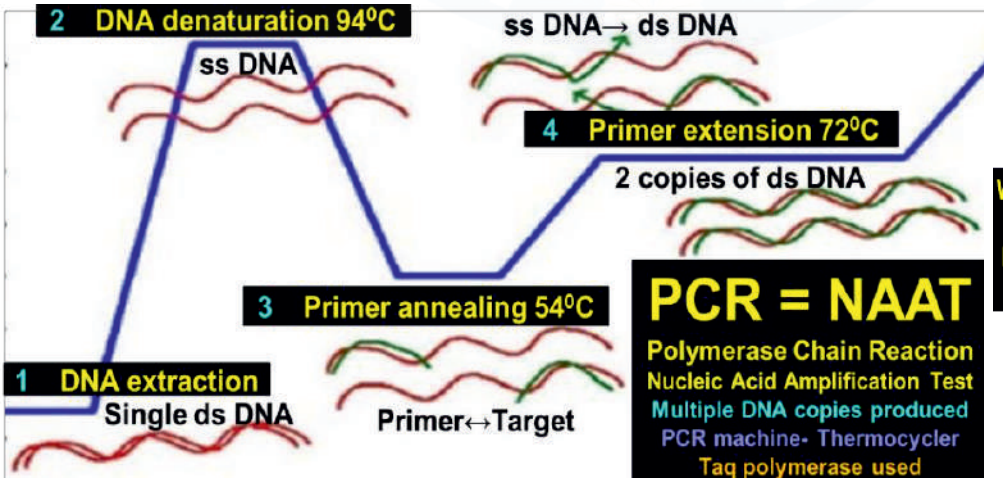
- **Central dogma of molecular biology**

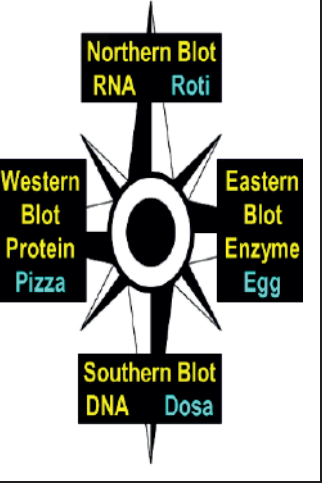
= DNA→ RNA→ Protein - All except RNA viruses and prions

- **4 types of PCR= Polymerase Chain Reaction= NAAT**

1. DNA- PCR→ All bacteria, fungi, parasites, DNA viruses
2. RNA- RTPCR= Reverse Transcriptase PCR→ All
3. Quantitative Real Time PCR= qPCR (PCR or RTPCR)→ HIV, Hep B/C viral load
4. Multiplex PCR=mPCR -N24/25- Unknown pathogens-Mening







- **Bacterial Genetics**

1. = ♀♂ change after gender reassignment Rx
2. = Cross dresser= Boy wears girls cloth
3. = Bacteria wears other bacteria DNA
4. = Jumping gene=
5. = Chromosome breaks/attaches to other

a. t (8;14)- cmyc gene-

b. t (9;22)= Philadelphia chromosome→BCR-ABL gene-

6. =Sexduction- Sex pilus=Conjugation tube

. Plasmid transfer-F23/24 resistance gene

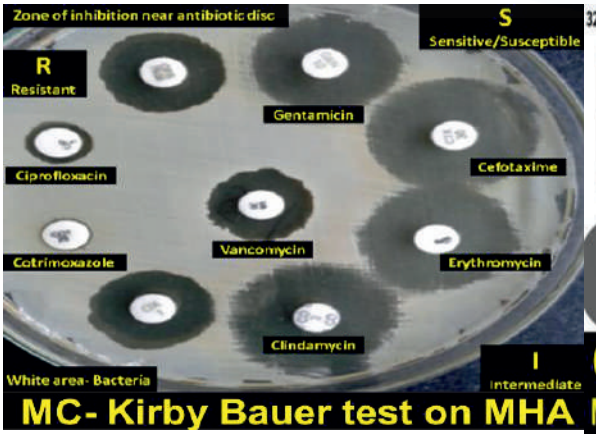
7. =Seduction -F21,N18 Lysogenic/ Lytic cycle

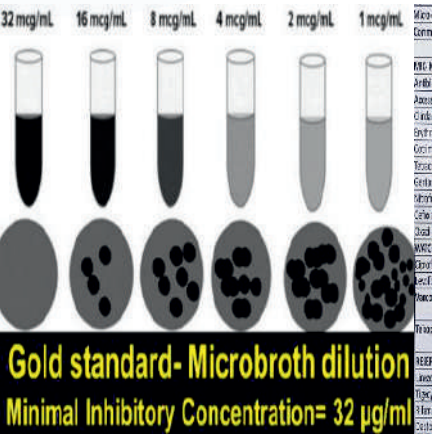
. =virus flirts/kills bacteria

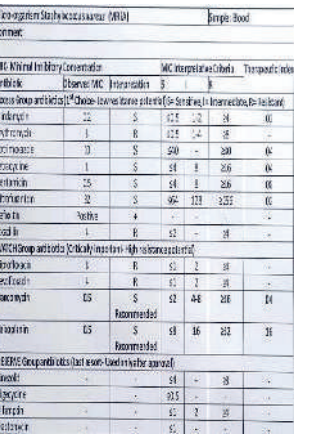
8. Transition- Purine↔Purine,Transversion- Purine↔Pyrimidine

- **Antimicrobial Resistance and Rx** -N19/20/21

1. Rx- known bacteria
2. Rx- unknown bacteria







Section II : Bacteriology

60 bacteria, 32 pages, 130 PYQs

1. Gram +ve cocci (6)- Staphylococcus (S. aureus, S. epidermidis), Streptococcus (S. pneumoniae, S. pyogenes, S. agalactiae), Enterococcus, Abscess, Food Poisoning, Osteomyelitis, Endocarditis, Pneumonia, Epiglottitis, Meningitis, Encephalitis

2. Gram positive bacilli (11)- Tetanus, Botulism, Listeria, Gas gangrene, Clostridium difficile diarrhea, Anthrax, Mycetoma (Madura foot), Diphtheria, Acne, Tuberculosis, Leprosy

3. Gram negative cocci (7)- Bordetella pertussis, Brucella, Legionella, Hemophilus (H. influenzae, H. ducreyi), Neisseria (N. meningitides, N. gonorrheae), Sexually Transmitted Infections (STI), Bacterial vaginosis, Trichomonas, Urinary Tract Infection (UTI)

4. Gram negative bacilli (14)- Escherichia coli, Klebsiella (K. pneumoniae, K. ozaenae, K. rhinoscleromatis, K. granulomatis), Plague, Shigella, Non typhoidal Salmonella, Typhoid, Helicobacter, Campylobacter, Vibrio (V. cholerae, V. parahemolyticus), Pseudomonas

5. Spirochetes (4)- Non venereal Treponema, Syphilis, Borrelliella, Leptospira- Weils dis

6. Misc(6)-Chlamydia (Trachoma, STI, LGV), Rickettsia (Typhus), Scrub typhus, Mycoplasma

GRAM POSITIVE COCCI (Round bacteria)

Staphylococcus (S. aureus, S. epidermidis)
Streptococcus (S. pneumoniae, S. pyogenes, S. agalactiae), Enterococcus

A. ABSCESS = Pus filled lesion -F 20/21/7-25

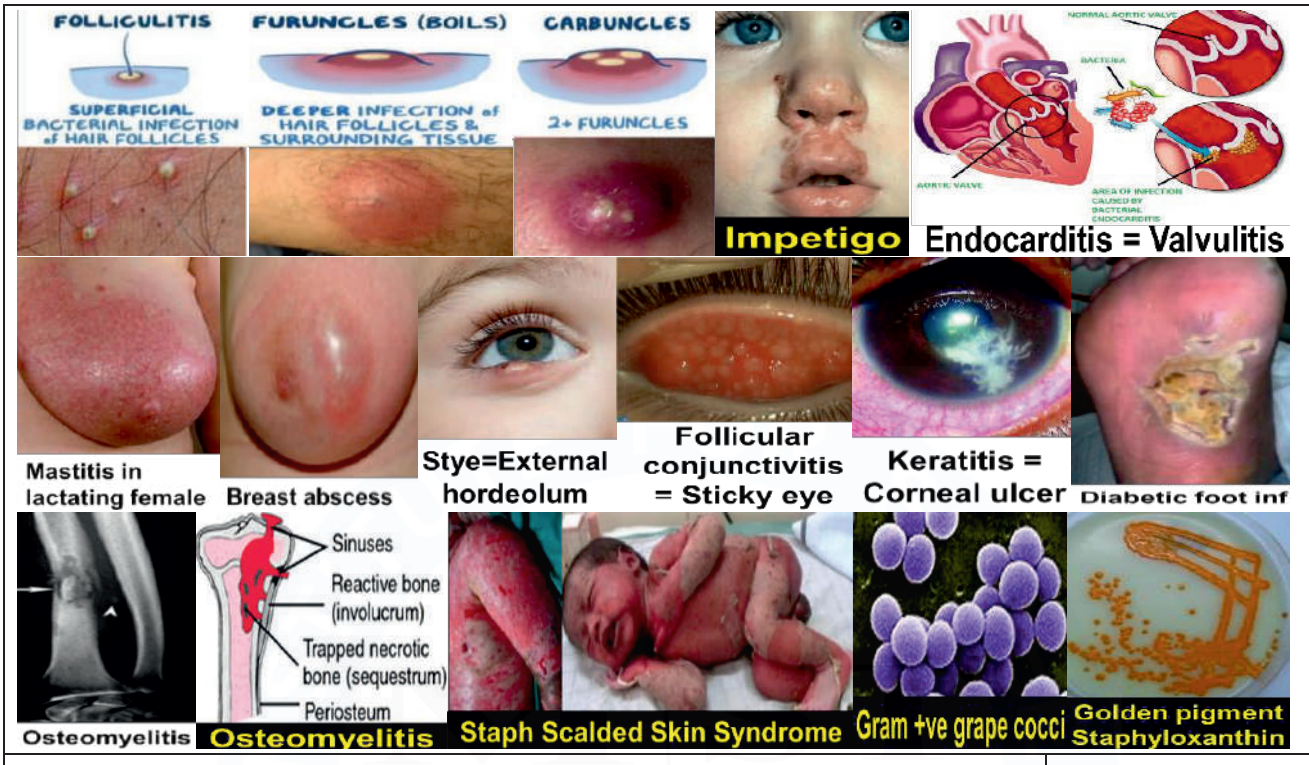
- MC overall
- MC abdomen/ perineum MC appendicitis
- MC lung

. Rx- USG-CT guided/ unguided incision and drainage -F7/25

B. STAPHYLOCOCCUS AUREUS Catalase + coagulase +

. 2 types of infections- Pyogenic and toxigenic -F1/7/24/25,22

- Pyogenic Inf = Abscess Staph aureus is MC cause of skin, breast, bone, joint, eye, heart, lymph node, surg wound, bloodstream, CABG, skin graft, hemodialysis infections



- **Toxigenic Inf** (3 toxins→ 3 infections)

1. **Staph enterotoxin** -F7/25/22,N18/20/21

2. **Staph Toxic Shock Syn Toxin**= STSST- Vaginal tampon-F7/25

3. **Staph Scalded Skin Syn Toxin**= SSSST= Exfoliatin -F7/25

N -F1/7/22/24,N18/20/21

- **Resistance in Staph aureus**

1. No resistance→ = Methicillin sensitive S. aureus-F6/19

2. = Methicillin resistant S. aureus → gene in bacterial DNA →

3. = Vancomycin resistant S. aureus → gene in plasmid →

. MC antibiotic resis transfer in S. aureus- Conjugation/

- **Rx of Staph aureus** -F23,24,1/25,N19

1. MSSA infec (e.g. Mastitis)
2. MRSA DoC & Empirical Rx
3. MRSA Carrier (Doctors) -F6/19
4. VRSA
5. Staph Toxic Shock/ Scalded Skin syndrome (Both lethal)

A

6. Food poisoning

C. FOOD POISONING - 3 types (<6h, 8-16h, >16h) -F1/7/25,21/22

- **<6 hr**- Preformed heat resistant toxin-

Stomach- Gastritis- Vomiting→ No antibiotic, supportive Rx

1. **Staph aureus**-F7/25,N20 Enterotoxin-

Sandwich, burger, cake, mayo, salad, dairy, poultry (momos)

2. **Bacillus cereus**-F7/25 Cereulide toxin-
3. **Clostridium botulinum**-F1/26 Botox-

- **8-16 hr** Small intestine- Enteritis- Diarrhea- No antibiotic

1. Bacillus cereus- Diarrheagenic toxin
2. Clostridium perfringens- α toxin

- **>16 hr**-F1/25 Small+Large- Enterocolitis-Diarrhea+Dysentery

1. Diarrhoea- Cholera, E. coli, Rotavirus NSP_4

.Rx -N23/24

2. Diarrhea- E. coli EHEC, Campylobacter

.Rx- A

3. Dysentery (↓stool ↑blood ↑mucus ↑pain ↑tenesmus)

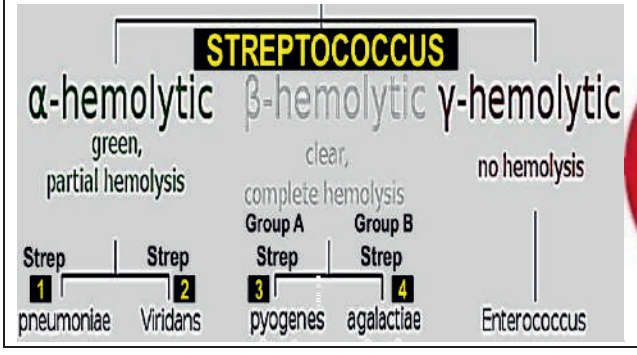
.Rx- Shigella- Ciprofloxacin, Entamoeba-N19-M

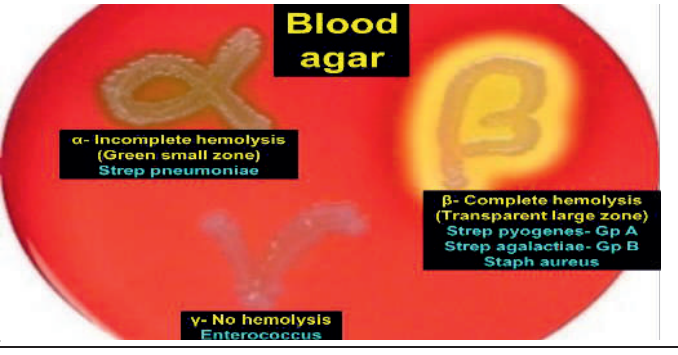
D. OSTEOMYELITIS .World Sickle cell capital

1. MC osteomyelitis/Brodie abscess/septic arthritis-F7/25
2. MC overall bone-F6/20,N18/22
3. MC site in long bone → Hair
4. MC in sickle cell disease
5. Dead necrotic bone =
6. New reactive bone =

E. STAPH EPIDERMIDIS Catalase +ve Coagulase -ve

1. Commensal in -F7/23/24
2. Makes → surface infec -F1/25
3. MC Stitch / Suture
4. MC CSF / shunt
5. MC catheter, implant, prosthetic joint/valve/ dialysis infec





F. ENDOCARDITIS = VALVULITIS (3 sets of blood culture)

1. C/F- Fever + new murmur/ emboli -N22 2. MC valve

3. Empirical Rx

4. criteria -N19 . Splinter/conjunctival hemorrhage

. Janeways lesion . Oslers nodes . Roth spots in retina -N18

5. MC overall/ native valve endocarditis

6. MC iv catheter/drug abuse endocarditis

Right/Tricuspid- Left/Bicuspid- Enterococcus

7. MC Prosthetic valve endocarditis

<12m- Staph epidermidis >12m-Strep viridans

8. MC dental extraction- Strep viridans/sanguinis -F21/23/24/25

9. MC endocarditis in child with CHD- Strep

G. STREPTOCOCCUS PNEUMONIAE = Pneumococcus

1. Gram +ve capsulated diplococci→ Quellung rxn -F1/24,IN25

2. Rx

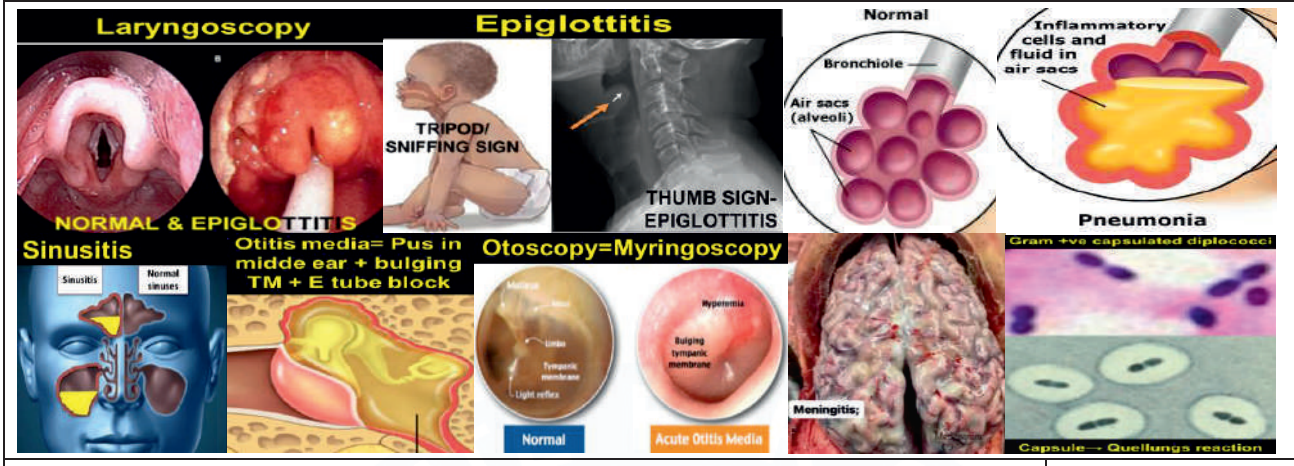
3. Vaccine made from (PPSV 23, PCV 13)

. Vacc in -F7/24, implant -INI25

4. Strep pneumoniae is the MC cause of EPSOM

.

H. EPIGLOTTITIS -F2/22 5 Ds- Drooling saliva, Dysphagia, Distress (resp), Don't examine, Dysphonia-Hot



I. PNEUMONIA (LRTI) -F7/25 Infec→ water/ pus filled alveoli

. World pneumonia=pn capital/max infant-child death

1. Type 1- Typical = Lobar = Alveolar pneumonia -F1/26,N20/21/24

a. One lung lobe full of pus- consolidation/opacity, ↑dyspnea

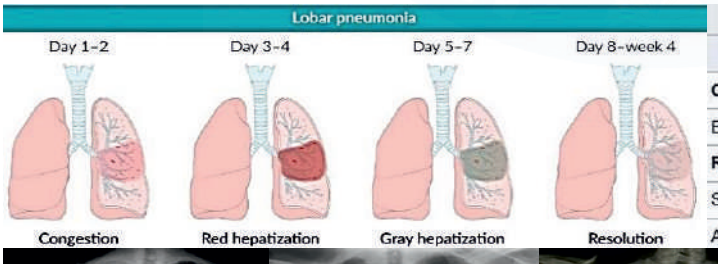
b. Empirical Rx- Ceftriaxone + Azithromycin ± Vancomycin

c. MC lobar pn -Rusty sputum=sp

d. MC pn alcoholic/diab -Red currant jelly sp

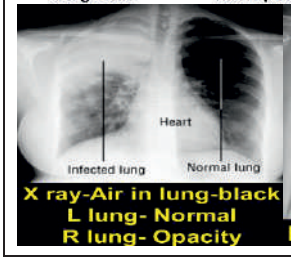
e. MC broncho pn (B/l basilar multifocal pn)- Staph aureus

f. MC lung abscess-N24 (Air-fluid level)

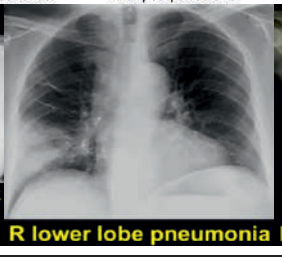


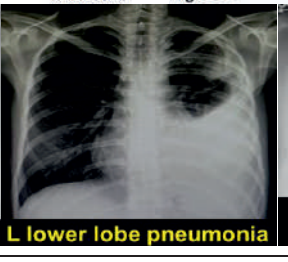
CURB-65	
Symptom	Points
Confusion	1
BUN>7 mmol/l	1
Respiratory rate≥30	1
SBP<90mmHg, DBP≤60mmHg	1
Age≥65	1

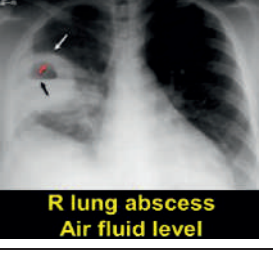
SCORE 0-1 = OUTPATIENT
SCORE 2 = INPATIENT
SCORE 3+ = ICU



X ray-Air in lung-black
L lung- Normal
R lung- Opacity


R lower lobe pneumonia


L lower lobe pneumonia


R lung abscess
Air fluid level

2. Type 2- Atypical = Walking = Interstitial pneumonia -F1/24

a. No pus in alveoli, ↓dyspnea

b. MC atypical pn

c. MC fungal pn in HIV- Dry cough

d. MC Loeffler pn e. MC viral pn

3. MC pn overall/child/elderly/COPD/smoker/community

a. MC pn in hosp/ VAP/ cystic fibrosis-N21

b. MC neonatal pn/ sepsis/ meningitis E. coli India > S. ag

c. MC bronchitis (cough)

d. MC bronchiolitis (small airway infec + wheeze)

J. MENINGITIS -F7/25 3 layers- Dura, pia, arachnoid mater

1. C/F- Triad FSH (Fever+ stiff neck/nuchal rigidity+ headach + vomit + K -N18/19

2. MC overall/ >20yr/skull base fracture/CSF leak

a. MC neonate/0-2 mnth E. coli India > S. agalactiae=GBS

b. MC 2-20 yrs

c. MC CSF shunt -F1/24

d. MC HIV (CD_4<100) neoformans

e. MC after neurosurgery

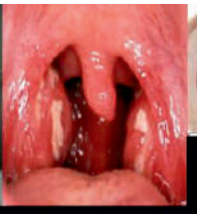
K. ENCEPHALITIS - FSH + confusion + seizure

a. MC India- b. MC West-N23

c. Most lethal (100%)

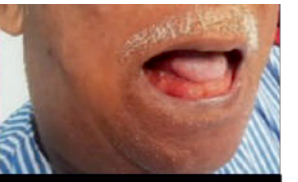
Cellulitis

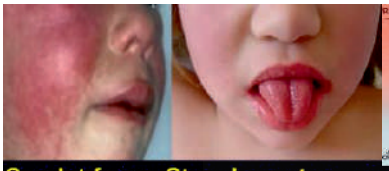
Erysipelas

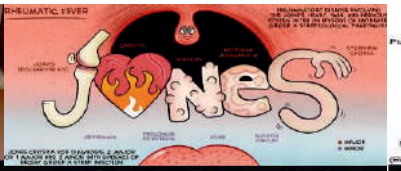

Tonsillitis

Quinsy=
Peritonsillitis
Hot potato voice
Deviated uvula

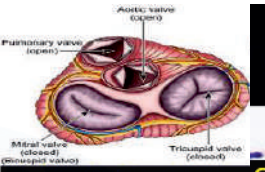
Pharyngitis

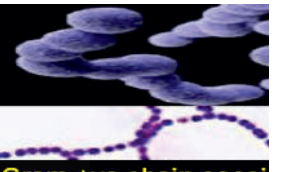

Ludwig angina
Double tongue sign


Scarlet fever- Strawberry tongue



Jones Criteria- Rheumatic fever


Heart-4 valves


Gram +ve chain cocci

1. Bacterial= Pyogenic meningitis -F7/23/25/26 (PNPG)

↑Pressure- ↑Protein- ↑Neutrophils=PMN cells- ↓

. Empirical Rx -F1/7/25/24/26,N19

2. TB meningitis -F1/24/26,N22 (LPC)- Rx

↑Protein- ↑Lymphocyte= Mononuclear cell- C

3. Viral= Aseptic meningitis -F1/21/26- Empirical Rx

↑Protein- ↑Lymphocyte-

L. STREPTOCOCCUS PYOGENES = Gp A β-hemolytic Strep

. 3 types of infec- Pyogenic, Toxigenic, Immunogenic -F1/24,7/23

1. Pyogenic Inf- Strep pyogenes is MC cause of -F6/19/20, 7/24

.

2. Toxigenic Inf- Strep pyrogenic = erythrogenic toxin

a. Necrotising fasciitis (MC) -F21 F

b. Scarlet fever- Rash + Straw

3. Immunogenic Inf- Molecular mimicry→ Shared antigen

. Strep antigen similar to heart/ kidney/ skin antigens

. 3 immun inf- Rheumatic heart dis, PSGN, Guttate psoriasis

1. Rheumatic fever = = =

Rheumatic Heart Dis (RHD)→ MC ♀ 5-15yr -F23/24/25,N19/22/24

a. Modified (5 Major + Minor)

b. Diagnosis of RHD→ 2 major or (1 major + 2 minor) criteria

+ Strep pyogenes in throat culture or ↑ASO antibody titre

c. Minor criter- Fever, arthralgia ↑WBC/ESR/CRP ↑PR interval

d. 5 Major = Maj Criteria

MC

MC

S

E

S (Most specific)

2. Post Strep Glomerulonephritis= -F19/20/21/22/23

3. Guttate psoriasis- sign negative -F19/21/22/23/24,N22

- Rx of Strep pyogenes = GABHS = GAS

1. Pyogenic inf

2. Toxigenic inf

3. Immunogen inf- Pharyngitis→ Rheumatic Fever, PSGN

.

4. RHD prophylaxis (5 yr/ 10 yr/ Lifetime)

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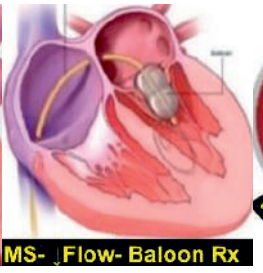
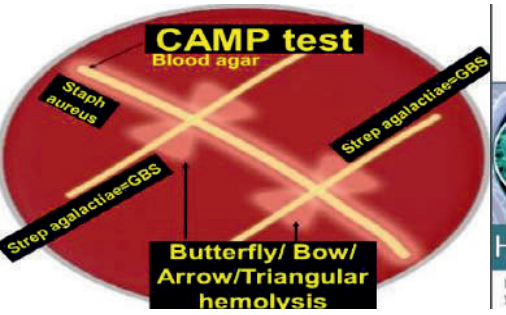




- **Heart valves damaged in RHD→ MC MR > AR > MS > AS**

1. MC MS/MR . MS more severe than MR in preg
2. Valve ↑ diameter ↑ blood back flow
3. Valve ↓ diameter ↓ blood forward flow

M. STREP AGALACTIAE Gp B β-hemolytic Strep

1. Lives in vagina -F19/21/23
2. S. agalactiae- MC Neonatal
3. India and preterm baby- E. coli/ Klebsiella > S. agalactiae

N. ENTEROCOCCUS UTI, abdominal infec

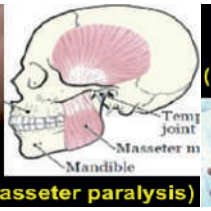
. VRSA→ van gene → Vancomycin resistant = VRE

GRAM POSITIVE BACILLI (Long bacteria)

Acne, Tetanus, Botulism, Gas gangrene, C. difficile, Anthrax, Diphtheria, Listeria, Mycetoma

A. CUTIBACTERIUM ACNES (Acne) -F6/22,1/25- Propionibacterium

1. Whitehead closed , blackhead open
2. Hair follicle + sebaceous gland infec (pilosebaceous inf)
3. Rx



B. CLOSTRIDIUM TETANI → T — Lives in soil

1. Neurotoxin- ↓GABA ↓Glycine

2. Upper

3. C/F-

. Maternal-Neonatal Tetanus- day diseas- Eliminated 2015

3. Rx + Penicillin + Diazepam- Muscle relax

4. Vacc Valid for 5 yr -N23/24

C. CLOSTRIDIUM BOTULINUM Spore→ B

1. Neurotoxin- Bioterrorism category → 5 Ds

Paralysis, Dysphagia, Dyspnea,

Diplopia (double vision), Dysarthria (difficult articulation)

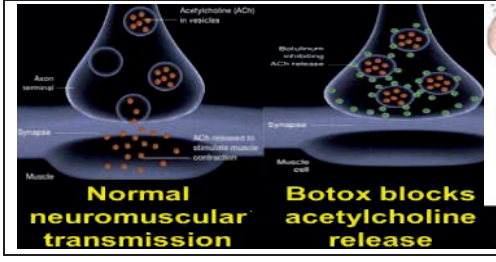
2. Botox = Wonder drug -F1/25/26,N22 >100 uses- Cos

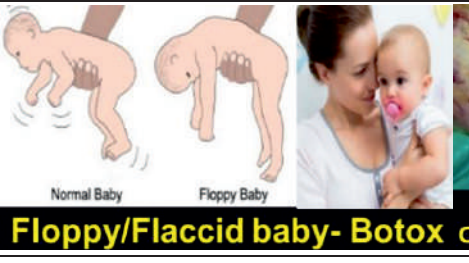
squint -N19/20, migraine, torticollis,

fissure, vaginismus, erectile dysfunc, premature ejaculation

3. Botox ↓acetylcholine from -F1/26 Dose 3-6 mnthly

4. Canned food poisoning, Pacifier-





D. CLOSTRIDIUM PERFRINGENS=WELSCHII (G

1. Road accident- Spore- α toxin=Lecithinase=Phospholipase

2. Gas in tissue ( ) + Muscle gangrene (

+ Necrotising fasciitis + Compartment Syndrome→ Death

a. N- Naeglers= α toxin= Lecithinase= Phospholipase test

b. T- Target = -F 1/7/23/24,N18

c. R- Reverse

3. Rx + Penicillin

+ Fas +Myo + Hyperbaric O_2

E. ANTIBIOTIC ASSOCIATED DIARRHEA -F1/22,N18,22

1. Prolonged antibiotics→ gut commensal

killed→ Toxins→ Diarrhea +

Pseudomembranous enterocolitis

2. Rx

+ Good bacteria→ MC

+ Fe microbiota transplant (FMT)- capsules

F. BACILLUS ANTHRACIS →Anthrax Zoonosis→ Bioterror

1. Polypeptide/glutamate capsule→

2. Lung anthrax= Wool sorters dis= H

3. Skin anthrax= Hide porters dis= M

Eschar-F21/23/24/25/26,I25

4. Colony- Medusa head, Christmas/ inverted fir tree -N18/19/20

G. CORYNEBACTERIUM DIPHTHERIAE -F19/20/21/22/24,N19

1. Human reservoir → Transmission

2. Biotypes- Gravis (Severe), intermedius,

3. Diphtheria= neck- Cervical

a. Fever + True/ P in throat -F20/21/22

b. Diphtheritic croup= Lar

-F6/22 Barking + S

4. Gram +ve bacilli-

a. Albert stain- Polar/ Bipolar/ Energy/ ATP (polyphosphate)/ Metachromatic/ Volutin/

b. Enriched media -F8/20

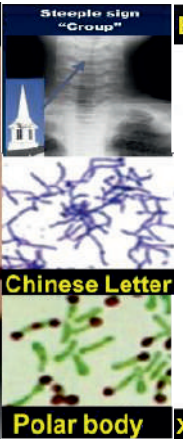
5. Diphtheria toxin-F7/24 ↓

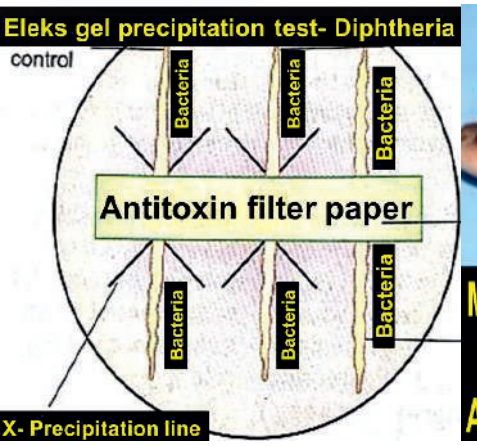
→ Damages ribosome→ ADP ribosylation of EF_2

6. Schick test (old) and Eleks

7. Rx + Penicillin/ Azithromycin

8. Vacc-N19 - Park William 8 strain





H. MYCETOMA = Maduramycosis = Madura Foot→ Madurai

1. Bacterial = > Fungal= Eumycetoma

2. SSP triad-F1/26- Swelling+ Sinus+ Pus

3. Rx- Amputation (30%) + Cotrimoxazole +

a. Actinomyces -F22,N18/22 Cotton-wool ball sunray fungus

b. Nocardia -F21/22,N22/25 Infec in HIV- Rx- Cotrimoxazole

I. LISTERIA- Meningitis- Rx- Ampicillin

. Tumbling motility at 25^0C, non motile at 37^0C

J. Non TB Mycobacteria- M. avium, M. marinum, M. ulcerans

K. MYCOBACTERIUM TUBERCULOSIS and M. BOVIS (TB)

1. MC- Resp transm-

. Unpasteurized milk-

2. World TB capital (26%) Infective dose 1-10

a. India- 1 sec→ 1 infec, 1 min→ → mask

3. MC HIV opportunistic infec/ MC HIV death -F1/23,6/20,N22

4. Doctors at high risk→ Isolate patient for post Rx

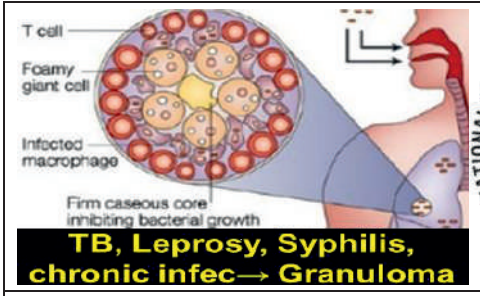
5. - TB (caseating), Leprosy (non-caseating)

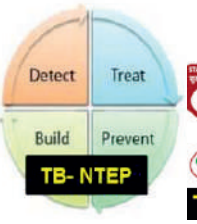
6. National TB Elimination Program (Old- RNTCP) -N18/22

a. Elimination target -F7/24

b. TB reporting mandatory/punishable- app

c. Free diag, Rx, Rs 1000 monthly→ Nikshay Poshan Yojana





1. RRTB = Rifampicin resistant TB (MC) -F1/22, N22/24

2. MDRTB= Multidrug res ( ) TB

3. Pre XDRTB= Pre Extensively Drug resistant TB

MDRTB + Fluoroquinolone (

4. XDRTB= Extensively Drug resistant TB

PreXDRTB + Gp A 2nd line (

1. Latent TB -F6/20 Asymp, No transmiss,

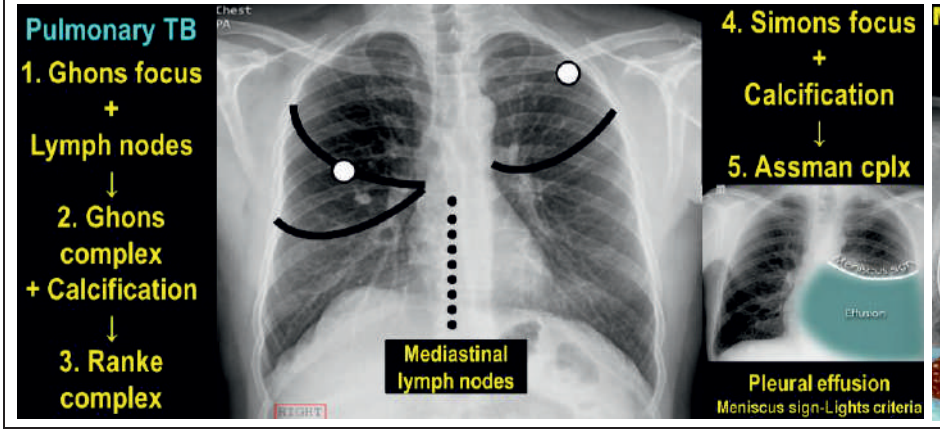
Diag- Quantiferon TB gold IGRA -N18/23 Not for active TB test

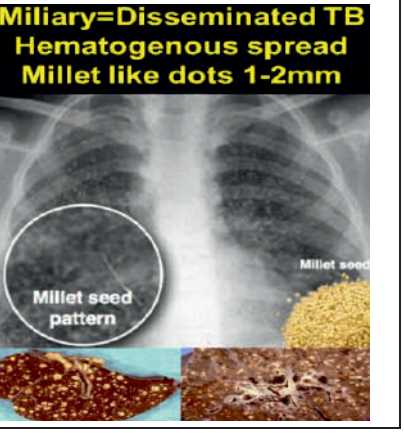
2. Active TB- Pulmonary 70%, Extrapulm 20%, Miliary-F7/24,N18

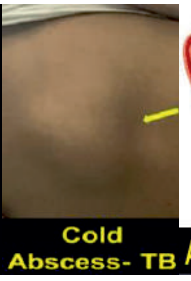
C/F of TB- Cough x 2 wks, hemoptysis→ MC hemoptysis

1. ↓Weight, night sweats, low

2. MC Fever of Unknown Origin (FUO = PUO)







- **Extrapulm TB→ All organs except**

1. **MC overall TB** -F8/20,1/21
2. **2nd MC TB** **ADA positive**
3. **TB Cold** **No rubor,calor,dolor,tumor,functio lasea**
4. **TB Meningitis** -F1/20/24
5. **MC osteoarticular TB** -F1/25,20/22,INI25

Narrow disc space→ Vertebra destruction→ Kyphosis→ Gibbus

6. **MC joint TB** **Internal rotation lost 1st**
7. **TB Pu** -N21,23,F19, **TB** **bladder**
8. **MC ileocaecal TB- Transverse constricting intestinal ulcer**
9. **MC PID in virgin** ♀-F25 **MC PID overall**-N23
10. **MC Genital TB** -N20/24 ♀ ♂
11. **Uterus- Synechiae** -F7/25,21,N23
12. **Skin TB=** **vulgaris> Scrofuloderma** -F22,1/24,N25

1. **TB BCG live vacc- Day 0 of birth** -F7/25 **strain**
2. **BCG protects only Extrapulm TB x 15 yr (Not pulm TB)**
3. **BCG turns PPD/ Mantoux/ Tuberculin test**
4. **BCG vacc used for Rx of** -F1/20/24

- **Diagnosis of TB (Rule of 2)**

1. 2 sputum sample
2. 2 stain a. Screening- Auramine Rhodamine stain

   b. Confirm- stain 20% H_2SO_4 → bacilli - NTEP grade 3+ 2+ 1+
3. 2 TB culture -F1/25- Biosafety level 2+, N95 mask

   a. Liquid culture (Takes 1-3 wk)

   b. Solid culture (Takes 2-8 wk)
4. 2 PCR test -F21,23,1/24/25, N25

a. 1st line test- UDST- Universal drug susceptibility test- 2 hrs

CBNAAT/ GenXpert/ TrueNaat finds

b. If UDST +ve- CDST- Cascading drug susceptibility test

Line probe assay/ MGIT/

5. Gold standard/ Most specific test
6. Fastest test (2 hrs)/ Most sensitive test
7. Test of choice in TB + HIV
8. India banned test- TB IgG 2012→ Positive in all exposed

IgG +ve- Typhoid, Covid, Dengue, Malaria IgG banned 2018

- **Rx of TB (ATT = Anti Tubercular Therapy)**

1. TB New / old case -F1/23

Duration Intensive phase → Continuation phase

2. 2025 USA (Not India)- 4 (PHZM) + 2 (PHM), P=Rifapentine

3. MDR-XDRTB -N25- 6m- BPaL or

Be

4. Best sterilizing drug in TB

5. DOTS (WHO)- Directly Observed Treatment Short course

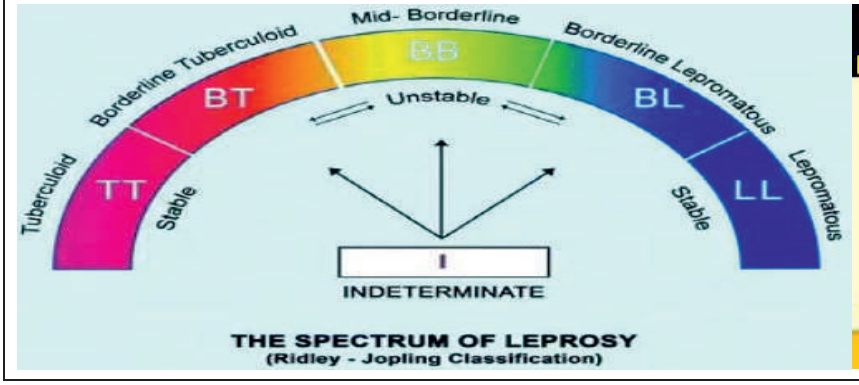
a. DOTS plus-

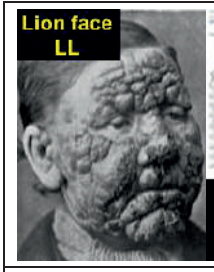
b. - Missed call telemonitoring

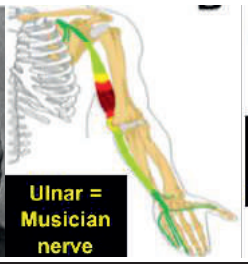
6. TB in preg→ ATT same dose (If <20wks→ MTP advised)

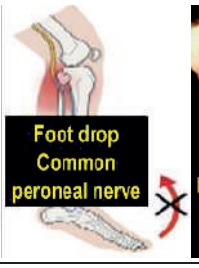
7. Breastfeed unsafe- TB mastitis, safe in pulm TB on Rx -INI25

Anti-TB -N20,24	OD dose	Bacter	Liver	Preg	Adverse effects -F1/25,23,7/24
R	600 mg	Cidal	Toxic	Safe	
H	450 mg	Cidal	Toxic	Safe	
Z	1500mg	Cidal			
E	1200mg			Safe	
B	400 mg	Cidal	Toxic	-	↑QT interval -F23
Pretomanid	200 mg	Cidal	Toxic	-	↑QT interval
Linezolid	600 mg	Static	Not	-	↓Platelets
Moxifloxacin	400 mg	Cidal	Not	Uns	↑QT interval









L. LEPROSY = HANSENS DISEASE -F20/22/23/24/25,N22

1. Mycobacterium > M. lepromatis
2. World Leprosy Capital (36%) ← Eliminated
3. Human→ Transmiss- Prolonged direct contact +
4. Leprosy affects skin + nerve (Motor, sensory, autonomic)

a. Pure neuritic Leprosy- No skin lesion, slit skin smear –ve

b. MC nerve -N20,F7/25- Overall/peripheral Cranial

c. Loss of sensations- Cold > Hot > Pain > Touch > Pressure

d. MC Type C unmyelinated nerve > myelinated nerve

e. Leprosy resistant organ

- Ridley Jopling Classification- TT, BT, BB, BL, LL, I -F1/24

1. TT- Tuberculoid Lep -N24- Mild- U/I- Hypoaesthesia, hair loss

. Cell mediated immunity, Lepromin test , Lepra rxn

2. LL- Lepromatous Leprosy -F1/21/23/26- Severe-B/I-Lion face

. CMI, Virchow foamy cell, Lepromin test , Lepra rxn

- Sample → X-shape 6-site

. Acid fast ZN stain 5% H_2SO_4 → Grade 6+5+4+3+2+1+

. Fite Faraco stain- Tissue/ Histoid leprosy

- Leprosy Rx→ Bacteria killed→ Allergic Lepra rxn -F21/26- Rx

a. Monthly 600 mg + 300 mg

b. Daily 100 mg + 50 mg

1. WHO Multibacillary (↑bacteria)-F23/25 Red pack

>5 skin lesions + nerve affected or slit skin smear

2. WHO Paucibacillary (↓bacteria) Green pack

1-5 skin lesions, no nerve affected, slit skin smear

. Child Packs- Multibacillary Orange, paucibacillary Blue

3. WHO Single lesion leprosy

ROM 1 dose- Rifampin + Ofloxacin +

4. Leprosy disability Rx- Reconstructive surg + rehabilitation

5. LCDC- Leprosy Case Detection Campaign by ASHAs

- NLEP = National Leprosy Eradication Program Goals

1. Zero leprosy transmission by

2. Reduce Grade II disability <1 per million population

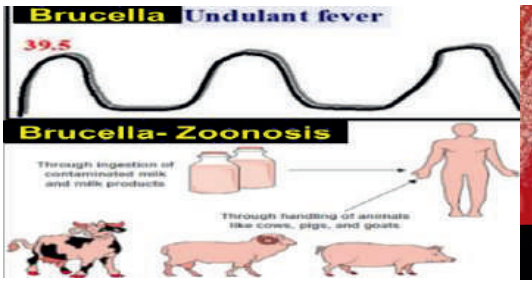
3. Eliminate stigma and discrimination

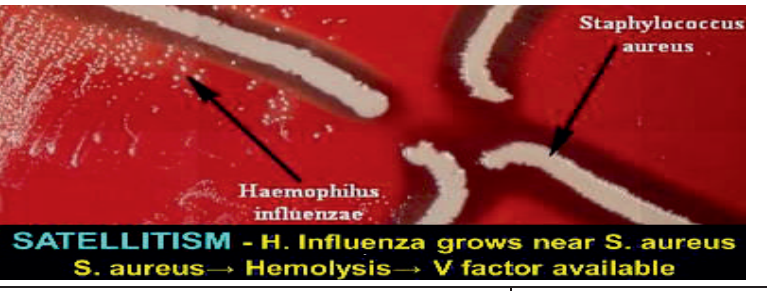
GRAM NEGATIVE COCCI/COCCOBACILLI (6) Bordetella, Brucella, Legionella, Hemophilus, STI, Neisseria (gonorrhea, meningitis), Bacterial vaginosis, Trichomoniasis, Urinary Tract Infection (UTI)

A. BORDETELLA PERTUSSIS (W

1. Whole cell vacc < 6yr . Rx

2. Acellular





B. BRUCELLA (Undulant/ Malta Fever in farmers) -F26/25,N21/25

. Zoonosis→ Touching animal secretions, unpasteurized milk

C. LEGIONELLA -N19 (Atypical Pneumonia- Airconditioner)

D. HEMOPHILUS INFLUENZAE (Never causes Influenza)

1. Gram neg capsulated bacilli -N23- Quellung rxn, capsul vacc
2. Culture -F7/25- Only Not blood agar
3. factor= Hemin-F1/26- Found in blood outside RBC
4. factor= NADPH+ -F1/26/25,N19,INI25 Found inside RBC
5. Influenza/ Flu virus→ Rx -N19→ Flu vacc

E. SEXUALLY TRANSMITTED INFEC(STI= STD) -F1/19/24/25/26

. STI C/F- Penis/vagina/anal discharge/rash/ulcer/wart, bubo

a. ♀ Vagina discharge-F1/25,N20/23 Normal- Thin clear + no smell

1. Bact vaginosis (MC)- Thin white/ gray + fishy smell
2. Trichomonas- + fishy smell
3. Candida- Cheesy + curdy smell
4. Neisseria gonorrheae- + foul smell
5. Pelvic Inflam dis= PID (MC Chlamydia)- Purulent/ frothy

Suraksha Clinic NACO

Sex partners Rx

1. Gonorrhea urethritis
2. Chlamydia urethritis
3. Chlamydia LGV
4. Mycoplasma/Ureaplasma
5. Syphilis- Chancre
6. Chancroid- H. ducreyi
7. Donovanosis- Klebsiella
8. Trichomonas (Protozoa)

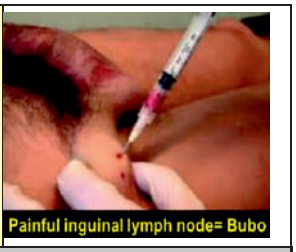
Sex partners not Rx

9-10-11. HIV, Hepatitis B/C
12-13. Herpes, HPV-MC STI
14. Molluscum contagiosum
15. Mpox, Ebola, Zika, Covid

b. ♂ **Urethritis-** MC Gonorrhea>Chlamydia, Trichomonas,HSV

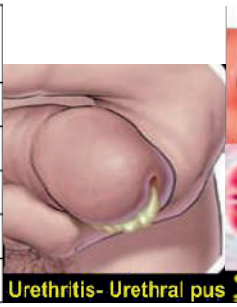
c. ♂♀ **Genital wart = Condyloma-** HPV acuminata, Syphilis lata

d. ♂♀ **Inguinal Bubo** = ↑size painful lymphnode-


Painful inguinal lymph node= Bubo

Kit 1	GREY/Gray	Urethritis (Gono> Chlamydia), Cervicitis
1 dose- Scrotum infec		Cefixime 800 mg + Azithromycin 1 gm
Kit 2	GREEN -F1/26	Vagina discharge- Candida, BV, Tricho
1 dose		Secnidazole 2 gm + Fluconazole 150 mg
Kit 3	WHITE	Genital ulcer (Syphilis, Chancroid) -F1/26
1 dose (injectable)		Benz penicillin 2.4 MU + Azithromycin 1 gm oral
Kit 4	BLUE -F23,N19	Genital ulcer- Penicillin allergy, donovanosis
14 day		Doxycycline 100 mg + Azithromycin 1 gm
Kit 5	RED	Herpes genital ulcer/ blister
7 day- Partners not Rx		Acyclovir 400 mg TDS
Kit 6	YELLOW	Pelvic Inflam Disease, Lower abdomen pain
Cefixime 800 1dose + (Doxycycline 100 BD+ Metronidazole 400 BD x 14 day)		
Kit 7	BLACK	Inguinal bubo (lymph node)- Chancroid, LGV
21 day		Doxycycline 100 BD x 21 day -N20,F25 (~~Azithro 1gm~~)
Kit 8- Brown- Anorectal discharge- Cefixime 800mg 1dose +Doxy 100bd x 7day		

	STI	Ulcer	Lymph Node
Come on	Chancroid	Painful	Painful
Sweetie	Syphilis	No pain	No pain
Don't	Donovanosis	No pain	No Pain
Have	HSV	Painful	No pain
Liquor	LGV	No pain	Painful


Urethritis- Urethral pus

Gram neg kidney shape diplococci

Liver

Fitz Hugh Curtis Syndrome
Violin String Pattern- Adhesions after PID

F. HEMOPHILUS DUCREYI (STI-) -F23/24/25/26

1. Chancroid (H. ducreyi) . Rx or Kit 7

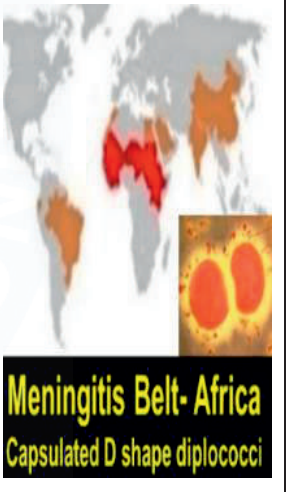
. (soft ulcer + unilateral lymphadenitis)

2. Chancre (Syphilis- T. pallidum) . Rx or Kit 3

. (hard ulcer + bilateral lymphadenitis)

G. NEISSERIA MENINGITIDIS = Meningococcus -N19/21/24

1. Human→ Transmission . Rx Ceftriaxone
2. Gram neg capsulated D- shape diplococci- Quellung
3. Capsular vacc (A, C, Y, W135)- Given in splenectomy -F20,21
4. Post Exp Prophy -N25 All exposed→


Meningitis Belt- Africa
Capsulated D shape diplococci

H. NEISSERIA GONORRHOEAE = Gonococcus

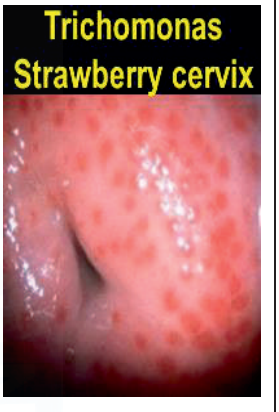
1. ♀ → Yellow vagina discharge + foul smell
2. Gonorrhea→ PID→
3. Neonate -N21,23
4. ♂ Urethritis -N19/21 Gonorrhea (IP<1wk) >
5. ♂♀ Proctitis, Pharyngitis . Rx Ceftriaxone/ Grey kit 1
6. Gram neg -N23

I. BACTERIAL VAGINOSIS (BV) (MC vagina infec overall)

1. Criteria >7- ↓ Growth of Doderlein= Lactobacillus Overgrowth of Gardenerella, Mobiluncus -F26,20/23/24,N24
2. Criteria- Vagina pH>4.5→ Clue cell >20%→ Thin white/ gray vagina discharge+ 10%KOH→W

J. TRICHOMONAS VAGINALIS (STI-

1. Lives in vagina→ Green vagina discharge→ W
2. Infec + diag stage- Trophozoite- Twitching/ motility
3. Strawberry cervix . No Cyst in Trichomonas -F22/23/24,N19/20/23



Bacterial vaginosis (BV) +Trichomoniasis→ Same test / Rx

1. Test of both- Vaginal discharge + 10% KOH = (Whiffamine test = Whiff test) . Rx of both- Green kit 2
2. Rx BV+ Trichomonas+ Anaerobes+ Entameoba+ Giardia-

K. URINARY TRACT INFECTION = UTI -F7/25,N24,INI25

1. MC overall/community/hosp/honeymoon/uncomp/complic
2. Lower (MC)=Ascending UTI- Cystitis- Urgency, frequency,
3. Upper=Descending UTI- Pyelonephritis-Kidney pain+ fever
4. Sex active ♀- MC cystitis- Rx
5. Sex active ♀- No vaginitis- Thick keratin vagina epithelium

GRAM NEGATIVE BACILLI (14) Escherichia coli, Plague, Helicobacter, Campylobacter, Shigella, Salmonella (2), Vibrio (2), Pseudomonas -N23
Klebsiella (K. pneumoniae, K. ozaenae, K. rhinoscleromatis, K. granulomatis)

A. ESCHERICHIA COLI (Gut commensal) -F1/23/24,N22/24

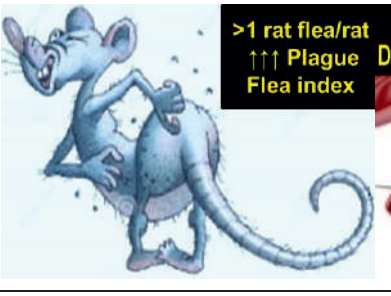
1. MC abdomen abscess (Liver, renal, biliary, appendix, pelvi pancreatic, peritoneal, mesentric, diverticular exc splenic)
2. MC perineal inf (UTI, anorectal, gangrene)
3. MC triad (Pain-fever-jaundice)

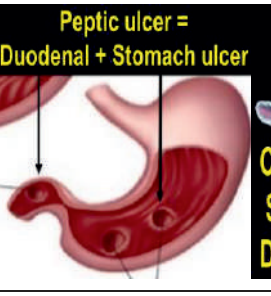
\+ pentad (Charcot triad + confusion + ↓ BP)

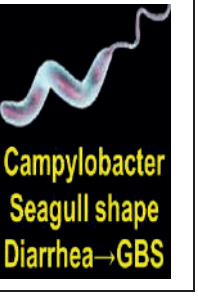
E. coli pathotype	Toxin	Diarrhea	Rx
1. EPEC	No toxin		Rehydration
2. EIEC	No toxin		Ciprofloxacin
3. ETEC	Labile toxin		A
4. EHEC=VTEC (O157:H7)	Verotoxin	Hemolytic uremic syndrom-F1/25	A

B. YERSINIA PESTIS (Black Death =

1. Bioterror Cat A→ Pandemics→ 1994 Surat epidemic
2. Vector
3. >1 rat flea per rat- ↑Plague-







C. HELICOBACTER PYLORI (+ Gastric ca)

1. Urea gun → -F1/7/25,N18/21

2. India Rx (2 antibiotics+ Omeprazole PPI) x 14day /Amoxycillin/Metronidazole+ Omeprazole

3. West Rx- Bismuth + Vonoprazan PCAB + Metronidazole + Tetracycline (↑ Clarithromycin resistance)

D. CAMPYLOBACTER JEJUNII (D

1. Diarrhea (Rx Azithromycin) → Pune outbreak 2025 syndrome 30% (Rx) -F1/23,7/25

2. GBS= AIDP=

Starts tingling- numbness- weakness in foot → Arreflexia → Bilateral symmetric

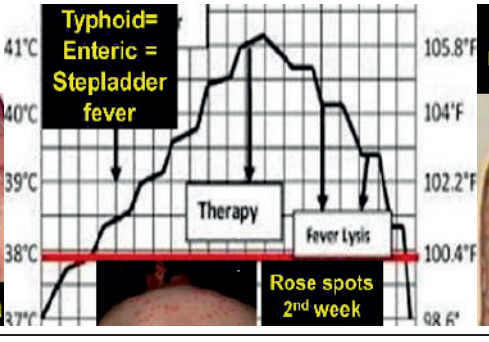
→ Motor-sensory-autonomic dysfunc→ Resp paralysis→ D

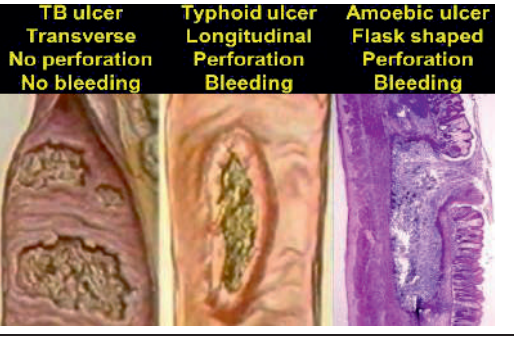
E. SHIGELLA (Lives in colon- D) -F1/20/21/25

1. MC India 2. MC world- S. sonnei

3. Most severe- S. dysenteriae 4. Rx

F. Non Typhoidal Salmonella typhimurium (Food poisoning)





G. SALMONELLA (Typhoid= Paratyphoid= E

1. Salmonella enterica serotypes (4)-

2. Human reservoir- Biliary/ urinary

3. Orofecal transmiss→ Intestine Microfold cell with brush border epithelium→ Blood bacteremia→ Sepsis→ Death

Week	C/F -F1/19/21,N22/23/24,F7/24 Weekly test- B→A→S→U	Test of choice
I	Step relative bradycardia	
II	R pea soup green stool	
III-IV	Perf peritonitis	

1. Temperature: Pulse Rate: Resp Rate :: ↑ : ↑ : ↑

2. Relative bradycardia = -F7/24- Typhoid, Brucella, TB, Dengue

- 3 ulcers in ascending colon / ileocaecal region

1. TB transverse constricting non-perforating non-bleeding
2. Typhoid longitudinal
3. Amoebic flask shape

- Salmonella transport media- Cary Blair

1. Salm enrichment med- Selenite F, Tetrathionate broth -F6/19
2. Salm selective media- DCA, XLD, Wilson Blair -F1/23,N21

- Gold standard test for Typhoid

. Gold std test post antibiotic

- 2 vacc- Oral Typhoral Ty21a (5 yr valid), im Typhim-Vi

- Widal test gives IgM titre = dilution (cut off used) -F1/25

Widal IgM O > IgM H >

- Rx of choice/ Empirical Rx- Total duration days-F1/26

iv → Oral Cefixime + Azithromycin BD

H. VIBRIO PARAHEMOLYTICUS -F7/23,N24,I25

I. VIBRIO CHOLERA (ICD Code 0001- C

- Human reservoir→ Orofecal→ Cholera toxin →

Tight junction/Zonula occludens→ Watery/secretory diarrhea

→ Ri (No blood) -F19/20/21/24/25 + Vomit →

Loss of electrolyte (Na-K-Cl-HCO_3) + Dehydration→ Death

- Rx→ No antibiotics→ Rehydration→ Oral (ORS- Vomits)→

iv (Veins not seen)→

- Serogroups (many) O1 ..... O139 Bengal

. Biotypes of O1 Classical, -F6/21

. Serotypes of O1 Ogawa,

- 1-6 pandemics

. 7^{th} pandemic

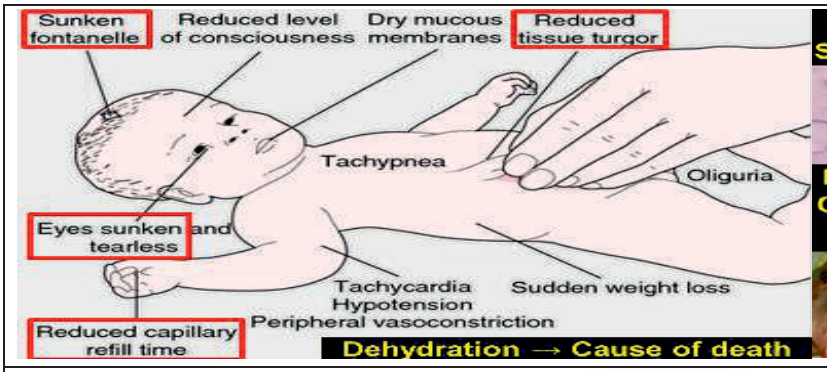
. 8^{th} pandemic (ongoing)

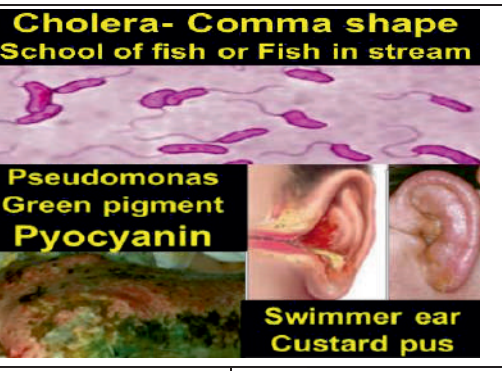
- Stool Hanging drop→ motility -F7/24

1. Cholera Transport media- Venkatraman Ramakrishnan

2. Cholera Enrichment media- Alkaline peptone water

3. Cholera Selective media- Monsurs GTTA, -F6/19/24





J. PSEUDOMONAS (Opportunistic infec)

1. MC BECCS→ Burns, ear, contact lens, cystic fibrosis, sav
2. Unsafe ear= Chr otitis media ( ) -N20,21,22
3. /Singapore/Cauliflower ear- Yellow custard pus
4. /Malignant otitis externa- Diabetics/ HIV
5. Blue green pigment- Kings A/B media
6. Fruity smell -F1/25 .Rx

K. KLEBSIELLA PNEUMONIAE (Pneumonia, UTI)

L. KLEBSIELLA OZAENAE (♀ Ozaena) -F23/24/25/26,N21

1. Rhinitis sicca/ atrophic/ dry crust= E
2. Nosebleed= , foul smell→ Mer
3. Rx

M. KLEBSIELLA RHINOSCLEROMATIS (Nasal polyp)

. Rx- Streptomycin long term + Rhin -F1/26

N. KLEBSIELLA GRANULOMATIS (STI) -F21/22/23,N19/20/21

1. Donovanosis=
2. Donovan body- Not related to

SPIROCHETE = 4 SPIRAL BACTERIA -N19,20,21,22,23/26

Treponema (Skin infec, Syphilis), Borrelia (Lyme dis), Leptospira (Weils dis)

A. Non-venereal Treponema (Skin infec)→ Bejel, Yaws, Pinta

B. TREPONEMA PALLIDUM (Syphilis) 1^0,2^0,Latent,3^0=4^0,Congenital

- 1^0 Syphilis- Painless chancre=ulcer + b/l lymph node

1. Site of chancre -F1/22/26 External genitalia, lips, finger

2. Blood transfusion/ mother to child

- 2^0 Syphilis- Painless palmoplantar rash -F21/23/26, INI25

. Painless (moist gray warts) -F1/7/23

- Latent Syphilis- Asymptomatic, no transmission

- 3^0 = 4^0 = Late Syphilis- Gumma, Aortitis, Meningitis

. Argyll -F7/20/23,N18

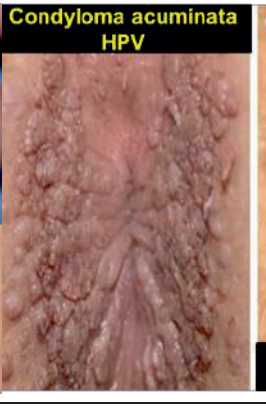
Light pupillary reflex Accommodation reflex

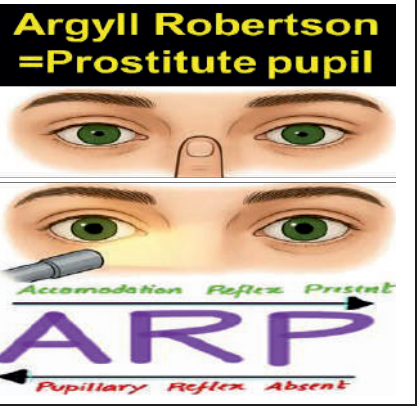
- Congenital Syphilis- triad→ >20 wk preg

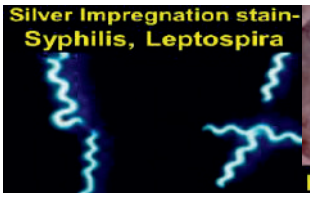
1. Eye-

2. Ear- VIII nerve sensorineural deafness -N22

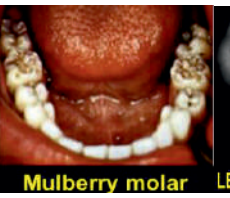
3. Mouth-







Interstitial keratitis

Hutchison teeth

Mulberry molar

LEPTOSPIRA- Umbrella handle

- **Syphilis→ No growth on agar→ Antigen-Antibody tests**

1. **Step1- Non-specific/treponemal= Standard= Screening test**

a. **Non-specific antigen from beef heart**

b. **- Venereal Dis Research Lab test (Microscope need)**

c. **(MC)- Rapid Plasma Reagin=VDRL + charcoal** -F1/22,N19

2. **Step2- Specific = Treponemal = Confirmatory test**

a. **Specific Treponema pallidum antigen**

b. **(MC)- Treponema pallidum haemagglutination**

3. **MC test overall/ screen/ prognosis**

4. **MC confirmatory test**

- **Rx- or Kit 3 white**

C. BORRELIELLA BURGDORFERI (Lyme disease)

D. LEPTOSPIROSIS (Leptospirosis = disease)-N23

1. **Andaman icterohemorrhagic fever (Jaundice+bleed)**-F1/26/25

2. **MC India zoonosis- Rice farmer, sewer/mine worker, floods**

3. **Triad-RRR-**

MOLLICUTES = EATON AGENT = PPLO -N18,IN (STI- Urethritis)

1. **Mycoplasma (MC Atypical pneumonia), Ureaplasma**

2. **Smallest bacteria 0.1 µm→ No cell wall→ No Gram stain**

ATYPICAL (Uncommon) GRAM NEGATIVE BACTERIA - 5 bacteria

Chlamydia, Rickettsia, Orientia, Bartonella, Coxiella- Q fever

. Obligate intracellular virus-like bacteria

. Rx **or Black kit 7**

A. CHLAMYDIA TRACHOMATIS (3) **(Many serotypes)**

1. Chlamydia **A, B, C serotype (T**

a. Corneal Keratitis→ **→ Eliminated**

b. strategy -F1/21/23/24,7/25,N24 Surgery- Corneal transplant, antibiotic- Azithro/ Doxy, Facewash, Environment improve

2. Chlamydia **DEFGHIJK serotype (STI) -F1/22**

. MC **. MC- Non gonococcal urethritis (STI)**

3. Chlamydia trachomatis L1L2L3L4 serotype-F1/26 (STI-

. Lymphogranuloma venereum→ G

B. RICKETTSIA **-N18,22,INI25 (Typhus/ Spotted Fever/ Pox)**

1. R. prowazekii	Louse	Epidemic
2. R. typhi/mooseri	Flea	Endemic
3. R. akari	Mite	Rickettsial
4. R. rickettsiae	Tick- Rocky mountain	

C. ORIENTIA TSUTSUGAMUSHI -F1/7/24/25 (S

. RRSET- Rain + Rodent/ Rat + Scrub grass/Bush + Eschar (Black necrotic ulcer) + Trombiculid chigger baby mite

Section III : Virology

30 viruses, 19 pages, 84 PYQs

A. BALTIMORE CLASSIFICATION -F7/25

Antiviral drug name has

- ssDNA dsRNA
- dsDNA HAPP- Herpes (Herpes simplex, Chickenpox + Shingles, EBV, CMV, HHV6, Kaposi sarcoma), Adenovirus, Pox (Smallpox, Mpox), HPV = Human Papilloma virus, Hepatitis B
- +ssRNA Japanese Encephalitis, Dengue, Chikungunya, Rubella, Zika, Yellow Fever, West Nile virus, Rubella, Corona, Polio, Rhinovirus, Enterovirus, Coxsackie, Hepatitis A-C-E viruses, KFD, Chandipura virus
- -ssRNA Influenza, Measles, Mumps, RSV, hMPV, Nipah, Ebola, Rabies, HepD
- Retrovirus (Reverse transcriptase)- HIV, HTLV

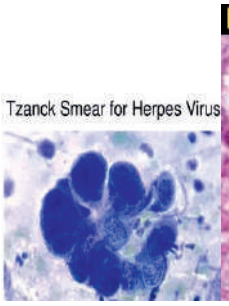
B. VIRUS

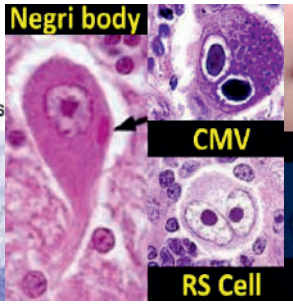
No cell→ Protein + DNA or RNA (never both)→

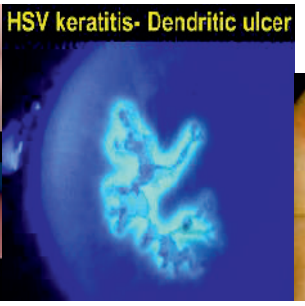
Non-living→ Lives inside human cell (Obligate intracellular)

Smallest Largest Bullet Noodle

1. Non enveloped (Tough to kill, orofecal)- Rotavirus, Hepatitis A,E, Polio
2. virus (Easy kill)- HIV, Hep BCD, Covid, Influenza
3. Diag a. DNA virus b. RNA virus







B. INCLUSION BODY (IB)= Virus growth in human cell -F7/25

- **Intranuclear IB- DNA virus- Cowdry A- Lipschultz body**

Herpes simplex, Varicella zoster- -F/23/24/25

- **Intracytoplasmic IB- RNA virus .Negri body-**
- **Both Intranuclear + Intracytoplasmic IB** -F1/23,7/24

1. Owl eye IB (Basophilic)-

2. Reed Sternberg cell (Eosinoph)- Hodgkins lymphoma -N18

DNA VIRUSES (12 viruses) → Diagnosed by

A. PARVOVIRUS B19 (ssDNA) = 5th dis -F1/22,N19- **Erythema infectiosum = Slap**

B. HUMAN HERPES VIRUS=HHV (7 type- Only human reservoir) -N18/19

1. HHV1- Herpes Simplex virus-1= HSV-1- Rx

2. HHV2- Herpes Simplex virus-2= HSV-2- Rx

3. HHV3- Varicella zoster VZV= Chickenpox Rx

4. HHV4- Epstein Barr virus = EBV Rx

5. HHV5- Cytomegalovirus- TORCHES -F21/23,N23 **Rx**

6. HHV6- 6th dis=Exanthem subitum -F6/19-

7. HHV8- virus- Rx

Skin ca in HIV CD4 <200- Purple red lesions -F23/24/25,N23

C. HSV = HERPES SIMPLEX VIRUS -F1/22,7/24,N19

1. HSV2 ↓waist Genital herpes (ulcer/blister) Rx Red Kit 5-

2. HSV1 ↑waist (Mouth, lip, skin, eye, CNS) Rx

a. Oral- MC viral infection of mouth

b. Lip (MC) -N20/25- Herpes labialis= Fever blister=

c. Skin- Herpetic = Felon -F7/20/25 ↑dentist

d. Eye- MC corneal blindness

MC viral keratitis- =epithelial ulcer-N23,F20/25/26

e. CNS- Meningitis, encephalitis -N23

D. VZV = VARICELLA ZOSTER VIRUS Rx

- Human reservoir→ Resp / direct contact transmission → 1st infec →

Chickenpox heals→ Becomes latent in Dorsal root ganglia or (V1) branch of ganglia -N19/21→

Reactivates→ 2nd infec

- Chickenpox in child- mild, adult- severe, neonate/ infant-

1. ↑↑Infectious→ Outbreaks in school/ward→ mask
2. Infective period- days before to days after rash
3. Home isolation/quarantine in school child- days
4. Rash starts from face/trunk (Centripetal distribution) → spreads to hand/feet (Centrifugal spread) ≈ Rash/force (inside to outside)→ Overall rash is
5. P rash- Macule, papule, vesicle, pustule, crust, scar altogether→ MC bacteria infec -F19/20/21/24/26,N18/23/24
6. Centrifugal pleomorphic itchy dew

- **Shingles = Herpes zoster** -F22/23- **Rx**

1. Painful itchy u/l rash→ Thoracic> Cranial

2. -Area supplied by spinal cutaneous nerve

.Vertebra .Spinal nerves pairs .Dermatome pairs

3. ↑↑Infectious→ Exposure causes chickenpox in unexposed

4. MC complication- Post

. Rx

5. Eye- Herpes zoster op -F1/23

6. Ramsay Hunt syndrom= Shingles oticus+ Facial palsy-F1/23

- Live vacc .Chickenpox .Shingles
- Post Exp Prophy (Chickenpox/Shingles)

E. EBV = EPSTEIN BARR VIRUS -F20/21/23/25/26,N18

1. Infectious mononucleosis=

= Glandular Fever (Lymphadenopathy)- Virus in s

. Diag- Paul Bunnel/ M Rx

2. Precancerous lesion

3. EBV Cancers- MC

. > Hodgkin> Non-Hodgkin lymphoma

F. POXVIRUS → Smallpox, Mpox, Molluscum contagiosum

1. Mpox- PHEIC 2022/2024 -F7/23/24 ↑↑Ho

2. Smallpox = Variola- Eradicated 1980 worldwide

a. Virus stored in US/Russia -F1/23- Bioterrorism cat →

b. Rash outside (Centrifugal distribution)→Inside (Centripetal)

Rx Vacc Jynneos/Imvanex/Imvamune stockpile

3. Molluscum contagiosum = Water warts- Rx Nil -F20/23/24/25

. Chickenpox/ Shingles→ Herpes virus-3 (Not pox virus)

G. HPV = HUMAN PAPILLOMA VIRUS 3 diseases

. Human reservoir→ Direct contact / Sexual transmission

1. MC viral skin infec- HPV = =

Rough benign hard skin growth without fluid inside

a. HPV2 Common wart -

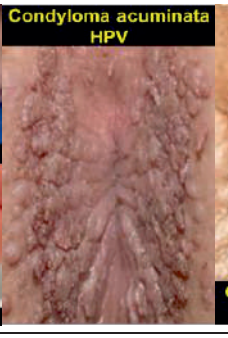
b. HPV1 Painful plantar wart -

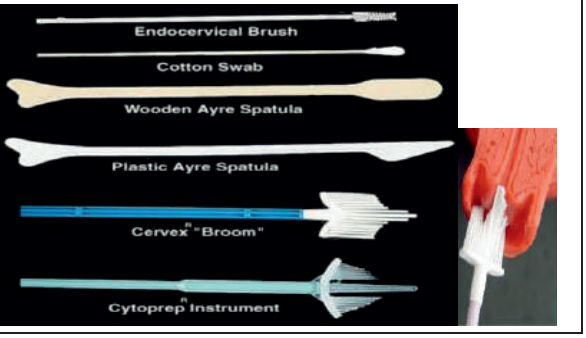
2. MC STI- HPV 6,11 Anogenital wart =

-FMGE 6-19/20/21/22/24/7-25, NEET 18/20/22/24/25, INICET 25

3. Cancer- HPV 16,18 is MC + oropharyngeal ca

. Ca cervix FIGO classification→ WHO Elim target





- Diagnosis of HPV cervical ca- 4 tests

1. VIA/VILI- Visual Inspection by Acetic acid/Lugol iodine(old)

2. Pap Smear- Mod Bethesda system- CIN 1/2/3, Invasive ca

Screen ♀ 21-65 yrs 3 yrly (HIV yrly) ACOG guideline -F1/24,N25

3. HPV DNA PCR- Screen ♀ 30-65 yrs 5 yrly (Gold standard)

4. Co-Test → P -F25/26

- Rx Wart- , cautery- Cryo/Laser/chemical/electro

. Ca cervix -N20 (cervix removed) +

(Platinum chemoRx) + radiation brachytherapy

- Vacc ♀♂ > 9 yrs-N23

RNA VIRUSES (25 viruses) → Diagnosed by

A. ROTAVIRUS -F1/24,N18/23/24 MC child & weaning diarrhea

1. toxin- ↑cAMP- Vomit +

2. Rx 3. Live vacc- Rotarix/tec/sil/vac

B. JAPANESE ENCEPHALITIS MC encephalitis in India

1. Reservoir- Ardeid bird/ wild duck→ Amplifier

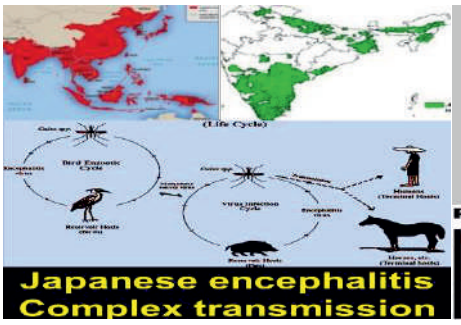
2. Dead end = Accidental host

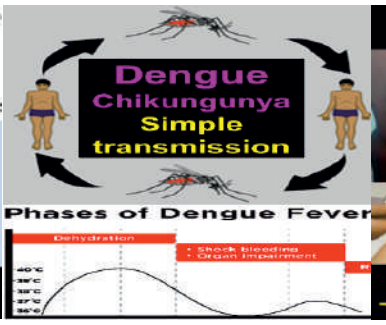
3. Vector-N19 vishnui, C. tritaeniorhynchus

4. Rx Nil . Vacc 2 dose (UIP Endemic areas only)

5. Live vacc strain = Imojev

6. Killed vacc Indian strain= JEEV, JENVAC





♀**AEDES Tiger mosq- CDZY-**

C. CHIKUNGUNYA Joint pain- Rx Nil Vacc

D. ZIKA -N20 Mild fever- Microcephaly- Rx/Vacc

E. YELLOW FEVER -F7/25,N22 Africa/ South America

1. Rx Nil Live vacc valid after 10 days of vacc till

2. Aedes aegypti/Breteau Index-No Aedes in 400m near airpo

F. DENGUE 5 types- DEN1/2/3/4/5 .Live vacc

- **Dengue/ Fever (1st infec- Mild- OPD) -F21**

1. Blanching Herman pin-point rash, retro

2. 1st week- antigen→ 2nd week

- **Dengue Hemorrhagic/ Fever**

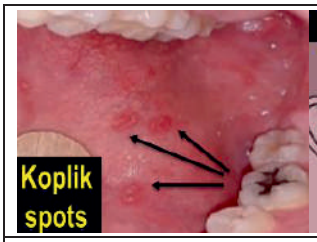
(2nd time infec- Severe- Ward- NS1 antigen +)

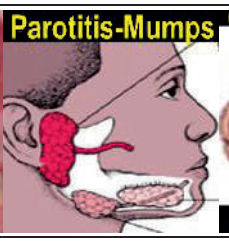
1. Bleeding- MC , gut, hidden occult internal bleeding

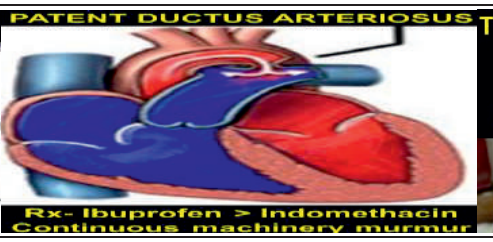
2. ↑ESR/ hematocrit >20%, ↓platelets, test +ve

→ petechiae/inch²→ Internal bleed -F25,N22/24,I19

- **Dengue Hypovolumic Shock Syndrome (Severe- ICU)**
- **Rx Nil, Supportive- Crystalloid fluids + p**







G. MEASLES = RUBEOLA -F19/20/23/25,N18/19/21/23 **Rx** Nil

1. **Live vacc-**
2. Human reservoir→ Resp + direct contact transmission
3. **3C** -N18,21 , coryza= catarrh,
4. spots + rash
5. ↓**Vitamin** → spots→ Vitamin A 2/3 dose -N19
6. Untreated- **SSPE**= Subacute sclerosing panencephalitis-N21
7. Otitis media (MC), pneumonia (severe), diarrhea (recovery)
8. **Catch up, keep up, follow up**→ Home isolation for 7 days
9. **Zero measles-rubella elimin campaign 2025-26**→N95 mask

H. MUMPS-F21/22/23/24,N22 **Rx** Nil **Vacc**

1. Human reservoir→ Respiratory + saliva transmission
2. MC- B/l non-suppurative > u/l orchitis -F19/20/21

I. RUBELLA = 3-day GERMAN MEASLES -N19/20 **Rx** Nil

1. Itchy rash behind ear, spreads downward-
2. spots -F1/23/25,I23 **Live Vacc**
3. **Congenital = Fetal Rubella** -N19 (TORCHES)- **DCH Triad**

a. **Deafness** (VIII nerve sensorineural)-N22 b. **Cataract**

c. **Heart** -N19 MC >PAS>VSD,TOF

J. POLIOMYELITIS Rx Nil Live vacc Sabin Killed vacc Salk

1. Human reservoir→ transmission→ Intestine→ anterior horn cell→

a. Wild polio (PV1, PV2, PV3)- World polio free exc

Inapparent polio 90%, Non paralytic 5%, Paralytic polio <1%

b. Vacc assoc paralytic polio (VAPP/VDPV)- OPV- 2024 Meghalaya

K. RABIES World rabies capital- Rx Nil

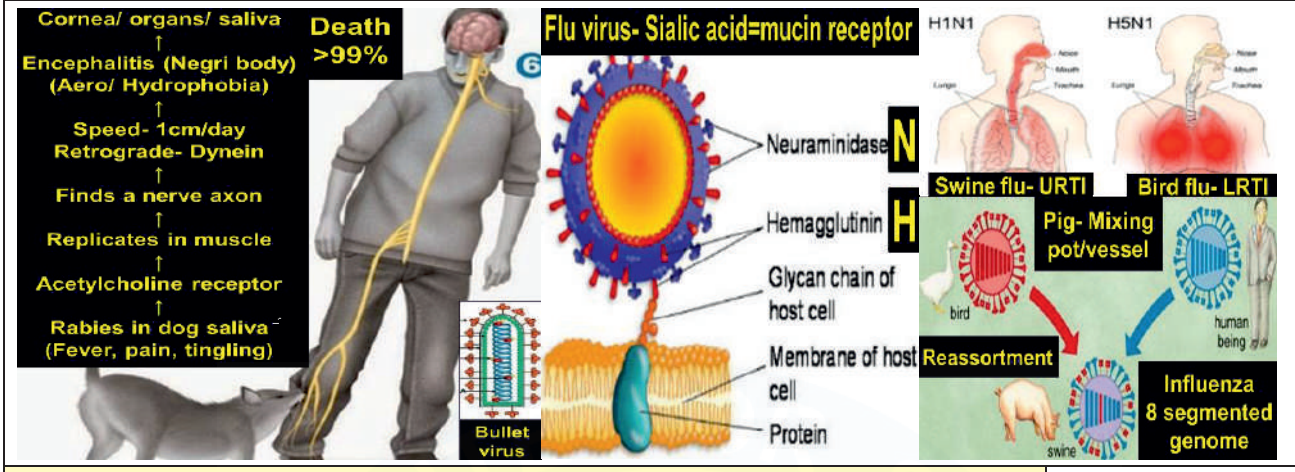
1. Transmission- Bite- Dog 99%, cat, bat, squirrel (Rat safe)
2. Rabies vacc
3. Rabies antisera= antibody= immunogloblin- Human/ horse
4. Rabies Post Exp Prophylaxis (Class Of Bite=C)-F1/21/23/25,N22

C1- Touch, feed, lick on intact skin

C2- Animal scratch- No bleed

C3- Broken skin/ mucosa/ head- neck- face- hand- genitalia (↑nerve)/ aerosol/ wild animal/ bat

5. intramusc (1 ml- Deltoid 5dose) 0-3-7-14-28 day
6. Thai Redcross/Oxford(0.1ml intraderm 2site 8dose) 0-3-7-28
7. Vaccinated pet/stray puppy-Full vacc-Dog lives 10day-stop
8. Re-exposure <3m- Wash, >3m- 2dose- 0,3 days
9. Rabies pre exp prophy- 3dose-0-7-14/21 day valid for 2-3yr
10. Rabid dog-Sudden bite,ataxia,↑saliva→Dont suture wound
11. Nat Rabies Cont Prog 2023→ Zero rabies→ Free PEP



11. Nat One Health Mission 2025→ NAPRE

12. Rabies free- Islands, Andaman-Nicobar, Lakshadweep -F20

13. Diag postmortem- Brain autopsy- Negri body- MC

14. Diag antemortem-IN24,F7/25 Cornea/nape of neck- DFAT,

L. INFLUENZA = FLU Diag Rx

1. 1920- H1N1 Swine/Spain flu→Bombay flu- Resp tranmiss

2. 2009- H1N1 Swine flu pandemic/PHEIC 2009-N20→Now mild

3. 2025- H5N1 Bird flu India 9 states→ Chicken/ eggs killed

4. Sialic acid receptor→ Hemagglutinin and Neuraminidase

a. Influenza A- 3 flu types- Seasonal, Swine, Bird = Avian flu

→ H3N2 Mild URTI

→ H1N1 Mild URTI

→ H5N1 Severe LRTI Antarctica penguin

b. Influenza B- Seasonal flu

. Vacc Annual live/ killed seasonal + swine flu (Not bird flu)

. Post Exp Prophy

. Antigenic Drift -F1/6/22 . Antigenic Shift -N19 → Minor change @ 2-3yrs Major change @ 10-20 yrs → Point in pig → Seasonal influenza/ flu influenza/ flu	
M. CORONAVIRUS = Cov- 7 types- Reservoir Rx Nil . 4 Common Cold Cov- OC43, NL63, HKU1, 229E . Severe Acute Resp Synd (SARS) Cov- China 2002-03 . Middle East Resp Syn MERS Cov- Saudi Arabia camel 2012 . Cov= SARS Cov2 → Dec 2019 Wuhan, China (Sporadic→Epidemic)→ Jan 20 PHEIC→ Mar 20 Pandemic -F20/24 → India wave 1st 2020- 2nd 2021 α+δ variant- 3rd 2022 Omicron → May 2023 not PHEIC→ 2025 mild wave India ----- Endemic . 5 Variants of concern (WHO) -F19 . Never had Covid→ . MC → URTI→ LRTI- Pneumonia→Type2 pneumocyte . Gold std , Rapid Antigen Test RAT, IgM/IgG- No role . Asym/Mild- Rx Home isolation + Covid appropriate behavior . Moderate- Acute ward + all tests + oxygen + oral/iv steroids . Severe- ICU + Ventilator + LMWH + Remdesevir- No proof . No live vacc- Killed mRNA- Pfizer, Moderna Adenovirus vector- Sputnik, DNA plasmid- ZyCovD Nasal spray- Universal Covid vacc	

N. RHINOVIRUS MC Common cold Rx/Vacc

O. RESP SYNCYTIAL VIRUS → MC

Rx Vacc

P. COXSACKIE A16 VIRUS -F23/25 Hand foot mouth disease

= flu/ fever 2022 outbreak in India

Q. EBOLA HEMORRHAGIC VIR -F1/24 Rx- Inmazeb- Vacc rVSV

. Bioterrorism Cat → Level 4 Biosafety→ Death 70%

. Reservoir bat → only Africa outbreaks → 2014/18

R. NIPAH ENCEPHALITIS -F1/24,N24 Rx/Vacc Nil- Death 70%

. Kerala >5 outbreaks→ Fruit bat saliva

. Transmiss- Eating bat eaten fruit, date palm sap= toddy

S. ENTEROVIRUS -N18/19 MC conjunctivitis

T. KYASANUR FOREST DIS= MONKEY FEVER Hard Tick

Only South India- Outbreak Feb 24- Rx Nil- Ring vacc

U. HMPV= HUMAN METAPNEUMOVIRUS Resp- Rx/Vacc Nil

V. WEST NILE ENCEPHALITIS All mosq- Gujarat- Rx/Vacc Nil

W. CHANDIPURA ENCEPHALITIS- Gujarat 2024- Rx/Vacc Nil

X. MARBURG HEMORRHAGIC VIRUS- Africa- Rx/Vacc Nil

Y. HUMAN IMMUNODEFICIENCY VIRUS → MC HIV-1 > HIV-2

1. Thai 1982→ Chennai1986→ Human reservoir (Not monkey)

2. India prevalence 0.2%→ Max Mizoram > Maharashtra

3. HIV Transmission	HIV	Hepatitis C	Hep B
a. Needle stick injury			
b. Sexual/ mucosal		0.1%	5-50%
c. Mother to Child -F20,N19,22		10%	90%
d. iv drug abuse	2-5%	20%	20%
e. Unsafe blood transf		95%	95%
f. Breastfeed -F22,I25	No Rx 35% Rx 10%	7%	15%

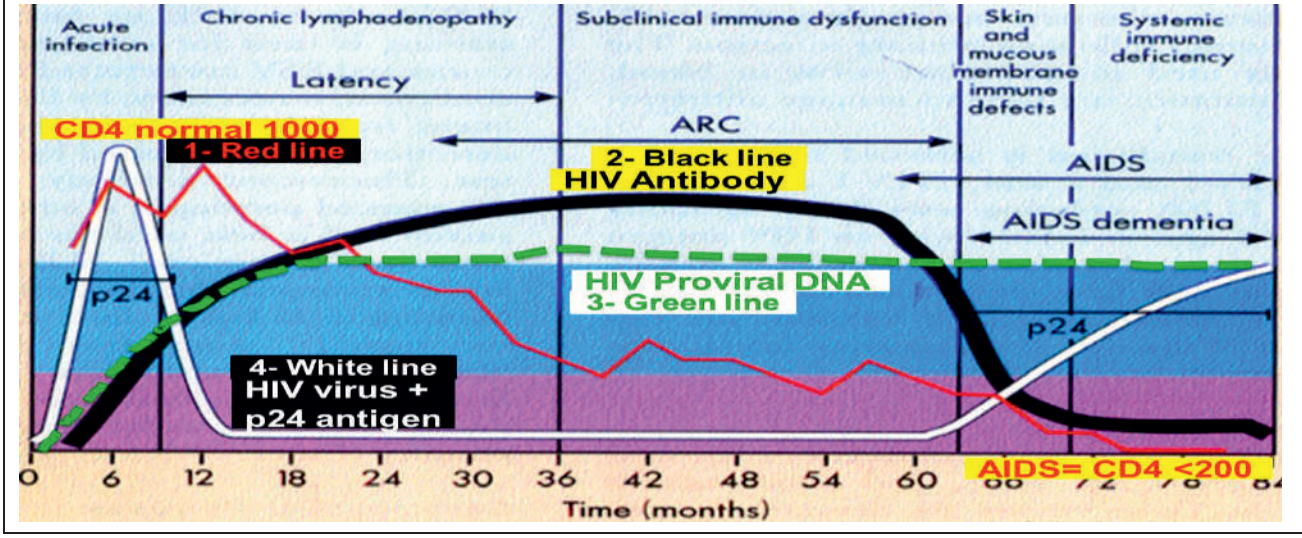
g. Kiss- 2 litre saliva→ Not from urine, sweat, tear, feces, air, orofecal, vector

h. Max transmiss chance → MC

1. Window period=WP- No test is +ve- Least WP with

2. Incubation period=IP- No clin features-N18- IP HIV→AIDS

Fast progressor 3yr, Longterm no progress/ Elite controller



HIV 4 Clinical Stages- People Living with HIV-AIDS = PLHIV =

Acute HIV synd→ Latent HIV→ AIDS related complex→ AIDS

1. **Acute HIV retroviral syndrome** (2-4 weeks after infec)→

Non-specific C/F- Fever, headache, rash, sore throat, cough

diarrhea, lymphadenopathy (Patient doesn't report to doctor)

2. **Latent HIV in CD4 cell**- Asymptomatic

3. **AIDS related complex**- Opportunistic infec- MC

a. MC Fungus in HIV

b. MC Fungal pneumonia

c. MC Meningitis

d. MC Diarrhea

. CD_4 500 (Asymp=Cryptic TB), pneumonia

. CD_4 400 =Thrush, HSV, Shingles

. CD_4 300 = Thrush -F1/7/24/25,N18/23,I25

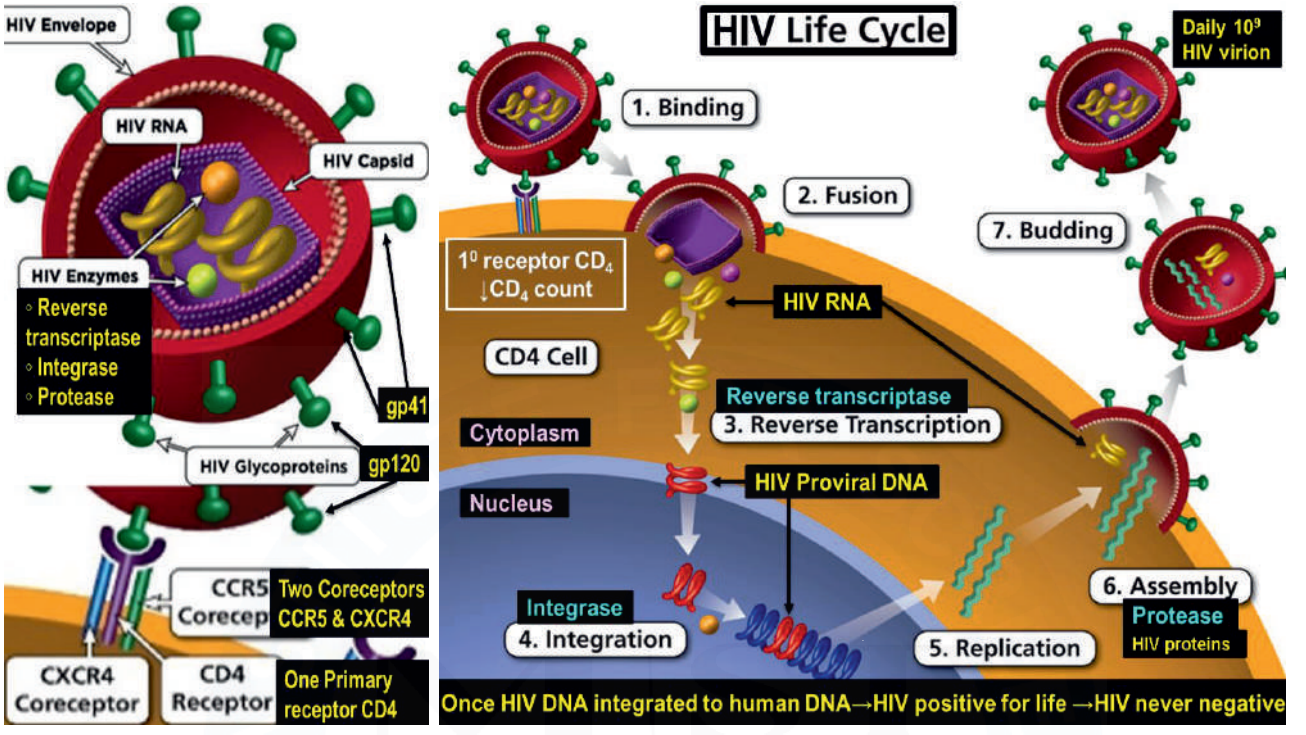
. CD_4 200 Pneumocystis jirovecii, K (skin)

. CD_4 100 -N19 Cryptococcal meningitis, Cryptosporidium diarrhea, CMV, Mycobacterium avium

4. **AIDS**- 3 AIDS defining cancers + Many AIDS assoc cancers

MC Non Hodgkin lymphoma>Kaposi sarcoma-F23/24/25>Cer

AIDS def illness-N19 Opportun infec+ cancer+ wasting+ encephalopathy



HIV Diagnosis -F20- **Rule of 3** ICTC/ VCCTC- Integrated Voluntary Confidential Counselling and Testing Centre

1. **Consent**, **Counselling**, **Confidentiality** (Doctor ↔ Patient)
2. **3 Screening Tests** (IgG/ p24 antigen) -N20- ERS- ELISA 3rd/4th gen, Rapid test- MC (Tridot/ ICT), Simple (Coomb Test)
3. **3 Confirmatory Tests** (MC/Most specific- HIV RNA RTPCR)

a. HIV RNA RTPCR (Viral load) done 3-6 monthly

b. HIV DNA PCR- DBS sample child 6wk- Final diag 18month

c. Western Blot (old)- Both confirmatory/ supplemental test

Old- 3rd gen ELISA (screen) + Western blot (Confirm/gold std)

New- 4th gen ELISA (screen) + HIV RTPCR (Confirm/gold std)

NACO- 3test +ve (Elisa/Rapid/Simp)→HIV +ve→ Western blot

4. 3 strategies

Strategy 1- 1 test- Mandatory- exc

Strategy 2A- 2 test- Anonymous epidemiological test

High risk/ Bridge population- Commercial sex worker CSW, homosexual, truck driver, iv drug user

Strategy 2B- 2 test- Test in

Strategy 3- 3 test- Test in asymp- Voluntary, Pre surg,

HIV HAART Rx- Highly Active Anti Retroviral Rx -F1/26,7/24,N22

1. Tenofovir 300mg+ Lamivudine 300mg+ Dolutegravir

Start asap→ Lifetime→ No cure, only control, normal life expectancy→ Death . UNAIDS 95-95-95 strategy

2. HIV suppression- Viral load< WHO 1000, US 100, ideally nil

3. HIV U=U→ If HIV is Undetectable→ HIV is Untransmissible . HIV +ve U-mother can breastfeed, HIV +ve↔HIV neg couple

4. IRIS -F1/25 Immune Reconstitution Inflam Synd after HAART

HAART→ ↑CD_4 ↑Immunity→ ↑C/F of hidden opportunistic inf

1. HIV + Infection→ Rx Infec→ After 2-8 wks→ TLD

2. HIV + Hepatitis B -N18/25 Only

3. HIV + TB- RHE (Rif-Iso-Eth)→HAART Dolutegravir 50mgBD

4. HIV + Cryptococcal meningitis -N22 Amphotericin B + Flucytosine→ After 4 wks→ HAART

1. HIV + Preg -N18 TLD → Vaginal delivery (India)/ LSCS (USA)

→ Breastfeed unsafe but allowed in India (WHO/NACO)

→ Breastfeed safe if mother U=U or mother + child on Rx

2. HIV Baby ↓Risk- Mother on regular Rx & no breastfeeding

Syp 2mg/kg ODx 6wk + delay cord clamp -N18/25

↑Risk- Mother not on Rx or breastfeed/ mixed feed

Syp Nevirapine+ Zidovudine 4mg/kg BDx 12wk+ early cord cl

HIV Prophylaxis -N19 Needle Stick Injury (NSI)

1. Pre-exp PrEP daily or

2 mnthly Cabotegravir or 6 mnthly injectable

2. Post-exp PEP asap→ Hollow > Solid suture needle

Test doctor- Baseline, 6-12-24 wks, Barrier protection- 12 wk

. OCP safe in HIV -F 7/25

BREASTFEED has IgA_2- protects baby from infec -F22,N18,I25

1. Breastfeed unsafe in breast infec (Breastfeed from healthy breast or EBF = expressed breast feed) - TB mastitis, herpes, chickenpox, shingles, abscess, burns/trauma of breast

2. Breastfeed allowed in systemic infection-

. HIV- TLD to mother and Nevirapine ± Zidovudine to baby

. Pulmonary TB- ATT + mask to mother and Isoniazid to baby

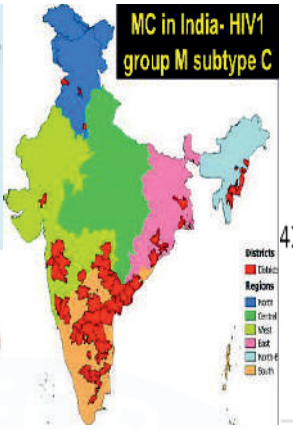
. Hepatitis B- Hep B vacc + Hep B immunoglobulin to baby

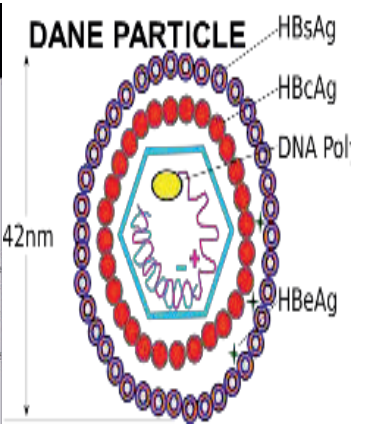
. Chickenpox/Shingles/Herpes- Acyclovir to mother and baby

2. Breastfeed absolutely contraindicated-

. Mother- ChemoRx, RadioRx, Lithium, ergot, drug abuse

. Baby- Classic galactosemia





Z. VIRAL HEPATITIS (MC Jaundice overall)- 5 types- Hep A, E, B, C, D

	Hep A	Hep E	HepB	HepC	Hep D
1	N		En		
2	Or		Sexual/BI		
3	Only		Ac, chr, cirrhosis, ca		
4	No Rx		Rx		
5	RNA-		PCR	RNA-	
6	Vacc	Vacc	Vacc	No vaccine	
7	-	-	Carrier state		

1. MC acute, MC epidemic- Hep
2. MC death in preg- Hepatitis
3. MC child- Hepatitis
4. Max chance of chronic/cirrhosis/ Liver ca (↑AFP)-
5. MC chr, cirrhosis, liver ca- Hep
6. MC post transfusion-F6/19
7. MC bilirubinemia-F19/22/23 viral he .

. Hepatitis C virus (HCV) → Rx

. Hepatitis D virus (HDV) → Satellite virus dependent on

. Hepatitis B virus (HBV) → Rx TLE

. Hepatitis B recombinant vacc 3/4 dose- 0,1,6 / 0,1,2,12 mnth

. Post Exp Prophy- Hep B immunoglobulin + Hep B vaccine

. Hep B complete virus + All components→

HepB 3 gene	Hep B 3 Antigens = Ag	3 Antibodies = Ab	3 Mutations
Surface gene = S gene	Australia Ag Comes 1st (Active infec)	Vaccination marker	Surface/Escape mutant-No HBsAg
Core gene - core part	Core antigen Only in liver- Not blood	IgM-Ac inf, IgG-Chr inf	No mutation Not in blood
Core gene -precore part	Precore Ag High inf- Super carrier	↓ inf- Simple carrier	Pre-core mutant No HBeAg

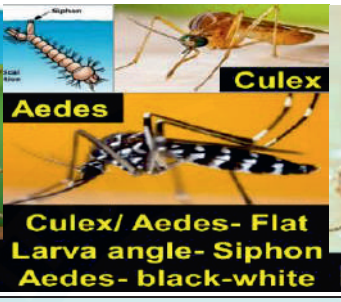
	HBsAg	Anti HBs	Anti HBc	HBeAg	Anti HBe	Interpretation -F 21/23/24/25, N 18/22/23/24 HBsAg- Both acute & chr Hep B
1	+	-	IgM	+	-	Hep B + infectivity
2	+	-	IgM	-	IgG	Hep B + infectivity
3	+	-	IgG	+	-	Hep B + infectivity
4	+	-	IgG	-	IgG	Hep B + infectivity
5	-	IgG	IgG	-	IgG	
6	-	IgG	-	-	-	
7	-	-	IgM	+	-	Surface = Escape mutant -NEET/INICET
8	+	-	IgM	-	-	Precore mutant -NEET PG/ INICET level

	Memory booster	Anti HBc	HBeAg	Anti HBe
1	Ac Hep B + high infectivity			
2	Chr Hep B + high infectivity			
3	Ac Hep B + low infectivity			
4	Chr Hep B + low infectivity			
5	Vaccinated FMGE doctor			
6	Cured/ Old Hepatitis B			

Section IV : Parasitology

30 parasites, 9 pages, 56 PYQs

1. Vector Borne Infections (12)- Mosquitoes, Flies, Bug, Lice, Flea, Mite, Larva, Cockroach
2. Protozoa (10)- Entamoeba, Giardia, Acanthamoeba, Naegleria, Kala azar, Toxoplasma, Cryptosporidium, Cyclospora, Cystoisospora, Malaria
3. Helminths (3)- Tapeworm=Cestode, Fluke=Trematode, Roundworm=Nematode



VECTOR BORNE INFECTIONS

-F20/21/23/24/25, N21/25, INI 25

. Vector (Insect) transmits infec, pathogen causes infection

A. ♀ Aedes Tiger Mosq- Clean stagnant/ rain water

Lives indoors, flies 100-200 metre, multi-person day biter

DCZY→ Dengue, Chikungunya, Zika, Yellow fever

B. ♀ Culex nuisance mosq- Dirty stagnant water, flies 10km

JF→ Japanese

C. ♀ Anopheles sophisticated mosq- Clean slow flow water

D. ♀♂ Mosq Control- Anti-bite- Odomos, bed net, pyrethrum

. Anti-adult- Indoor Residual Spray-

. Anti-larva-F1/24 =Guppy fish,Temephos=Abate

. Anti Sandfly-N24,F19 Cypermethrin 5%

E. Sandfly Phlebotomus→ Leishmaniasis- Leish. donovani

F. Tse Tse Fly Glossina→ African

राष्ट्रीय वेक्टर जनित रोग नियंत्रण कार्यक्रम
NVBDCP

G. Reduviid Kissing Bug Triatoma→ South Am Chagas dis-

H. Body Louse Pediculus→ Trench fever, Epidemic Typhus

I. Rat Flea Xenopsylla cheopis→ Endemic Typhus

J. Mech Vector (Housefly, Cockroach)- Orofecal infec

K. Trombiculid Mite→ -Orientia tsutsugamushi

L. Itch Mite Sarcoptes scabei- Rx

Severe nodular crusted Norwegian scabies -F18/19/22/23/24, N19

M. Myiasis (Maggot=Fly larva)- Rx- Surg removal+ Ivermectin

N. Insect in ear- Put olive oil/lignocaine to suffocate- Remove

1. Nat Vector Borne Disease Control Program 2002 -N20-

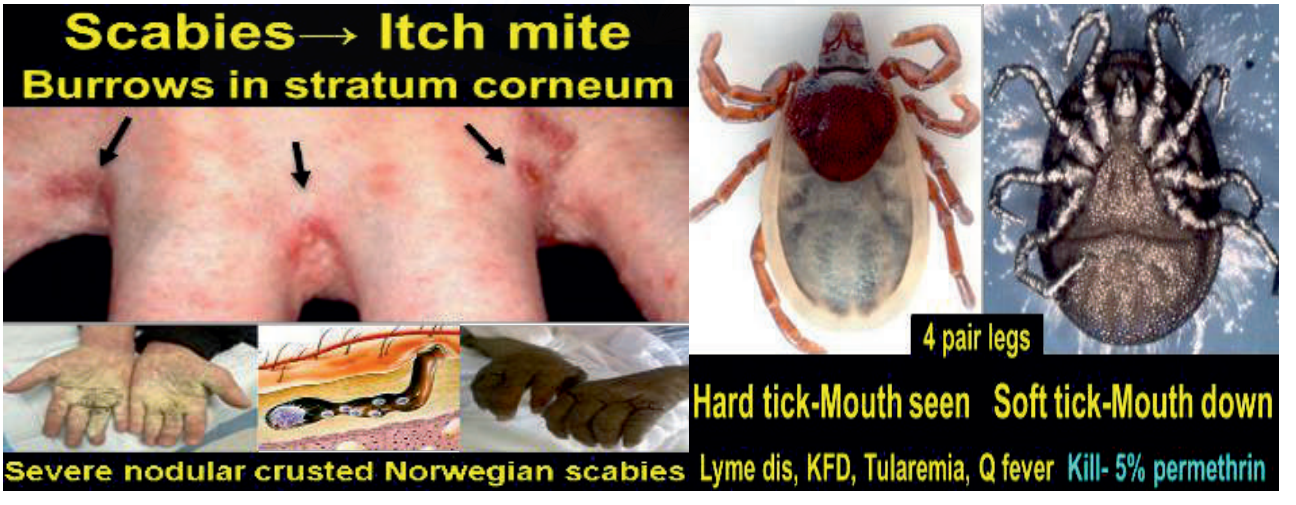
6 inf- Malaria, Filaria, Leishmania, Dengue, Chikungunya, JE

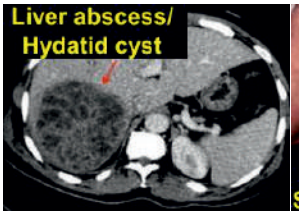
2. Human Host- Definite- Sexual cycle, Intermediate- Asexual

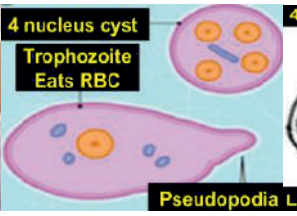
Man interm host- Toxoplasma, Malaria, Cysticercus, Hydatid

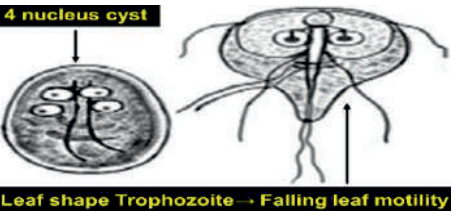
3. Infective stage- Enters body . Diag stage- Seen in sample

4. Zoite- Stage of parasite







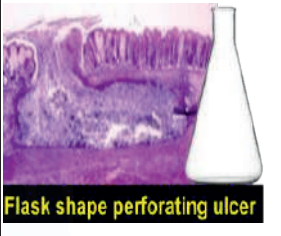


PROTOZOAL INFEC Unicellular, microscopic **Rx**

1. Entamoeba, Giardia, Trichomonas-
2. Leishmania
3. Cryptosporidium
4. Toxoplasma
5. Plasmodium

A. ENTAMOEBA **Rx** M + Paromomycin

1. Lives colon- Amoebic — Flask shape perf ulcer
2. MC Rt liver- Amoebic - Anchovy sauce pus
3. Human reservoir- Orofecal -F21/24,N23 Infec stg- 4-nucleus
4. Diag stg- Trophozoite- Pseudopodia- Eats


Flask shape perforating ulcer

B. GIARDIA -F7/21/23/24,N20 **Rx**

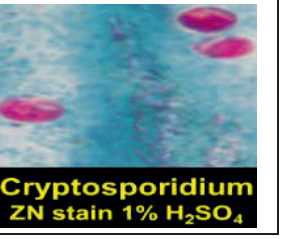
1. Lives duodenum- Watery , steatorrhea=fat stool
2. Human reservoir- Orofecal- Infective stg- 4-nucleus
3. Diag stg- Leaf/pear/monkey/tennis racket trophozoite

C. ACANTHAMOEBA & NAEGLERIA

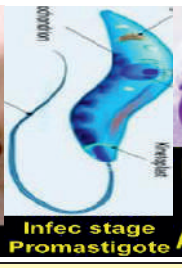
1. Meningoencephalitis- B
2. Contact lens keratitis (rare) -F7/25

D. CRYPTOSPORIDIUM, CYCLOSPORA, CYSTOISOSPORA

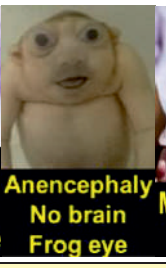
1. MC diarrhea in HIV- Cryptosporidium -F21/22/23/25, N19
2. Infec & Diag stg- Oocysts- Kinyoun cold ZN stain 1%H_2SO_4


Cryptosporidium
ZN stain 1% H_2SO_4

Kala azar

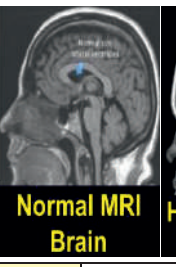

Infec stage Promastigote

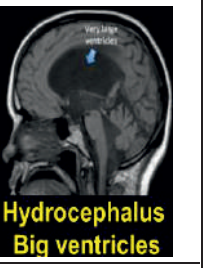
Diag stage LD body= Amastigote


Anencephaly No brain Frog eye

Microcephaly Small head

Hydrocephalus


Normal MRI Brain


Hydrocephalus Big ventricles

E. LEISHMANIASIS = KALA AZAR -F22/23/24,N18/23

1. Vector Antigen Rx

2. Reservoir- World- Zoonosis, India-

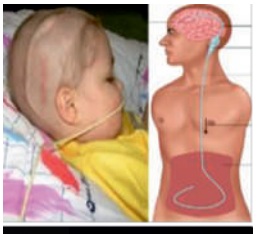
3. NVBDCP- Kala azar elimination target 2030

4. Inf stg- Promastigote, Diag stg-

5. Leishmania donovani- Visceral Leishmaniasis (VL) =

= Black/Dum dum fever + ↓CMI + Hep

6. Leish tropica- Cutaneous Leishm (CL)=


Hydrocephalus CSF= VP shunt

F. TOXOPLASMA (TORCHES) -F23/24/25,N18/19/25

1. Triad- HIT- Hydrocephalus + Intracerebral calcification + torchlight chorioretinitis 2. Tachyzoite, bradyzoite, oocyst

3. Cat feces 4. Sabin Feldman dye test 5. Rx

G. HYDOCEPHALUS ↑CSF build up ↑ventricle size -F 21/24,N20

MC Child- Neonatal sepsis MC Congenital-

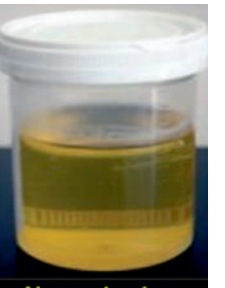
H. MALARIA = JUNGLE/ MARSH/ SWAMP FEVER -F1/25,IN19

1. India- stage→ Elim target 2030

2. Malaria mild- Old malaria, ↓G6PD, thalassemia,

3. Vacc Mosquirix=RTSS, Matrix M=R21, India AdFalcivax 25

Plasmodium	Malaria	Fever	Dots
P. falciparum Most common (53%) Most severe/virulent Most complicated Most resistant Infects all RBCs	Malignant tertian = Severe complicated = Pernicious malaria Banana gametocyte Accole=Applique form Artemisinin combo Rx	High fever + chills+ rigors Every 3rd day / Alternate day Gap 48hrs Day1- Fever Day2- No Day3- Fever	Maurers dots Recrudescence due to drug resistance
P. vivax 2nd common (47%) Infects young RBCs	Benign tertian = Simple uncomplicated = Oval gametocyte Chloroquine+ Primaquine		Schuffner dot Relapse due to liver hypnozoite. Cyclopropagative malaria
P. ovale	Ovale tertian malaria		~~James dot~~
P. malariae Max incubation period	Quartan malaria	Every 4th day Gap 72 hrs	~~Ziemann dot~~
P. knowlesi Monkey malaria	Quotidian malaria	Everyday Gap <24 hrs	~~Sinton and Mulligan dot~~



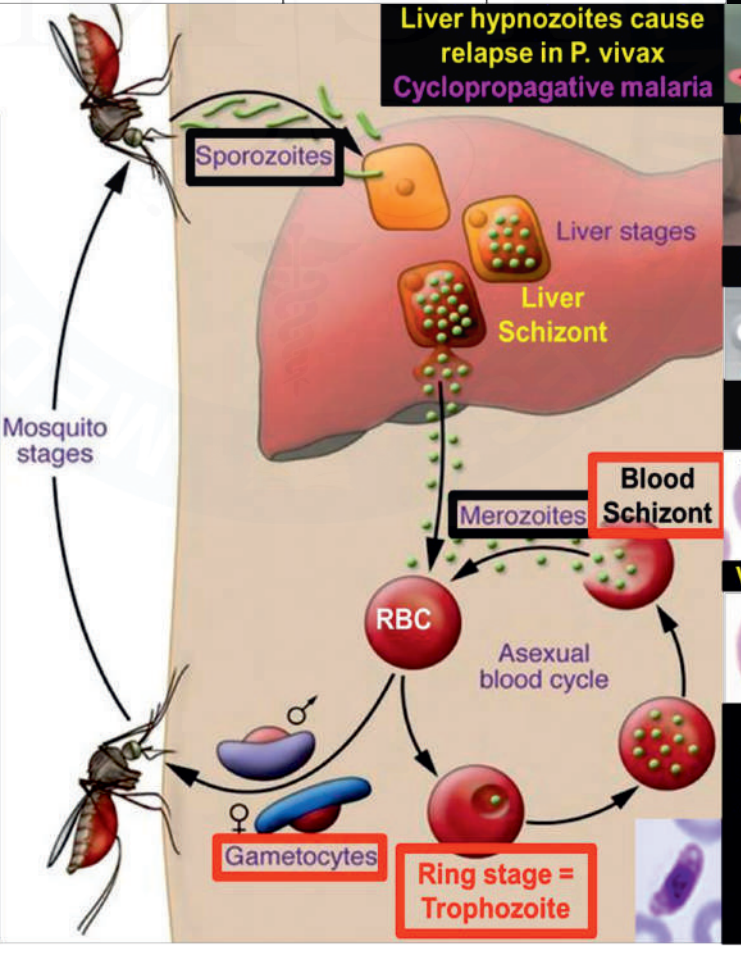
Normal urine

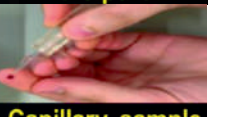
Hemoglobinuria
Cola color urine
Black water fever
P. falciparum

VECTOR- ♀ Anopheles
1. Urban A. stephensi
2. Rural A. culicifacies

INFECTIVE STAGE
1. Mosquito to man Sporozoite
2. Man to mosquito Gametocyte
3. Blood txn / Vertical Merozoite/Trophozoite

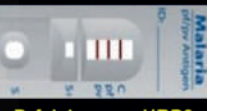
DIAGNOSTIC STAGE
1. Schizont- Not in falci
2. Ring form / stage = Trophozoite
3. Gametocyte



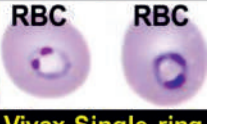


Capillary sample

Peripheral blood smear



P. falciparum- HRP2
All plasmodia- pLDH



Vivax-Single ring

Falciparum-
Multiple rings
Accole forms
Gametocyte
1. Vivax- Oval
2. Falciparum- Banana shaped

1. → Target

Most important indicator- Indicates malaria burden

2. - Child splenomegaly- Indic endemicity

3. Infant parasite rate -F21- Indic recent malaria transmission

4. Annual blood exam rate -N20- Indicate operational efficiency

C/F Malaria -F1/24 3 stage- Cold 1hr Hot 1-4hr Sweating 1hr

1. High + chills=shivers + rigors alternate/3rd day

2. Hemolytic

3. Coma + seizure -F1/23

4. Hemoglobinuria- Blackwater = Cola/dark color urine

Diag of Malaria Capillary sample- Peripheral blood smear

1. Leishman-Giemsa, Jaswant S Bhattacharya stain (MC India)

2. Thick smear- Diagnostic, Thin smear- Identify

Rx 1. Drug of choice/ Clinical/ Symptomatic Rx

2. Radical Rx/ Relapse Rx/ Hypnozoiticidal

3. Stops transmission/ Gametocidal

4. Uncomplicated P. vivax relapsing cyclopropagative benign tertian malaria (Hypnozoitical)

5. Complic P. falci malignant tertian pernicious recrudescent Malaria- Artemisinin Combo Rx ACT + (Gametocidal)

North-East ACT-AL -F1/26

Others- ACT-ASP- Artesunate + Sulfadoxine-Pyrimethamine

-F 21/23/24/25/26, N20/21/23/24, I22

6. All trimester- Chloroquine, Artesunate, Quinine + Clindam

7. 1st trim falciparum- ACT AL (WHO), Quinine + Clinda (India)

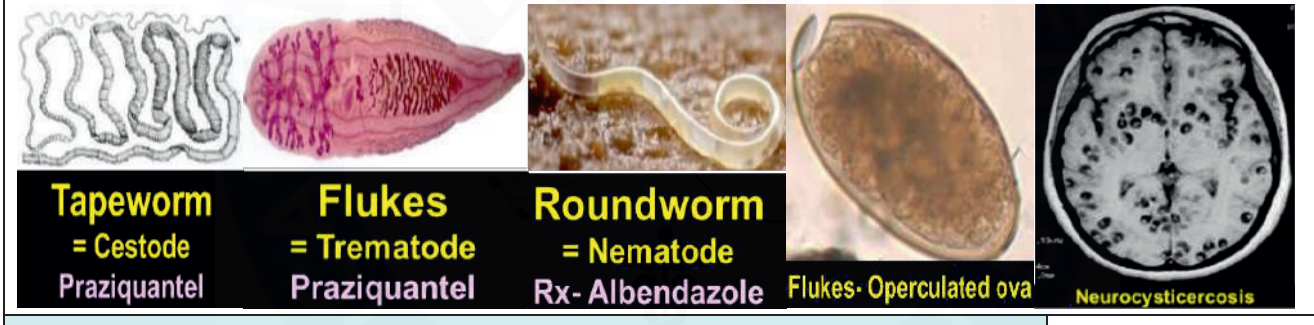
8. Rx failure- Quinine+ Doxycycline

9. Chemoprophylaxis -F19/20

10. Travel prophy- Doxycycline 100mg 2day before- 4wks aft

.P falci- Severe recudescence pernicious resistant, all RBC, Maurer dot, HRP2 antigen, multiple signet ring, accole/applique form, banana gametocyte, ACT

.P vivax- Mild, cyclopropagative, relapse liver hypnozoite, Schufner dot, pLDH + aldolase antigen, single ring, oval gametocyte, Chloroquine + Primaquine



HELMINTHIC INFEC Multicellular, macroscopic Rx

1. Dog tapeworm (tw) = Hydatid cyst 2. Beef tw = Taenia saginata

3. Fish tw = Diphyllobothrium- Megaloblastic anemia -F7/25

4. Dwarf tw = Hymenolepis nana 5. Rat tw = Hymenolepis diminuta

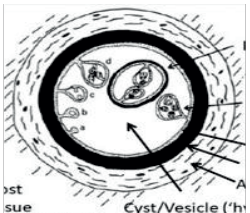
6. Pork tw = Taenia solium (Adult-Taeniasis, Larva- Neurocysticercosis)

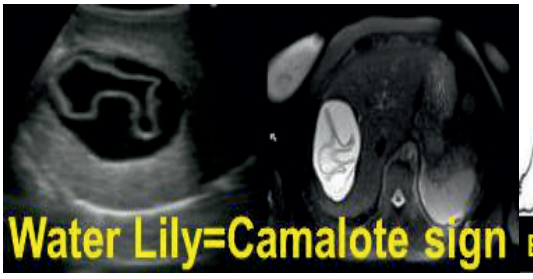
7. Liver fluke, Intestinal fluke, Lung fluke, Blood fluke -F6/19

8. Round worm=Ascaris Strongyloides 9.Guinea worm 10.Trichinella

11. Hook worms = Necator > Ancylostoma 12. Pin worm=Enterobius

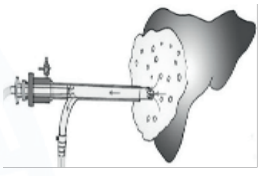
13. Filaria worms = Wuchereria > Brugia 14. Whip worm = Trichuris





A. NEUROCYSTICERCOSIS=TAENIA SOLIUM- MC C/F

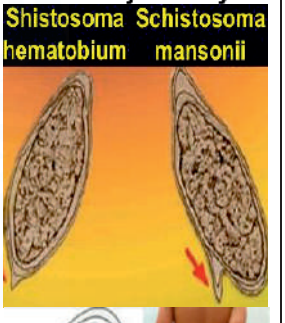
1. MC CNS parasite-F21/24,N19/24 . MC adult onset seizure= epil
2. Inf Stg- Stool ova→ Soil (not pork)- Veg- Cabbage, Lettuce
3. Diag Stage- Brain cysts- CT/MRI-
4. Rx



B. HYDATID CYST = ECHINOCOCCUS = DOG TW -F20/21/24

1. MC 75% -Single cyst, Multiple Alveolar= Metastatic
2. Brood capsule, hooklet, most active
3. Rx -N18/23

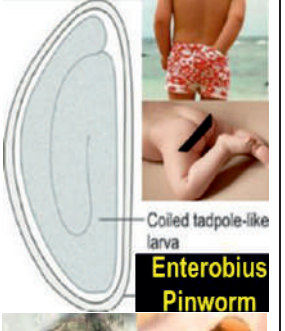
Puncture + Aspirate + Inject saline scolicidal + Reaspirate



C. LIVER FLUKE = LF -N23 a. Sheep LF= Fasciola hepatica

b. China LF= Clonorchis c. SE Asia LF= Opisthorchis

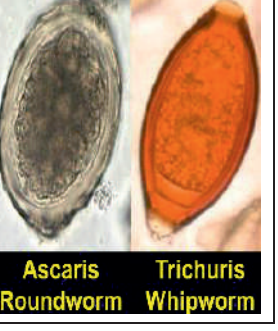
. China + SE Asia LF-



D. BLOOD FLUKE = SCHISTOSOMA -F1/23 Snail

E. ENTEROBIUS = PIN/ THREAD WORM -F1/21,N24,I25

1. Nocturnal eneuresis = Bed wetting +
2. Sample- NIH swab, Scotch/
3. Infec & Diag stage- D shaped planoconvex ova -N25



F. ASCARIS = ROUNDWORM- MC & largest- Orofecal

G. TRICHURIS = WHIPWORM -F6/20 Barrel shape ova

H. HOOKWORM = Necator small (MC) > Ancylostoma large

1. anemia→ Anemia Mukt Bharat Program

→ Deworming albendazole 400mg HS + iron folic acid in-F1/26

preg+ lactating+ child (blue/pink tab) + delay cord clamp

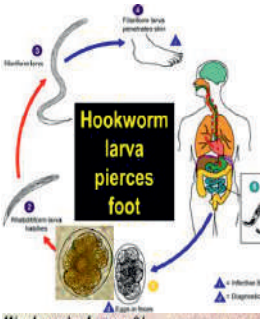
2. >300 ova/gm stool- ↑problem

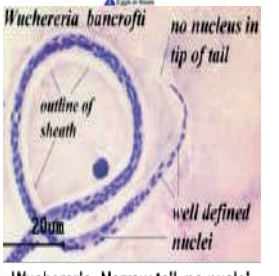
I. **STRONGYLOIDES** No ova-F1/26,N21 Only Larva- Rx

. Both hookworm + Strongyloides pierce

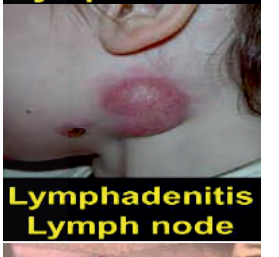
K. LYMPHATIC FILARIASIS -F21/23/24,N21/23

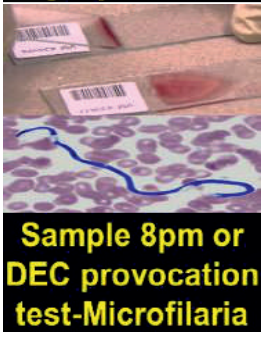
1. MC (99%)
2. Vector injects infective stage = L3 larva
3. Diagnostic stage
4. C/F- Hydrocoele(MC), lymphangitis/adenitis, Elephantiasis, Chyluria (milky lymphatic urine), chylous ascites
5. India 40%world filaria- Elimination target 2027
6. Microfilaria rate- Indicates prevalence- Most sensitive
7. Sample after otherwise
8. Mass drug Rx 6 mnthly-N20 DEC + Albendazole + Ivermectin
9. DEC = DEC fortified salt
10. Elephantiasis- Disability/social stigma Rx-Debulking surg + lymphovenous anastomosis + compression stockings





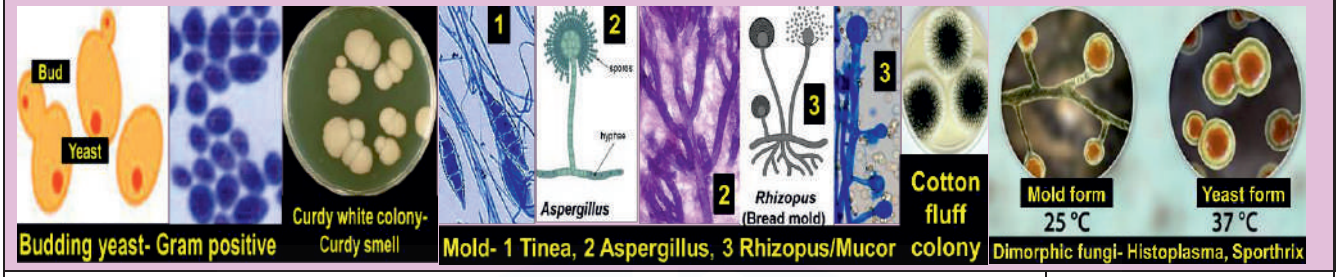
Wuchereria- Narrow tail, no nuclei
Brugia- Broad tail, many nuclei





Sec V : Fungology = Mycology

16 Topics, 3 pages, 28 PYQs



A. YEAST- Unicellular + microscopic + Gram +ve- C

B. MOLD- Multicellular (branching hyphae) + macroscopic

. 4 stains- KOH, Lactophenol cotton blue, PAS, Grocott -N18,19

1. Thin septate hyphae- -F7/24,1/21

2. Thick sep hyp- Acute angle branch -F23/25

3. Thick asep hyp- Right ang br -F6/21/22

C. DIMORPHIC FUNGI- Sporothrix, Histoplasma (Bird excreta

37^0C in human-F21,23,N25 25^0C in

D. Deuteromycetes- Asexual Fungi imperfecti- Most

E. Fungal Culture -F25,N19 SDA pH5.6= S

1. Yeast- Curdy white colony 2. Mold- Cottony fluffy colony

F. Rx of fungi-F1/23,N19 . Severe . Mild

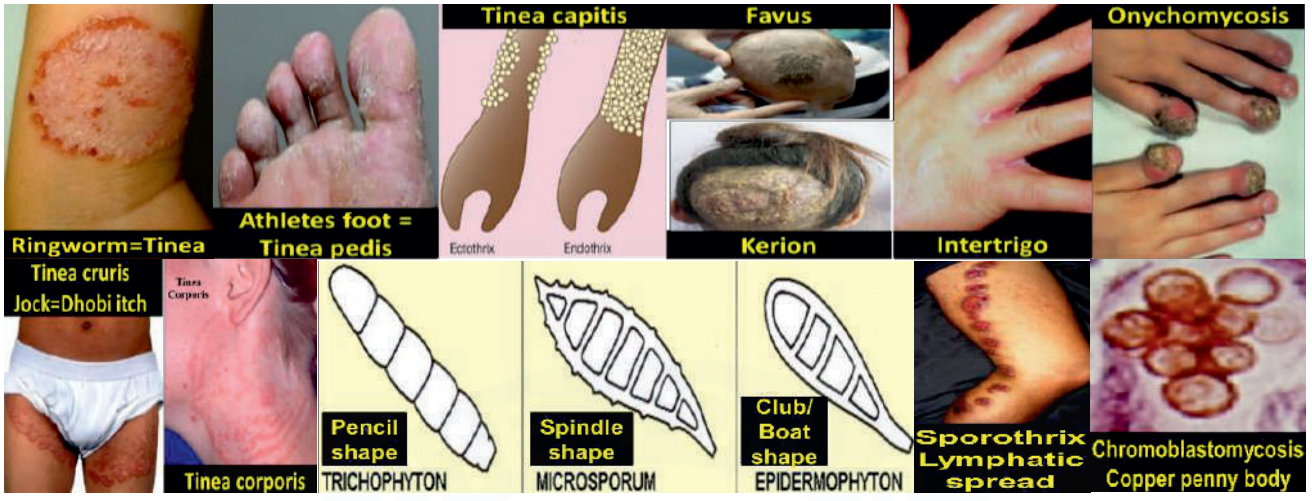
Flu/ Keto/ Mi/ E/ Vori/ Lulli/ Itraconazole tab/ cream/ shampoo

Not antifungal- Metronidazole,Albendazole,Cotrimoxazole,Pantoprazole,Aripiprazole

1. Mucormycosis, Cryptococcal meningitis, Leishmania -N22/23

2. Aspergillus -F25,N20

3. Pneumocystis 4. Rhinosporidium



G. RINGWORM = Tinea = Dermatophytosis -F20/22/23/24,N21/22/24

1. MC superficial mycosis-
2. MC Adult- Tinea
3. MC Child- Tinea
4. Steroid Rx- Tinea

a. Pencil shape (Thin) Skin- Hair- Nail

b. Spindle shape (Mid fat) Skin- Hair ---

c. Club/Boat shape (End fat) Skin ---- Nail

H. MALASSEZIA -F1/6/22- Tinea versicolor,

I. SPOROTRICHOSIS -F23/24/25- Rose Thorn/

Sporothrix- Lymphocutaneous linear infection

J. CHROMOBLASTOMYCOSIS- Warts- Verrucous dermatitis -N22

Wood workers- Sclerotic= Medlar=

K. CRYPTOCOCCUS - MC in HIV

. Gold std- Cryptococcal antigen test -F20/21/24/25,N19

L. PNEUMOCYSTIS JIROVECII - MC fungal pneumonia in HIV

. stain- Cup-saucer, pingpong ball -F21/24/25,N23

M. MUCORMYCOSIS = Zygomycosis = BLACK FUNGUS

. Mucor, Rhizopus . ↑Covid, -F21/22/23/24/25,N22,I25

N. CANDIDA ALBICANS- Thrush- Curdy white curdy smell

1. MC fungal inf, MC opportunistic fungus-F20/22/24/25,26,N19/20/22

2. Germ tube test- Gram +ve budding yeast + pseudo

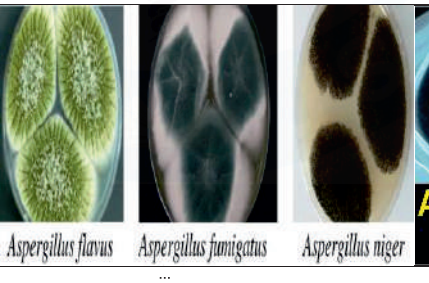
O. ASPERGILLUS = Storage fungus -F7/25/24/21/22,N20/23

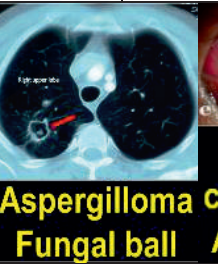
1. MC otomycosis, MC fungal corneal ulcer (Tree twig injury)

2. Aspergilloma= Fungal ball . Diag- IgG +

3. Groundnut/ maize/ spices- - Liver ca ↑AFP

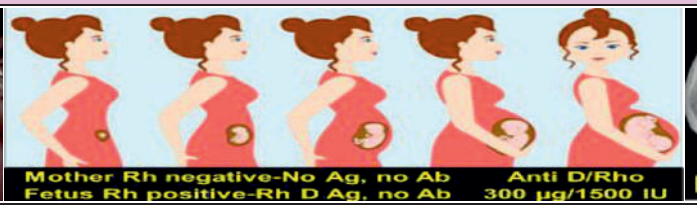
P. CLAVICEPS = Field fungus-F21/25 Wheat, rye, millet

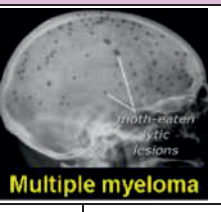




Sec VI : Immunology & Vaccinology

9 Topics, 6 pages, 56 PYQs





A. ANTIGEN = Ag- Comes from pathogen

1. Self antigen- No Attack, Non-self Ag- Attacked by Antibody

2. Heterophile=Shared Ag-F24/25 Cardiolipin Ag- VDRL test- Sy

3. Fetal Ag-N20 Erythroblastosis fetalis in

Rx post delivery, abortion, trauma -F1/24

4. Sequestered= Hidden Ag-F19,22

a. Lens crystallin Ag- Trauma to 1 eye- Sympathetic

b. Testis Ag- Trauma to 1 testis-

B. ANTIBODY = Ab- Formed by body -F19/21/22/23/24/25,N19/20

. Ab 5types- G>A>M>D>E- Production/half-life/concentration

1. IgG= Lightest= MC antibody-80%- ABO/Rh incompatibility

. Cross . IgG_3 most potent complement activ

2. IgA . Serum IgA_1- Monomer- valency -F6/21,IN25

. Secretory IgA_2-Saliva, milk- Dimer-valency2-Local immunity

3. IgM=Millionaire=Heaviest Ab-1st Ab made in infec and fetus

Valency . Opsonization-IgG (MC,↑specific)>IgM (↑Effective)

4. IgD- Acts as B cell receptor, IgD/ IgE dont fix complement

5. IgE= Reagin/Homocytotropic Ab- hypersensiti

6. **Multiple Myeloma** -F21/22/23/24/25 MC 1^0

a. **Moth eaten** b. IgG+IgA light chain, no heavy chain

c. **Bence Jones** proteinuria d. Protein electrophoresis-

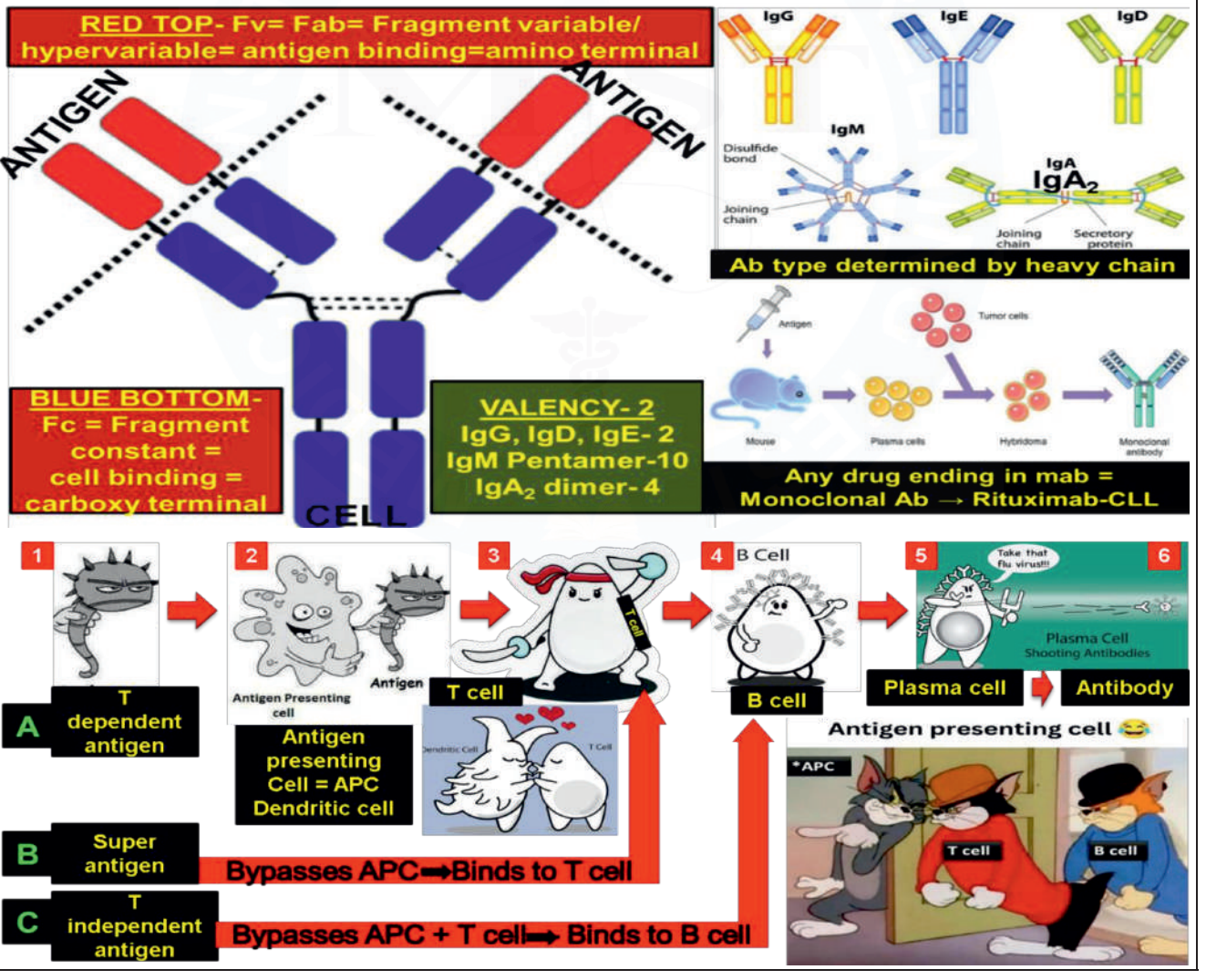
C. IMMUNE RESPONSE (IR) **Prozone** ↑↑Ab, **Postzone** ↑↑Ag

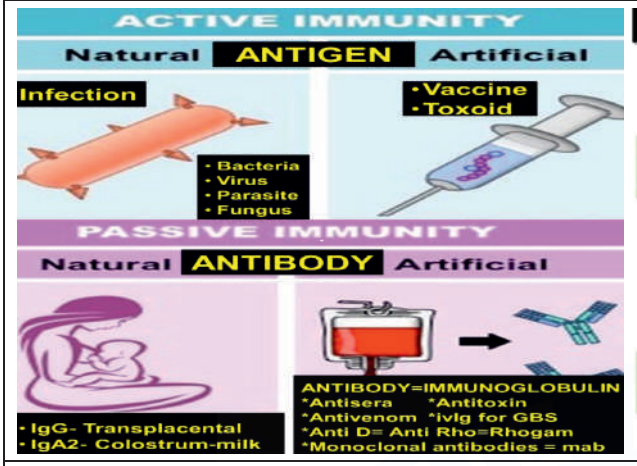
. 1^0 IR- IgM- Naïve B cell- Slow, Sluggish, Short lived -F19,20

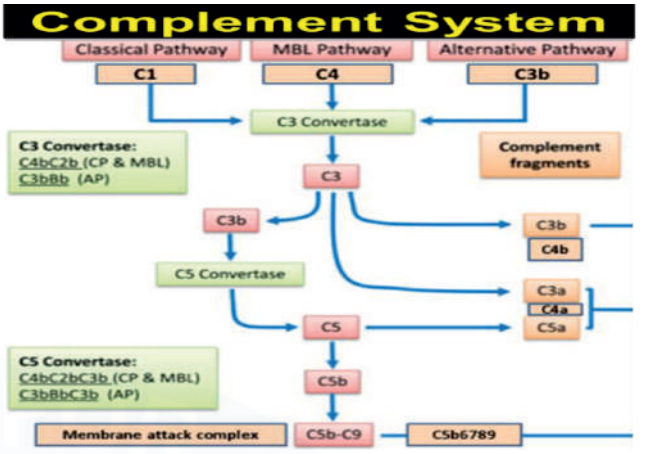
. 2^0 IR- IgG- Memory B cell- Prompt, Powerful, Prolonged

D. CYTOKINE = Cell signal molecule -F7/23,N19/20 Interferon,

Interleukin, TNF α- Key cytokine in sepsis, diabetes, inflamm







E. COMPLEMENT Supports immune system -F6/19,N18/25

1. Opsonins- C3b, C4b 2. Anaphylotoxins- C3a, C4a, C5a

3. Last step- MAC = Membrane Attack complex = C5b6789

4. ↓MAC- 4. No C3

F. ORGAN TRANSPLANT= Tx -F19/20/21/22/23/24/25, N18/19/20

1. Solid organ Tx (Donor graft→ recipient) MC Kidney > liver

2. USA2024-1st kidney xenotransplant . Skin graft > Kidney Tx

3. Matching ↓Tx/Graft rejection=failure Rx- Immunosuppress

4. Bone marrow Tx= BMT/HSCT- Leukemia, Lymphoma, SCID

5. Graft vs Host Disease in BMT Rx- Immunosuppression

G. IMMUNODEFICIENCY=id -F7/24 1. 1^0 id- From birth 2. MC ↓IgA

3. Most severe- SCID (X linked)= No T & B cell- No Ab

4. 2^0 id- HIV, splenectomy, ca, pancytopenia, steroid, organ tx

5. Physiological id - Neonate, elderly, pregnant

6. 1^0 lymphoid organ - Thymus, bone marrow -F7/24

7. 2^0 lymphoid organ - Lymph node, spleen, tonsil

H. HYPERSENSITIVITY ↓↑response by normal immun system

4 type ACID- 1 Allergy, 2 Cytotoxic, 3 Immune cplx, 4 Delayed

Allergy Type1 IgE→ Mast cell→ Histamine→ Allergy, asthma, atopy, urticaria, vasodilation, bronchoconstriction

1. Food/pollen/dust/drug allergy MC -N20/21
2. Anaphylaxis→5min death-F22/23 Rx- Adrenaline(1st), cetrizine

Cytotoxic Ty2 IgM/IgG→Blood- Anemia-Hemolytic/pernicious

1. Penicillin allergy- . Penicllin induced anemia-F7/23
2. ABO/Rh incompatibility/Erythroblastosis -F19/22/24
3. Stimulatory Ty5- Subset of Ty2-F7/24 Graves dis,
4. RHD-F23,N19/22, Goodpasture syndrome, pemphigus vulgaris

Immune Ag-Ab Complex Ty3 -F19,22,23,26

1. Hyperacute graft reject (Type 2/3), PSGN, Lepra reaction
2. Arthus rxn- Farmer lung, Serum sickness, autoimmune dis

Delayed/ Cell mediated Ty4 Immune Inflamm-F21/23/26 CD_4/CD_8

1. Test- Tuberculin, Lepromin . Granuloma-TB
2. Ac/subac/chr graft rejection- Type 2/4, GVHD
3. SJS, IBD, contact dermatitis, Elephantiasis, Lepra rxn

. BRIDGE (Boosting Routine Immunization Demand Generation & Expansion)- 5 Saal 7 Baar

I. VACCINOLOGY India- World largest vacc program

1. Universal Immunization Program 1985- 12 infec→ Latest PCV

2. Intensified Mission Indradhanush 2014 5.0 -N20- 12th phase

3. MC child vacc failure- Persistent maternal Ab

4. Vacc C/I a. Immunocomprom/Preg/SCID -F1/25

b. CNS dis- Pertussis c. ↑Fever d. Age >1yr

e. Anaphylaxis- Next dose f. Lactation-

6. Vacc given in disaster-

1. Open vial 4wk policy- PPPJRV- Pentavalent, Polio, PCV, JE killed, Rotavac, Vit A -F7/25 2. No open vial policy - BJMRY- BCG, JE live, Measles-Rubella, Rotasil, YF -F23,7/25,N22

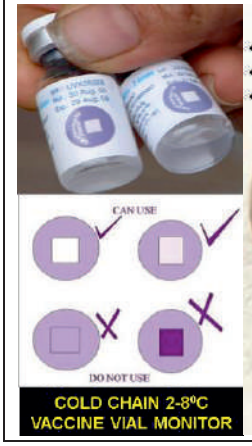
1. Vaccine sensitive to both heat and cold

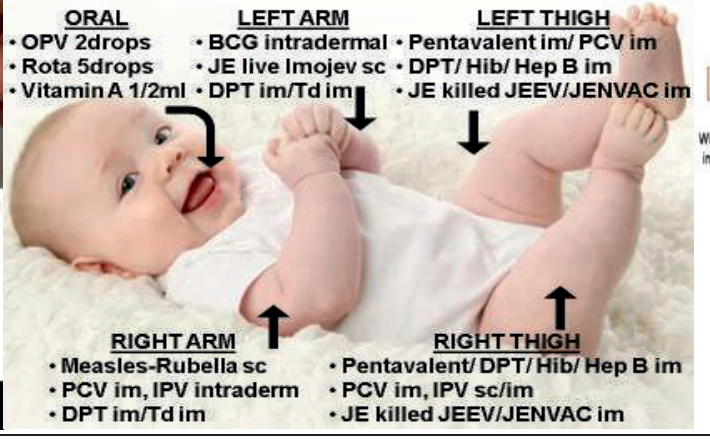
2. Heat

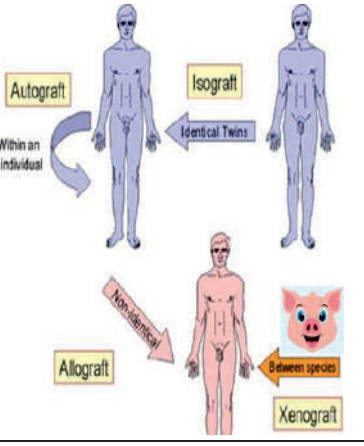
3. Cold Hepatitis B > Pentavalent

1. Child received all vacc by 1yr- FIC plus

2. Child not given pentavalent by 1yr







Vacc	Birth	6w	10w	14w	9m	18m	5y	10y	16y	Preg
Vit K	•									-
HepB	•									Rec
BCG	•									C/I
OPV	•	•	•	•		•				C/I
Penta valent		•	•	•						C/I
Rota		•	•	•						C/I
fIPV		•		•	•					-
PCV		•		•	•					-
MR					•	•				C/I
JE					•	•				-
Vit A					•	•				-
DPT						•	•			DaPT
Td								•	•	••
Rubella in teenage girls to ↓Congenital Rubella									•	C/I
HPV & TCV under planning, All live vacc C/I in preg exc Yellow fever										
Pentavalent vaccine = DPT + Hepatitis B + Hemophilus influenza b										
OPV 2drop, Rota 5drop, VitA 2 lakh IU 6m till 5y (9dose/17 lakh U)										

Pathogen	Vaccine	Virus	Vaccine
Neisseria meningitides	Capsular A,C,Y,135	Chickenpox	Oka
Hemophilus influenza	Capsular	Shingles	Shingrix
Strep pneumoniae	Capsular PCV13, Capsular PPSV23	Smallpox Mpox	Jynneos, Imvanex, Imvamune
Diphtheria	Toxoid Park William 8	Measles	Edmonston-Zagreb
Pertussis	Toxoid, acellular	Mumps	Jeryll-Lynn
Tetanus	Toxoid	Rubella	RA 27/3
Tuberculosis	BCG Danish 1331	Polio	Sabin oral, Salk killed
Salmonella	Ty21a oral, Typhim Vi	HPV	Cervarix, Gardasil, Cervavac Indian
Malaria	Mosquirix, Matrix M	Influenza	Live/killed, Nasovac
Japanese encephalitis	Imojev SA 14-14-2 Killed Jeev, Jenvac	Rotavirus	Rotavac, Rotatec, Rotarix, Rotatec
Hepatitis A,E	Live	Dengue	Dengvaxia
Hepatitis B	Recombinant vacc	Yellow fever	17D
Rabies	Rabipur, Verorab	Chikungunya	Ixchiq
RSV- Arexvy, Abrysvo, m-Resvia			

7 Feb - Rose Day
8 Feb - Propose Day
9 Feb - Chocolate Day
10 Feb - Teddy bear Day
11 Feb - Promise Day
12 Feb - Hug Day
13 Feb - Kiss Day
14 Feb - Valentine's Day
15 Feb - Slap Day
16 Feb - Gift Return Day
17 Feb - Kick Day
18 Feb - Quarrel Day
19 Feb - Try to Convince Day
20 Feb - Breakup Day
21 Feb - Find Another-one Day

ROSE DAY 1. Rose spots- Typhoid = Enteric fever -N24
2. Rose Gardener/Thorn disease-F23/24/25,N20- Sporothrix
3. Roseola infantum = Exanthem subitum- HHV6
4. Rose Bengal Card test- Brucella
5. Rose position- Adenoidectomy/Tonsillectomy after 6wk
6. Rose pink rash- Erysipelas (Strep pyogenes)
7. Rose sign- Deep Vein Thrombosis = DVT
8. Rose Thorn skip serpiginous ulcer- Crohns disease
9. Rose/ Hay fever- Allergic rhinitis (Type1 hypersens)
10. Sabouraud DextROSE agar- Fungus culture-F7/25
11. Dewdrops on Rose petal- Chickenpox

CHOCOLATE Agar- Hemophilus influenzae -F26/25,N23
2. Chocolate cyst- Ovarian endometriosis -N24,INI25

KISS ing disease/fev- Infectious mononucleosis
2. Kissing Reduvid Assassin bug- Chagas dis
3. Kissing ulcer/lesion/spine/tonsil-When they touch

V-DAY-F1/21- Panton-Valentine Leucocidin- Staph aureus

SLAPped cheek appearance- Parvovirus 5th dis -F22,N19

MASTER MIND MAP - 90 LOGOS

1. GENERAL MICROBIOLOGY 62 PYQs My Score ____/ 62, ____/62

1. Tumbling and falling leaf motility -F 7/23 A. Treponema, Proteus B. Cholera, E. coli C. Listeria, Giardia D. Protozoa, prion	2. India Ink=Nigrosin stain -F1/23/24/25, -N19,I24 A. Tuberculosis B. Cyptococcal meningitis C. Candidiasis D. Cryptosporidium
3. Eradicated worldwide -N18 A. Polio, B. Trachoma, C. Leprosy, D. Smallpox	4. TORCHES transmission -N18 A. Intranatal B. Transplacental C. Post natal D. Breastfeeding
5. Chlorine demand, end user chlorine? -N18 A. Horrock's test, 0.5 ppm B. Anton test, 1 ppm C. Weil Felix test, 5 ppm D. Neils test, 10 ppm	6. Spill, linen, clothing treated by -F6/20,N20,22 A. Dettol B. 1% Sodium hypochlorite X 20min C. 1%Silver nitrateX10m D. 1%ChlorhexidineX1hr
7. Syringe and needle in -F 1/24 A. Red, white B. Red, Red C. White, Blue D. Yellow, yellow	8. Insulin & tuberculin syringe in -F 1/24 A. Red B. Yellow C. White D. Blue
9. Snakes and scorpions have -NEET 2019 A. Cytotoxins B. Superantigen toxins C. Neurotoxins D. Bacteriophage toxins	10. Antimicrobial susceptibility test? -N19,20,21 A. Chick Martin test B. Neil Mooser test C. ONPG test D. Kirby Bauer test
11. Selective media for tuberculosis -NEET 2019 A. Blood agar B. Lowenstein Jensen C. Chocolate agar D. Loeffler's serum slope	12. Test for sterilization -N18,19 A. Standard agglutination B. Monospot B. Sabin Feldman Dye D. Rideal Walker
13. Exotoxin& spore produced in phase-F19,21 A. Lag B. Stationary C. Log D. Decline	14. Culture media sterilized-F18,22 A. Autoclave B. Hot air oven C. Ethylene oxide D. Gamma rays
15.Left over food,metal implant-F1/24A.Red,Red B. Black, Blue C. Green, green D. Red, White	16. Blood bag, vacutainer-F7/22/25 A.Red, Red B. Black, Blue C. Yellow, Red D. Yellow, Yellow
17. N95, oxygen and surgical mask discarded in A. Red, Red, Red -F20 B. Yellow, Red, Yellow C. White, red, blue D. Yellow, Yellow, Yellow	18. Cotton, gauze, plaster in -F 7/24, 8/20 A. Blue, compost B. Black, municipal landfill C. Yellow, incineration D. Red, microwave
19. Red, blue container is for ____ -NEET 22 A. General and sharp B. Rubber and metal C. Plastic and glass D. Both B and C	20. Diagnosis of DNA and RNA viruses-F1/25,7/24 A. Compound microscopy B. India Ink stain B. Culture on blood agar D. PCR & RTPCR
21. Foley catheter, abortus -N18 A.Blue, White B. Black, Blue C. Green, Yellow D. Red,Yellow	22. First test to be done from CSF sample -N 21 A. Cytology, B. Stain, C. Glucose, D. Culture
23. White, yellow for __ waste -F1/24 A. Dry, wet B. Sharp, incinerable C. Plastic, rubber D. None	24. Gloves with & without blood -N19 A.Red, Red B. Black, Blue C. Yellow, Red D. Yellow, Yellow
25. Quellung reaction except A. Meningococcus B. Staph C. Pneumococcus D. Hemophilus -F24	26. Transduction is bacterial attack by A. Fungus B.Protozoa C.Macrophage D.Bacteriophage -F1/26
27. Condom -Fut Q A. 1^0 B. 2^0 C. 3^0D. Primordial	28. iPill -Fut Q A. 1^0 B. 2^0 C. 3^0 D. Primordial
ANSWERS 1C, 2B, 3D, 4B, 5A, 6B, 7A, 8C, 9C, 10D, 11B, 12D, 13B, 14A, 15B, 16C, 17B, 18C, 19D, 20D, 21D, 22D, 23B, 24A, 25B, 26D, 27A, 28B	

29. Expired medicines, disinfectants in-F1/26/24 A. Red, microwave B. Yellow, incineration C. White, compost D. Blue, secured landfill	30. Gram negative cell wall has -F 6/22 A. Mycolic acid B. Thick peptidoglycan C. Teichoic acid D. Lipopolysaccharide
31. RNA made from DNA -N 20 A. Transversion B.Transcription C.Transformation D.Translation	32. Protein made from RNA -N 20 A. Transversion B.Transcription C.Transformation D.Translation
33. Intra-operative infection prophylaxis -N19 A.Linezolid B.Cefazolin C.Vancomycin D.None	34. Plasmid transfer -F 1/24,6/19 A. Conjugation B. Transformation C. Transduction D. None
35. Antibiotic inhibiting cell wall A. Penicillin B. Ciprofloxacin C. Dapsone D. Rifampicin -N17	36. Not neurotoxin -F7/24,1/26 A. Tetanospasmin B. Botox C. Krait venom D. Diphtheria toxin
37. WHO labelled PHEIC in 2024 A. Covid B. Ebola C. Nipah D. Mpox -FutureQ	38. Toxins increasing cAMP are all except -F 19 A. Cholera, B. Rotavirus, C. Tetanus, D. ETEC
39. Lactose fermenter seen A. Chocolate agar B. Blood agar C. CLED agar D. LJ agar -N20	40. Most potent disinfectant (sporicidal) -N22 A. Ethanol B. Dettol C. Soap D. H_2O_2
41. Vibrio cholera has _____ motility -F 7/24 A. Swarming B. Darting C. Gliding D. None	42. Not included for transfusion safety -NEET18 A. Malaria B. Hepatitis-B C. Hep-C D. Hep-D
43. Antibiotic in penicillin allergy -F 1/25 A. Azithromycin B. Ciproflox C. Both D. None	44. Western blot detects -NEET17 A. DNA B. RNA C. Proteins D. Cell wall
45. Cytotoxic drug in syringe -F A.Yellow B. Red C. White D. Bue	46. Wax,oil,grease sterilized-F6/22 A. Autoclave B. Ethylene oxide C.Hot air oven D. Inspissation
47. Cruetzfeldt Jacob disease -NEET 18 A. Fungus B. Protozoa C. Prion D. Virus	48. H2SO4 %age-Cryptosporidium, Nocardia-N19 A. 40% B. 20% C. 5% D. 1%
49. Pregnant females screened for -NEET 18 A. Fibroid B. Uterine TB, endometriosis C. PCOD D. HIV, Hep B, Hep C, Syphilis	50. Breastfeed is absolutely contraindicated A. Galactosemia B. Prematurity -N18,N24 C. Low birth weight D. Neonatal meningitis
51. MC transmission of HAI -IN20 A. Fomites B. Hands C. Surgery D. Blood transfusion	52. Eliminated from India in 2024 -Future Q A. Polio, B. Malaria, C. Trachoma, D. Smallpox
53. Swarming motility A. Proteus on blood agar B. Cholera in liquid media C. Both -N18 D. None	54. 1st line test for TB & typhoid -N18 A.Both PCR B. Genxpert, blood culture C. Both IgM D. None
55. N95 not recommended for -N19 A. Measles B. TB C. Bird flu D. Diphtheria	56. Thalidomide causes -N18 A. Phocomelia B. Phenylketonuria C. Polycythemia D. Poilo
57. Not contraindicated in pregnancy -INI25 A. CT B. MRI C. Thyroscan D. Angiography	58. Autoinfection -INI25 A. Strongyloides B. Ascaris C. Hookworm D. Filariasis
59. Tetanus toxoid after injury is ___ prevention A. 1^0 B. 2^0 C. 3^0 D. Primordial -N22/23	60. Malaria doxycycline before travel is ___ prev A. 1^0 B. 2^0 C. 3^0 D. Primordial -Future Q
61. Screening -FQ A. 1^0 B. 2^0 C. 3^0 D. Primordial	62. Rehab -Fut Q A. 1^0 B. 2^0 C. 3^0 D. Primordial
ANSWERS 29B, 30D, 31B, 32D, 33B, 34A, 35A, 36D, 37D, 38C, 39C, 40D, 41B, 42D, 43C, 44C, 45A, 46C, 47C, 48D, 49D, 50A, 51B, 52C, 53A, 54B, 55D, 56A, 57B, 58A, 59B, 60A, 61B, 62C	

2. GRAM POS BACTERIOLOGY 78 PYQs My Score ____/ 78, ____/78

1. MC cause of PID in virgin female -N19 A. E. coli B. Staph aureus C. TB D. Leprosy	2. Staph aureus food poisoning in -F19, 7/25 A. 6h B. 8-16h C. >16h -N20/21 D. >24h
3. MC mastitis& breast abscess -F 1/7/22/24/25 A. Staph epidermidis -INI25 B. Strep pyogenes C. Pseudomonas -N19,24 D. Staph aureus	4. Nikolsky and Auspitz sign -F 7/20/24,1/21/25 A. SSSS, Psoriasis B. Endocardits, meningitis C. Both Tuberculosis D. Leprosy, TB
5. Spastic and flaccid paralysis -N19 A. Plague, anthrax B. C. difficile, Erysipelas C. Kerion, favus D. Tetanus, Botulism-F1/25/26	6. MC cause of lobar pneumonia -F1/23/25,N21 A. Strep pneumoniae B. Klebsiella pneumoniae C. Mycoplasma pneumoniae D. Legionella
7. MC and rarest leprosy in India -F 1/24,N22 A. LL, TT B. BT, BB C. LL, BL D. TT, LL	8. MRSA-Drug of choice-N18 A. Penicillin B. Azithromycin C. Ciproflox D. Vancomycin
9. Millet like dots in liver, lungs, spleen A. Pumonary TB B. Potts spine -F 7/24, 6/20 C. Renal TB D. Miliary=Disseminated TB	10. First line test for TB in HIV with TB -F 1/24 A. CBNAAT B. Chest X ray C. Bronchoscopy D. Culture on LJ medium
11. Gram +ve capsulated diplococci -F 1/24 A. Staph aureus B. Neisseria gonorrhoeae C. Strep pneumoniae D. Strep pyogenes	12. Related to vaginal tampons -F7/25,N19 A. Staph toxic shock synd B. Gas gangrene C. Necrotising fasciitis D. Bacterial vaginosis
13. Drugs killing intracellular MTB, killing fast growing MTB, causing orange secretions, causing optic neuritis, causing gout -F 7/24 A. R, E, H, E, Z B. H, Z, R, E, H -F1/25 C. Z, H, R, E, Z D. E, R, H, E, Z	14. Drugs contraindicated in pregnancy, safe in pregnancy, most hepatotoxic, not hepatotoxic, causing pyridoxine deficiency -F 1/25, 22 A. R, Z, R, H, Z B. Z, E, Z, E, H C. Z, H, R, E, Z D. E, R, H, E, Z
15. Cobweb coagulum in TB of -F1/24 A. Uterus B. Spine C. Meninges-N22 D. Kidney	16. Lupus vulgaris-F1/24,N20/25 A. HPV lesion B. STI C. Lesion in acne D. TB of skin
17. All in rheumatic fever except -F 7/23,N19/22 A. Polyarthritis B. Pancarditis C. Hepatitis D. Syndenhams chorea	18. Strawberry tongue, tripod sign in -F 1/23 A. Scarlet fever, epiglottitis B. Orf, TB C. Leprosy, Anthrax D. Dengue, HIV
19. All are beta hemolytic except -N18 A. Staph aureus B. Strep pyogenes C. Strep agalactiae D. Strep pneumoniae	20. Vaccine in splenectomy & cochlear implant A. Diphtheria toxoid -F 7/24 B. Tetanus toxoid C. Pneumococcal capsular vacc D. BCG
21. Oculotoxic A. R B. H C. Z D. E -N19	22. ↑QT interval A. R B. BDQ C. H D. Z -F23
23. Rifampicin dose in TB & Leprosy (mg) -F1/25 A. 100, 100 B. 200, 300 C. 450, 450 D. 600, 600	24. MC nerve damage in leprosy A. Median B. Musician C. Radial D. Tibial -F1/23/25

ANSWERS 1C, 2A, 3D, 4A, 5D, 6A, 7B, 8D, 9D, 10A, 11C, 12A, 13C, 14B, 15C, 16D, 17C, 18A, 19D, 20C, 21D, 22B, 23D, 24B

25. Mascot Sapna -F 1/25, N 8/24 A. Safe sex B. Hansens disease C. Kochs disease D. STI	26. Virchow cell, Fite Faraco stain -IN 22 A. TB B. Impetigo C. Hansens disease D. Anthrax
27. Prosthetic valve infection- Etiology, criteria A. Staph epidermidis, Dukes criteria -F 1/7/24/25 B. Strep pyogenes, Jones criteria -N 8/24/19 C. Strep agalactiae, Ransons criteria D. Staph aureus, Mantrel criteria	28. Rheumatic fever, diagnostic criteria? A. Staph epidermidis, Dukes criteria -N19/22 B. Strep pyogenes, Jones criteria C. Strep agalactiae, Ransons criteria D. Staph aureus, Rose criteria -F 7/23, 1/24
29. Glomerulonephritis-F19/20/21/22/23 A.E.coli B. Chlamydia C. Neisseria D. Streptococcus	30. TB reporting app -F 7/24 A. Aarogya saathi B. Nikusth C. Nikshay D. Nischay -N18
31. Necrotizing fasciitis+myonecrosis+crepitus A. Acute limb ischemia B. Fournier gangrene C. Meleney gangrene -N18 D. Gas gangrene	32. MC neonatal pneumonia and sepsis -N19 A. Strep pneumoniae B. Strep pyogenes C. Strep agalactiae D. Strep viridans
33. MC native valve endocarditis, osteomyelitis A. Staph aureus, Strep pyogenes -N22 B. Staph epidermidis, Strep agalactiae -F 1/7/25 C. Clostridium, Bacillus D. Both Staph aureus	34. Hot potato voice and barking voice-N20,22 A. (Ludwigs angina, tonisllitis) and laryngitis B. (Epiglottitis and quinsy) and croup C. Both epiglottitis D. Both croup
35. Anaerobic bacterial myonecrosis -NEET 21 A. Clostridium perfringens B. Clost botulinum C. Enterococcus fecalis D. Citrobacter koseri	36. Toxin-ADP ribosylation- Elongation Factor-2 A. Cholera B. Diphtheria -F 7/24 C. Tetanus D. Gas gangrene
37. Pseudomembranous enterocolitis -N18,22 A. Clostridium difficile B. Clostrid perfringens C. Clostridium botulinum D. Corynebacterium	38. Pseudomembrane in throat -F19 A. Clostridium difficile B. Clostrid perfringens C. Clostridium botulinum D. Corynebacterium
39. Rx of C. difficile diarrhea -N23 A. Ciproflox B. Azithromycin C. Doxycycline D. Fidoxamicin	40. MC cause of MS -NEET 21 A.RHD B. VSD C. Staph endocarditis D. Strep viridans
41. Black necrotic ulcer -F21/23/24/25/26,I25 A. Keloid B. Eschar C. Wart D. Vitiligo	42. Category A bioterrorism -F22 A. HIV, Nipah B. Anthrax, Botox C. Diphtheria, Tetanus D. All
43. CBNAAT=GenXpert=UDST detects -F 1/24 A. TB only B. TB + Rifampicin resistance C. TB + Leprosy D. TB + RHZE resistance	44. Commensal bacteria in conjunctiva -F 7/24 A. Staph aureus B. Staph epidermidis C. Strep pyogenes D. E.coli
45. Strep pyogenes causes -F1/24 A. Carbuncle B. Food poisoning C. Guttate psoriasis D. None	46. Not in diphtheria -F20/21/22 A. Bull neck B. Pseudomembrane C. Eleks test D. Quinsy
47. MC cellulitis & erysipelas -F20 A. S. aureus B. Strep pyogenes C. E.coli D. Klebsiella	48. MC skin abscess -F20 A. Staph aureus B. TB Cold abscess C. E.coli D. Bacillus
ANSWERS 25B, 26C, 27A, 28B, 29D, 30C, 31D, 32C, 33D, 34B, 35A, 36B, 37A, 38D, 39D, 40A, 41B, 42B, 43B, 44B, 45C, 46D, 47B, 48A	

49. MC in RHD -N18 A.AS B.MR C.MS D.AR	50. TB H_2SO_4% -F22 A.20 B.5 C.1 D.50
51. MC stitch abscess -N18 A.Staph epidermidis B. Staph aureus C. E.coli D. Corynebacterium	52. Babinski sign in___lesion -N21/24 A. UMN B. LMN C. Cerebellar D. Peripheral nerve
53. Bacteriostatic A. R B. H C. Z D. E -N21	54. Unsafe in pregnancy A. R B. H C. Z D. E -N22
55. Chinese fried rice food poisoning -F1/25 A. Staph aureus B. Clostridium perfringens C. Bacillus cereus D. Clostridium botulinum	56. Exfoliatin toxin. Sign +ve -F1/25 A. Auspitz sign B. Nikolsky sign C. Finger sign D. Lhermitte sign
57. Syndrom-Conjunctivitis+Urethritis+Arthritis A. Horners B. WPW C. Reiters D. Serotonin-N19	58. Resistant gene in MRSA,VRSA A. mecA, van A. rpoB, katG C. hox, cmyc -F 1/25 D. ras, Bcl-2
59. Rx-Staph aureus food poisoning-FQ A.None B.Clindamycin C. Cloxacillin D. Coamoxiclav	60. Sydenhams chorea seen in A. Anthrax B. Gas gangrene C. Psoriasis D. RHD -N22
61. Peritonsillar abscess -F17,N20 A. Croup B. Quinsy C. Furuncle D. Cellulitis	62. Acute Laryngotracheobronchitis -F17,N23 A. Croup B. Quinsy C. Furuncle D. Cellulitis
63. Blackhead and whiteheads A. Rosacea B. Acne C. Pityriasis D. Eczema -F 7/23, 1/25	64. Slit skin smear negative leprosy -N22 A. LL B. BB C.TT D. Pure neuritic leprosy
65. Austin Flint murmur A.MS B.AR C.AS D.PS	66.Graham Steel murmur A.MR B.AR C.AS D.PR
67. Asherman syndrome -F7/25/21,N23 A. Spine B. Uterus C. Liver D. Spleen	68. New TB Rx regimen in 2024 -Future Q A. DOTS 2.0 B. ATTK C. BPaLM D. None
69. CAMP and reverse CAMP test -F 7/23 A. Strep pyogenes, Clostridioides difficile B. Strep agalactiae, Clostridium perfringens C. Strep pneumoniae, Clostridium tetani D. Strep viridans, Corynebacterium diphtheriae	70. Double tongue, thumb, steeple sign A. Gas gangrene, quinsy, tonsillitis -F 1/24,N23 B. Ludwigs angina, epiglottitis, croup C. Sore throat, vestibulitis, labyrinthitis D. Otitis media, sinusitis, laryngitis
71. TB pleural effusion, which criteria -N8/24 A. Light B. Amsel C. Ranson D. CRAB	72. Millet like dots in ____ TB -F1/23 A. Cystitis B. Miliary C. Genital D. Spinal
73. MC pneumonia & meningitis -F1/25 A. None B. Klebsiella pneumoniae, Neisseria meningitis C. Mycoplasma penumoniae, E. coli D. Strep pneumoniae, Strep pneumoniae	74. MC otitis media & sinusitis-N19 A. None B. Staph aureus, Clostridium perfringens C. Strep pyogenes, Strep agalactiae D. Strep pneumoniae, Strep pneumoniae
75. Endodarditis, RHD criteria A. Midas, MELD B. Dukes, Jones C. NYHA, CRAB D. None -N19	76. All related to TB except -Future Q A. Putty B. Thimble C. Charcot D. Cobweb
77. Resistant genes for R and H A. mecA, van B. rpoB, katG C. hox, cmyc -F 1/25 D. ras, Bcl-2	78. Panton Valentin leucocidin A. Streptococcus B. Staphylococcus C. Diphtheria D. Clostridium
ANSWERS 49B, 50A, 51A, 52A, 53D, 54C, 55C, 56B, 57C, 58A, 59A, 60D, 61B, 62A, 63B, 64D, 65B, 66D, 67B, 68C, 69B, 70B, 71A, 72B, 73D, 74D, 75B, 76C, 77B, 78B	

3. GRAM NEG BACTERIOLOGY 52 PYQs My Score ____/ 52, ____/ 52

1. Blood culture in typhoid in ___ week A. 1st B. 2nd C. 3rd D. 4th -F 7/24	2. Widal test in typhoid in ___ week A. 1st B. 2nd C. 3rd D. 4th -F 1/20/25, 7/24
3. MC Fournier gangrene-F 1/23/24 A. S. aureus B. Pseudomonas C. E. coli-N22,23 D. Klebsiella	4. Argyll Robertson pupil-F7/23,N18 A. 1^0 syphilis B. Sec syphilis C. Tertiary syphilis D. Glaucoma
5. Not for Helicobacter -N18 A. Clarithromycin B. Dapsone C. Metronidazole D. Bismuth	6. Inguinal bubo -F 1/24/25/26 A. Chancroid, LGV B. HCV, HSV C. HIV, HBV D. Trichomonas, HPV
7. Acute watery secretory diarrhea -F 1/25, 7/24 A. Cholera, ETEC B. Shigella, EHEC C. Salmonella, EIEC D. All of the above	8. Barking cough, whooping cough -F1/21,IN22 A. Croup, pertussis -N20 B. Tonsillitis, quinsy C. Zenker diverticulum D. Pneumonia
9. MC confirmatory syphilis test -F 1/21/22,N19 A. VDRL B. TPHA C. TPI D. DFA	10. Scrub typhus transmitted by -F 1/7/24, INI25 A. Tick B. Fly C. Flea D. Mite
11. Modified Youngs operation-F 7/24 A. Ozaena B. Rhinosporidiosis C. Quinsy D. Rhinorrhea	12. Selective media only for Salmonella A. DCA B. XLD C. TCBS D. Wilson Blair -F 1/23
13. Seafood poisoning -F 7/23,N24, INI25 A. Staph aureus B. Vibrio parahemolyticus B. Bacillus cereus D. Salmonella typhimurium	14. First line of intervention in rice water stool? A. Doxycycline B. Albumin Infusion C. Oral rehydration D. Blood transfusion -N19
15. Painful ulcer+lymphadenopathy-F24/25/26 A. Mycoplasmosis B. Syphilis = Chancre C. Chancroid D. Donovanosis -F 1/22/23	16. Painless ulcer+lymphadenopathy-F24/25/26 A. Mycoplasmosis B. Syphilis = Chancre C. Chancroid D. Herpes simplex -F 1/22/23
17. SAFE strategy is for -F 1/21/24 A. Glioma B. Glaucoma C.Trachoma D. Myeloma	18. Bacteria without cell wall -F19,N18 A. E.coli B. Toxoplasma C. Plasmodium D. Mycoplasma
19. Vacc must for Hajj and Africa travel -N18 A. Covishield, Cholera B. Oka, RA 27/3 C. Meningococcal, 17D D. Influenza, Ebola	20. Wrong about Donovanosis -F22/23, N19,21 A. Leishm donovani B. Granuloma inguinale C. Klebsiella granulomatis D. STI INICET21
21. Epidemic typhus and Scrub typhus-N19,I25 A. Rickettsia typhi, Bartonella henselae B. Rickettsia rickettsiae, Coxiella burnetti C. Chlamydia trachomatis, Mycoplasma D. Rickettsia prowazekii, Orientia tsutsugamus	22. Not seen in syphilis -F1/22,6/21,N18/23 A. Hunterian chancre, palmoplantar rash B. Hutchisons teeth, mulberry molar C. Condyloma acuminata, painful ulcer D. Interstitial keratitis, gumma
23. MC zoonosis in India affects-N21 A. Trucker B. Rice farmers C. Fishermen D. Shepherds	24. MC infec in burns and unsafe ear A. E.coli B. Klebsiella C. Pseudomonas D. Proteus -N19
ANSWERS 1A, 2B, 3C, 4C, 5B, 6A, 7A, 8A, 9B, 10D, 11A, 12D, 13B, 14C, 15C, 16B, 17C, 18D, 19C, 20A, 21D, 22C, 23B, 24C	

25. Typhoid and typhus -N19,INI25 A. Salmonella typhimurium, Chlamydia B. Salmonella enteritidis, Bartonella quintana C. Salmonella paratyphi, Rickettsia D. None	26. Enrichment media in cholera & typhoid A. TCBS, DCA B. Both Cary Blair C. Alkaline peptone water, Selenite F broth D. Venkatraman Ramakrishnan, XLD -N23
27. MC Syphilis test overall, screening, confirm A. TPHA, VDRL, RPR B. RPR, RPR, TPHA C. TPI, DFA, ELISA -N21 D. All FTA-ABS	28. Pyocyanin and Staphyloxanthin pigment A. E. coli, Klebsiella B. Salmonella, Vibrio C. Pseudomonas, S. aureus D. None -F1/25,23
29. Ogawa, Inaba, Hikojima-N18 A. Proteus B. Vibrio C. Camplobacter D. Legionella	30. Hemolytic uremic syndrome A. Typhoid A. Pseudomonas C. Brucella D. E.coli -F1/25,I24
31. Selective media for cholera, shigella A. Cary Blair, VTM -F21,7/24 B. TCBS, DCA C. Cetrimide agar, LJ D. None of above	32. All seen in typhoid except -F 7/24 A. Transverse intestinal ulcers B. Rose spots C. Stepladder fever D. Relative bradycardia
33. Ceftriaxone + Vancomycin is Empirical Rx of A. Pneumonia B. Meningitis -F1/7/25 C. Endocarditis D. Peritonitis	34. MC pneumonia in cystic fibrosis -F1/23 A. Klebsiella pneumoniae B. Strep pneumoniae C. Mycoplasma penumoniae D. Pseudomonas
35. Rats can transmit all except-F1/25 A. Plague B. Leptospira C. Scabies D. Scrub typhus -N24	36. Bloody stools seen in -F1/25 A. Cholera B. Shigella C. Giardia D. Rotavirus
37.O157:H7-N24 A.EPEC B.EIEC C.ETEC D.EHEC	38. Trachoma A.ERCP B.FAST C.SAFE D.None
39. Doderlein bacillus reduced in -N24 A. Peptic ulcer B. Dermatophytosis C. Emphysema D. Bacterial vaginosis	40. Amsel- Nugent related to -F7/231/24/25,N24 A. Bartholin cyst B. Bacterial vaginosis C. Salpingitis D. Trichomoniasis
41. Pseudomonas produces ____ pigment -F1/25 A. Red B. Pink C. Green D. Yellow	42. Widal test related to ____ sign A. Cullen B. Faget C. Babinski D. Raccoon -F1/25
43. Hemophilus influenzae causes -F1/26,7/25 A. Influenza B. Meningitis C. Gastritis D. None	44. Hemophilus ducreyi causes A. Chancre B. Donovanosis-F7/24/26 C. Chancroid D. Piles
45. Painful inguinal swelling after sexual contact A. Yaws-F7/24 B. Bubo C. Chancre D. None	46. TCBS and XLD for-F7/24 A. Cholera,Typhoid B. Shigella,Proteus C. E .coli,Klebsiella D. None
47. Ozaena all except -F24/25/26 A. Empty nose B. Epistaxis C. Merciful anosmia D.DNS	48. Triple and quadruple Rx-F1/7/25 A. E.coli B. Osetomyelitis C. H. pylori D. Proteus
49. A,C,Y,W135 vaccine -N21 A. Gonorrhea B. Meningitis C. Pneumonia D. Infleunza	50. Satellitism -N19 A. Bordetella pertussis B. Neisseria C. Hemophilus influenza D. None
51. Waterhouse Friderichsen syndrome -FutureQ A. Hemophilus B. Neisseria C. Brucella D.None	52. C-13 urea breath test-F1/7/25 A. Mycoplasma B. Klebsiella C. H. pylori D. Alcoholism
ANSWERS 25C, 26C, 27B, 28C, 29B, 30D, 31B, 32A, 33B, 34D, 35C, 36B, 37D, 38C, 39D, 40B, 41C, 42B, 43B, 44C, 45B, 46A, 47D, 48C, 49B, 50C, 51B, 52C	

4. VIROLOGY 84 PYQs

My Score _____/ 84, _____/ 84

1. Owl eye inclusion body, ReedSternberg cells A. HSV, RSV B. EBV, HIV -F 7/24 C. HBV, HCC D. CMV, Hodgkins lymphoma	2. Tzanck smear positive for-F23/24/25,N24 A. Herpes simplex virus B. Epstein Barr virus C. CMV D. Human papilloma virus
3. Drug(s) common for HIV and HepB -F 1/26,7/24 A.Tenofovir B.Lamivudine C.Abcavir D.Both AB	4. Pap smear -F 7/24/26,N24 A. HSV, every 10 yrs B.EBV,every yr C.HPV,every 3y D.CMV,every 5y
5. Burkitts lymphoma cymc gene -F21/24/25/26 A. HHV-8 B. HPV C. HTLV-1 D. Epstein Barr vir	6. Water warts- central umbilication -F7/24/25 A. HSV B. Molluscum contag C. Syphilis D. HPV
7. Diag of Mpox, Influenza, EBV, Ebola -F 7/24,25 A. PCR, RTPCR, PCR, RTPCR B. PCR for all C. Antigen test for all D. Antibody test for all	8. Acute Hep B + high infectivity-F 1/25,23,N24 A. IgM Anti HBc+, HBeAg+ B. HBcAg+, HBeAg- C. IgG Anti HBc+, HBeAg+ D. HBxAg+, HBsAg-
9. Chronic hepatitis negative for HBsAg, HBeAg but positive for Anti HCV. Rx? -F 7/23, 7/24, 1/25 A. Entecavir B. Sofosbuvir, Ledipasvir C. Oseltamivir D. Remdesevir	10. Viral anogenital warts. Diagnosis? -F1/22/24, 7/23, 6/20/21, N18/20/22 A. Chancroid B. Verruca vulgaris C. Condyloma lata D. Condyloma acuminata
11. Virus causing orchitis, vaccine -F 1/7/23,N18 A. Measles, Edmoston Zagreb B. Varicella, Oka C. Rubella, RA 27/3 D. Mumps, Jeryll Lynn	12. Correct Rx except -N21,23 A. CMV- Ganciclovir B. Rotavirus- Ribavirin C. RSV- Ribavirin D. Influenza- Oseltamivir
13. Encephalitis after eating palm fruit in Kerala A.Hanta B.Chikungunya C.Nipah D.Ebola -F1/24	14. Human Metapneumovirus transmission-F Q A. Orofecal B. Resp C. Vector borne D. Vertical
15. Vaccine preventing ca cervix -N22,23 A. Ixchiq B. Shingrix C. mResvia D. Gardasil	16.Rx-Patent ductus arteriosus in Rubella-F1/25 A. Ribavirin B. Dopamine C. Ibuprofen D. None
17. All are correct spots except -F1/24/25/26, N18 A. Roth- Yellow fever B. Koplik- Measles C. Forcheimmer- Rubella D. Rose- Typhoid	18. Hep B marker of vacc & high infectivity -N21 A. HBsAg, HBcAg -F 1/25 B. Anti HBs, HBeAg C. Anti HBe, HBsAg D. Anti HBc, HBeAg
19. Marker seen in both acute and chronic HepB A. HBsAg B. HBxAg C. Anti HBs D. All -F21	20. Kaposi sarcoma is caused by-F7/23/24/25,N23 A. HSV-2 B. HBV C. HIV D. HHV8
21. Amplifier host in Japanese encephalitis -F19 A. Bird B. Man C. Culex D. Pig	22. Imiquimod & podophyllin related to -N22/24 A. EBV B. HSV C. HPV D. HIV
23. Mother with active genital herpes -N21/24 A. FTND B. LSCS C. Forceps D. Ventouse	24. Ramsay Hunt syndrome -INICET 11/24 A. HPV B. HSV C. VZV D. CMV
25. Hematocrit = ESR is important in -N 8/24 A. HIV B. Dengue C. Chikungunya D. Zika	26. MC childood and weaning diarrhea -N23,24 A. Rhino B. Rota C. Adeno D. HIV

ANSWERS 1D, 2A, 3D, 4C, 5D, 6B, 7A, 8A, 9B, 10D, 11D, 12B, 13C, 14B, 15D, 16C, 17A, 18B, 19A, 20D, 21D, 22C, 23B, 24C, 25B, 26B

27. Petechaie in cubital fossa in ____ test -F1/25 A. Schick B. Tourniquet C. Naeglers D. Patch	28. Rehydration is Rx for ____ toxin -N8/24 A. NSP4 B. Botox C. Verotoxin D. None
29. Swine flu pandemic 2009 caused by -N20/21 A. H1N1 B. H3N2 C. H5N1 D. H7N9	30. SA 14-14-2 is for -N24 A. Yellow fever B. Covid C. Nipah D. Japanese encephalitis
31. Ribavirin treated bronchiolitis -N19 A. CMV B. HHV6 C. RSV D. Polio	32. Haemagglutinin, Neuraminidase -N18 A. AIDS B. Influenza C. Pox D. Papilloma
33. Post Exposure Prophylaxis of HIV -N19 A. Tenofovir, Lamivudine, Dolutegravir B. Tenofovir, Lamivudine, Abacavir C. Tenof, Nevira, Efavirenz D. Tenofovir only	34. Wrong disease-etiology pair -N19,22 A. Chickenpox- Pox virus B. Roseola- HHV6 C. Common cold - Rhinovirus D. Erythema infectiosum - Parvovirus
35. Lysogenic and lytic cycle -N18,F1/26 A. HIV B. HPV C. HCV D. Bacteriophage	36. Cephalocaudal rash in -N18,21 A. Measles B. Rubella C. Both A & B D. Mumps
37. Acute flaccid paralysis + Tripod sign -N23 A. Rabies B. Hep D C. Polio D. Varicella	38. Vitamin supplemented in measles -N19,23 A. A B. B complex C. C D. None
39. HIV strategy for pregnant, blood transfusion A. I, IV B. IIA, IIB C. IIB D. III, I -N18	40. Genetic reassortment due to antigenic -N23 A. Drift B. Shift C. Epitope D. Valency -N19
41. Cold sores, dendritic ulcer -F7/24/25/26,N18 A. Chandipura B. Polio C. Zika D. HSV1	42. Herpes zoster rash -F1/26,N23 A. Centripetal B. Centrifugal C. Cephalocaudal D. Dermatomal
43. Post herpetic neuralgia Rx-N22 A.Tramadol B. Ibuprofen C. Paracetamol D. Pregabalin	44. Rotavirus toxin-N19,23 A. Shiga like Toxin B. Verotoxin C. NSP4 D. Labile toxin
45. Aedes transmits all except -N20/22 A. Zika B. Chikungunya C. Filaria B. Dengue	46. Dengue antigen and yellow fever vacc -N18 A.pp65,Oka B.HRP,MMR C.rk39,R21 D.NS1,17D
47. Virus in Africa, not in India -N23 A. Ebola B. Yellow fev C. Both D. None	48. Dolutegravir prevents -N23 A. Integration B. Fusion C. Reverse Transcription D. None
49. MC mortality in pregnancy -N23 A. HepA B. Hepatitis B C. Hepatitis D D. Hepatitis E	50. Bilirubinemia in viral hepatitis A. Direct B. Unconjugated C. Conjugated D. Mixed-N19
51. Sofosbuvir for -INICET23 A. HIV B. Hepatitis C C. Hep E D.Hep A	52. Satellite virus -N23 A. Hepatitis A B. Hep B C. Hep D D. Hep E
53. Skin warts= verruca vulgaris -N18/19 A. HPV B. EBV C. HSV D. HIV	54. Pleomorphic rash seen in -F1/26,N24 A. VZV B. EBV C. HPV D. HSV
55. Chickenpox becomes latent in -F19 A. Dorsal root ganglia B. Spiral ganglia C. Geniculate ganglia D. Ciliary ganglia	56. Chickenpox after reactivation causes -F19 A. Varicella zoster B. Herpes zoster= Shingles C. Herpes simplex D. Herpes oticus -N19
ANSWERS 27B, 28A, 29A, 30D, 31C, 32B, 33A, 34A, 35D, 36C, 37C, 38A, 39D, 40B, 41D, 42D, 43D, 44C, 45C, 46D, 47C, 48A, 49D, 50D, 51B, 52C, 53A, 54A, 55A, 56B	

57. Tenofovir prevents -F1/26,N23 A. Integration B. Fusion C. Rev Transcription D. Translation	58. Lamivudine prevents -N23 A. Integration B. Fusion C. Rev Transcription D. Translation
59. Mpox vacc and Rx -F1/25 A. Oka, acyclovir B. Ixchiq, ribavirin C. 17D, ganciclovir D. Jynneos & Tecovirimat	60. Burkitts lymphoma exc -F21/24/25/26 A. CD20/CD21 positive B. t(8:14)- cymc gene C. Diag by RTPCR D. MC EBV lymphoma
61. HIV primary receptor, coreceptors -N18 A. CD4, CXCR4, CCR5 B. CD8, CD19, CD20 C. CD56, HACE2, ACE D. CD4, CD4, CD4	62. Cryptopsoridium diarrhea and Cryptococcal meningitis in HIV at CD4 A. 600, 600 B. 400, 200 C. 300, 500 D. 100, 100 -F6/20,N19
63. Gold standard for diag of ca cervix -F1/7/25 A. Visual Inspection by acetic acid B. Visual Inspection by Lugols iodine C. Trachelectomy D. HPV PCR	64. Rx of ca cervix -F1/25 A. Hystero- salpingo -oophorectomy B. Pelvic exentration C. Marsupialization D. Trachelectomy + Cisplatin (Platinum based)
65. Immune Reconstitution Inflam synd after A. ATT B. HAART C. PCR-F1/25 D. Pair Rx	66. Measles and Chickenpox ____ isolation A. Home B. Acute ward C. ICU D. None -F1/25
67. Ramsay Hunt syndrome ganglia involved A. Sacral B. Otic -N8/24 C. Ciliary D. Geniculate	68. Centrifugal rash -F1/26,N21 A. Rubella B. Variola C. Varicella D. Rubeola
69. Centripetal rash -F1/26,N21 A. Pemphigus B. Smallpox C. Chickenpox D. Roseola	70. Multinucleate giant cells seen in___ smear A. Slit skin-F7/24/25 B. PBS C. PAP D. Tzanck
71. Noodle and bullet shape virus A. CMV, HPV B. HSV, VZV-F19 C. Ebola, Rabies D. Adeno, Pox	72. Ardeid bird related to -F19 A. Anopheles B. Aedes C. Culex D. Mansonia
73. Condyloma acuminata-F20/21/22/23/24, N20/22 A. HSV B. CMV C. HPV D. EBV	74. Most lethal influenza=flu -Future Q A. H3N2 B. H1N1 C. H5N1 D. All
75. Cataract, glaucoma, nerve deafness, PDA A. Congenital CMV B. Congenital Rubella C. Congenital HIV D. Congenital HSV -N21	76. Chronic Hep B + low infectivity-F 1/25,23,N24 A. IgM Anti HBc+, HBeAg+ B. HBcAg+, HBeAg+ C. IgG Anti HBc+, HBeAg- D. HBxAg+, HBsAg-
77. Cured/ recovered Hep B positive for-Future Q A. All antigen B. All IgM C. All IgG D. None	78. 95-95-95 and 90-70-90 strategy -Future Q A. HSV,HPV B. CMV,EBV C. HIV,HPV D. None
79. Hep B marker not seen in blood-Future Q A. Anti HBc B. HBcAg C. Anti HBs D. HBeAg	80. Mass killing of poultry + egg destruction A. H3N2 -Future Q B. H1N1 C. H5N1 D. All
81. Precore and escape mutants in hepatitis___ A. A B. B C. C D. D -N22	82. Viral hepatitis diagnosed by PCR A. A B. B C. C D. D -N23
83. Vaccinated doctor positive for -Future Q A. Anti HBc B. HBcAg C. Anti HBs D. HBeAg	84. Nevirapine in HIV given to -Future Q A. Mother B. Baby C. Both D. None
ANSWERS 57C, 58C, 59D, 60C, 61A, 62D, 63D, 64D, 65B, 66A, 67D, 68C, 69B, 70D, 71C, 72C, 73C, 74C, 75B, 76C, 77C, 78C, 79B, 80C, 81B, 82B, 83C, 84B	

5. PARASITOLOGY 56 PYQs

My Score ____/ 56, ____/ 56

1. Malaria clinical & radical cure -F 7/24,NEET A. Chloroquine, Primaquine B. Quinine C. Artemisinin combo Rx D. Doxycycline	2. Anchovy sauce pus, flask shape ulcer A. Trichomoniasis B. Amoebiasis -F 1/21,7/24 C. Fascioliasis D. Enterobiasis -N19/20/21
3. All correct about Kala azar except -F 1/24,N18 A. Phlebotomus B. Infective- Amastigote C. rk 39 antigen D. Rx- Amphotericin B	4. Quadrinucleate cyst in -F 7/24 A. Entamoeba, Giardia B. Hookworm, Pinworm C. Trichomonas, Strongyloides D. None
5. Wrong about Trichomoniasis-F1/24,6/22,IN24 A. Strawberry cervix B. Whiff test negative C. Rx- Metronidazole D. Male partner treated	6. Hydrocoele, chyluria, lymphangitis -F21/23/24 A. Pinworm B. Rx-Mebendazole C. Cello tape mount D. DEC provocation test
7. Ova- D-shaped, mucus plugs -F 7/23,20 A. Guinea worm B. Hookworm,Tapeworm C. All roundworms D. Pinworm, Whipworm	8. Rx of falciparum in Manipur/Mizoram A. Chloroquine B. Quinine C. Artesunate D. Artemether+Lumefantrine -F1/26/23
9. All seen in P. falciparum except -N 23, 8/24 A. Relapse B. HRP2 antigen -F1/21 C. Accole forms D. Banana gametocytes	10. Sleeping sickness and Chagas dis -N19 A. Flea, Tick B. Tse tse fly, Reduviid bug C. Mite, Aedes D. Lice, Culex
11. Fish eats mosq larva+cyclops A. Tuna B. Salmon C. Gambusia D. Rohu -F 1/24	12. Congenital infec assoc with cats A. Rubella B. Toxoplasma C. CMV D. Herpes -F 7/23,1/24/25
13. Aedes transmits A.Malaria B.Kalaazar -F7/23 C.Chikungunya,Yellow fev,Dengue,Zika D.All	14. Banana gametocyte -F7/24,1/21 A. P. ovale B. P. malariae C. P. knowlesi D. P. falciparum
15. Cholangiocarcinoma -N23 A. Lung fluke B. Blood fluke C. Clonorchis D. None	16. HIV + with larva in stool -F1/26,N22 A. Giardia B. Dracunculus C. Strongyloides D. Entamoeba
17. Accole forms -F 7/21/23/24 A. P. ovale B. P. malariae C. P. knowlesi D. P. falciparum	18. Itch mite -F7/20/24 A. Scabies, 5%permethrin B.Giardia, dapsone C.Typhus, doxy -N19 D. None
19. MC diarrhea in HIV -F7/21/23,N19 A. Ascaris B. Gardia C. Taenia solium D. Cryptosporidium	20. Water lily sign -F 1/20/21/24 A. Giardia B. Chagas dis C. Hydatid cyst D. Larva migrans
21. Mosq breeding in dirty water -N22 A. Aedes B. Anopheles C. Culex D. Mansonia	22. Single ring, oval gametocyte A. P. falciparum B. P. vivax C. P. knowlesi D. P. ovale -F7/21/23/24
23. Erythrophagocytosis -N19/21 A. Entamoeba B. Acanthamoeba C. Giardia D. Leishmania	24. Duodenal string test -N23 A. Entamoeba B. Acanthamoeba C. Giardia D. Leishmania
25. Vagina secretion + KOH=Fishy odour -N23 A. Naeglers test B. Whiffamine test C. Casoni test D. Sabin Feldman test	26. Wrong about tertian malaria -N18 A. Alternate day fever B. Fever comes on 3rd day C. Periodicity of fever 48 hours D. Gap 3 days

ANSWERS 1A, 2B, 3B, 4A, 5B, 6D, 7D, 8D, 9A,10B, 11C, 12B, 13C, 14D, 15C, 16C, 17D, 18A, 19D, 20C, 21C, 22B, 23A, 24C, 25B, 26D

27. Elephantiasis most commonly affects -F7/23 A. Rt leg B. Lt leg C. Both legs D. Scrotum	28. Compression stockings for A. Mycetoma B. Hydrocele C. Elephantiasis D. CTEV -F 7/23
29. MC adult onset epilepsy -N19 A. Meningitis B. Encephalitis C. Neurocysticercosis D. Stroke	30. Mite borne nocturnal itch A. Scrub typhus B. Plague -F20/24 C. Leishmaniasis D. Scabies
31. Anemia Mukt Bharat Program -F1/26,N23 A. Wuchereria B. Ancylostoma C. Strongyloides D. Trichinella	32. Wrong about tertian malaria -N18 A. Alternate day fever B. Fever comes on 3rd day C. Periodicity of fever 48 hours D. Gap 3 days
33. Reservoir of Kala azar in India -F1/24 A. Soil B.Only human C.Only animal D.Human+animal	34. Leishmania tropica A. Delhi=Oriental sore B. Kala azar C. Donovanosis D. None -N 8/24
35. Most important indicator in malaria -N19 A. Spleen rate B. Annual Parasite Index C. IRS coverage D. Slide positivity rate	36. All correct about Primaquine except -N23 A. Prevents recrudescence B. Radical cure C. Prevents transmission D. Kills hypnozoites
37. Jaswant Singh Bhattacharya stain -N18 A. Toxoplasma B. Filaria C. Malaria D. Giardia	38. Malaria Rx in 1st trimester -F1/23 A. Quinine B. Mefloquine C. Primaquine D. Amodiaquine
39. Hemoglobinuria=Black water fever -F21/23 A. Filaria B. Malaria C. Leishmania D. Ascaris	40. Rx of tapeworms& flukes -N18 A. Albendazole B. Mebendazole C. Ivermectin D. Praziquantel
41. BCG Rx for bladder ca-F 7/23 A.Trypansoma B. Schistosoma C. Toxoplasma D. Mycoplasma	42. Not related to malaria relapse A. Merozoite B. Tafenoquine C. Hypnozoite D. Primaquine -N24
43. Burrows in stratum corneum -N18/19 A. Mite B. Tick C. Flea D. Reduviid bug	44. Cyclopropagative malaria -N 23, 8/24 A. P. vivax B. P. falciparum C. Both D. None
45. P. falciparum coma-F1/23/24 A. Algid malaria B. Cerebral malaria C. Quartan malaria D. None	46. Torchlight chorioretinitis -F1/25,N 8/24 A. CMV B. HSV C. Toxoplasma D. Endocarditis
47. Twitching, fallling leaf motility -F7/23,N8/24 A. Toxoplasma, Entamoeba B. Fluke, Pinworm C. Trichomonas, Giardia D. Malaria, Filaria	48. Seizures + brain cysts after eating lettuce A. Cystic hydatid B. Neurocysticercosis C. Cystic fibrosis D. Cystolithaisis -N8/24
49. Malaria is not transmitted-F A. Vector borne B.Transplacental C.Fomite D.Blood transfusion	50. Malaria traveler prophylaxis-F A. Mefloquine B. Chloroquine C. Quinine D. Doxycyline
51. NVDCP doesn't cover-N20 A. Chikungunya B. Filaria C. Typhus D. Japanese encephalitis	52. Perianal itch + D shape ova -F21,N24/25 A. Hookworm B. Pinworm C. Filaria D. Giardia
53. Indicator of recent malaria -F21,N20 A. API B. Infant parasite rate C. ABER D. Spleen rate	54. Malaria radical Rx A. Mefloquine B. Quinine B. Chloroquine+Primaquine D.Doxycyline-N24/I22
55. Signet ring in RBC -F7/24,N8/24 A. Merozoite B. Trophozoite C. Hypnozoite D. Sporozoite	56. Unsuccessful parasitism -F7/24 A. Filaria B. Plasmodium C. Hookworm D. Hydatid cyst
ANSWERS 27B, 28C, 29C, 30D, 31B, 32D, 33B, 34A, 35B, 36A, 37C, 38A, 39B, 40D, 41B, 42A, 43A, 44A, 45B, 46C, 47C, 48B, 49C, 50D, 51C, 52B, 53B, 54B, 55B, 56D	

6. FUNGOLOGY = MYCOLOGY 28 PYQs My Score ____/ 28, ____/ 28

1. Tinea incognito -N19 A. Epidermophyton B. Microsporum C. Trichophyton D. Steroid Rx	2. Cryptococcal meningitis Rx-N19 A. Dapsone A.Econazole C.AmphotericinB+Flucytosin D.All
3. Copper penny=Sclerotic body -F7/22/23, N22 A. Mycetoma B. Chromoblastomycosis C. Sporotrichosis D. Rhinosporidiosis	4. Curdy white vaginal discharge -F 1/24/26 A. Bacterial vaginosis B. Trichomoniaisis C. Vaginal Candidiasis C. Cervical cancer
5. Ectothrix, endothrix, kerion, favus -N20/21/24 A. Tinea capitis B. Onychomycosis C. Tinea cruris D. Athletes foot	6. Rx of invasive fungal infection -F 1/23/25 A. Fluconazole B. Metronidazole C. Albendazole D. Amphotericin B
7. Club shaped -F7/23/24 A. Trichophyton -N24 B. Epidermophyton C. Microsporum D. Mucor	8. Grocotts silver stain -F7/24/25 A. Sarcocystis B. Pneumocystis C. Blastocystis D. None
9. MC fungal pneumonia in HIV-F7/24 A.Candida B. Histoplasma C. Aspergillus D. Pneumocystis	10. Black fungus in diabetics -N21 A.Candida B. Histoplasma C. Rhizopus D. Coccidioides
11. Most medical fungi -N19 A. Deuteromycete B. Zygomycete C. Basidiomycete D. Ascomycet	12. Ringworm -F1/23/24/26 A. Trichosporon B. Trichophyton C. Trichuris D. None-N21,22
13. India Ink stain for -F 1/24 A. Cryptococcus B. Candida C. Aspergillus D. Cryptosporidium	14. Dhobi=Jock itch Rx -N18 A. Cotrimoxazole B. Lulliconazole C. Dapsone D. Amphotericin B
15. Multiple sinuses in foot with black granules A. Eymycetoma B. Chromoblastomycosis C. Botryomycosis D. Blastomcyosis -N18	16. Aflatoxin from groundnuts causes –N23 A. Cholangiocarcinoma B. Lymphoma C. Hepatocellular carcinoma D. None
17. Fungal ball in lung cavity-F7/24/25 A.Myxoma B. Mycetoma C. Aspergilloma D. Chondroma	18. Tinea versicolor&Seborrheic dermatitis -N19 A.Candida B.Trichophyton C.Malassezia D.None
19. Fungus Rx-Cotrimoxazole-F23/25 A.Candida B. Aspergillus C. Pneumocystis D. Mycetoma	20. Fungal antigen test -INICET A. HRP2 B. p24 C. NS1 D. Galactomannan
21. Septate hypha-acute angle branch A. Mucor B.Aspergillus C.Trichophyton D.Candida -F21	22. Aseptate hypha-right angle branch A. Mucor B.Aspergillus C.Trichophyton D.Candida -F21
23. Rose gardener dis & fungal keratitis -N20 A. Candida, Mucor B. Both Mycetoma C. Sporothrix, Fusarium D. None -F 23/24/25	24. Gold standard test-Cryptococcus meningitis A. Nigosin stain B. Crytptococcal antigen C. RTPCR D. Culture on Bird seed agar -N21
25. Fungal culture media/agar-F7/25,N23 A.TCBS B. CLED C. Sabourauds dextrose D. BCYA	26. All fungal stains except -N21 A. Sudan black B. KOH C. Grocott D. Lactophenol cotton blue
27. Onychomycosis Rx-N20 A. Nail lacquer B. Nail removal C. Artificial nail D. None	28. Pessary for Rx of ___ Candidiasis A. Oral B. Esophageal C. Dermal D. Vaginal -F20,N18

ANSWERS 1D, 2C, 3B, 4C, 5A, 6D, 7B, 8B, 9D, 10C, 11A, 12B, 13A, 14B, 15A, 16C, 17C, 18C, 19C, 20D, 21B, 22A, 23C, 24B, 25C, 26A, 27D, 28D

7. IMMUNOLOGY-VACCINOLOGY 56 PYQs My Score ____/ 56, ____/ 56

1. Bence Jones protein-F20/21/22/23/24 A.Prion B.Multiple myeloma C.Apoptosis C.Granuloma	2. Superantigens bind to -N20 A. Plasma cells B. Ag presenting cells C. T cells D. B cells
3. Primary lymphoid organ -F 7/24 A. Spleen B. Peyers patches C. Liver D. Thymus	4. Moth eaten skull -F 1/25 A. Multiple myeloma B. Immunoblastoma C. Leukemia D. Lymphoma
5. Ratio of T cell to B cell -INICET 22 A. I:2 B. 2:1 C. 3:1 D. 1:3	6. ABO/ Rh incompatibility is due to -F 1/24 A. IgA B. IgG C. IgM D. IgE
7. Slow, sluggish, short lived is__response -F21 A. Primary B. Secondary C. Tertiary D. All	8. Primary & secondary immune response -N20 A. IgD, IgC B. IgG, IgM C. IgM, IgG D. IgA, IgE
9. First antibody formed in fetus -NEET 21 A. IgA B. IgG C. IgM D. IgE	10. Erythroblastosis fetalis is prevented by -N17 A. Antitoxin B. Anti Rho C. Antigen D. Antidote
11. Penicillin allergy is __ hypersens -NEET 20 A. Type I B. 2 C. 3 D. 4	12. Penicillin induced anemia is____hyp -F 7/23 A. Type I B. 2 C. 3 D. 4
13. MC organ transplant -F 1/24 A. Liver B. Kidney C. Heart D. Intestine	14. Most suitable for xenotransplant -N20,INI 22 A. Monkey B. Chimpanzee C. Gorilla D. Pig
15. Given 1st in anaphylaxis -N21 A. Steroids B. Antihistaminics C. Oxygen D. Adrenaline	16. Open vial policy -F7/25N22,INI23 A. BCG B. PPSV23 C. Pentavalent D. Measles
17. No herd immunity -N18/22/23 A. Measles B. Mumps C. Polio D. Tetanus	18. Sympathetic ophthalmia due to ___ antigen A. Iris B. Lens C. Cornea D. Conjunctiva -N17,22
19. Toxoid is ___ immunity ? -N18 A. Active B. Passive C. Local D. Herd	20. Antivenom is ___ immunity ? -N18 A. Active B. Passive C. Local D. Herd
21. Reversal (1:2) of Albumin:Globulin ratio A. Multiple myeloma B. Chronic viral hepatitis C. Guillain Barre syndrome D. All -F 1/7/25	22. Vaccine routinely given to infants -F 2019 A. Cholera B. Meningococcal vacc C. 17D vaccine D. Measles vaccine
23. Cervavac and Jenvac for -F 1/23 A. Chickenpox, Juvenile RA B. HCV, HBV C. HPV, Japanese encep D. Herpes, Shingles	24. Vaccines C/I for pregnancy -F1/25,N19, IN 21 A. All live vacc B. Recombinant vacc C. Killed vaccines D. DNA plasmid vaccines
25.Prompt,powerful,prolonged is_response-F21 A. Primary B. Secondary C. Tertiary D. All	26. Absolute C/I to live vacc -N18/23/24 A. HIV B. Pregnancy C. SCID D. Splenectomy
27. Fused plasma and tumour cell, use -INICET A. Hybridoma cell, monoclonal Ab B.Tcell, imm C. Dust cell, allergy D. Popcorn cell, atopy	28. All heterophile antigen except -F 1/25,19 A. Widal test B. Weil-Felix test in typhus C. VDRL in syphilis D. Cold agglutination

ANSWERS 1B, 2C, 3D, 4A, 5C, 6B, 7A, 8C, 9C, 10B, 11A, 12B, 13B, 14D, 15D, 16C, 17D, 18B, 19A, 20B, 21D, 22D, 23C, 24A, 25B, 26C, 27A, 28A

29. Antigen presenting cell except -N19 A. T cell B. B cell C. Dendritic cell D. Macrophage	30. Antibody produced from -N22 A. T cell B. Plasma cell C. Megakaryocyte D. Macrophage
31. Most potent complement activator-N 23 A. IgA2 B. IgG3 C. IgM D. IgE	32. Antibodies not fixing complement -INICET 22 A. IgG, IgM B. IgD, IgE C. IgA D. None
33. Rx of Cervical ca and breast ca -F1/25 A. Imatinib Ibrutinib B. Cisplatin, Tamoxifen C. Rituximab, Vinblastin D. None -N19/20/23/24	34. Vaccine schedules correct except -F 1/25, 21 A. fIPV- 6,14 wks B. MR, JE- 9, 18 months C. PCV- 6,14 wks D. Vitamin A- 6,10,14 weeks
35. Vacc vial monitor ensures -N18 A. Density B. Surface tension C. Volume D. Cold chain	36. No open vial policy -F7/25INI23,N22 A. Polio B. Pneumococcal conjug C. BCG D. Pentavalent
37.Match Type1,2,3,4 hypersensitiv-F22/23/24/26 P. Rheumatoid arthritis Q. Rheumatic fever R. Seafood allergy S. Granuloma A. 1P- 2Q- 3R- 4S B. 1S- 2Q- 3P- 4R C. 1R- 2Q- 3P- 4S D. 1Q- 2R- 3S- 4P	38.Match Type1,2,3,4 hypersensitiv-F22/23/24/26 P. Lepromin test Q. Urticaria R. Rh incompatibility S. Arthus reaction A. 1P- 2Q- 3R- 4S B. 1S- 2Q- 3P- 4R C. 1R- 2Q- 3P- 4S D. 1Q- 2R- 3S- 4P
39. Colostrum -N21 A. IgA B. IgG C. IgM D. IgE	40. Widal Test -F19/24 A. IgA B. IgG C. IgM D. IgE
41. Valency of MC antibody, reaginic antibody, heaviest antibody, secretory antibody -N18 A. 1 for all B. 2,2,10,4 C. 4,4,10,2 D. 10,4,2,4	42. Anaphylotoxins and opsonins -N19 A. (C3a,C4a,C5a),C3b,C4b B. C3b,C4b,C3a,C4a C. C3b,C4b,(C3a,C4a,C5a) D. C1a,C2a,C3a,C4a
43. Allergy/itch -N18 A. IgA B. IgG C. IgM D. IgE	44. Antigen binds-N21 A. Fv B. Fab C. Both D. Fc
45. Cross match in blood transfusion for -N18 A. Antigen in donor B. Clotting factors in donor C. Antibody against donor RBC D. None	46. Ptosis, weakness, easy fatiguability-F7/24/25 A. Lambert Eaton B. Tetanus -N18,21 C. Botulism D. Myasthenia gravis
47. _____ kills viruses and cancer cells -INI23 A. Plasma cell B. B cell C. NK cell D. CD4 cell	48. Main cytokine in sepsis, diabetes, chr inflam A. IL-1 B. TNF alpha C. IFN alpha D. SCID -N18
49. All lymphocytes except -NEET 23 A. Reed Sternberg cell B. Mesangial cells B. Downey cells D. Skirt cells	50. All correct vaccines except -N19,23 A. RSV- Shingrix B. Chikungunya- Ixchiq C. Malaria-Mosquirix D. Influenza – Live/killed
51. Vacc at birth-F17 A.OPV B.IPV C.HPV D.DPT	52. TT valid for -F17 A. 6m B. 1 yr C. 2 yr D. 10 yr
53. All live vacc C/I in pregnancy except -F1/25 A. MMR B. 17D C. Oka D. None -N18	54. Most recently added vacc in UIP -N24 A. Rotavirus B. PCV C. JE D. Measles-Rubella
55. VDRL uses ___ antigen for diagnosis -F1/25 A. Heterophile Ag B. Sequestered Ag C. Tumour Ag D. Incomplete Ag=Hapten	56. Infec shows cross reactivity with Widal, Zika, Dengue, Chik, Kala azar, Leptospira, Typhus-F25 A. Filaria B. Malaria C. JE D. Chickenpox
ANSWERS 29A, 30B, 31B, 32B, 33B, 34D, 35D, 36C, 37C, 38D, 39A, 40C, 41B, 42A, 43D, 44C, 45C, 46D, 47C, 48B, 49B, 50A, 51A, 52D, 53B, 54B, 55A, 56B	

8. CLINICAL INFECTIOLOGY 76 PYQs My Score ____/ 76, ____/ 76

1. Wrong about blackwater fever-F19/20/23,N20 A. Anopheles B. Blackish discoloration of skin C. High fever with chills D. Hemoglobinuria	2. Drug of choice for anerobes and protozoa A. Albendazole B. Cotrimoxazole -N21 C. Lulliconazole D. Metronidazole
3. MC urethritis & non gonococcal urethritis A. Syphilis, HIV B. Gonorrhea, Chlamydia C. E. coli, HPV D. HBV, HCV -N19,20	4. Xenopsylla & Chandler index -N21 A. Plague, Ancylostoma B. Malaria, Filaria C. Anthrax, Schistosoma D. Zika, Dengue
5. Clue cells seen in -F 26,1/24,N24 A. Vaginal Candidiasis B. Bacterial vaginitis C. Neisseria gonorrhoeae D. Bact vaginosis	6. Dendritic ulcer seen__keratitis -F26/25/19,N20 A. Bacterial keratitis B. Fungal keratitis C. Viral keratitis D. Parasitic keratitis
7. All seen in endocarditis except -F 1/24,N22 A. Oslers nodes B. Janeways lesions C. Splinter hemorrhages D. Assman focus	8. Ac inflam demyelinating polyneuropathy A. Lesch Nyhan syn B. Felty syn -F7/25,IN24 C. Guillain Barre syndrome D. Reiter syn
9. ↑Lymphocytes-↑Protein-Cobweb coagulum A. TB meningitis B. Bac menin -F1/23/25/26 C. Viral meningitis D. Fungal meningitis-N22	10. ↑Pressure↑Neutrophil↓glucose -F23/24/25/26 A. TB meningitis B. Bacterial meningitis C. Viral meningitis D. Fungal meningitis
11. Eschar except -F21/23/24/25/26 A. Burns B. Scrub typhus C. Anthrax D. Kala azar	12. Ebola bioterror cat, biosafety level-F7/24, N20 A. A,4 B. B, 3 C. C,2 D. D,1
13. Burrow lesions seen in-N18,19 A.Scabies B. Herpes C. Chickenpox D. Measles	14. Hypopigment, hypoaesthesia leprosy -F7/23 A. LL B. BB C. TT D. None
15. Wrong food poisoning -F1/7/25, 7/23, 6/22 A. Mayo- S. aureus B. Fried Rice- B. cereus C. Canned f- Strep D. Seafood- V. parahemol	16. Identify incorrect murmur -F1/7/25/26, N19 A. PDA- Machinery B. M. stenosis- Mid-diastolic C. Mitral regurg-Pansystolic D. All correct
17. All flaccid paralysis except -F1/20,N19 A. Krait poisoning B. Organophosphate pois C. Botox poisoning D. Strychnine posoning	18. All spastic paralysis except -F1/20,N18/19/23 A. Cerebral palsy B. Myasthenia gravis C. Tetanus D. Strychnine posoning
19. MC typical and atypical pneumonia -N22 A. Klebsiella, Chlamydia B. Covid, Influenza C. Pneumococcus, Mycoplasma D. None	20. Verruca vulgaris, Lupus vulgaris -F 7/23 A. Cutibacterium, HIV B. HPV, TB C. HSV, CMV -N18/20/25 D. Anthrax, Plague
21. Infective stage of malaria is -F19,20 A.Trophozoite B.Merozoite C.Sporozoite D. No	22. Primary acne lesion -F 1/24/25 A. Comedone B. Papule C. Pustule D. Nodule -N23
23. Chocolate cyst is seen in -N20 A. Brain B. Heart C. Peritoneum D. Ovary	24. Sexual partners Rx except A. Trichomoniasis B. Syphilis C. Gonorrhoea D. Orchitis -F 1/23/24

1B,2D,3B,4A,5D,6C,7D,8C,9A,10B,11D,12A,13A,14C,15C,16D,17D,18B,19C,20B,21C,22A,23D,24D

25. CURB-65 & SOFA related to -F1/21/23, N 23 A. Enteritis, cystitis C. Pneumonia, Sepsis D. Croup, Quinsy A. Hyphaema, hypopyon	26. Hematuria, hemoglobinuria -F19/20, N 23 A. Prostatitis, urethritis B. Balanitis, orchitis C. Glomerulonephritis, malaria D. None
27. MC bone affected in osteomyelitis-F19,N22 A. Cranium B. Vertebrae C. Clavicle D. Pelvis	28. Trachelectomy related to cancer of ___-Fu Q A. Breast B. Cervix C. Liver D. Lung
29. MC site of osteomyelitis in long bone A. Diaphysis B. Articular cartilage C. Metaphysis D. Epiphysis -N18/21,F7/25	30. Breastfeeding contraindicated in all except A. TB mastitis -F 6/21 B. Pulmonary TB on Rx C. Galactosemia -N21/24 D. Lithium Rx
31. Vacc required in splenectomy -F 7/24 A. Hemophilus, meningococus, pneumococc B. Cholera, typhoid, Hepatitis A, Hepatitis E C. Covid, Influenza, Resp syncytial virus D. Ebola, Malaria, Dengue, Chikungunya	32. MC travelers diarrhea, toxin and treatment A. ETEC, labile toxin, azithromycin -N19 B. EPEC, No toxin, no treatment C. EHEC, verotoxin, ciprofloxacin D. EAEC, NSP4 toxin, no treatment
33. Charcots triad & Reynolds pentad -F 1/24 A. Hepatitis B. Oesophagitis C. Cholangitis D. Lymphangitis	34. Risk of ___for married girl on isotretinoin Rx A. Photosensitivity B. Neural tube defect C. Melasma D. Teratogenicity -F 1/24, 1/25
35. MC menigitis- All correct except -N19 A. Overall- Pneumococcus B. >60yr-Neisseria C.HIV-Cryptococcus D.Aseptic-Herpes simple	36. Fried egg colony -F 1/26 A. Mycoplasma B. Malassezia C. Both D. None
37. Iron deficiency anemia -N19 A. Chandler index B. Breteau index C. Cheopis index D. Annual parasite index	38. Liver made protein ↑ in inflammation except A. C-Reactive Protein B. Albumin C. Fibrinogen -F 7/24,N24 D. Globulin
39. Most important drug in TB, Leprosy -N 20 A. Pyrazinamide, Clofazimine B. Bedaquiline, Moxifloxacin C. Isoniazid, Dapsone D. Both Rifampicin	40. All correct smear except -F23/24/25/26, N24 A. Tzanck- HHV8 B. Pap smear- HPV C. Thick smear-To find any Plasmodium D. Thin smear-To identify Plasmodium species
41. Ransons criteria, Cullens sign A. Lymphangitis B. Cholangitis -F 1/21,N24 C. Appendicitis D. Pancreatitis	42. McBurneys point tenderness, Mantrel criteria A. Lymphangitis B. Cholangitis -F 1/22 C. Appendicitis D. Angioedema
43. Vacc contraindicated during lactation A. Smallpox B. Yellow fever C. Both D. None	44. Given to Bombay blood gp child -F 7/24 A. AB B. B C. Oh D. O
45. Black fever, Black death, Blackwater fever A. Botulism, Anthrax, Filaria B. HIV, HPV, HSV C. Kala azar, Plague, Malaria D. None -N18	46. 21yo female with urinary urgency, frequency, hesitancy after honeymoon. Diagnosis? -N17 A. Pyelonephritis B. Cystitis C. Urethritis D. All
25C,26C,27B,28B,29C,30B,31A,32A,33C,34D,35B,36C,37A,38B,39D,40A,41D,42C,43C,44C,45C,46B	

47. Centripetal, centrifugal, cephalocaudal rash A. Smallpox, Varicella, Measles B. HBV,HCV,HIV C. Dengue, Chikungunya, typhoid D. None -N18	48. 77yo with confusion, uremia, tachypnoea, hypotension, Rt lower zone consolidation -N19 A. OPD B. Acute Ward C. ICU D. Palliative care
49. Painless genital ulcer except -N19,21 A. LGV B. Syphilis C. Chancroid D. Donovanosis-F25/26	50. Painful genital ulcer A.Syphilis,gonorrhea-N21 B.HIV,LGV C.HBV,HCV D.Herpes,chancroid -F1/26
51. Virus with highest mortality -IN20, F1/25 A. HIV B. Rabies C. Ebola D. Nipah	52. Presence of macules,papules,vesicles,crusts A. Varicella B.Smallpox C.Shingles D.None -F20
53. Filaria affects__system -F7/23 A. Endocrine B. Lymphatic C. Respiratory D. Hepatobiliary	54. Kernig,Brudzinski sign -F1/26,N18 A.Meningitis B. Endocarditis C. Encephalitis D. Mononeuritis
55. Vaginal ph increases in A. Trichomoniasis B. Bacterial vaginosis C. Both D. HIV -N23	56. Dysentery+trophozoite with pseudopodia-N19 A. E.coli B. Entamoeba C. Shigella D. Giardia
57. MC encephalitis in India -N19 A. West Nile B. Japanese encephalitis C. Nipah D. Ebola	58. Ventriculoperitoneal shunt for A. Cerebellitis B. Microcephaly C. Hydrocephalus D. None -F19
59. MC contact lens keratitis -N18 B. Herpes B. Staph aureus C. Pseudomonas D. E.coli	60. Urgency, frequency, hesitancy -N19 A. Hepatitis B. Pyelonephritis C. Cystitis D. None
61. Croup affects all except -N21,22 A. Tongue B. Larynx C. Trachea D. Bronchus	62. MC pneumonia in smokers -N19 A. E.coli B. Pneumococcus C. Gonococcus D. Klebsiella
63. Indirect bilirubinemia in ___ jaundice -N19 A. Hemolytic B. Obstructive C. Both D. None	64. Kit for inguinal bubo, PID -INI 23 A. Red, red B. Green, grey C. Black, yellow D. Blue, Brown
65. Helminth associated with stepwells -F1/26 A.Ringworm B.Guineaworm C.Pinworm D. None	66. Which live vacc can be given in pregnancy A. Oka B. 17D C. DtaP D. RA 27/3 -N18
67. Sequestrum, involucrum are __bone -F6/20 A. Dead and live B. Live and dead -F7/25 B. Cancellous and cortical D. Flat and long	68. Strawberry tongue seen in all except -N18 A. Scarlet fever B. Kawasaki disease C. Vit B12 deficiency D. Typhus fever
69. Pel Ebstein fever + Reed Sternberg cells A. Burkitts lymphoma B. Hodgkins lymphoma C. Non Hodgkins lymphoma D. None -F1/23	70. All are correct indices except -N18 A. Spleen- HIV B. Chandlers index- Hookworms C. Cheopis- Plague D. Breteau- Aedes
71. MC mode of HIV transmission A. Intranatal B. iv drug C. Blood transfusion D. Sexual -F19	72. Highest chance of HIV transmission A. Sexual B. iv drug C. Blood transfusion D. Intranatal -N18
73. NSAID used for Rx of congenital ____ -N21 A. Measles B. Mumps C. Rubella D. CMV	74. CRISPR-cas9 mechanism -F7/24 A. RTPCR B. Genome editing C. Conjugation D. Transposon
75. HAART started at CD4 level -N19 A. < 500 B. < 350 C. <200 D. Any CD4 level	76. Sex partner Rx except -F1/23/24 A. Herpes B. Chancroid C. Chlamydia D. Donovanosis
ANSWERS 47A, 48C, 49C, 50D, 51B, 52A, 53B, 54A, 55C, 56B, 57B, 58C, 59C, 60C, 61A, 62B, 63A, 64C, 65B, 66B, 67A, 68D, 69B, 70A, 71D, 72C, 73C, 74B, 75D, 76A	

9. RAPID FIRE CLINICOVISUAL 50PYQ (180pics) My Score ____/ 50,____/50

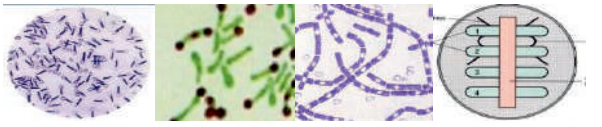
1.

A. Eleks test, CAMP test, hemolysis, swarming

B. CAMP test, swarming, satellitism, hemolysis

C. Swarming, Quellungs, CAMP test, Eleks test

D. Satellitism, CAMP test, Eleks test, Quellungs

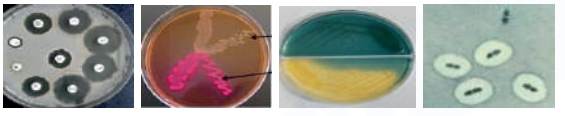
2. 

A. Tetanus, Diphtheria, Anthrax, Diphtheria

B. Anthrax, Tetanus, Diphtheria, Plague

C. Diphthera, Tetanus, Anthrax, Pertussis

D. Diphtheria, Anthrax, Tetanus, Botulism

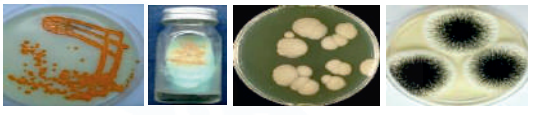
3. 

A. CLED, McConkey, Kirby Bauer, Quellung-N20

B. CAMP test, hemolysis, CLED, satellitism

C. Quellungs, CLED, McConkey, Kirby bauer

D. Kirby Bauer, McConkey, CLED, Quellungs

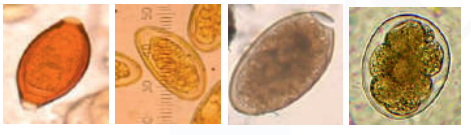
4. 

A. Colonies of yeast, Staph aureus, mold, TB

B. Colonies of TB, mold, yeast, Staph aureus

C. Colonies of Staph aureus, TB, Yeast, Mold

D. Colonies of Mold, yeast, TB, Staph aureus

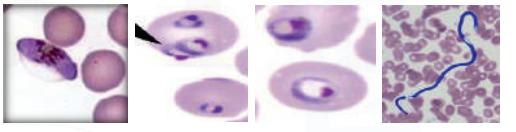
5. 

A. Ova of tapeworm, hydatid, Trichinella, fluke

B. Ova of filaria, malaria, Leishmania, Giardia

C. Ova of roundworm, fluke, whipworm, filaria

D. Ova of Trichuris, Pinworm, fluke, hookworm

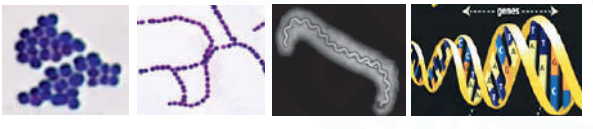
6. 

A. Falciparum, falciparum, vivax, microfilaria-N23

B. Amoeba, vivax, vivax, falciparum-F1/26

C. Giardia, filaria, Kala azar, malaria

D. Sleeping sickness, fluke, tapeworm, hydatid

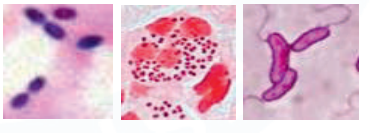
7. 

A. Falciparum, microfilaria, Entamoeba, Borrelia

B. Amoeba, Neisseria, falciparum, Treponema

C. Staph aureus,Streptococcus,Leptospira,DNA

D. Streptococcus,Staph aureus,Anthrax,E. coli

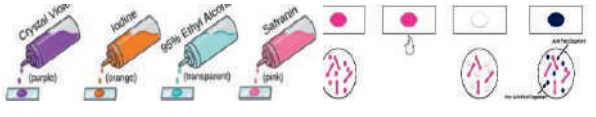
8. 

A. Strep pyogenes, Strep pneumoniae, Anthrax

B. Strep pneumoniae, Neisseria, Vibrio cholera

C. Staph aureus, Strep pyogenes, Leptospira

D. Diphtheria, Pertussis, Tetanus

9. 

A. Gram stain, Acid fast ZN stain

B. Alberts stain, Ponders stain

C. Acid fast ZN stain, Gram stain

D. Capsule stain, Silver stain

10.

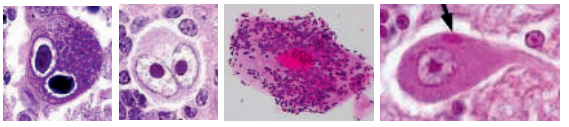
A. NVBDCP, NLEP, NTEP, PPTCT -N23

B. NRCP, NHM, NIP, NPCDCS

C. NRHM, NOTP, NMHNSP, NIDDCP

D. NHM, ABY, PMJAY, NUHM

1B,2A,3D,4C,5D,6A,7C,8B,9A,10A---- 2025 antibiotic- Cefipime+ Zidebactam (Gram negative HAI)

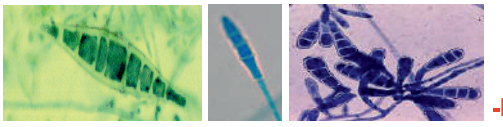
11. 

A. Eleks test, CAMP test, satellitism-F1/23/26,7/24

B. Macrophage, neutrophil, lymphocyte, RBC

C. CMV, Hodgkins lymphoma, clue cell, Rabies

D. EBV, CMV, rabies, clue cell, Tzanck cell

12. 

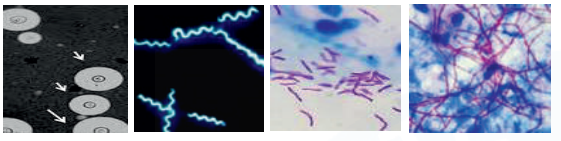
-N21,22

A. Trichophyton, Microsporum, Epidermophyton

B. Microsporum, Trichophyton, Epidermophyton

C. Epidermophyton, Microsporum, Trichophyton

D. Epidermophyton, Trichophyton, Microsporum

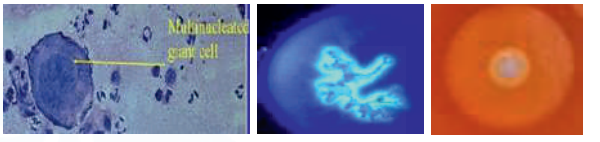
13. 

A. Quellungs, Staph, Strep, Listeria

B. Cryptosporodium, Cryptococcus, TB, TB

C. Cryptococcus, Syphilis, TB, Nocardia -N24

D. Syphilis, Leptospira, Diphtheria, TB

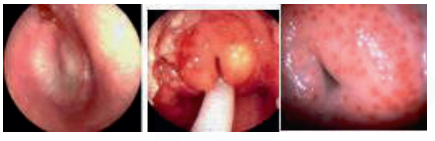
14. 

A. Falciparum, Chromoblastomycosis, Fusarium

B. Amoeba, Sporothrix, filaria

C. Giardia cyst, Nocardia, malaria

D. Varicella, Herpes simplex, Target hemolysis

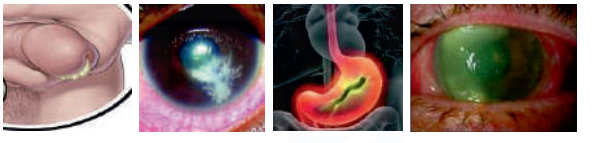
15. 

A. Strawberry cervix, otitis media, epiglottitis

B. Otitis media, epiglottitis, strawberry cervix

C. Epiglottitis, otitis media, strawberry cervix

D. Tonsillitis, quinsy, mastoiditis

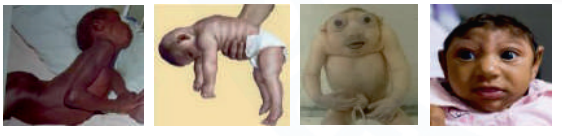
16. 

A. Urethritis,Keratitis,Helicobacter,Pseudomonas

B. Balanitis, conjunctivitis, E. coli, Clostridium

C. Balanoposthitis, retinitis, Klebsiella, Orientia

D. Epididymitis, iridocyclitis, Proteus, Chlamydia

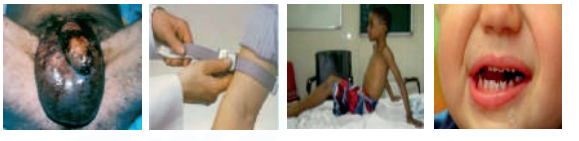
17. 

A. Cysticercosis,sepsis,acephaly,oxycephaly

B. Syphilis,polio,macrocephaly,acrocephaly

C. Botulism,tetanus,otocephaly,hydrancephaly

D. Tetanus,botulism,anencephaly,microcephaly

18. 

A. Orchitis, myositis, sacroiliitis, epiglottitis

B. Fournier gangrene, Dengue, Polio, Syphilis

C. Gas gangrene, malaria, botulism, sore throat

D. Hydrocele, marasmus, tetanus, otitis -N22,23

19.

A. Entamoeba, Anchovy sauce pus, Giardia

B. Giardia, Hydatid cyst, Entamoeba

C. Toxoplasma, Schsitosoma, Cryptosporidium

D. Trichomonas, Hepatitis B, Leishmania

20.

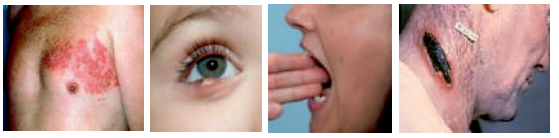
A. Gloves, Placenta, Lancet, Metal implant, food

B. Bone, oxygen mask, syringe, Copper T, blood

C. Food, ampoule, Ryles tube, Scalpel, medicine

D. Cotton, bottle, vial, blood bag, sharps

11C,12B,13C,14D,15B,16A,17D,18B,19A,20A --- 2025 antibiotic- Gepotidacin (Gonorrhea)

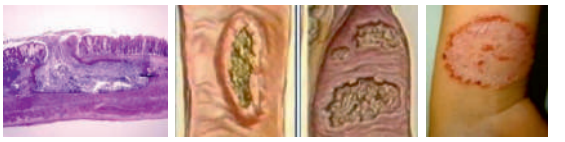
21. 

A. Roseola, dacryocystitis, quinsy, dermatitis

B. Chickenpox, keratitis, mastoiditis, pulpitis

C. Shingles, Stye, Trismus, Eschar -F1/26

D. Measles, chalazion, botulism, stomatitis

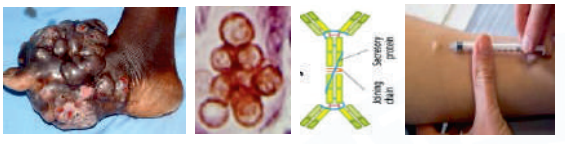
22. 

A. Entamoeba, Typhoid, TB, Ringworm

B. TB, Entamoeba, Typhoid, Sporothrix

C. Typhoid, TB, Entamoeba, Kerion

D. Helicobacter, EBV, Kaposi sarcoma, Favus

23. 

A. Diabetic foot, Mucor, IgE, Measles vaccine

B. Osteoma, Sporothrix, IgM, Hepatitis B vacc

C. Aspergillosis, Histoplasma, IgD, IPV

D. Mycetoma,Chromoblastomycosis,IgA2,BCG

24.

A. Glossitis, pulpitis, osteitis, rubella

B. Epiglottitis, quinsy, felon, measles

C. Gingivitis, esophagitis, tomato fever, syphilis

D. Tonsillitis, Laryngitis, Osler node, Janeway les

25.

A. Spina bifida, chemotherapy, HPV, tonsillitis

B. Hydrocephalus,CSF shunt,SSSS, epiglottitis

C. Hyperthroidism, Coma, Botulism, croup

D. Anencephaly, Microcephaly, Tetanus, Polio

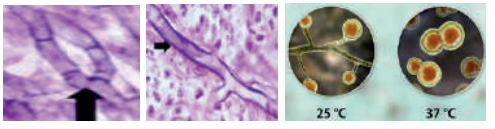
26.

A. ADPKD, mitral stenosis, thyroiditis, Zenkers d

B. Peptic ulcer, PCOD, stomatitis, esophagitis

C. Liver abscess, hydatid, epiglottitis, croup

D. Splenic abscess, cystitis, tonsillitis, laryngitis

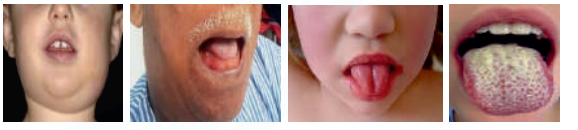
27. 

A. Pneumocystis, Fusarium, Cryptococcus

B. Malassezia, Trichophyton, Candida

C. Nocardia, Actinomyces, Staph aureus

D. Aspergillus, Mucor, dimorphic fungus -INI25

28. 

A. Mumps,Ludwings angina,scarlet fever,thrush

B. Mastoiditis, tonsillitis, stomatitis, gingivitis

C. Petrositis, laryngitis, pharyngitis, quinsy

D. Labyrinthitis, epiglottitis, ozaena, sinusitis

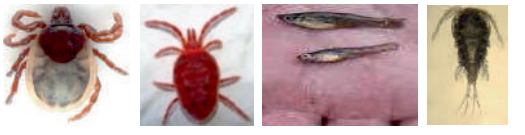
29.

A. Sandfly, Aedes, Flea, Armadillo -N18/24

B. Hard tick, Mite, Gambusia, Cyclops

C. Flea, Mite, Aedes, Soft tick

D. Anopheles, Mansonia, Aedes, Culex

30. 

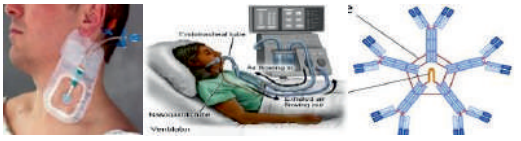
A. Falciparum, microfilaria, Entamoeba

B. Amoeba, Neisseria, falciparum

C. Mite, Flea, Cyclops, Gambusia

D. Hard tick, Mite, Gambusia, Cyclops

21C,22A,23D,24D,25B,26C,27D,28A,29A,30D --- 2025 antibiotic- Zosurabalpin (Acinetobacter)

31. 

A. Intraatrial catheter, bronchoscopy, IgA

B. Central line, mechanical ventilation, IgM

C. Carotid artery line, esophagoscopy, IgG

D. Carotid vein line, duodenoscopy, IgE

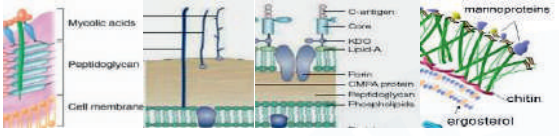
32.

A. Condom catheter,autoclave,suprapubic aspir

B. Foley catheter, hot air oven, CCMS sample

C. Suction catheter, HEPA, Salpingoscopy

D. Balloon urethroplasty, vortex, hysteroscopy

33. 

A. Cell Wall- Mycoplasma, TB, Candida, Vibrio

B. Cell wall- Histoplasma, TB, Mucor, Proteus

C. Cell wall- Strep, Candida, Leprosy, Staph

D. Cell wall- TB, Staph, E. coli, Aspergillus

34.

A. Nerve- Radial, ulnar, facial, ulnar + median

B. Nerve- Tibial, median, trigeminal, cutaneous

C. Nerve- Brachial plexus, humeral, facial, digital

D. Nerve- Abducens, Vagus, ciliary, circumflex

35.

A. BMW- Red, blue, red, yellow, yellow, red, red

B. BMW- Yellow, blue, red, red, red, red, red

C. BMW- Yellow,black,yellow,red,red,red,yellow

D. BMW- Yellow,green,black,black,black,red,red

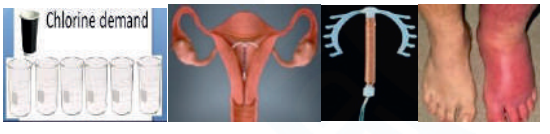
36.

A. Atelectasis, Emphysema, COPD

B. Pneumothorax, lung abscess, bronchiectasis

C. Lobar pneumonia, Miliary TB, Normal X ray

D. Pleural effusion, Cannon Ball, Dextrocardia

37. 

A. VDRL test, Fibroid, Silver curve, Tinea pedis

B. Casoni test, Mirena, Diaphragm, Whitlow

C. Rapid test,Tampon,ring med,onychomycosis

D. Horrocks test,Cu T380A,ML Cu 375,cellulitis

38.

A. Vibrio, Conjugation, Growth curve, PCR steps

B. Proteus, Transduction, PCR steps, Gram stain

C. E. coli, Translation, TB diagnosis, autoclave

D. TB, Transcription, PCR steps, Growth curve

39. -F 1/25

A. Folliculitis,carbuncle,mastitis,breast abscess

B. Boil,cellulitis,hypopyon,mastoiditis,breast ca

C. Impetigo, erythema, retinitis, allergy, abscess

D. Erysipelas, scald, iritis, nipple ca, adeno ca

40.

A. Parotitis, laryngitis, otitis interna, otitis media

B. Glossitis, stomatitis, labyrinthitis, petrositis

C. Gingivitis, sinusitis, typhilitis, intervillitis

D. Quinsy, pharyngitis, otitis media, otitis externa

31B,32A,33D,34A,35A,36C,37D,38A,39A,40D-- 2024 antibiotic- Lolamicin, Nafithromycin (Indian)

41.

A. Vacc vial monitor, inguinal bubo, Molluscum

B. Pedigree chart, Sebaceous cyst, measles

C. Growth chart, Second degree burns, Herpes

D. Snellens chart, Lipoma, Scabies

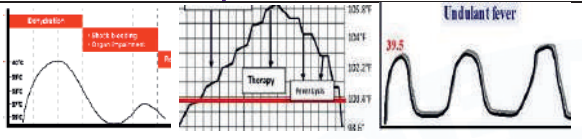
42.

A. Cellulitis,measles,Transposition of great vess

B. Furuncle, shingles, dextrocardia

C. Ramsay Hunt syndrome, plantar warts, PDA

D. Dermatitis, chickenpox, VSD

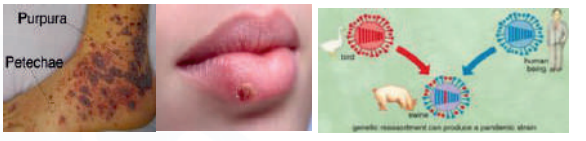
43. 

A. Fever- Typhus, Dum dum, Ebola

B. Fever- Dengue fever, typhoid, Brucella

C. Fever- Scarlet, glandular, breakbone fever

D. Fever- Blackwater fever, Weils, Black fever

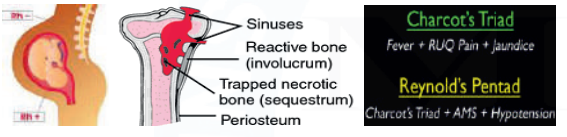
44. 

A. Foot drop, Shingles, Recalcitrance

B. Diabetic foot, HPV, Recrudescence

C. Mycetoma, Tomato fever, Relapse

D. Hemorrhagic fever, Cold sore, Reassortment

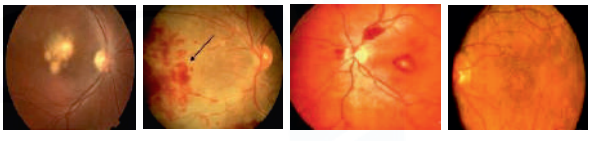
45. 

A. Erythroblas. fetalis,osteomyelitis,cholangitis

B. Lupus vulgaris, osteosarcoma, appendicitis

C. Verruca vulgaris, osteoclastoma, hepatitis

D. Sezary syndrome, periosteitis, cholecystitis

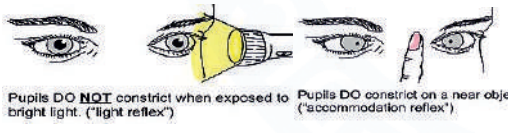
46. 

A. Chorioretinitis (CR)- Cataract, HPV, HSV, HIV

B. CR- Rubella, Measles, Adenovirus, Influenza

C. CR- Varicella, Mycoplasma, Molluscum, Zika

D. CR- Toxoplasma, CMV, Roths spots, Rubella

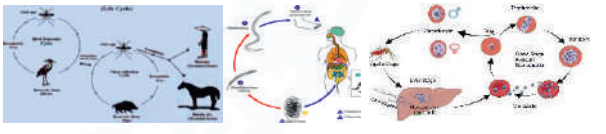
47. 

A. Hutchisons pupil, uncal herniation

B. Marcus Gunn pupil, optic neuritis -N18

C. Argyll Robertson pupil, 30 syphilis -F7/23,N18

D. Colobomatous pupil, congenital coloboma

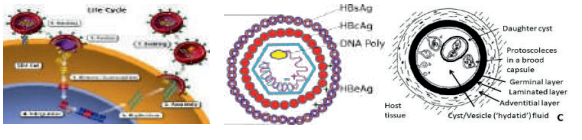
48. 

A. Japanese encephalitis, Hookworm, Malaria

B. Dengue, Filariasis, Blood Fluke

C. Chikungunya, Plague, Liver Fluke

D. Zika, Hydatid Cyst, Kala azar

49. 

A. HSV, Influenza, Cryptosporidium oocyst

B. HPV, Hepatitis C, Giardia cyst

C. HIV, Hepatitis B, Hydatid cyst

D. CMV, EBV, Entamoeba cyst

50.

A. Vaginal speculum, Woods lamp, Proctoscope

B. Cystoscope, Capillariscope, Uteroscope

C. Laparascope, Dermoscope, Otoscope

D. Endoscope, Nephroscope, Myringoscope

41A, 42C, 43B, 44D, 45A, 46D, 47C, 48A, 49C, 50A * There you are! The other side of 200 is not far!

10. SCENARIO BASED 191 PYQs

My Score _____/ 191, ____/ 191

1. Young female with fever, sunburn like rash, low BP after prolonged use of tampon -F7/25 A. Staph scalded skin syndrome B. Steven Johson syndrome C. Staph toxic shock syndrome D. Waterhouse Friderichsen syndrome	2. Infant with fever, desqauamation, Nikolsky sign positive. Diag and Rx -F7/25 A. Staph scalded skin syndrome, Cephalexin B. Steven Johson syndrome, steroids C. Staph toxic shock syndrome D. Waterhouse Friderichsen syn, steroids
3. 6yr old with fever, sore throat, drooling and dyspnea due to Gram negative coccobacilli growing on chocolate agar -F1/26,7/25 A. Hemophilus influenzae B. E. coli C. Strep pneumoniae D. Listeria	4. Aliens from Andromeda galaxy with vomiting + diarrhea after 3 hours of eating sandwich/ salad. Food posioning -F1/7/25 A. Salmonella enterica B. Bacillus cereus C. Clostridium perfringens D. Staph aureus
5. 54 yr with b/l lower limb flaccid paralysis + arreflexia. CSF shows albumino-cytologic dissociation. Diagnosis -F7/25 A. Polio, no Rx B. Cauda equina syndrome C. Guillain Barre syndrome D. None	6. Patient with painful black eschar with surrounding edema due to mite bite after forest trip. Diag and Rx -F7/25 A. Kala azar, Amphotrericin B. Plague, Genta C. Scrub typhus, Doxycycline D. None
7. Chest indrawing (retractions) + nasal flaring A. Severe COPD B. Severe bronchiectasis C. Emphysema -F7/25 D. Severe pneumonia	8. Painless corneal ulcer with branching -F1/26/25 A. Fungal keratitis B. Interstitial keratitis C. Dendritic keratitis D. Bacterial keratitis
9. Painful swelling of eyelid with pus point. Which gland involved -F7/25 A. Zeis gland B. Meibomian gland C. Lacrimal gland D. Bartholin gland	10. 5 yr old with retracted tympanic membrane with air fluid levels. Diagnosis? -F7/25 A. Acute otitis media B. Choleasteatoma C. Serous otitis media d. Otitis externa
11. 15 yr old with well defined lytic lesion with sclerotic margins in metaphysis of tibia. Diag A. Bone cyst -F7/25 B. Chondroblastoma C. Osteochondroma D. Brodie abscess	12. 25 yr old with rheumatic fever with loud first heart sound and mid diastolic murmur. Diag A. Aortic regurgitation -F7/25 B. Mitral stenosis C. Mitral regurgitation D. Aortic stenosis
13. 45 yr old farmer with 3-wk undulant fever after drinking unpasteurized milk. Diagnosis? A. Bartonella B. Brucella -F 7/25 C. Bordetella D. Borrelia	14. Secondary amenorrhea after dilation and currettage. Diagnosis? -F 7/25 A. Sheehan syndrome B. Reiter syndrome C. Asherman syndrome D. Reye syndrome
15. Child with fever + lesions on palms, soles and oral mucosa. Diagnosis? -F7/25 A. HHV8 B. Coxsackie A16 C. VZV D. Parvovirus B19	16. HIV positive with multiple skin cancerous lesions. Best description of lesions/patches A. White flat B. Red smooth -F7/25 C. Painful ulcer D. Purple red elevated
ANSWERS 1C, 2A, 3A, 4D, 5C, 6C, 7D, 8C, 9A, 10C, 11D, 12B, 13B, 14C, 15B, 16D	

17. Diabetic patient with white plaque in mouth. Choose culture media -F7/25 A. Chocolate agar B. Cetrimide agar C. Sabourauds dextrose agar D. CLED agar	18. Eye pain with ring shape ulcer in a girl using contact lenses after swimming in pond water A. Adenovirus B. Acanthamoeba C. Anaplasma D. Acinetobacter -F7/25
19. Family consuming Bajra roti presents nausea, vomiting and diarrhea. Likely cause A. Ergotism B. Aflatoxicosis C. Fusiformis D. Sanguinarine -F7/25	20. Chest X ray with radiodense mass in lung cavity with fever, cough, hemoptysis. Rx -F7/25 A. Griseofulvin B. Amphotericin B C. Voriconazole D. Flucytosine
21. HIV patient with dry cough, dyspnea. Giemsa stain shows crushed ping pong ball A. Cryptococcus B. Histoplasma -F7/25 C. Pneumocystis D. Mycobacterium	22. Rose gardener with rose thorn prick with painless firm ulcerated nodule. Likely spread A. Hematogenous B. Lymphocutaneous C. Transcoelomic D. Continguous -F7/25
23. Blackish discoloration in diabetic ketoacidosis. CT shows sinus involvememt A. Bacterial sinusitis B. Nasal polyposis C. Mucormycosis D. Candidiasis -F7/25,1/26	24. Antibody coming 1st (in infection, fetus, allergy, colostrum, crossing placenta) -F7/25 A. IgG,IgA,IgM,IgD,IgE B. IgA,IgD,IgE,IgG,IgM C. IgM,IgM,IgE,IgA,IgG D. IgM,IgG,IgE,IgD,IgA
25. 60 yr old female with weakness, back pain, repeated infections, Moth eaten skull, M spike in electropheresis. Prognosis by -F7/25 A. IgG + WBC count B. Procalcitonin C. CRP D. Albumin + β2 microglobulin	26. 26 yr old ♀ with ptosis, diplopia. Ptosis improves with ice pack test. Pathology -F7/25 A. Blocked acetylcholine receptor by antibodies B. Destruction of voltage gated calcium channel C. Demyelination of motor neurons D. None
27. Skin grafting from another individual resulted in contractures. Cause -F7/25 A. GVHD B. Delayed wound healing C. SJS D. Graft failure	28. Cotton swab, placenta, vaccine, medicines, surgical mask, N95 mask discarded in -F7/25 A. Red chlorinated B. Yellow non chlorinated C. Blue chlorinated D. Yellow chlorinated
29. DNA virus with multinucleate cells A. HIV B. HSV C. HCV D. None -F7/25	30. Dentist with vesicular lesions in index finger A. HIV B. HSV C. HPV D. None -F7/25
31. Rabies sample before death obtained from A. Brain B. Spinal cord -F7/25 C. Nails D. Nape of neck/ cornea	32. HBsAg + HBeAg + IgG anti HBc +. Rx? -F7/25 A. Lamivudine B. Ritonavir C. Abacavir C. Dolutegravir
33. Pap smear uses which fixative -F7/25 A. 50% formaldehyde B. 95% ethanol C. 10% buffered saline D. 95% propanolol	34. MR and Pentavalent open vial -F7/25 A. Use both B. Use MR, discard Pentavalent C. Discard both D. Use Pentavalent, discard MR
35. MC head and neck cancer by HPV -F7/25 A. Squamous cell ca B. Orbital ca C. Oropharyngeal ca D. Astrocytoma	36. Co-test in cervical ca -F7/25 A. HPV DNA + LBC B. VIA + VILI C. PAP smear + HPV D. HPV RNA + DNA
ANSW- 17C,18B,19B, 20C,21C,22B,23C,24C,25D,26A,27D,28B,29B,30B,31D,32A,33B,34D,35C,36C	

37. Person shifts from Nigeria to USA. Vacc? A. Oka B. 17D C. RA27/3 D. Ixchiq -F7/25	38. All transmitted by saliva except A. Scabies B. Nipah C. Rabies D. Kissing disease
39. 70 year old with fever, breathlessness, confusion, uremia, hypotension. Left lung consolidation seen. Best patient position A. Supine B. Trendelenburg C. Left lateral D. Right lateral	40. 80 yr old positive for IgG Anti HBc developed esophageal varices.All are related except A. C14 urea breath test B. Beta blocker C. Sclerotherapy D. Transjugular intrahepatic portosystemic shunt
41. Kerala One Health Centre for Nipah has advised high index of suspicion for A. Palm toddy B. Meat from wild animals C. Beef and pork D. Street food- Momos	42. 42 yr old tribal from Shimoga, Karnataka was bitten by a tick leading to hemorrhagic fever A. Dengue B. Chikungunya C. Malaria D. Kyasanur Forest Disease
43. A nursing superintendent was auditing wards. MC drug allergy -NEET 21 A. Penicillin B. Vancomycin C. Streptomycin D. Dapsone	44. A school headmistress was coordinating vaccination for only teenage schoolgirls -INI23 A. Measles B. Mumps C. Rubella D. DTaP
45. 45 yr old snake charmer bitten by snake and got flaccid paralysis. Pathogenesis A. Cholinesterase B. Neurapraxia C. Demyelination D. Axonotmesis	46. 46 yr old from Nagaland donated blood. All can be seen in her blood except A. Gametocyte B. Hypnozoite C. Trophozoite D. Ring forms
47. 2 FMGE doctors Bunty and Babli went to North Korea to investigate bioterrorism. Bunty got exposed to rat flea and Babli got exposed to ardeid bird. They will be admitted in A. Both in ICU B. Isolation and general ward C. Both isolation ward D. Home Rx only	48. 35 yr old G3P2L1, undergoing laparoscopic sterilization, was found to have perinephric abscess while creating pneumoperitoneum with Veress needle. Most common cause A. Staph aureus B. Klebsiella C. Entamoeba D. E. coli
49. 3 yr old with a deep 5x5 cm laceration on face due to cat bite. Not to be done A. Wound wash woth soap and water B. Clean wound with 70% ethanol C. Suture wound with catgut suture D. Instill rabies antisera around the wound	50. Bacteriophages in 2025 Mahakumbh ganga water would have killed __ by transduction-F1/26 A. Salmonella, Shigella, E. coli, Klebsiella B. Rotavirus, Hepatitis A and E C. Entamoeba, Giardia, Cryptosporidium D. All of the above
51. 3 FMGE doctors developed rhinorrhea, epiphora, itching and dyspnoea after touching a plant. Which antibody is involved and Rx A. IgG, dapsone B. IgM, NSAIDs C. IgE, adrenaline D. IgA, soframycin	52. Bird flu pandemic in 2030 led to lockdowns, border closures and unavailability of hosp beds. Which phenomenon is responsible A. Antigenic sin B. Antigenic shift C. Antigenic drift D. Antigenic exchange
53. Waterhammer = Corrigan pulse seen in A. MR B. AR C. MS D. AS	54. Gibsons train in tunnel murmur + bounding pulse in A. MS B. MR C. ASD D. PDA
ANSWERS 37B,38A,39D,40A,41A,42D,43A,44C,45A,46B,47B,48D,49C,50A,51C,52B,53B,54D	

55. 44 yr old traveler with painful inguinal twin ulcer on thigh and scrotum touching each other, after unprotected contact. Ulcer type A. Trophic B. Marjolin C. Kissing D. Curling	56. 45-year-old male sewage worker with hemorrhagic fever, jaundice and acute renal failure. Diagnosis -NEET 25 A. Fabry dis B. Weil dis C. Hansen dis D. RHD
57. Handshake with a girl having burrow lesions in stratum corneum can lead to A. Nodulocystic lesions B. Eschar C. Nocturnal itching in finger webs D. Painless ulcer + painless lymphadenopathy	58. HIV CD4 count 80. Fever, headache, neck stiffness. India Ink stain- capsulated yeasts. Rx A. 1st HAART, after 1 month Am B + flucytosine B. 1st Am B + flucytosine, after 1 month HAART C. Both together D. All correct
59. 25 yr old with chronic viral hepatitis with high infectivity is due to positivity of A. Surface antigen B. Core antigen C. Australia antigen D. Precore antigen	60. 60 year old ♀ with SLE. Pathology A. Immune inflammation mediated B. Immune complex mediated C. Caspase mediated D. Autophagy mediated
61. Love/cuddle hormone, reward hormone A. Dopamine, serotonin B. Both Oxytocin C. Serotonin, dopamine D. Both Endorphin	62. 62 yr old with acute MI underwent PCI with doubling of diameter of LAD. Flow will increase A. 2X B. 4X C. 8X D. 16X
63. Rh incompatibility will be seen in A. Rh positive mother and Rh negative fetus B. Rh negative mother and Rh positive fetus C. Both D. None	64. 41 yr old G1P1 with HIV viral load >1 lakh presented in labour. PEP for newborn -NEET 25 A. Nevirapine for 6 wk B. TLD for 24 wk C. Nevirapine + Zidovudine for 12 wk D. None
65. Anti D = Anti Rho = Rhogam given to A. Mother B. Baby C. Both D. None	66. Vaccines planned to be added in UIP A. HSV, HIV B. HPV, TCV C. IPV, Ixchiq D. None
67. Two 18 yr old admitted in ICU for Japanese encephalitis and Tetanus. Masks used A. Both surgical mask B. Both N95 mask C. Any mask is ok D. None	68. Best PCR to diagnose meningitis, pneumonia and gastrointestinal infections -NEET 25 A. Conventional PCR B. RTPCR C. Nested PCR D. Multiplex PCR
69. Dapsone used for all except -NEET 25 A. Syphilis B. Deramatitis herpetiformis C. Leprosy D. Rhinosporidium	70. Larva at acute angle and larva flat to water A. Anopheles, Aedes B. Both Culex -NEET 25 C. Aedes, Anopheles D. Masonia, sandfly
71. C5-C9 complememt deficiency -NEET 25 A. Brucella infection B. Staph infection C. Neisseria infection D. E. coli infection	72. McFadyean in Anthrax due to -NEET 25 A. Flagella B. Polypeptide capsule C. Plasmid D. Bacterial capsule
73. Exclusion criteria for BPaLM Rx -NEET 25 A. TB in pregnancy B. Genitourinary Tb C. Miliary TB D. None	74. 65 yr old with HIV + TB for ART and ATT A. ART followed by ATT B. Both together C. ATT followed by ART D. Only ATT, skip ART
ANS 55C,56B,57C,58B,59D,60B,61C,62D,63B,64C,65A,66B,67D,68D,69A,70C,71C,72B,73D,74C	

75. A doctor couple travelled to Mars through Elon Musk SpaceX spaceship to investigate bioterrorism. Bf got exposed to rat flea and gf to ardeid bird. They will be admitted in A. Both in ICU B. Both isolation ward C. Bf Isolation and gf general ward D. Gf Isolation and bf general ward	76. A youtuber with history of Krait bite at a tertiary care hospital in Noida. All are responsibilities of doctor except A. Raise MLC B. Inform police C. Notify Government of India D. Written informed consent before giving antivenom
77. Tribal boy ate street food and developed helminthiasis. All are possible except A. Tapeworm B. Roundworm C. Pinworm D. Ringworm	78. 6 year old was stung by red scopion while playing in the garden. All are correct except A. Neurotoxic B. Pain at bite site C. Hemotoxic D. Rx by Prazosin
79. MBBS student bitten by water snake while swimming in a pond. She caught the snake and reached hospital emergency. Doctor will A. Administer antivenom immediately B. Take vitals and establish iv line C. Tie a tourniquet above the bitten area D. Barbeque the snake for party after stabilizing the patient	80. New viral infection emerged in 2030 in Nepal and Bhutan. After 5 yrs Nepal reduced incidence to <1 per 10000 population, Bhutan reduced incidence, prevalence, transmission to zero. A. Nepal passive, Bhutan active B. Nepal controlled, Bhutan removed C. Nepal Eradicated, Bhutan Eliminated D. Nepal Eliminated, Bhutan Eradicated
81. 55 year old on 2(HRZE) + 4(HRE) presented with pain in right great toe. Which crystals would be seen A. Monosodium urate B. Struvite C. Cholesterol D. Oxalate	82. Two teenagers presented with urethritis after Bangkok trip (a) Incubation period < 1 week (b) Incubation period > 1 week A. Syphilis, LGV B. Gonorrhea, Chlamydiasis C. Donovanosis, HSV D. Scabies, Candidiasis
83. House lizard fell over an 18 year old MBBS student. In a frenzy, she rubber her hand over the lizard to throw it away. She started crying and alled her boyfriend. What is the treatment A. Antivenom + Antitoxin B. Antibiotics C. Antitoxin + Antidote D. None	84. 21day old neonate brought with fever, tachycardia, vomiting and bulging fontanelle. due to Listeria infection. CSF will reveal A. Neutrophilic leukocytosis, no glucose B. Lymphocyte predominance, normal glucose C. High proteins + cobweb coagulum D. All
85. 55 year old rice farmer presented with Weil's disease with jaundice and hemorrhage. Stain will reveal _____ shaped bacteria A. Gull wing shape B. Comma shape C. Spirally coiled D. Metachromatic granule	86. Saturn high prevalence of HIV-AIDS. Alien spaceship from Jupiter should take _____ as pre-exposure prophylaxis A. Isoniazid+Rifapentin B. Lenacapavir B. Oseltamivir + Ribavirin D. Doxycyline
87. Early and delayed cord clamping except A. PPH, LBW B. HIV, Preterm infant C. Rh incompatibility, anemia D. None	88. Avian influenza can be controlled by al except A. Rapid diagnosis B. Antiviral Rx C. Vaccine for humans D. Chemoprophylaxis
ANSWERS 75C, 76D, 77D, 78C, 79B, 80D, 81A, 82B, 83D, 84A, 85C, 86B, 87D, 88C	

89. Bioterrorism attack happened between in World War III in 2026. Drones dispersed spores over military camps leading to widespread respiratory illness. Likely agent A. Yersinia pestis B. Bacillus anthracis C. Smallpox D. Ebola	90. FMGE doctor established clinic in Pattaya. He was proposed by many Thai girls. He tested them all before accepting their proposal. In which case he would be safe A. Donovanosis B. Bacterial vaginosis C. Trichomoniasis D. Condyloma acuminata
91. FMGE doctor invited his Swiss gf to Gautam Nagar. She ate street food A. EPEC B. EIEC C. ETEC D. EHEC	92. Swiss gf gifted canned food to FMGE doctor A. Staph aureus B. Clsotridium perfringens C. Bacillus cereus D. Clostridium botulinum
93. 44 year old presented with anchovy sauce pus. Trophozoite can be identified by A. Leaf shape B. Peritrichous flagella C. 4 nucleus cyst D. Erythrophagocytosis	94. FMGE doctor got cyclopropagative malaria in Nigeria while meeting his Nigerian gf. Radical Rx A. Chloroquine only B. Chloroquine + Primaquine C. Artemisinin combo Rx D. Doxycycline
95. Vaccine- Danish 1331 strain and Vaccine-Park William 8 strain. Routes of administration A. Intradermal and intramuscular B. Subcutaneous and oral C. Intraarticular and Intrathecal D. Intracameral and Intraventricular	96. 45 year HIV positive male with CD4 count 350 has larva in stools. Diagnosis, treatment? A. Ancylostoma, albendazole B. Strongyloides, ivermectin-F1/26 C. Toxoplasma, cotrimoxazole D. Schistosoma, praziquantel
97. 39 year old female with green vaginal discharge after unprotected contact. To check for fishy odour, which test will be done A. Schick test B. ONPG test C. Whiff test D. String test	98. Sequentially arrange diagnostic antigen targets for pan-malaria, Dengue, Leishmania, and malignant tertian malaria A. HRP2, NS1, IgG, IgG B. IgA, IgM, IgG, IgE C. pLDH, NS1, rk39, HRP2 D. Hg, Rt, pp65, MPT64
99. 82 yr old ICU patient was given adrenaline. Effects of all can be antagonized except A. Histamine B. Dopamine C. Acetylcholine D. Insulin	100. 3 year old rescued from a drain after Punjab floods in 2025. All are possible infections except A. Hepatitis B cirrhosis B. Typhoid bacteremia C. Poliomyelitis D. Giardia steatorrhea
101. 45 year old multigravida delivered a child with teratogenic infections. Identify A. HACEK B. TORCHES C. ESKAPE D. None	102. Bear Grylls got infected by a zoonotic virus in Amazon rainforest. All are possible except A. Rabies B. Varicella zoster C. Mpox D. Flu
103. 16 year old bitten by Naja naja. All other conditions have similar manifestation except A. Strychnine poisoning B. Floppy baby syndrome C. Organophosphate poisoning D. Guillain Barre syndrome	104. 38 yr old social butterfly went for botox. Botox will not be useful for A. Achalasia cardia, anal fissure, vaginismus B. Migraine, wrinkles, chronic pain C. Strabismus, myoclonus, blepharospasm D. Foot drop, wrist drop, myasthenia gravis
ANSWERS 89B, 90B, 91C, 92D, 93D, 94B, 95A, 96B, 97C, 98C, 99D, 100A, 101B, 102B, 103A, 104D	

105. 17 year old MBBS student with dandruff. Diagnosis, shampoo treatment A. Aspergillosis, fluconazole B. Mucormycosis, eteseconazole C. Seborrheic dermatitis, ketoconazole D. Tinea capitis, miconazole	106. 43 year old HIV patient with desert rheumatism. Etiology, treatment A. Histoplasma, voriconazole B. Mucormycosis, itraconazole C. Coccidioides, Amphotericin D. Tinea imbricata, miconazole
107. PLWHA with fungal pneumonia treated by cotrimoxazole. Diagnosis, stain used? A. Sarcocystis strainii, PAS stain B. Aspergillus flavus, lactophenol cotton blue C. Blastocystis hominis, Giemsa stain D. Pneumocystis jirovecii, Grocott silver stain	108. Pruritic rash affecting fingerwebs after sexual contact. Diagnosis and topical treatment A. Scabies, 5% permethrin B. Dermatitis herpetiformis, acyclovir C. Eczema herpeticum, ganciclovir D. Rubella, lulliconazole
109. Anaphylotoxins, opsonins and membrane attack complex are A. (C1, C2, C3), TNF alpha, IFN alpha B. (C3a, C4a, C5a), (C3b, C4b) and C5b6789 C. (C3b, C4b), C5b6789, (C3a, C4a, C5a) D. C5b6789, IgD, p24	110. Helminths which can pierce the sole of foot and cause soil transmitted helminthiasis are A. Necator, Dracunculus B. Wuchereria, Babesia C. Ancylostoma, Strongyloides-F1/26 D. Trichuris, Schistosoma
111. 6 year old with fever, cephalocaudal rash on fourth day. All are correct except A. Vitamin B6 deficiency B. No treatment available C. Edmonston-Zagreb strain vaccine D. Koplik's spots-F1/26	112. Two patients matched Mantrels criteria and Ransons criteria. Identify signs sequentially A. Mc Burneys sign, Grey Turners sign B. Murphys sign, sunset sign C. Kehrs sign, Blumbers sign D. Boas sign, sunrise sign
113. Fever and tomato sized red blanching rash on palms and soles. Diagnosis and Rx A. Chickenpox, acyclovir B. Hand foot and mouth disease, no treatment C. Monkeypox, tecovirimat D. Rocky Mountain Spotted fever, no Rx	114. Mandatory blood donor screen for infectious diseases in India includes A. HIV, Hepatitis B, Hepatitis C, syphilis, malaria B. HIV, Hepatitis B, Hepatitis C, Hepatitis D, toxo C. HIV, Hepatitis B, Hepatitis C, dengue, malaria D. HIV, Hepatitis B, Hepatitis C, parvovirus, WNV
115. Violin string appearance between liver and diaphragm. Diagnosis? A. Miller Fischer Guillain Barre syndrome B. Fitz Hugh Curtis syndrome C. Ehler Danlos syndrome D. Asherman syndrome	116. FMGE doctor won a kissing contest in Russia and kissed many girls for 8 hrs. Risk of all except A. HIV, Syphilis, HSV, EBV B. Gonorrhea, Gonorrhea, Trichomoniasis C. Covid-19, Influenza, RSV, hMPV D. Scabies, Rabies, Impetigo, Ringworm
ANSWERS 105C, 106C, 107D, 108A, 109B, 110C, 111A, 112A, 113B, 114A, 115B, 116B	

117. FMGE doctor got Staph aureus food poisoning. She would have been safe with A. Sandwich B. Burger C. Samosa D. Momos	118. FMGE doctor got Vibrio parahemolyticus food poisoning. She would have eaten A. Mutton B. Bacon C. Chilly chicken D. Tuna
119. Abdominal pain after eating food. Test? A. Barium swallow B. ERCP C. C13 urea breath test D. Colonoscopy	120. Flask shape intestinal ulcer on colonoscopy A. Blue green pus B. Serosanguinous pus C. Anchovy sauce pus D. Pautrier microabscess
121. 33 year old with steatorrhea. Falling leaf motility seen in stool. Infection due to A. 4 flagella trophozoite B. 4 nucleus cyst C. Rhabditiform larva D. Planoconvex ova	122. 22 year old Fashion model refused Rx of Shigella dysentery. She is at risk of A. Flask shaped ulcer B. Volvulus C. Lead pipe colon D. Rectal prolapse
123. 46 year old bollywood actress drank plenty of water, avoided spicy/processed food, took isabgol and lactulose. Diagnosis A. GERD B. Esophgeal varices C. Hemorrhoids D. Anorectal abscess	124. 21 year old FMGE doctor saw his gf cheating with another guy. He had angina, ↑Troponin, ECG ↑ST ↓P, due to severe stress A. Pericaditis B. Complete heart block C. Arrythmia D. Takotsubo cardiomyopathy
125. 64 yr old CBNAAT positive on Rx was found with mutated rpoB gene. Colour of urine A. Milky B. Orange C. Cola D. Green	126. Post cholecystectomy, fatty fertile female of forty should avoid A. Egg B. Butter C. Rice D. Spinach
127. Devdas developed fever with CURB65 score 3 in memory of Anarkali A. Strep pneumoniae B. Legionella C. Klebsiella pneumoniae D. Mycoplasma	128. 18 yr old with Faget sign, bleeding ulcer and rose spots. IgM O 1:100, IgM H 1:200. Type of fever A. Saddleback fever B. Pel Ebstein fever C. Stepladder fever D. Undulant fever
129. 5 yr old Rx with ACT-AL brought with cola colored urine. Banana gametocyte and accole form seen but schizont not seen. Antigen? A. HRP2 B. NS1 C. p24 D. rk39	130. 55 yr old with brown shrunken nodular fibrotic hard firm liver with bile duct proliferation and hepatocyte ballooning. Marker of active infection A. HBsAg B. Anti HBs C. HBeAg D. Anti HBe
131. HIV negative couple joined swingers club. Safest sexual perversion A. BDSM B. Sodomy C. Frotteurism D. Voyeurism	132. HIV positive teen sodomized by trans-sexuals. Test to detect sperm in anal canal A. Benzidine B. Barberio C. Balamuth D. Bismuth
133. HIV transmission prevented by A. Diaphragm B. Coitus interruptus C. Abstinence D. Sexual promiscuity	134. New antibiotic having serial concentrations 1000→ 800→ 600→ 400. Which order kinetics A. 1st B. 2nd C. 3rd D. Michelis Menten zero order
135. Repeat blood transfusion required except A. Thalassemia B. Aplastic anemia C. Hemophilia D. von Willebrand diseas	136. Blood bank did all tests and blood was HIV negative, but recipient became HIV positive A. HIV test faulty B. Donor in window period C. Recipient immunocompromised D. All
ANSWERS 117C, 118D, 119C, 120C, 121B, 122D, 123C, 124D, 125B, 126B, 127C, 128C, 129A, 130A, 131D, 132B, 133C, 134D, 135D, 136B	

137. Sheep wool carrier with painless black scar on leg-F1/26 A. Scrub typhus B. Anthrax C. Rickettsia D. Staph aureus	138. 30yr old foreign traveler with rice water -F1/26 diarrhea (no blood/pus) + dehydration. Toxin? A. Cholera B. Shigella C. E.coli ST D. E. coli LT
139. 24yr old female with dysuria, frequency. No fever, flank pain, vomit, vagina discharge. Not preferred as first line empirical Rx-F1/26 A. Nitrofurantoin B. Cotrimoxazole C. Ciprofloxacin D. Fosfomycin	140. 4yr old with LRTI. Sputum culture- Fastidious Gram negative bacillus needing X + V factor grows only on chocolate agar and not blood agar. Identify A. Influenza virus -F1/26 B. Strep pneumoniae C. Hemophils influenzae D. Staph aureus
141. 28yr old HBsAg +ve pregnant female with high HBV DNA. No liver failure. Rx -F1/26 A. Pegylated interferon B. Tenofovir C. Zidovudine D. Acyclovir	142. 32yr old returns with high fever, drowsiness, confusion from falciparum malaria endemic area A. iv Artesunate B. iv Artesunate + SP -F1/26 C. Oral ACT AL D. Oral Chloroquine
143. 44yr old from endemic area with resp problems. Small oval yeast with narrow bud seen inside macrophages. Diagnosis -F1/26 A. Blastomyces B. Histoplasma C. Cryptococcus D. Candida	144. 62yr old with fever, headache, neck stiffness, vomiting, photophobia, confusion, Kernig sign. CSF low glucose. No papilledema. Empirical Rx A. Meropenem B. Ceftriaxone+ Vancomycin C. Vancomycin D. Clindamycin -F1/26
145. 22yr old with typhoid Rx by ciprofloxacin. No improvement after 5 days. Next Rx -F1/26 A. Chloramphenicol B. Ampicillin C. Ceftriaxone D. Doxycycline	146. 34yr old with rapidly moving, serpiginous, itchy rash. Only larva in stool, no va, no cyst-F1/26 A. Ancylostoma B. Ascaris C. Enterobius D. Strongyloides
147. 20yr old with fever, sore throat, cervical lymphadenopathy, atypical lymphocytes. Paul Bunnell heterophile antibody +ve. Diagnosis A. CMV B. EBV C. HIV D. Influenza -F1/26	148. Opportunistic infec in HIV +ve. Curdy white vaginal discharge shows oval yeast+ narrow-F1/26 budding on KOH mount. Identify characteristic A. Pseudohyphae B. Clue cell C. Flagella D. None
149. Canned food poisoning in a family lead to botox flaccid paralysis. Transmission -F1/26 A. Vector B. Direct C. Indirect D. Vehicle borne	150. 32yr old with eye pain, redness, photophobia. Dendritic ulcer on fluorescin stain. Rx -F1/26 A. Ciproflox B. Acyclovir C. Fluconazole D. Steroid
151. 51yr old fever, myalgia after eating pork. Muscle biopsy shows helminthic infec -F1/26 A. Taenia solium B. Trichinella spiralis C. Toxoplasma D. Trichuris trichura	152. 28yr old with painless genital ulcer + painful unilateral inguinal swelling suggestive of bubo, after unprotected sexual contact. Chlamydia strain A. D-K -F1/26 B. A-C C. L1-L3 D. M-O
153. Hypersensitivity- Rheumatoid arthritis, Lepra reaction I and II -F1/26 A. I,II,III B. II,III,IV C. III,IV,III D. IV,III,II	154. Main cytokine in granuloma activating macrophages in Langerhan cell histiocytosis -F1/26 A. IFN-α B. Interferon-γ C. TNF-α D. TGF-β
155. DNA fingerprinting based on -F1/26 A. VNTR B. Blots C. SNP D. Start codons	156. 7 yr old with EBV related jaw lymphoma A. t (14:18) B. t (11:14) C. t (8:14) D. t (9:22) -F1/26
ANSWERS 137B, 138A, 139D, 140C, 141B, 142A, 143B, 144B, 145C, 146D, 147B, 148A, 149D, 150B, 151D, 152C, 153C, 154B, 155A, 156C	

157. 52yr old diabetic with fever, face pain, black necrotic lesion over nasal turbinate. Maxillary black sinus. Rx -F1/26

A. Only surgery B. Surgery + Antibiotics

C. Only antibiotics D. Surgery + Antifungals

158.27yr old low socioeconomic status female with foul smell nasal discharge, epistaxis,crust,external nose, pale granular nodular mass blocks nose. Rx

A. Only surgery B. Nose decongestion+ Antifungal

C. Only antihistaminic -F1/26 D. Surg + Antibiotic

159. 28yr old with BT leprosy had fever, tender erythematous lesion after 2mnth of WHO-MDT. No new nerve problem. Next step -F1/26

A. Continue MDT and start oral prednisolone

B. Continue MDT and start oral doxycycline

C. Stop MDT and start thalidomide

D. Stop MDT and start oral prednisolone

160. 25yr old returns with high fever, chills of 3day from falciparum malaria endemic Manipur. Rx

A. Artemether-Lumefantrine (ACT-AL) x 3days + single dose Primaquine on Day 2

B. ACT-AL x 3days + Primaquine for 14 days

C. Chloroquine x 3days + Primaquine single dose

D. Oral quinine for 7 days -F1/26

161. 11yr old diabetic with itchy eye, redness, watering every summer with large papillae on upper tarsal conjunctiva. Rx -F1/26

A. Only surgery B. Only antihistaminics

C. Antibiotic D. Mast cell stabilizer+ Steroid

162. Level of prevention for rehab, regular meds for heart disease, screening for cervical cancer, reducing salt intake, maternal nutrition -F1/26

A. 3^0, 2^0, 2^0, Primordial, 1^0 B. 4^0, 3^0, 2^0, 1^0, 1^0

C. 4^0, 3^0, 3^0, 1^0, Primordial D. 3^0, 3^0, 2^0, 1^0, 1^0

163. Dispose urobag, expired medicines, TPN

A. Red, yellow, red B. Yellow, red, yellow

C. Red, black, white D. Red, blue, white-F1/26

164. 27 yr old with discoid rash + photosensitivity. Worst prognosis antibody in SLE-F1/26 A. Anti Ro

B. Anti histone C. Anti Smith D. Anti dsDNA

165. 42yr old with fever, headache, confusion. CSF shows WBC 150/µl (Lymphocyte), normal glucose, protein 81mg/dl. MRI brain temporal lobe lesion. Diagnosis -F1/26

A. Viral meningitis B. HSV encephalitis

C. Mollaret meningitis D. Bacterial meningitis

166. 36yr old with headache, neck stiffness. CSF 90%Lymphocyte 10%neutrophils, glucose 40mg/dl, (plasma glucose 100 mg/dl), protein 1mg/dl. Diagnose meningitis (m) and Rx -F1/26

A. Viral m + Acyclovir B. Fungal m + Secnidazole

C. Bact m + Ceftriaxone D. TB m + ATT

167. 34yr old with shortness of breath, effort intolerance, opening snap, mid diastolic-F1/26 murmur. Echocardiograhy EF 38%. Diagnosis

A. AS B. MR C. MS D. AR

168. 35 yr old with high fever, calf pain, red eyes after walk in flood water. No jaundice, renal lesion. Which phase of leptospirosis A. Weils disease

B. Leptospiremic C. Septicemic D. Immune -F1/26

169. Add Vit A to butter to prevent defi -F1/26

A. Enriched B. Fortified C. Adulterated D.None

170. Triage category for crush injury limb -F1/26

A. Red B. Yellow C. Green D. Black

171. 37 wk preg, Hb <8 gm/dl. Anemia Mukt Bharat program Rx (Elemental iron) -F1/26

A. 60 mg iron + 500 µg folic acid wkly x 6m

B. 60 mg iron + 500 µg folic acid daily x 6m

C. 100 mg iron + 500 µg folic acid daily x 100d

D. 60 mg iron + 500 µg folic acid BD x 6m

172. 7yr old with pleomorphic centripetally distributed rash. Diagnosis -F1/26

A. Variola B. Varicella C. Rubeola D. Rubella

173. 2yr old with bluish spot on buccal mucosa. Diagnosis -F1/26

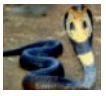
A. Variola B. Varicella C. Rubeola D. Rubella

157D,158D, 159A,160A, 161D,162A, 163A,164D, 165B,166D, 167C,168B, 169B,170A, 171D,172B, 173C

174. 5yr old with dyspnea, cough, fever. Chest Xray right upper lobe consolidation -F1/26 A. Pneumothorax B. Pneumonia C. Lung abscess D. Pleural effusion	175. 50yr old Tamil Nadu farmer with multiple foot sinuses with black granule. KOH fungal hyphae.Rx A. Only amputation B. Anti tubercuar Rx (ATT) C. Only antibiotics D. Surgery + Antifungals
176. Truck driver with painless genital ulcer + painless inguinal lymph node. Diagnos -F1/26 A. LGV B. Chancroid C. Syphilis D. HSV	177. Truck driver with painful genital ulcer + painful u/l inguinal lymph node. Diagnos -F1/26 A. LGV B. Chancroid C. Syphilis D. HSV
178. 40 yr old with left temporal head injury speaks fluently but meaningless. Unable to comprehend spoken language. Affected area A. Broca B. Primary auditory cortex C. Wernicke D. Arcuate fasciculus -F1/26	179. Kidney transplant recipient on long term Rx with non healing ulcer/blisters on sun exposed skin. Diagnosed squamous cell ca. Reason -F1/26 A. Immunosuppressants B. Steroids C. UV exposure D. Viral infection
180. 22yr old with chronic anemia,fatiguability, intermittent jaundice, episodic bone pain due to exertion/dehydration, pallor, splenomegaly, Hb 8gm/dl, normal MCV, indirect bilirubin, reticulocyte 6%. PBS- Flat curved RBC -F1/26 A. Iron def anemia B. Sickle cell anemia C. Aplastic anemia D. Thalassemia trait	181. 32yr old chokes suddenly, develops stridor and turns cyanotic. Heimlich maneuvre failed, entotracheal intubation failed, oxygen saturation falling rapidly. Next step to secure airway -F1/26 A. Repeat endotracheal intubation B. igel C. Laryngeal mask airway D. Cricothyroidotomy
182. 3yr old with flat nasal bridge, up-slant palpebral fissure, developmental delay, palamar crease. Diagnosis of syndrome -F1/26 A. Edward B. Down C. Turner D. Noonan	183. 27yr old with high intracranial pressure after head injury. Cushing reflex corresponds to -F1/26 A. Tachycardia+ HTN B. Bradycadia+ hypotension C. Bradycardia+ HTN D. Tachycardia+ hypotension
184. Coadministered with iron folc acid -F1/26 A. Vitamin A B. Vit B1 C. Vit C D. Vit D	185. Mycoplasma and Malassezia colony -F1/26 A. Pin point B. Mucoid C. Daisy head D. Fried egg
186. Rinne test +ve both ear (AC>BC), Weber test lateralizes to left ear in 45yr old -F1/26 A. Left Conductive deafness (CD) B. Right CD C. Left Sensorineural deafness D. Right SNHL	187. 28yr old with still born baby at 38wks POG. On 3rd day postpartum, she develops breast fulllness and engorgement. Rx to suppress lactation -F1/26 A. Oxytocin B. Misoprostol C. Bromocriptine D. Nil
188. 50yr old industry worker with chronic loud noise exposure reported gradual b/l deafness. PTA characteristic dip at 4000Hz, lower frequencies preserved -F1/26 A. Presbyacusis B. Otosclerosis C. Noise induced deafness D. Meniere disease	189. 30yr old flies Delhi- Ladakh at 15000ft altitude, develops throbbing headache, nausea, dizziness, fatigue, insomnia. Pulse 104/min, resp 22/min, room air SpO_2 88%. No pulmonary edema. Best Rx A. Acetazolamide B. Descent to low altitude C. Dexamethasone D. Aspirin -F1/26
190. Alcoholic with chronic liver dis. Rx -F1/26 A. Vitamin A B. Vit B1 C. Vit C D. Vit D	191. 48yr old, biconvex hematoma after head injury A. EDH B. SDH C. SAH D. ICH -F1/26
ANSWERS 174B, 175D, 176C, 177B, 178C, 17A, 180B, 181D, 182B, 183C, 184C, 185D, 186B, 187C, 188C, 189B, 190B, 191A	

11. MICROBIOLOGY HAPPY 34 PYQs

My Score _____/ 34, _____/34

1. Farmer in a coma after neurotoxic snake bite  A. Cobra B. Viper C. Adder D. Python	2. Biohazard-B,Chemical haz-C,Radiation haz-R A. BCR B. RCB C. CBR D. CRB
3. Orchitis,oophoritis -N16 A. Scrotum, Dartos B. Testis, Ovary C. Prostate, cervix D. None	4. HIV is not transmitted by -N18 A. Breastfeed B. iv drugs C. Needle D. Vectors
5. Acute infection-F20 A. IgG B .IgA C.IgE D .IgM	6. Ringworm A.Parasite B.Virus C.Fungus D.Bact
7. Thrush -F7/24 A.HIV B.Candida C.HPV D.HSV	8. Tenofovir-F24/26 A. HIV B. Hep B C.Both D.HSV
9. Most dangerous bioterrorism (Cat A) -N18/20 A. Plague B. Covid C. HIV D. Rabies	10. Malaria Rx, vacc -N21 A. Ribavirin, RA27/3 B.Niacin, Oka C.Chloroquine, Mosquirix D.None
11. Kala azar Rx -F21/23,N18 A. Metronidazole B. Albendazole C. Neomycin D. Amphotericin B	12. Measles-Rubella vaccine schedule -F1/19,N18 A. 6-10-14 wks B. At birth C. 9 months D. 5 yrs
13. Widal and VDRL test for -F1/25 A. TB, HIV B. Plague, RHD C. Typhoid, Syphilis D. UTI, JE	14. Transmission of diphtheria -F24 A. Orofecal B. Vertical C. Respiratory route D. Vector borne
15. Rh-mother, Rh+fetus, anti ___ given -N19 A. A B. B C. C D. D	16. Main Rx of cholera -N21 A. Azithromycin B. Amikacin C. Doxycycline D. Rehydration
17. Keratitis, salpingitis A.Keratin, saliva gland B. Skin, scalp C.Cornea, fallopian tube D. None	18. Lymph node TB is ___ TB -F23/24 A. Latent B. Resistant C. Extrapulmonary D. Miliary
19. Fever, headache, neck stiffness -F 23/24/25 A.Torticollis B.Paraplegia C.Meningitis D.None	20. Dengue is transmitted by -F1/19 A. Culex B. Virus C. Aedes D. Bacteria
21. Filaria Rx-N19 A. DEC B. TLD C. DCA D. TLE	22. Smallpox eradicated-F A.1960 B.70 C.80 D.90
23. Rubella vacc A.DPT B.17D C.Oka D.RA 27/3	24. Orofecal Hepatitis-F1/6/21 A. A B. B C. C D. D
25. MDRTB Rx-N19 A.RHZE B.RHE C.BPAL D.All	26. Trachoma eliminated-F A.2020 B.22 C.24 D.18
27. Shingles-N17 A.VZV B.HSV C.CMV D.HPV	28. Swine flu. A. H3N2 B. H1N1 C. H7N9 D. None
29. HIV→AIDS (years)-F21 A. 1 B. 2 C. 5 D. 10	30. Vaccine is __ prevention A. 1^0 B. 2^0 C. 3^0 D. 4^0
31. Rx is __ prevention A. 1^0 B. 2^0 C. 3^0 D. 4^0	32. Condom is __ prevention A. 1^0 B. 2^0 C. 3^0 D. 4^0
33. i-pill is ___prevention A. 1^0 B. 2^0 C. 3^0 D. 4^0	34. Antidote is ___prevention A. 1^0 B. 2^0 C. 3^0 D. 4^0

ANSWERS 1A, 2B, 3B, 4D, 5D, 6C, 7B, 8C, 9A, 10C, 11D, 12C, 13C, 14C, 15D, 16D, 1CD, 18C, 19C, 20C, 21A, 22C, 23D, 24A, 25C, 26C, 27A, 28B, 29D, 30A, 31B, 32A, 33B, 34B

PREVIOUS YR TOPICS (PYTs)- GEN-Cell wall, capsule, motility, culture, growth curve, antibiogram, toxin, steriliz, BMW. BACT-Gas gang, TB, Leprosy, Typhoid, Cholera, Syphilis, Staph, Strep, E. coli, Diphtheria, Rickettsia, Chlamydia. VIR-Inclusion body, JE, Herpes, Dengue, Covid, Influenza, HIV, Hepatitis, Polio, Rabies, MMR. PAR-Vector, Ova, Amoeba, Kala azar, Giardia, Trichomonas, Malaria, Filaria. MYC-Tinea, Cryptococcus, Pneumocystis, Mucor, Candida. IMM- Antibody, hypersen, vacc, transplant, mul myeloma. CLIN-RHD, meningitis, endocarditis, pneumonia, UTI, STI, fever, food pois

12. BOLLYWOOD PYQs Non-Hindi folks, turn the page No Score, have fun

1. Ye shaam mastani, madhosh kiye jaye A. Dengue B. Malaria C. Typhoid D. TB	2. Ye chand sa roshan chehra, which syndrome A. Conn B. Cushing C. Reiter D. Horner
3. Lag ja gale ke phir ye hasin raat ho na ho A. Brucella B. Typhoid C. Typhus D. Diphtheria	4. Aajkal paon zamin par nhi padte mere A. Mycetoma B. Kyphosis C. Gibbus D. UTI
5. Hothon se chu lo tum, mera geet amar kar do A. CMV B. EBV C. MCV D. HPV	6. Panchhi banu udti firun mast gagan me A. Covid B. Influenza C. TB D. Diphtheria
7. Dil jigar nazar kya hai, main to tere liye jaan A. Left heart failure B. Heart failure with low EF C. Congestive heart failure D. Tamponade	8. Blue eyes hypnotize kardi hai mainu A. Cataract B. Colour blindness C. Glaucoma D. Corneal ulcer fluorescin stain
9. Ye reshmi zulfein, ye sharbati aankhein A. Seborhheic dermatitis B. Atopic dermatitis C. Nummular dermatitis D. Contact dermatitis	10. Gori hai kalaiyaan, la de mujhe hari chudiyan A. Carpal tunnel syn B. Tarsal tunnel syn C. Rotator cuff syn D. Trigger finger syn
11. Ye jo halka halka suroor (Euphoria) hai A. Alcohol B. Opioid C. Cannabis D. GHB	12. Zindagi ek safar hai suhana A. Quinine B. Mefloquine C. Chloroquine D. Doxycycline
13. Mujhe neend na aaye, mujhe chain an aaye A. Kerion B. Epistaxis C. Scabies D. Flu	14. Beedi jalaile, jigar se piya, jigar ma badi aag A. Asthma B. COPD C. Empyema D. Sarcoidosis
15. Neele neele amber pe chand (Neonatal) A. Jaundice B. Meningitis C. Sepsis D. VSD	16. Bheege hoth tere, pyaasa ye badan A. Cirrhosis B. Croup C. COPD D. Cold sore
17. Sun raha hai na tu, ro rahi hun main, Rx? A. Grommet B. Holter C. Drain D. Stent	18. Ban ke titli dil uda uda uda hai kahin door A. Lichen planus B. Acne C. Rosacea D. SLE
19. Ayi chikni chameli, chup ke akeli A.Osteoma B. Leiomyoma C. Lymphoma D. Melanoma	20. Aankh surme se bhar ke taiyar ki (Fracture?) A. Mandible B. Vertebra C. Skull D. Alveolar
21. Jiya dhadak dhadak jaye A. Asystole B. Heart block C. Fibrillation D. Wenkebach	22. Barsaat ke din aaye, mulaqat ke din aaye A. Scrub typhus B. TB C. Measles D. None

ANSWERS 1D (Shaam ~Low grade evening fever- TB), 2B (Chand~ Moon face- Cushing syn), 3D (Gale ~Bull neck- Diphtheria), 4A (Paon~Mycetoma foot), 5B (Hoth~Lip, Kissing dis- EBV), 6B (Bird flu~Panchi), 7C (Jigar~Liver, hepatomegaly CCF), 8D (Fluorescin stain~Blue cornea), 9A (Dandruff ~Seborrheic dermatitis- Hair+ eylash), 10A (Kalaiyan~Wrist), 11C (Alcohol, opioid, GHB- Depress), 12D (Safar~Malaria travel prophylaxis- Doxycycline), 13C (Neend~Sleep, Scabies- Night itch), 14B (Beedi~Smoking- COPD), 15A (Neele~Blue light Rx- Neonatal jaundice), 16D (Cold sore on lip- HSV) 17A (Sun~Listen, MC deafness- Glue ear-Rx Grommet), 18D (Titli~Butterfly rash- SLE), 19B (Chikni~ Smooth muscle tumour- Leiomyoma), 20C (Surma~Black/Raccoon eye- Skull/Orbit/Maxilla fracture) 21C (Dhadak~Fibrillation), 22A (Barsaat~Rain- Scrub typhus)

23. Ik pal ka jeena, phir to hai (Shortest half life) A. IgA B. IgE C. IgG D. IgD	24. Sandese aate hain, mujhe tadpate hain A. Complement B. Cytokine C. CRP D. None
25. Raat kali ek khwaab me aayi A. Candida A. Clostridium B. Campylobacter D. None	26. Naina barse rimjhim rimjhim A. Epilepsy B. Epistaxis C. Epiphora D. None
27. Main na bhoolunga, tu na bhoolegi A. Parkinsons B. Korsakoffs psychosis C. Migraine D. Tourette syndrome	28. Chalte chlate mere ye geet yaad rakhna A. Strep pneumoniae B. Klebsiella pneumoniae C. Mycoplasma pneumoniae D. None
29. Chingari koi bhadke (Chingari= Flame cell) A. Burkitts lymphoma B. Hodgkins Lymphoma C. Multiple myeloma D. Ac myeloid leukemia	30. Chura liya hai tumne jo dil ko A. Coronary steal phenomenon B. Arrhythmia C. Torsades de pointes D. Myocarditis
31. Main saans leta hoon, teri khushbu aati hai A. Rhinovirus common cold B. Klebsiella ozaenae atrophic rhinitis C. Strep pneumoniae sinusitis D. All	32. Saans me teri saans mili to, mujhe saans aayi A. Hypoxia + Hypercapnia/ Hypercarbia B. Hyperoxia + Hypercapnia/ Hypercarbia C. Hypoxia + Hypercarbia D. None
33. Nach meri rani rani A. Asterixis B. Chorea C. Tremor D. None	34. Jhoom barabar jhoom sharabi-Rx A.Letrozole B. Fluconazole C. Anastrazole D. Fomepizole
35. Aankhon me base ho tum, image formed at A. Cornea B. Lens C. Choroid D. Retina	36. Tip tip barsa pani, pani ne aag lagai A. None B. Proteus C. Pseudomonas D. Pasteurella
37. Ye moh moh ke dhaage (All except) A. Giardia B. H. pylori C. Strongyloides D. All	38. Hum tum ek kamre me band ho (Phobia) A. Acro B. Agora C. Claustro D. Ophidio
39. Zara sa jhoom lun main A. Achalasia B. Thrush C. Sciatica D. Meniere disease	40. Mere mehboob qayamat hogi (Delusion) A. Grandeur B. Narcissism C. Nihilism D. None
41. Dum maro dum, mit jaye ghum all except A. Bhang B. Ganja C. Charas D. Tequila	42. Khaike paan banaras wala, khul jaye band A. Oral ca B. Gastric ca C. Intestinal ca D. All
43. Sara zamana hasino ka diwana A. Endemic B. Epidemic C. Sporadic D. Pandemic	44. Nashe si chadh gyi oye, kudi nashe si chadh A. Ketamine B. Lidocaine C. Vecuronium D. N_2O
45. Dil diyan gallan, karange naal naal beh ke A. Fremitus B. Murmur C. Stridor D. Tinnitus	46. Tuj ko kitna chahta hun tu kabhi soch na sake A. Dementia B. Delirium C. Both D. Narcolepsy

ANSWERS 23B (Pal~Moment, IgE shortest life), 24B (Sandese~Signal-Cytokine), 25A (Kali~Flower Bud- Candida), 26C (Naina barse~Watery eye-Epiphora), 27B (Bhool~Memory loss-Korsakoffs), 28C (Chalte~Walking pneumonia-Mycoplasma), 29C (Chingari~Flame cell-Multipl myeloma), 30A (Chura ~Steal), 31B (Ozaena- Bad smell), 32A (Saans- Breath exhaled air), 33B (Nach~Dance-Chorea), 34D (Jhoom~Alcohol overdose-Rx Fomepizole), 35D, 36A (Aag~Burns-MC Pseudomonas), 37D (Dhaage ~String test), 38C, 39C (Jhoom~Vertigo), 40C (Qayamat~Pralay~Nihilism), 41D (Dum~Smoke), 42A (Paan~Oral ca), 43D (Zamana~World), 44A (Nashe~Ketamine), 45B, 46C (Soch~Thought block)

13. Molly/Kolly/Tolly/Sandalwood/Punjabi/Marathi/Bhojpuri PYQs Have fun

MALYALAM 1. Mundiri paadam poothu nilkana A. Staph aureus B. E. coli C. Proteus D. Shigella	MALYALAM 2. Kiliye kiliye, mani mani meghathoppil A. Nipah B. Ebola C. Yellow fever D. JE
TAMIL 3. Kodi Nilavugal Varum Azhage A. Conn B. Cushing C. Turner D. Down syndrom	TAMIL 4. Nallaru po, Enna seithalum nallaru po A. E. coli B. Proteus C. Shigella D. Lactobacillus
TELUGU 5. Chuttamalle chuttestandi tuntari A. Squint B. Asterixis C. Sciatica D. Heart burn	TELUGU 6. Chikiri chikiri Aa Chandrullo Mukka A. Conn B. Cushing C. Turner D. Down syndrome
KANNAD 7. Mungaru Maleyalli Tandalle Neenilli A. Scrub typhus B. TB C. Measles D. None	KANNAD 8. Hot'Tu Mamma A. Tetanus B. Malaria C. Rabies D. TORCH
MARATHI 9. Komal kaya ki moh maya A. Eczema B. Acne C. Psoriasis D. Ehler Danlos syndrome	MARATHI 10. Kevadyach,aturlya dolyach sapaan tu A. Nystagmus B. Tenesmus C. Trismus D. None
GUJARATI 11. Jagthi door ne kyaank manmooki A. Psychosis B. Bipolar dis C. Mania D. PTSD	GUJARATI 12. Haye kya baag maa thaodi lau phul A. Candida B. Sporothrix C. Histoplasma D. None
ODIYA 13. Dela marla jahara, tate deuche ishara A. Antigen B. Antidote C. Anthelmintic D. None	PAHADI 14. Chunri teri chamakni, gulabi sharara A. Asthma B. COPD C. Emphysema D. ARDS
BANGL 15.Keno emon ratri namche janle bojhena A. ↓Vitamin A B. ↓Vit B C. ↓Vit C D. ↓Vit D	BANGL 16. Dak shune ku na ase tobe ekla cholo re A. DNA B. RNA C. Both D. None
BHOJPURI 17. Toot jayi rajaji, palang sagwan ke A. Hiccup B. Decubitus ulcer C. Hives D. Acne	BHOJP 18. Kamariya kore lapalop, lollipop lage lu A. Lordosis B. Ptosis C. Miosis D. Otosclerosis
KHASI 19. Jalieh jalieh oh jadoh jadoh A. Plague B. Anthrax C. Brucella D. Cholera	ASSAM 20. Mayabini ratir bukut dekha palu tumar A. ↓Vitamin A B. ↓Vit B C. ↓Vit C D. ↓Vit D
PUNJABI 21. Same time same jagah agle o din A. Acne B. Fix drug eruption C. SJS D. SSSS	PUNJABI 22. Lakh 28 kudi da, 47 weight kudi da A. Typhoid B. Malaria C. TB D. Hypothyroidism
KASHMIRI 23. Yeth lola bagas zakhmi dil gul A. Metformin B. Metoprolol C. Morphine D. Nil	MEITEI 10. Eikhoi Nungpaibi, nungai nongmaibi A. Candida B. Sporothrix C. Histoplasma D. None

ANSWERS 1A (Mundiri~Grape-Staph), 2D (Klli~Bird-Japan enceph), 3B (Nilavugal~Moon face - Cushings), 4D (Nallaru~Good bacteria-Lactobacillus), 5A (Chuttamalle~Twisted-Squint), 6B (Chandrullo~Moon face-Cushings), 7A (Mungaru~Rain- Scrub typhus), 8D (Mama~Mother to child-TORCH), 9B (Komal kaya~Soft skin-Ehler Danlos syn), 10A (Aturlya dolyach~Unstable eye-Nystagmus), 11D (Manmooki~Mind agree- PTSD, others no insight), 12B (Baag~Gardener- Rose Gardener dis= Sporothrix), 13B (Jahara~Poison- Antidote), 14C (Gulabi~Pink puffer- Empysema), 15A (Ratri~Night blindness- ↓Vit A), 16B (Ekla~Alone-RNA single stranded), 17B (Palang~Bed-sore=Decubitus ulcer), 18A (Kamariya~Waist-Lumbar lordosis), 19D (Jadoh~Rice-Cholera), 20A (Ratri~ Night blind- ↓Vit A), 21B, 22C (↓Weight- TB), 23C (Zakhmi dil-Morphine), 24B (Nungpaibi~Rose Gardener is=Sporothrix)

ANTI- CONFUSION ANTIVENOM FOR SCORE >200

1. Tinea= Ringworm→ Fungal infection(Not worm) 2. Taenia solium-Tapeworm
3. Typhoid→ Salmonella typhi/paratyphi A-B-C 4. Typhus→ Rickettsia -INI25
5. Typhoid = Paratyphoid 6. Salmonella typhimurium→ Food poisoning
7. Donovanosis (STI) → Klebsiella granulomatis, donovan body (7-8 unrelated)
8. Kalazar=Leishmaniais-N23→ Leishmania donovani, Leishman donovan body
9. Hemophilus influenza (bacteria)-No influenza 10. Influenza→Influenza virus
11. Pox virus→ Smallpox= Variola, Mpox 12. German measles= Rubella
13. Herpes virus-3→ Chickenpox= Varicella-N21 14. Measles= Rubeola-N19,21,23
15. Mycosis fungoides= Cutaneous T-cell lymphoma (Not fungal infection)

1. RHEUMATIC fever/ carditis/ valvulitis/ heart dis- Strep pyogenes -N18/19/22
2. Rheumatoid arthritis-N22,F1/26 Non infec autoimmune dis like SLE, psoriaisis
3. Desert rheumatism/ Valley/ California fever- Coccidioides fungus -N24

1. LUPUS vulgaris -N20/25- MC skin TB Lupus ~ Wolf
2. Systemic LUPUS Ery (Lupus arthritis/nephritis)- Non infec autoimmune dis

MALIGNANT(Mlg)=Severe -F 1/25 1. Mlg pustule- Cutaneous anthrax
2. Mlg otitis externa- Pseudomonas 3. Mlg tertian malaria- P. falciparum
4. Mlg hypertension >180/120 5. Mlg hyperthermia- Scoline- Rx- Dantrolene

PERNICIOUS(Per)=Severe 1. Per anemia-↓V-B_{12} 2. Per malaria-F1/25-Falciparum

Fever→Day Of Rash=R (Moniker) -F19-25(7) .List as per discovery- Outdated
1st dis- Chickenpox, Rubella=German measles, Smallpox 4th dis- Measles
5th dis- Parvovirus 6th dis- HHV6=Roseola 8th day dis- Neonatal tetanus
1. Cephalocaudal R- Occiput/Nape of neck→Face→Trunk- Measles,Rubella
2. Centrifugal Pleomorphic R- Chickenpox-F1/26 3. Centripetal R- Smallpox
4. Dermatomal R- Shingles=Herpes zoster 5. Blanching Herman R- Dengue
5. Itchy/ Pruritic R- Ringworm, Scabies, Chickenpox, Rubella, Herpes
6. Butterfly/ malar R- SLE 7. Slapped cheek R- Parvovirus

8. Sandpaper R- Scarlet fever 9. Morbilliform R- MC drug reaction
9. Palmoplantar R- Smallpox, Mpox, Measles, Syphilis, Tomato fever=HFMD, Spotted fever, Meningitis, Kawasaki dis, psoriasis, Reiter syn, TSS, SSSS
10. R with lymphocutaneous spread- Sporothrix= Rose Gardener dis
11. Heliotrope R + Gottron papules- Dermatomyositis
12. Multilocular R- Chickenpox,smallpox,Mpox,measles,rubella- Multiple area
13. Unilocular R- Shingles, insect bite, con tact dermatitis- Single area

WARTS=VERRUCA (Hard skin, no fluid) 1. MC wart overall- HPV
2. Bluish wart with central umbilication- Molluscum contagiosum virus
3. Warty verrucous lesion with copper penny body- Chromoblastomycosis

FEVER OF UNKNOWN ORIGIN =FUO =PUO 1. MC FUO- TB
2. Undiagnosed fever >101^0F/ >38.3^0C for >3wk despite investigation
3. 4 Types- Classical, Neutropenic (<500/dl), HIV, nosocomial FUO

1. Low grade evening fever with weight loss- TB 2. Weils fev-N22-Leptospira
3. High grade fev + chills + rigors- Malaria 4. Typhus fever- Rickettsia
5. Typhoid= Enteric= Step-ladder fever- Salmonella typhi, paratyphi
6. Scarlet fever- Streptococcus pyogenes 7. Undulant= Malta fever- Brucella
8. Saddleback=Biphasic fev, Breakbone fev, Breakheart fever- Dengue -N21,22
9. Black/Dum dum fev=Kala azar-N18/23 10. Black death=Plague- Yersinia pestis
11. Blackwater fever= Hemoglobinuria- Cola colour urine- P. falciparum -N20
12. Kissing=Glandular fev- EBV 13. Pel Ebstein fev- Hodgkins lymphoma -N18
14. Tomato fever= Hand foot mouth dis- Coxsackie virus A16- Hox gene -F23/25
15. Yellow fev- Africa 16. Spotted fev- Rickettsia 17. Relapsing fever- Borrelia
18. Shanghai fev- Pseudomonas 19. Trench fev- Bartonella 20. Q fev- Coxiella
21. Valley fever- Coccidioides fungus 22. Viral fev- Flu, common cold, Covid

1. Fev + night sweats- TB, HIV, Covid, Lymphoma, endocarditis, Brucella
2. Fev + relative bradycardia (Faget sign)- Typhoid, Brucella, TB, Dengue
3. Fev + sore throat- Pharyngitis, laryngitis, tonsillitis, quinsy, kissing fev

4. Fev + cough- TB, pneumonia, bronchitis, Covid, Influenza, lung abscess
5. Fev + wheeze- Bronchiolitis, asthma 6. Fev + muscle pain- Myositis
7. Fev + ARDS- Covid, Influenza, Malaria, Dengue, pneumonia, sepsis, trauma
8. Fev + headache/seizures- Mening/encephalitis, brain abscess, sepsis
9. Fev + encephalitis- Jap encep, HSV, Nipah 10. Fev + burning urine- UTI
11. F + palmoplantar rash + oral rash- Hand foot mouth dis-Coxsackie A16 vir
12. Fev + eye sym- Conjunctivitis, uveitis 13. Fev + ear sym- Otitis, mastoiditis
14. Fev + abdom pain- Appendic/gastroent/cholecystitis, PID, food poisoning
15. Fev + abd distension- Peritonitis (perforation/rupture), intestine obstruct
16. Fev + tender hepatomegaly- Amoebic liver abscess, Hydatid cyst
17. Fev + pain at the back- Pyelonephritis, pancreatitis, nephrolithiasis
18. Fev + diarrhea + Hemolytic anemia- Hemolytic uremic syn, TTP -F1/25
19. Hemorrhagic fev-F1/25-INI25- Dengue, Ebola, Plague, Typhus, Leptospira-N19,23
20. Fev + murmur- Endocarditis, RHD 21. Fev + multiorgan dysfunc- Sepsis
22. Fev + arthritis- Chikungunya, Rheumatic, psoriatic, septic, SLE, gout
1 joint- Septic arthritis/gout, >1 joint- SLE/ Rheumatoid arthitis (symmetric) -F1/26
23. Fev + skin symptoms- Abscess, cellulitis, shingles, necrotizing fasciitis
24. Fev + lymphatic symptoms- Filariasis (lymphadenitis/angitis), lymphoma
25. Fev + testis pain/swelling- Epididymo-orchitis
26. F+ hemoptysis- TB, pneumonia, ca lung 27. F+ hematuria- UTI, renal stone
28. F+ rigidity + antypical antipsychotic Clozapine- Neuroleptic malignant syn

MULTIPLE MYELOMA 1. MC primary bone tumour
2. Bence Jones proteinuria, serum protein electrophoresis- M spike
3. IgG+IgA light chain, no heavy chain, Reverse (1:2) A: G ratio
4. Cells- Hodgkins + Flame (Plasma cell) + Mott cell + RBC rouleaux
5. Body- Russell + Dutcher body
6. CRAB criteria- ↑Ca, ↓Renal, Anemia, Bone lytic lesion (Moth eaten skull)

150 ONE-LINER BUZZWORDS - Delicious Dumplings A→Z

APPROACH TO A PATIENT Presenting complaints→ History of present illness → Past/ family/ allergy/ Rx history→ General (T-P-R-BP-SpO_2) + systemic exam→ Provisional diagnosis→ Investigations→ Final diag→ Rx→ Prevention→ Follow up

STAGES OF INFECTION Pathogen entry- Incubation period- Illness- Convalescence or Complication - Recovery or death

ANTIMICROBIAL = Antibiotic/Antibacterial + Antifungal + Antiviral + Antiparasitic

.Chemotherapy= Antibiotic + Cancer ChemoRx

1. ↓cell wall- β lactam a. Penicillin = cillin- Ampi, amoxy, ticar, piperacillin

b. Cephalosporin = Cef- 5generations- 1-Cefazolin 2-Cefuroxime 3-Ceftazidime, Ceftriaxone, Cefoperazone, Cefixime 4-Cefipime 5-Ceftobiprole, Ceftaroline

c. Carbapenem = Penem -N19- Mero, Imipenem d. Vancomycin, Teicoplanin

e. Bacitracin f. Isoniazid g. Ethambutol h. Pretomanid

j. β lactamase inhibitors- Clavulanic acid, tazobactam, sulbactam, enmetazobactam

i. Amoxycillin + Clavulanic acid = Coamoxiclav ii. Piperacillin+ Tazobactam

iii. Cefipime+ Enmetazobactam (New Indian drug 2025 for UTI)

2. ↓cell membrane a. Colistin b. Isoniazid c. Antifungal conazoles d. Amphotericin B

3. ↓50s ribosome -F1/7/23- CCLM a. Clindamycin b. Chloramphenicol c. Linezolid,

d. Macrolide = thromycin- Clari, Ery, Roxi, Azithromycin

4. ↓30s -F6/19- DA a. Doxycycline b. Aminoglycoside- Streptomycin, Gentamicin

5. ↓DNA/RNA a. Rifampin b. Clofazimine c. Nitrofurantoin d. Metronidazole

e. Fidoxamicin f. Fluoroquinolone = floxacin -F1/23,N19/20/22- Cipro, O, Levo, Moxi

g. Clofazimine h. Antivirals- Acyclovir, Ribavirin, Remdesevir, Sofosbuvir

6. Protease- Lopinavir, Ritonavir 7. ↓ATP- Bedaquiline

8. ↓Folic acid -F1/23- Trimethoprim + Sulfamethoxazole = Cotrimoxazole

9. ↓Fatty acid- Pyrazinamide 10. Neuromuscular blockage- Botox-F1/26

1. Ceftriaxone- Strep pneumoniae, Gonorrhea, N. meningitides -N19/23/24/26 Hemophilus influenza, Typhoid, PID 2. Ceftazidime- Pseudomonas

3. Azithromycin- Diphtheria, Pertussis, Chancroid, Donovanosis, Mycoplasma Campylobacter, Penicillin allergy, Traveler diarrhea

4. Doxycycline- Rickettsia, Scrub typhus, Chlamydia, traveler prophylaxis malaria

5. Ciprofloxacin- Shigella, EIEC 6. Streptomycin- Brucella 7. Gentamicin- Plague

8. Penicillin -N20- RHD, Strep Group A/B, Syphilis, Leptospira, Clostridium
Penicllin can cause Type 1, Type 2, Type3, Type 4 hypersensitivity
9. MSSA/Mastitis/Breast abscess- Cloxacillin/ Dicloxacillin/ Flucloxacillin
10. MRSA- Vancomycin 11. VRSA- Linezolid/ Daptomycin/ Teicoplanin
12. TB- 6 months- 2 (RHZE)+ 4 (RHE) 13. MDRTB/XDRTB- 6 months BPaLM -N22
14. Leprosy- Rifampicin+Clofazimine+Dapsone 15. Acne- Isotretinoin
16. C.difficile-Metronidazole/Vancomycin/Fidoxamicin 17. H.pylori-COM/BOMT-F7/25
18. No antibiotic- Cholera, food poisoning (S. aureus, botox, Bacillus cereus)
19. Stat = Single dose- Rheumatic fever, Candidiasis, Bacterial vaginosis, STI
(Syphilis, Gonorrhea, Chancroid, Donovanosis, Chlamydia, Trichomoniasis)
20. Cotrimoxazole-NTP- Nocardia-N24, Toxoplasma, Pneumocystis jirovecii-N23,F7/25
21. Metronidazole-BACHPR- Bacterial vaginosis, Anaerobe, C. difficile, H. pylori, sepsis, PID, open fracture, intraabdominal/ bone/ joint/dental infections, endocarditis prophylaxis, skin infections caused by Demodex folliculorum mite (folliculltis, blepharitis, rosacea, seborrheic dermatitis, perioral dermatitis)
EGT- Entamoeba, Giardia, Trichomonas, Balantidium, Blastocystis
22. Dapsone- DLR- Dermatitis herpetiformis, Leprosy, Rhinosporidium
23. Amphotericin B- Mucormycosis, Cryptococcal meningitis, Leishmania
24. Conazoles (Flu/Itra)- Mild fungal infec 25. Voriconazole- Aspergillus
26. Acyclovir- Herpes, Chickenpox, Shingles, EBV 27. Ganciclovir- CMV
28. Tecovirimat- Smallpox, Mpox 29. Ribavirin- RSV 30. Oseltamivir- Influenza
31. Tenofovir + Lamivudine+ Dolutegravir- HIV, Hep B-F1/26 32. Entecavir- Hep B
33. Tenofovir+ Emtricitabine/ Cabotegravir/ Lenacapavir- HIV prophylaxis (PrEP)
34. Sofosbuvir/ Ledipasvir/ Velpatasvir- Hep C 35. Cryptosporidium- Nitazoxanide
36. Chloroquine 600mg 0,6,24,48h+ Primaquine 15mg x14days- Plasmodium vivax malaria
37. Artemisinin combo Rx 0,8,24,36,48,60hr+ Primaquin 45mg on day2- Falci malaria -F1/26
38. Mefloquine- Malaria prophylaxis 39. Praziquantel- Tapeworm, Flukes
40. Albendazole- Cysticercosis, Hydatid cyst, roundworm, pinworm,
Hookworm, whipworm 41. Ivermectin- Strongyloides-F1/26
42. DEC= Diethy Carbamazine + Albendazole + Ivermectin- Filaria

ANTIBIOTICS used for diag nosis (not Rx) 1. Methicillin (MSSA/MRSA)- Unstable
2. Optochin- Strep pneumoniae sensitive 3. Bacitracin- Strep pyogenes sensitive
4. Novobiocin- Staph epidermidis sensitive, Staph saprophyticus resistant

EMPIRICAL ANTIBIOTIC RX a. MRSA- Vancomycin or Linezolid/Teicoplanin
b. Typhoid- Ceftriaxone c. PID- Ceftriaxone + Doxycycline ± Metronidazole
d. UTI- Nitrofurantoin -N24 e. Mastitis/Cellulitis (MSSA)- Clox/ Dicloxacillin -N24
f. Community acquired pneumonia, Acute COPD- Ceftriaxone + Azithromycin ± Vancomycin
g. Meningitis, Endocarditis, Febrile neutropenia -F1/7/25,N19- Ceftriaxone + Vancomycin
h. Intra-abdominal infection (Cholecystitis)- Ceftriaxone + Metronidazole
i. Open fracture- Ceftriaxone + Metronidazole + Amikacin j. Traveler diarrhea- Azithromycin
k. Hosp acq pneumonia, Diab foot, Osteomyelitis, Sepsis- Piperacillin tazobactam ± Vanco

CHEMOPROPHYLAXIS 1. Post-exposure a. TB- Isoniazid+Rifapentin 900mg wkly x 12 wk
b. Influenza-Oseltamivir 75 mg OD x 10days c. Chickenpox/HSV-Acyclovir
d. Diphtheria, Pertussis- Azithromycin e. Neisseria meningitides, Anthrax- Ciprofloxacin
f. HIV- Tenofovir + Lamivudine+ Dolutegravir x 4 weeks
2. Pre-exposure a. HIV- Tenofovir+ Emtricitabine/ Cabotegravir/ Lenacapavir
b. Malaria- Mefloquine d. UTI- Nitrofurantoin
c. Toxoplasma, Pneumocystis, Nocardia- Cotrimoxazole in HIV
3. Pre-travel a. Malaria- Doxycline b. Traveler diarrhea- Azithromycin

ANTITOXIN for Tetanus, Botulism, Gas gangrene, Toxic shock syndrome,
Scalded skin syndrome (Not given in diarrhea/dysentery toxin, anthrax, pertussis)

ANTIGEN (Ag) for Diagnosis 1. NS1 Ag- Dengue 2. p24 Ag- HIV
3. HBsAg- HepB 4. rk39 Ag- Kala azar 5. Galactomannan- Aspergillus
6. HRP_2 Ag- P. falciparum 7. pLDH/aldolase Ag- All Plasmodium
8. C carbohydrate Ag/CRP- Strep pyogenes 9. pp65 Ag- CMV
10. Superantigens=SAg (5)- SAg activate 20% T cells→ Massive cytokine release
a. Staph Toxic Shock Syndrome toxin b. SARS Cov-2 c. Staph enterotoxin
d. Staph Scalded Skin synd toxin e. Streptococcal pyrogenic= erythrogenic toxin
e. HIV f. EBV g. CMV h. Mycoplasma i. Yersinia j. Malassezia

ANTIBODY (Ab) for Diagnosis 1. Rheumatic fever- ASO=Anti Streptolysin O
2. SLE- ANA=Anti nuclear Ab (Screening/most sensitive), Anti Smith Ab (Most specific),
Anti dsDNA (sensitive+specific)-F1/26, Anti Ro(Neonatal lupus) 3. RA factor- RA,SLE
4. Rheumatoid arthritis-F1/26- Anti CCP- Most specific
5. Severe dengue due to Ab Dependent/Immune Enhancement =Original antigenic sin

ABSCESS= Pus filled lesion 1. Cold/Potts- TB 2. **Lung**- **MC Strep pneumoniae**
3. **Overall**/Skin/breast/psoas/brain/spine/pleura/Brodie/bone- **MC Staph aureus**-F7/25
4. Peritonsil/pharynx-Strep pyogenes
5. **Intra-abdominal**/liver/pelvic/perineal/anorectal/prostate/ovary- **MC E. coli**
6. Stitch/suture/artificial surface abscess- Staph epidermidis
7. Munro micro abscess (Neutrophil in stratum corneum)- Psoriaisis
8. Pautrier microsbcess- Mycosis fungoides=Cutaneous T cell lymphoma
9. Collar stud abscess- TB lymphadenitis 10. Crypt abscess- IBD

PUS 1. Blue green- Pseudomonas -F1/25,N20 2. Anchovy sauce- Entamoeba -N19

WOUND 1. Clean surgical wound- Heals by primary intention
2. Large wound with granulation tissue- Heals by secondary intention
. Most important factor for wound healing- Blood supply
. Tetanus prone wound- Devitalized tissue + anerobic environment

SUTURES 1. Absorbable- Chromic catgut, Vicryl, Monocryl
2. Non absorbable- Silk, Nylon, Dacron, Prolene

HYPERTROPHIC SCAR Excess collagen within wound margins

KELOID Excess collagen beyond wound margins

ANAEMIA=An 1. **Iron deficiency** microcytic hypochromic an- Hookworm, thalassemia
↓Hb-MCV <80 fL-MCH-MCHC, ↓Iron-Ferritin-Transferrin sat, ↑**TIBC**, aniso/poikilocytosis
MC nutritional deficiency worldwide- Iron deficiency anemia
2. ↓**Folate/Vitamin B_{12} megaloblastic/ pernicious** macrocytic hypochromic an-N20-
Fish tape worm, ↓ Vit B_{12}-Folate
↓Hb-↑MCV>100fL- Homocysteine, macro/ovalocyte, hypersegmented neutrophil
3. **Hemolytic** normocytic normochromic an- Malaria, Kala azar, sepsis, rheumatoid arthritis
↓Hb, MCV 80-100 fL normal, ↑reticulocyte, ↑bilirubin-LDH, splenomegaly
4. **Aplastic** an= Pancytopenia= Bone marrow failure- Chr infec, HIV, EBV, alcohol
↓Hb-WBC-Platelet-Reticulocyte, hypocellular bone marrow, ↑bleeding
1. **An+ old age**- Malignancy 2. **An+ edema+ hypertension**- CHF/CCF, hypoproteinemia
3. **An+ Koilonychia**- Iron deficiency an 4. **An+ Hyperpigmentation**- Megaloblastic an-F7/25
5. **An+ jaundice+ splenomeg**- Hemolytic an 6. **An+ hypertension**- CRF/CKD

AGGLUTINATION=Agg TEST 1. Widal agg- Typhoid 2. Standard agg- Brucella
3. Haemagglutination- Syphilis TPHA 4. Micro/Macro agg- Leptospira
5. Cold agg 4^0C- Mycoplasma 6. Weil-Felix agg- Rickettsia/ Scrub typhus

7. Paul Bunnel agg-F1/26- Infectious mononuleosis EBV 8. ABO/Rh/Coombs agg
9. Coagglutination- Staph aureus protein A 10. Latex agg- ASO, CRP, RA factor

ANIMALS 1. Pig- dSwine flu, Taenia solium, JE, Trichinella-F1/26, best xenotransplant
2. Dog- Rabies 99%, Dog tapeworm = Hydatid cyst 3. Lion face- Leprosy
4. Cat- Rabies, Cat Scratch dis, Toxoplasma-N19 5. Elephantiasis- Filaria -F20/23
6. Rat- Plague, Leptospira, Scrub typhus, Rat tail pattern- Ca esophagus
8. Bat- Rabies, Coronavirus, Nipah, Ebola, Hanta, Bat wing + Kerley B lines- Pulm edema
9. Tiger mosq- Aedes 10. Chicken- Chickenpox, chicken wire erythema- RHD
11. Fish tapeworm 12. Gambusia = Guppy fish- Eats mosquito larva & cyclops
13. Fish smell- Bacterial vaginosis/ Trichomonas 14. Fish vertebra- Osteoprosis
15. Fish in stream/school of fish pattern- Vibrio cholera, Hemophilus ducreyi (Tram track)
16. Scrombroid tuna fish poisoning- E. coli-N18 17. Crocodile tear=Bogorad syndrome
18. Tapir nose- Leprosy, syphilis, rhinoscleroma -F1/26/23 19. Rabbit fever- Tularemia
20. Buffalo hump- HIV Rx, Obesity, Cushings syn, prolonged steroid Rx
21. Bird- Bird flu, JE, Cryptococcus, bird beak pattern- Achalasia cardia
22. Giant panda face-F7/25- Wilson dis 23. Turtle sign- Shoulder dystocia
24. Moth eaten- Multiple myeloma skull, Chondrosarcoma bone
25. Arbovirus=Arthropod borne virus-N20,22 Aedes (Dengue,Chik,Zika,YF), JE, KFD
26. Spider- Angioma/naevi- Chr liver dis, spider molar teethy colony- Actinomyces
27. Swan neck + deformity (Boutonniere + Z) + caput ulnaris + hand (Claw + opera glass) + Finger (Mallet+ trigger+ piano key+ telescope+ windmill) + ulnar drift- Rheumatoid arthritis
30. Monkey a. Fever- KFD b. Mpox c. Malaria- P. knowlesi d. Shape- Giardia
31. Chicken- Chickenpox, chicken wire- Chromoblastomycosis, oligodendroglioma, ALD
Neurotoxic- Snake (Cobra, Krait, Taipan, Mamba, Rattlesnake, Copperhead, Cottonmouth), Scorpion, Black widow spider, Honeybee, Wasp, Fire ant, Tick, Blue ring octopus, Stingray, Box Jellyfish, Pufferfish, Shellfish, Lionfish, Stonefish, Tuna (Scrombroid poison), Cocaine, Heroin, LSD, Plants (Belladona, Hemlock, Nux vomica, Mushroom), Botox, organophosphat

AGGLUTINATION=Agg TEST 1. Widal agg- Typhoid 2. Standard agg- Brucella
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9. Coagglutination- Staph aureus protein A 10. Latex agg- ASO, CRP, RA factor

ACID 1. ↑Uric- Gout 2. ↑Fatty- Obesity 3. ↓Folic (Given in preg)- Neural tube defect
4. Retinoic- Vit A 5. Pantothenic- Vit B_5 6. Ascorbic- Vit C 7. Nucleic- DNA/ RNA
8. Clavulanic- β lactamase inhibitor (Amoxy+Clavulanic acid 9. Boric- Urine preservative
10. H_2SO_4- ZN stain decolorizer (20% TB, 5% Leprosy, 1% Cryptosporidum-Nocardia)
11. Acetic acid- Ca cervix- VIA 12. Peracetic- Sterilization 13. Sialic- Influenza receptor
14. Teichoic- Gram positve cell wall 15. Mycolic acid= Cord factor- Mycobacterial cell wall

BACTERIA Name MONAS 1. Pseudomonas- Bacteria 2. Trichomonas- Protozoa
COCCUS=Round bacteria 1. Staphylococcus, Streptococcus, Enterococcus, Pneumococcus, Meningococcus, Gonococcus (All bacteria)
2. Cryptococcus- Yeast 3. Echinococcus= Hydatid cyst-N22- Tapeworm
ELLA=Gram neg bacteria 1. Klebsiella, Shigella, Salmonella, Brucella, Coxiella Bordetella, Legionella, Gardenerella, (All bacteria) 2. Flagella (Part of bacteria)
3. Rubella, Varicella=Chickenpox- Virus 4. Trichinella-F1/26- Helminth
PLASMA 1. Mycoplasma-Bacteria 2. Toxoplasma, Plasmodium-Protozoa
3. Histoplasma-Fungus -F21,23,N24,I25 4. Plasma cell pneumonia- Pneumocystis

BACTERIA (Extra) 1. Burkholderia pseudomallei-N18- Meliodosis→ Rx Ceftazidime
2. Bartonella quintana- Trench fever 3. Bartonella henselae- Cat scratch disease
4. Bartonella bacilliformis- Oroya fever 5. Streptobacillus moniliformis- Rat bite fever
6. Legionella-N19- Atypical walking interstitial pneumonia + Legionnaire disease + Pontiac fever- Airconditioner- Rx Azithro 8. Leptospira canicola- Canicola fever
9. Leptospira hebdomadis- 7 day fever 10. Leptospira autumnale- Pretibial fever
7. Listeria- Diarrhea, Meningitis- Tumbling motility- Rx- Ampicillin
8. Rickettsia prowazeckii- Gaol fever 9. Papatacci 3 day sandfly fever- Sandfly bite
10. Corynebacterium minutissimum- Erythrasma 11. Mycobacterium buruli- Buruli ulcer
12. Geobacillus stearothermophilus spores- Biological indicator (BI) for Autoclave, plasma Sterilization 13. Bacillus subtilis spores- Dry heat BI for hot air oven
14. Bacillus pumilis- BI for Gamma rays 15. Bacillus subtilis/globigii- BI for Ethylene oxide

BABY 1. Floppy-Botulism 2. Blueberry muffin-Rubella, CMV, ABO/Rh incompatibility
3. Happy-Angelman syn 4. Gray-↑↑Chloramphenicol 5. Brown-↑↑Phototherapy

BRUSH BORDER 1. Epithelium- Small intestine 2. Vertebra- Osteoporosis

BURNS 1. Wallace rule of 9 (Head 9%, chest 9%, back 9%, abdomen 18%, perineum 1%, 2 upper limbs 18%, 2 lower limbs 36%, total 100%) 2. Rule of palm = 1% body surface
3. Brooke/Parkland formula- Total RL required= 4ml x weight (kg) x %bisy surface area
4. 1^0- Superficial- Only epidermis burnt, painful red skin, no blisters, heals in 7day
5. 2^0- Partial- Epidermis + dermis burnt, painful red blisters, grafting required if severe
6. 3^0- Full - All skin layers + nerves burnt, no pain, black eschar, grafting, heals in months
7. 4^0- Deep full- All skin layers + nerves + muscles + bones burnt
8. MC in burns a. Infection- Pseudomonas b. Early death- Hypovoluemic shock
c. Late death- Sepsis due to infection

BODY -F22/23/24,N19/20/21,INI24 1. LD- Kala azar 2. Donovan- Donovanosis
3. Owl eye- CMV 4. Negri- Rabies 5. Cystoid- Degenerated nerve fibres
6. Copper penny/Sclerotic/Medlar- Chromoblastomycosis 7. Torres- Yellow fever
8. Councilman- Hepatitis/Yellow fever
9. Aschoff, Anitschkow (Caterpillar nuclei)- Rheumatic heart dis
10. Howell Jolly- Splenectomy, hemolytic/ pernicious anemia, thalasemia
11. Cowdry A/ Lipschultz (Tzanck)- Blisters- HSV, VZV, Pemphigus
12. Heinz-↓G6PD 13. Asteroid=star body- Sporothrix, Sarcoidosis 14. Nissl- Neuron
15. Gamma Gandy- Splenomegaly due to portal HTN 16. Caterpillar- Porphyria
17. Coccoid X- Psitacossis 18. Civatte- Lichen planus 19. Hirano- Alzheimer
20. Lewy- Parkinsons-N23, Alzheimer 21. Mallory-Denk- Alcoholic hepatitis
22. Pappenheimer- Splenectomy, sickle cell, sideroblastic anemia
23. Citron- Clostridium septicum 24. Masson- Pneumonia in rheumatism
25. Creole- Asthma 26. Bamboo/ ferruginous- Asbestosis
27. Michaelis Guttman- Malakoplakia 28. Russell, Dutcher- Multiple myeloma
29. Henderson-Peterson- Molluscum contagiosum 30. Guarnieri- Vaccinia virus
31. Psammoma- Papillary thyroid ca, meningioma, serous ovarian ca,
pancreatic serous cystadenoma
33. Bollinger- Pox 34. Brassy- Malaria 35. Highmore- Mediastinum testis
36. Polar/Bipolar/Metachromatic/ATP/Energy/Volutin/Babes Ernst- Diphtheria
37. Kimmelsteil Wilson nodule- Diabetic nephropathy 38. Auer rod- AML

BIOCHEMICAL TESTS 1. Coagulase=Coag +ve- Only Staph aureus
2. Catalase=Cat +ve- Staphylococcus, Pseudomonas, All enterobacteriaceae, Mycobacterium tuberculosis, Neisseria, Hemophilus influenza, H. pylori, Listeria
3. Staph aureus- Cat +ve Coag +ve, Staph epidermidis- Cat +ve Coag negative
4. Oxidase +ve- Pseudomonas, Vibrio, Helicobacter, Hemophilus, Neisseria
5. H_2S +ve- Salmonella, Brucella, Proteus, Helicobacter 6. ONPG +ve- Shigella sonnei
7. Urease +ve- Proteus, Brucella, H. pylori 8. IMViC- E. coli I+M+Vi-C- Klebsiella I-M-Vi+C+

BIOTERRORISM/WARFARE 1. Cat A- Highest risk- Major public health impact SBPAVT- Smallpox, Botox, Plague, Anthrax (MC), Vir hemorrhagic fev (Dengue,Ebola), Tularemia
2. Cat B Moderate risk Burkholderia, Brucella, Rickettsia, Chlamydia, Salmonella, Shigella, Vibrio, Coxiella, Coccidia, Viral encephalitis, Ricin toxin, Staph enterotoxin B, C. perfringens ε toxin, E. coli O157:H7 (Shiga toxin)
3. Cat C Emerging Pathogens Nipah, Hanta, Yellow fever, Rabies, SARS, Influenza, Arbovirus, MDRTB, Tick borne hemorrhagic fever, Tick borne encephalitis

CRITERIA/ SCORE/SCALE -F7/23,1/20/24,INI25 1. Dukes-N19/22- Endocarditis, colon ca
2. Jones- RHD 3. Alavardo/ Mantrel- Appendicitis 4. GCS- Brain dis-N18/23
5. Amsel/Nugent- Bac vaginosis-N24 6. Del Brutto (MRI- Starry sky)- Neurocysticercosis-N24
7. NYHA- Heart failure 8. Ranson/BiSAP/Manheim-Pancreatitis-N21/24 9. Bishop- Labour
10. Possum- Post-op 11. CURB65/PSI-Pneumonia 12. APACHE II- Critical illness
13. CRAB- Multiple myeloma (↑Ca, ↓Renal, Anemia, Bone lytic lesion) 14. Framingham- CHF
15. Light- Pleural effusion 16. Modified Berlin- ARDS 17. NIHSS/CHAD- Stroke
18. Midas- Migraine 19. APGAR- Newborn 20. Milan/UCSF- HCC 21. Rotterdam- PCOS
21. Child Pugh/ MELD- Liver disease 23. Westhaven- Hepatic encephalopathy
24. SOFA/q-SOFA (Resp>22/min, SBP<100, GCS<15- Sepsis) 25. Maamatti- Airway-F1/26
25. Downe- Resp distress syn (Neonatal resp distress syn- Hyaline membrane dis)
26. Glasgow Blatchford/ Rockall- Upper GI bleed 27. Paradise- Tonsillectomy indication
28. Parkland- Burns (4 ml x %TBSA x body weight (kg) 29. Wells- DVT/ Pulm embolism
30. Gleason- Prostate ca 31. Jackson- Penis ca 32. Centor/McIsaac Strep pharyngitis
33. Bloom Richardson- Breast ca 34. Butchart- Mesothelioma 35. Nevine- Gall bladder ca
36. Noguchi- Adenoca lung 37. Shimda- Neuroblastoma 38. Braden- Bedsore
40. Clavien-Dindo- Surg complications 32. Robson- Renal cell ca 33. Masaoka- Thymoma
33. mMRC- Dyspnea

CIRRHOSIS Brown shrunken nodular fibrotic hard firm liver, bile duct proliferation, allooning of hepatocytes, Metavir score, Ishak score, Child Turcott Pugh score
1. **Macronodular**- Viral hepatitis (Post necrotic) 2. **Micronodular**- Alcohol

CELL 1. **Cell cycle**- G1 (Gap1, cell growth)→ S (Synthesis of DNA)→ G2 (Gap2, ready for cell division)→ M (Mitosis) .G0 (Resting-dormant cell)
2. **Cell injury** a. Earliest change- Mitochondrial swelling b. Irreversible (Apoptosis through caspases, perforins, granzymes, Annexin V)- Cell shrinks, Mitochondria amorphous density c. Free radical scavenger- Superoxide dismutase
CYTO = Cell 1. Cytomegalovirus-N23 2. Cytotoxic 3. Cytoplasm 4. Cytokine 5. Cytophagy 6. Cytotrophoblast- Placenta 7. Cytogenetics 7. Cytometry

-CYTE 1. Leukocyte (WBC- High to low %age→ Neutrophil (never) > Lymphocyte> (let) Monocyte (monkey)> Eosinophil (eat)> Basophil (banana)- (Granulocyte- Neutrophil, Eosinophil, Basophil), (Agranulocyte- Lymphocyte, Monocyte), Thromobocyte, Megakaryocyte, Histiocyte = Macrophage 2. Podocyte- Kidney 3. CNS- Astrocyte, oligodendrocyte, myelocyte 4. GIT- Hepatocyte, Enterocyte 5. Skin- Keratinocyte, Melanocyte, Adipocyte 6. Koilocyte- HPV cervical ca 7. Peg cell- Fallopian tube 8. Spindle cell- Histoid Leprosy 9. Degmacyte=Bite cell (↓G6PD-Common in India) 10. Depranocyte-Sickle cell 11. Myo/ Cardiomyo/ Pneumo/ Thymo/ Dendro/ Spermato/ Osteo/ Chondrocyte 12. **Multinucleate giant cell**- Granuloma (TB), Tzanck smear (HSV, VZV) -F23/24/25 13. Hurthle cell- Hashimoto thyroiditis 14. Smudge cell- CLL
CYTOSIS= ↑cells 1. **Blood**- Erythrocytosis, Leuco 2. **Pleocytosis**- CSF/fluids 3. **Poikilocytosis**= ↑↓RBC shape- Hemorrhagic fever (Dengue -F25, Malaria) eg- Spherocyte, Acanthocyte= Spur/spike cell, Target cell codocyte - bulls eye- Thalassemia/ Iron def anemia/liver dis, Dacrocyte= Tear drop cell, Elliptocyte, Ovalocyte, Echinocyte= Burr cell, Schistocyte= Fragmented cell, Stomatocyte 4. **Reticulocytosis**= Immature RBC 5. **Phagocytosis**- Macrophage + Neutrophil 6. Anisocytosis- ↑↓RBC size- Microcyte, macrocyte 7. Pancytosis- ↑ all cells

COLD=C 1. C Abscess- TB 2. C Sores=Fever blister- HSV 3. Common C- Rhinovirus 4. C clammy skin- Sepsis 5. C agglutination- Mycoplasma 6. C enrichment- Listeria

COUGH 1. Barking seal like- Croup 2. Whooping- Pertussis-N23 3. Staccato- Chlamydia

CANCER 1. **MC** ca- Skin (UV light) 2. **MC**♀ca- Breast (HER2/neu, BRCA1/2)
3. MC child ca- ALL 4. MC skin ca- Basal cell ca 5. MC breast tumour- Fibroadenoma
6. MC breast ca- Invasive ductal ca (NST) 7. Mc oral ca- Squamous cell ca
8. MC benign bone ca- Osteochondroma 9. MC radiation induced bone ca- Osteosarcoma
10. MC bone ca- Metastasis 11. MC 1^0 bone ca- Multiple myeloma
12. MC chemoradio sensitive bone ca- Ewing sarcoma (Bone to bone metastasis)
13. MC thyroid ca- Papillary ca 14. Porcelain gallbladder- ↑Risk of gallbladder ca
1. **EBV** -F21/24/25/26,N18,INI25 Gastric ca/Nasopharyngeal ca > Burkitts > Hodgkins lymphoma
2. **Hep B&C**, Aspergillus **aflatoxin, alcohol, arsenic**- Liver ca (↑Alpha FetoProtein)-N23
3. **HPV, OCP**-F7/25-Cervical ca *HPV→MC ca oropharynx-F7/25, penis, vulva, vagina, anal ca
4. **HHV8**- Kaposi sarcoma-Purple red lesions -F7/23/24/25,N23
5. **Helicobacter**-N21 Gastric ca 6. **Chinese/SE Asian liver fluke**- Cholangio ca
7. **Schistosoma**= Blood fluke-N18/19/21- Bladder ca 8. Estrogen- Endometrial ca
9. **OCP, Tobacco**- Oral, lung, renal cell ca 10. **Desmoplasia**- Fibrosis around ca
11. **Asbestos**- MC Adenoca lung> Mesothelioma 12. **Arsenic**- Skin ca
13. **Dysplasia**- Premalignant reversible 14. **Anaplasia**- Hallmark of malignancy

CYST 1. Hydatid cyst- Brood capsule + hooklets+ membranes 2. Neurocysticercosis
3. Cystic fibrosis 4. Cholecystitis 5. Grade 4 nodulocystic acne 6. Fungus- Pneumocystis
7. Cystitis, cystolithiasis, cystinuria, cystoscope, cystourethrography, cystostomy
8. Dacryocystitis→ Dacrocystorhinostomy 9. Chocolate cyst- Ovarian endometriosis
9. Parasites→ cyst 9. 4nucleus cyst- Entamoeba, Giardia 10. No cyst- Trichomonas
11. Cystoisospora & Cyclospora-F25- Protozoa oocyst- Diarrhea

CORD CLAMPING 1. **Early**- Fetal distress, abruptio placenta, PPH, neonatal resuscitation, cord prolapse, Rh isoimmunization, HIV, Hep B/C
2. **Delayed**- Term infant, preterm, anemic, Caesarean section, LBW, HIV mother on Rx

Reverse A:G ratio 1:2 (Normal 2:1/1:1)	
My	Multiple myeloma
Hot	Hepatitis
GF	Guillain Barre syndrome
Never	Nephrotic syndrome
Reads	Rheumatoid arthritis
HIV-TB	HIV-TB

ANGINA (an) = Precordial chest pain (Non infec)- Levine sign
1. Angina pectoris 2. Prinzmetal variant angina
INFEC (Oral cavity) ANGINA (No chest pain)
1. Ludwig angina- S.pyogenes 2. Vincent angina 3. Herpangina

COLOURS- Red-Blue-Yellow-Orange-Green-Purple-White-Gray-Black-Brown/Cola-Rust

RED 1. Red eye= Hemorrhagic conjunctivitis- MC Enterovirus 2. Red algae- Agar
3. Thai Redcross intradermal regimen- Rabies 4. Reduviid= Kissing bug- Chagas dis
5. Red currant jelly a. Sputum- Klebsiella b. Stool- Intussusception
6. Tongue a. Beefy red- ↓Vit B_{12} b. Magenta red- ↓Vit B_2
7. Cherry red spot/skin 8. Red pulp- Spleen 9. Red pigment- Serratia
10. Red man syndrome- Vancomcyin 11. Red light/ Infrared Rx
12. Congo red stain- Amyloidosis-F7/25 13. Red Kit5 STI- Herpes simplex- Acyclovir
14. Red BMW- Plastic + rubber 15. Red- Blood agar, blood in body
16. Beefy red ulcer- Klebsiella granulomatis

BLUE 1. Blue BMW- Glass + metal 2. Blue Kit4 STI- Genital ulcer (Penicillin allergy)
3. Blue-green pigment- Pyocyanin- Pseudomonas 4. Blueberry muffin- Rubella
5. Blue light Rx- Neonatal jaundice 6. Blue cervix- Chadwick sign
7. Stains - Methylene blue- ZN stain, Lactophenol cotton blue- Fungus
8. Sclera- Osteogenesis imprefecta 9. Blue bloater- Chr bronchitis
10. Blue baby- Cyanotic CHD 11. Blue man syn- Amiodarone, silver (argyria)

YELLOW 1. Yellow BMW- Incinerable waste+ Blood bag 2. Yellow Kit6 STI- PID
3. Yellow fever (Africa and South America)- Aedes 4. Yellow vagina discharge- Gonorrhea
5. Custard yellow pus- Swimmer ear- Pseudomonas

PINK 1. McConkey agar 2. Carbol fuchsin 3. Pink puffer- Emphysema

VIOLET 1. Crystal violet- Gram stain 2. Ultraviolet C- Sterilization of OT
3. Crystal violet blood agar- Strep pyogenes

PURPLE 1. Purple straie- Cushings syndrome 2. Purple veins- Marbling
3. Purple red lesion- Kaposi sarcoma

ORANGE 1. Orange secretions- Rifampicin 2. Peau d orange- Breast ca
3. Orange wrinkled colony- Nocardia

GREEN 1. Green pus/urine- Pseudomonas 2. Pea soup green stool- Typhoid
3. Green vagina discharge- Trichomonas 4. Green H_2S poisoning- Rotten egg smell
5. Green kit2 STI- Bacterial vaginosis + Candida 6. Paris green kills Anopheles-F7/24
7. Smoky green- Aspergillus fumigatus 8. Yellow green- Aspergillus flavus

WHITE 1. White BMW- Sharps 2. White kit3 STI- Syphilis
3. White vaginal discharge-F1/26 a. Thick- Candida b. Thin- Bacterial vaginosis
4. Curdy white- Candida 5. Rice water stool- Cholera-F1/26 6. White= Leuko/Leuco

GREY/GRAY 1. Grey kit1 STI- Gonorrhea 2. Grey Turner sign- Pancreatitis
3. Gray baby syndrome- Chloramphenicol 4. Gray- Condyloma acuminata-HPV
BROWN 1. Brown Kit8 STI- Anorectal discharge 2. Brown- Povidone iodine
3. Brown baby- ↑↑Phototherapy 4. Rust colour sputum- Strep pneumoniae
5. Brown tumour of bone- Hyperparathyroidism 6. Bronze skin- Hemochromatosis
BLACK 1. Black BMW- General waste 2. Black kit7 STI- Bubo (Chancroid, LGV) -F1/26
3. Black necrotic ulcer= Eschar 4. Black death- Plague 5. Black fever- Kala azar
6. Black- Aspergillus niger 7. Black fungus- Mucormycosis-F20/21/22/23/24/25/26
8. Black water fever= Cola colour urine- Hemoglobinuria- P. falciparum malaria
9. Black hairy Tongue- ↑Tobacco, bismuth 10. Blackhead= Open comedo- Acne
11. Stain- India Ink=Nigrosin- Cryptococcus, Sudan black- Lipid
12. Skin- Kala azar, plague, addisons disease, melasma, diabetes acanthosis nigricans, lichen planus, heavy metal poisoning, malnoma, Minocycline

CRYSTAL=cy 1. Gram stain- Crystal violet 2. Cystinuria- Cysteine hexagonal cy
3. Staghorn calculus- Struvite= Mg ammonium triple phosphate coffin lid cy- Proteus infec
4. Nephrolithiasis- MC Ca oxalate cy a. monohydrate dumbell cy b. dihydrate envelope cy
5. Ca carbonate round wheel cy 6. Eye Lens- Crystallin protein (Sympathetic ophthalmia)
7. Intrasynovial steroid- Steroid cy 8. Laxative abuse- Ammonium biurate apple thorn cy
9. Gout (↑uric acid→Pain, swelling, redness great toe→ Podagra (Pyrazinamide)-
Neg birefringent monosodium urate needle-like cy 10. Pseudogout- CPPD rod like cy
11. Rheumatoid arthritis-Cholesterol cy 12. Oxaluria/ethylene glycol poisoning-Oxalate cy
13.Dengue Rx- Crystalloids (NS-0.9%, Ringer lactate, Dextrose5%=D5W, D10W, DNS)
14. Tyrosine thin needle cy 15. Leucine lamellated cy 16. Bilirubin pigmented thin needle cy

CRISPR cas9 - HUMAN HEALTH and EUGENICS - Applications
1. Clustered Regularly Interspaced Short Palindromic Repeats
2. CRISPR is bacterial immunity against bacteriophages, plasmids
3. GeneRx- HIV CCR5, flu, RSV, HSV, Down syn, World 1st baby Rx in US2025
4. Chimeric antigen receptor (CAR)-T cell- Cancer immunotherapy
For leukemia, lymphoma 2024- Tata Memorial Hosp, India
5. Precision Rx, personlized Rx, regenerative Rx- Non homologous end joining DNA repair

CLASSIFICATIONS=cl 1. Virus- Baltimore cl 2. Peptic ulcer- Johnson cl
3. Leprosy-Ridley Jopling, WHO, Indian cl 4. Gastric ca- Japanese/Bormann cl

DIARRHEA= Enteritis= Delhi Belly→ Small intestine infection (Loose motions)

1. MC child/wean- Rotavirus 2. MC travellers- ETEC 3. MC HIV- Cryptosporidium

DYSENTERY= Colitis→ Large intestine infection (Loose motions + fever + pain)

1. Ameobic dys (Entamoeba)- ↑Smelly no blood stool- ↓Pain/↓Tenesmus -N19

2. Bacillary dys (Shigella)- ↑Bloody sticky no smell stool- ↑Pain ↑ Tenesmus

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DAYS 4 Feb- Cancer 10 Feb- Deworming 24 Mar- TB 7 Apr- Health 25 Apr- Malaria

5 May- Hand hygiene 16 May- Dengue 31 May- No tobacco 14 Jun- Blood donor

1 Jul- Doctors 6 Jul- Zoonosis 28 Jul- Hepatitis 20 Aug-Mosquito 13 Sep- Sepsis

28 Sep- Rabies 3 Nov- One Health 24 Oct- Polio 12 Nov- Pneumonia 1 Dec-AIDS

Free diag+ Rx- TB, HIV, Leprosy, Hep B/C, malaria, filaria, Kalazar, JE, Dengue, Chikung, STI

DM COMPLICATIONS- ↑Infec-F7/24 (Mucormycosis, diabetic foot, otitis externa, skin infec, UTI, periodontitis, Fournier gangrene), MI, stroke, Neuro/nephro/retinopathy, DKA-HHS

MC death in DM- MI

DENTAL CARIES/PLAQUE→ Dental Abscess- MC Streptococcus mutans > S. sanguinis

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ERYTHEMA 1. Marginatum- RHD 2. Infectiosum- Parvovirus 3. Multiforme=Bulls eye -N19

ERYTHRO=Red 1. Erythrocyte/osis/penia 2. Erythroblast/blastosis fetalis 3. Erythroderma

4. Erythropoietin/poiesis 5. Erythrophagocytosis- Entamoeba 6. Erythromycin- Macrolide

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EYE INFECTIONS 1. MC overall- Staph aureus 2. MC stye (Zeis gland)- Staph aureus-F7/25

CONJUNCTIVITIS = Eye Flu 1. MC follicular conj (Sticky eye)- Staph aureus

2. MC hemorrhagic conjunctivitis (Red eye flu)- Enterovirus (Outbreak 2023 Delhi)-N18/19

3. Ophthalmia Neonatorum (Conjunctivitis)-N18/19 Gonorrhea, Chlamydia, HSV

KERATITIS-N20/23 1. MC bacteria- Staph aureus 2. MC virus- HSV (Dendritic ulcer -F7/25)

3. MC fungus- Aspergillus/ Fusarium=Field fungus 4. MC contact lens- Pseudomonas

5. MC protozoa-F7/25,N24 Acanthamoeba 6. Other- Candida, Strep pneumoniae, VZV

7. Trachoma= Keratitis+ Herbert pits+ Arlt line+ sago grain follicle+ pannus+ blindness-F7/25

CHORIORETINITIS 1. Congenital rubella- Salt and pepper retinopathy

2. Roth spot (retinal bleed)-N18- Bacterial=Infec endocarditis, leukemia, DM, HTN

3. Toxoplasma- Torchlight/ Headlight in fog/ snow chorioretinitis

4. CMV (MC in HIV)- Brushfire + pizza pie / Scrambled egg + ketchup chorioretinitis

5. Diabetic retinopathy (Argyll Robertson pupil)-N22 Cotton wool spots+ microaneurysm

6. Hypertensive retinopathy- Flame shape hemorrhage+ cotton wool spot+ AV nicking

EAR INFEC 1. MC otitis externa/Singapore ear/Cauliflower ear- Pseudomonas
2. MC acute/acute suppurative otitis media= ASOM- Strep pneumoniae
3. MC CSOM→ Pseudomonas→ Safe ear- Central perf, unsafe ear- Both ear attic/marginal perforation→ MC Incus eroded → Rx mastoidectomy
4. TB otitis media- Multiple/central perforation of TM, pale granulation, thin odorless fluid
5. Glue ear= Otitis media + effusion (Viral infec)→ Myringotomy + Grommet tube used

EPIDEMIOLOGICAL INDEX 1. Malaria-Annual Parasite Index,Spleen rate,Infant parasite rate
2. Plague- Cheopis/Flea/Xenopsylla index 3. Hookworm- Chandler index
4. Yellow fever/Dengue- Breteau/Aedes aegypti index, container/larva/pupa/ovitrap/house
5. Obesity- Quetlet/ Corpulence/ Ponderal/ Body mass index 6. Pearl- Contraception
7. Index case-F23,I22- First patient identified

FERMENTATION=Ferm 1. Lactose fermenter- McConkey/CLED agar
2. Stromy fermentation- Clostridium perfringens

FROZEN=F 1. Frozen Shoulder (Adhesive capsulitis) 2. Frozen Pelvis- PID
3. F Section- Intraoperativerapid histopathology diag of cancer 4. Fresh frozen plasma

FOOT 1. Madura foot -F1/26 → Actinomyces > Fungus 2. Diabetic foot
3. Athletes foot→ Tinea=Ringworm→ Foot powder used 4. Foot drop→ Leprosy
5. Hand foot mouth disease= Tomato fever/flu- Coxsackie A16 virus-F7/25
6. Hookworm and Strongyloides L3 filariform larva pierce foot to cause infection

	DISCIPLINE	FATHER		DISCIPLINE	FATHER
1	Microscopy	Leeuwenhoek	11	Molecular Biology	Linus Pauling
2	Microbiology	Louis Pasteur	12	Taxonomy	Carolus Linnaeus
3	Modern Bacteriolo	Robert Koch	13	Pathology	Rudolph Virchow
4	Anaerobic bacterio	Sidney Finegold	14	Epidemiology	John Snow
5	Virology	WM Stanley	15	Ancient Surgery	Susruta
6	Immunology	P. Ehrlich/E Jenner	16	Evidence based med	David Sackett
7	Vaccinology	Edward Jenner	17	Ancient Medicine	Charaka and Imhotep
8	Chemotherapy	Paul Ehrlich	18	Modern Medicine	Hippocrates
9	Antibiotics	Alexander Fleming	19	Tropical Medicine	Manson
10	Genetics	GJ Mendel	20	Antiseptic Surgery	Joseph Lister

FOOD 1. Icecream, chocolate, meat, egg, poultry- Salmonella food poisoning (FP)
2. Mayonnaise, sandwich, cake, salad, ham, poultry-F7/25,N20/21 Staph aureus FP <6hrs
3. Canned Food and Honey- Botulism 4. Chinese Fried Rice- Bacillus cereus FP
5. Seafood- Vibrio parahemolyticus FP -N24 6. Raw poultry- Campylobacter FP
7. Dirty water- E. coli FP 8. Cooked meat, leftover- Clostridium perfringens
9. Cigar/flower/rosette shape- Sporotrichosis= Rose thorn/gardener disease
10. Anchovy sauce pus- E. histolytica 11. Banana gametocyte- P. falciparum
12. Spaghetti + Meatball (or Banana + grapes) on KOH mount- Malassezia fungus
13. Olive shape mass- Pyloric stenosis 14. Sausage shape mass- Intussusception
15. Pea soup green stool- Typhoid 16. Custard ear discharge/pus- Pseudomonas
17. Red currant jelly a. Stool-Rotavirus, Intussuception b. Sputum- Klebsiella
18. Apple jelly nodule- Lupus vulgaris-N20/25 19. Apple core lesion- Ca colon
20. Coca Cola/Thumbs Up/Pepsi/Campa Cola (Dark)- Hemoglobinuria in P. falciparum
21. Mirinda/Fanta/Frooti/Slice/Maaza (Orange)- Rifampicin urine
22. Limca (White)- Rice water stool- Cholera -F1/26 23. Millet dots- Miliary TB
24. Yellow sauce- Gonorrhea 25. Thick white sauce (mayo/cheesy dip)- Candida
26. Green sauce- Trichomonas 27. Thin white sauce- Bacterial vaginosis
28. Pear shape- Giardia 29. Human milk deficient- Iron + VitaminC
30. Sheesh Kebab- Candida 31. Dinner fork deformity- Colles fracture
32. Popcorn calcification- Fibroadenoma breast, cavernous angioma, pulm hamartoma, chondroma 33. Nutmeg liver- Right heart failure
34. Egg vacc- Rabies, flu 35. Fried egg-F1/26- Mycoplasma, Malassezia, Yersinia, multiple myeloma, seminoma, dysgerminoma, oligdendroglioma, hairy cell leukemia
35. Onion skin- Ewing sarcoma, Peau d orange- Breast ca, Arteries in Malignant HTN, Synovium in Rheumatoid arthritis, Hyperplastic arteriosclerosis
BERRY 1. Strawberry a. Vagina- Trichomonas b. Infant hemangioma
c. Tongue (Beefy red large tongue)- Scarlet fever, Kawasaki disease, ↓Vitamin B_{12}
d. Nose- Rhinosporidiosis, Sarcoidosis e. Gum- Wegeners granulomatosis
f. Skull- Edward syn (Trisomy18)- Rocker bottom feet g. Gall bladder- Cholesterosis
h. Scrotum- Multiple sebaceous cyst 2. Berry aneurysm- MC ACA→SAH
3. Mulberry molar- Congenital syphilis 4. Blueberry muffin- Congenital rubella, CMV
5. Cherry red spot- Tay-Sachs, Niemann Picks dis 6. Cherry red skin- CO poisoning

SALT & PEPPER 1. Retinopathy-Rubella 2. Mucor on agar 3. Hyperparathyroidism
4. Pigmentation- Systemic sclerosis 5. Chromatin- Neuroendocrine tumour

HONEY 1. Honey crust- Impetigo 2. Honey in pacifier- Floppy baby- Botulism

Honeymoon=HM 1. HM cystitis- UTI after 1st sex- MC E. coli 2. HM muscle- Sartorius

Honeycomb=HC 1. HC- Interstitial lung dis 2. HC bone- Hemangioma

MILK 1. Unpasteurized milk- TB, Brucella, Campylobacter, Listeria 2. IgA_2 in milk
3. Salt milk agar- Staph 4. Milk producing hormone- Oxytocin 5. Methylene blue test

FACE 1. Lion/Leonine- Lepromatous Leprosy 2. Moon- Cushings syn
3. Bulldog- Congenital syphilis (Frontal bossing + saddle nose + mucous patches)
4. Chipmunk- Thalassemia, bulimia 5. Mouse- CKD 6. Monkey- Marasmus
7. Mask - Parkinson dis 8. Mitral- Mitral stenosis 9. Marshall Hall- Hydrocephalus
10. Coarse Gorilla- Acromegaly, hypothyroidism 11. Ashen grey- Myocardial infarction
12. Snarling- Myasthenia gravis 13. Bird-like- Progeria, Pierre Robin sequence
14. Potter- Oligohydramnios 15. Torpid- Myxedema 16. Adenoid- ↑Adenoid size
17. Plethoric- Cushings sy, polycythemia vera 18. Myopathic/Hatchet- Muscular dystrophy
19. Downs/Flat+ palmar crease-Down syn-F7/25 20. Gargoyle- Hurler syn
21. Hippocratic- Terminal illness, impending death 22. Cachectic- Malnutrition
23. Bovine/Cow- Craniofacial dysostosis 24. Elfin- Williams syn

GANGRENE 1. Gas- C. perfringens 2. Fournier scrotal- E. coli -F1/23/24,N22,23
3. Meleney- Post surg mixed infec-N22 4. Dry/ Ischemic- Diabetes 5. Wet ganagrene

GARDEN 1. Gardenerella- Bacterial vaginosis 2. Rose Gardener dis- Sporotrichosis
3. Garden spade deformity- Smith fracture

GROUP 1. Lancefield- Group A Strep pyogenes, Group B Strep agalactiae
2. HACEK group- Endocarditis 3. Serogroup- Vibrio cholera O1, O139 Bengal

GENES 1. Jumping- Transposon 2. MRSA- mecA 3. VRSA- van A/B gene
4. Gene Rx- SCID, HIV, CAR-T cell cancer Rx 5. Burkitts lymphoma- cmyc gene-F1/26
6. HIV- gag, env, pol, tat, rev, nef, vif, vpu, vpr 7. CML- BCR-ABL gene
8. Hepatitis B- Surface, core, X, polymerase 9. Coxsackie- hox gene -F23/25
10. Antibiotic resistance a. Rifampicin- rpoB b. Isoniazid -F1/25- inhA, katG
c. Pyrazinamide- pncA d. Ethambutol- embB e. Bedaquiline- atpE
11. Oncogenes- ras, myc 12. Oncoregulator- Bcl-2 13. Oncosuppressor- p53, pRB
14. CFTR- Cystic fibrosis -F7/25 15. Gene editing tool- CRISPR-cas9

GOLD STANDARD TEST 1. Any virus- PCR/RTPCR 2. Any bacteria- Culture
3. Any fungus- Culture/Antigen 4. Any parasite- Microscopy/antigen/antibody
5. Covid/Influenza/HIV- RTPCR 6. HPV- HPV PCR (Not Pap smear/ VIA/ VILI) -F1/7/25
7. Typhoid/Brucella/sepsis/endocarditis/invasive fungi/CLABSI- Blood culture
8. TB- MGIT Liquid culture 9. TB from tissue- Biopsy/FNAC + CBNAAT/ MGIT
*Quantiferon TB gold IGRA for diagnosis of Latent TB is Not gold standard test for TB
10. Aspergillus/Cryptococcus-Antigen test-F20/21/24,N19
11. Blood parasite (Malaria/ Filaria)- Peripheral blood smear
12. Stool parasite- Microscopy 13. Any cancer- Biopsy + Histopathology
14. TB kidney/ bladder- CT urography/ cystoscopy- Biopsy + CBNAAT/ MGIT
15. TB cold abscess/ lymph node/ pleural effusion/ intestine/ skin/ genital/ bone
Arthritis/Potts spine- Biopsy/FNAC + CBNAAT/ MGIT
STANDARD TEST- VDRL/RPR in syphilis, Brucella

GRANULOMA (IFN gamma)→ Cell surround bacteria to limit spread of chronic infec
1. Granuloma→ Type 4 hypersensitivity- Lymphocyte, slipper shape epitheloid cell
activated macrophage, Langhans multinucleate giant cell (necklace/horse shoe nuclei)
2. Caseating- TB, Syphilis gumma 3. Durck- P. falciparum cerebral malaria
4. Non-caseating- TT Leprosy, most infec, Sarcoidosis-N22, Crohn dis
5. Stellate- Cat scratch disease-N18 6. Doughnut- Q fever (Coxiella burnetti)

HEMO= Blood (RBC) 1. Hemoglobin, hemoglobinuria, hemoglobinopathy
2. Hemophilia-F7/25 3. Hematocrit = ESR 4. Hemostasis 5. Hemochromatosis- ↑Iron
6. Hemorrhoid,hemorrhoidectomy 7. Hemoconcentration-Shock 8. Hemolysis
hemolytic anemia, hemolytic transfusion rxn, hemolytic dis of newborn =
erythroblastosis fetalis 9. Hemangioma 10. Hemodynamic 11. Hemodialysis,
hemodialyzer 12. Hemodilution 13. Hemocytometer 14. Hematin, hemopexin
15. Hemophagocytosis 16. Hematopoiesis 17. Hematogenous- Miliary TB-N18
18. Hemorrhage (bleeding)- Hematoma/ turia/ emesis/ ochezia, Hemoglobinuria,
Hemo/ thorax/ tympanum/ pericardium/ peritoneum/ ptysis, Hemorrhagic shock/stroke
Hemarthrosis, hyphaema (eye-anterior chamber), Petechiae, Purpura, Ecchymosis= Bruise,
splinter hemorrhage- finger/toenails (endocarditis, vasculitis, Rheumatoid arthritis)
MENINGEAL BLEED 1. Epidural/Extradural- Lens shape 2. Subdural- Crescent shape
3. Subarachnoid- Curvilinear shape 4. Intracerebral- No fixed shape

INFLAMMATION 1. Cystitis- Urinary bladder 2. Cholecystitis- Gall bladder
3. Cellulitis- Skin 4. Cholangitis- Bile duct 5. Colpitis- Vagina 6. Mastitis- Breast
7. Adenitis- Gland 8. Sialadenitis- Salivary gland 9. Lymph node- Lymphadenitis
10. Lymphangitis- Lymph duct 11. Salpingitis- Fallopian tube 12. Phlebitis- Vein
13. Orchitis- Testis 14. Oophoritis- Ovary 15. Otitis- Ear 16. Osteitis- Bone
17. Gingivitis- Gum 18. Glossitis- Tongue 19. Gastritis- Stomach 20. Panniculitis- Fat
21. Enteritis- Small intestine 22. Colitis- Large intestine 23. Pneumonitis- Lung
24. Pyelonephritis- Kidney 25. Vasculitis- Blood vessel 26. Fasciitis- Fascia
27. Parotitis- Parotid 28. Prostatitis- Prostate 29. Proctitis- Rectum 30. Bursitis
31. Vestibulitis- Nose/ear 32. Radiculitis- Nerve root 33. Balanitis- Glans penis
34. Balanoposthitis- Glans+prepuce 35. Funiculitis- Spermatic cord 36. Vasitis
37. Hypophysitis- Pituitary 38. Labyrinthitis- Inner ear 39. Trabeculitis- Bone/eye
40. Tenosynovitis- Tendon+ joint 41. Costochondritis- Ribs+ Cartilage 42. Epiphysitis
43. Myositis- Muscle 44. Cementitis- Cementum-teeth 45. Osteomyelitis, periosteitis
46. Eye- Conjunctivitis, iridocyclitis, uveitis-N24, dacryoadenitis- Lacrimal gland, dacryocystitis- Lacrimal sac, keratitis-cornea, chorioretinitis, meibomitis, canaliculitis, epi/scleritis -N24, endoph/ panophthalmitis, blepharitis- Lid, orbital cellulitis -N18
47. Myelitis- Spinal cord 48. Arthritis- Joint 49. Vulvovaginitis- Vulva+ vagina
50. Carditis- Heart 51. Sinusitis- Nasal sinus (MC <3yr- Ethmoid, >3yr- Maxillary)
52. Diverticulitis 53. Periostitis 54. Cheilitis- Lips 55. Stomatitis- Mouth
56. Folliculitis- Hair follicle 57. Dermatitis- Skin 58. Pericarditis- MC Coxsackie B
59. Arteritis a. Giant cell arteritis- Giant cell granuloma in temporal artery
b. Takayasu arteritis- No pulse c. Polyarteritis nodosa- Necrotizing vasculitis
60. Endometritis- Uterus, placentitis, chorioamnionitis, intervillitis
Rx endometritis=Post partum fever- Clindamycin+ Gentamicin
Rx PROM=PPROM- Erythromycin ± Ampicillin
61. PID→ MC Chlamydia→ MC ectopic preg→ MC C/F- Backache→ Rx Yellow kit6
62. IBD= Inflammatory Bowel Dis- Crohns dis and ulcerative colitis (Lead pipe colon)
63. IBS= Irritable Bowel syndrome 64. Ascitis- ↑Fluid in peritoneal cavity
65. Inflammation- NADPH oxidase most important Decreased- Albumin, Transferrin
Increased- CRP, Procalcitonin, Fibrinogen, Prothrombin, Ferritin, Ceruloplasmin, Properdin

ICD-11- 2022- 22 chapters. C1- Infection, C2- Cancer, C17- STI, C22- Poisoning

IMAGING IN INFECTIONS

1. X ray Uses Ionizing radiation for 2D images C/I in pregnancy, safe in lactation

X ray types/ X ray based imaging- Plain radiograph, CT, DEXA scan, angiography, mammography, HSG, MCU, VCU, Barium study, fluoroscopy, arthrography, myelography, intravenous pyelography, sialography, dacryocystography

X ray appearance in infections

a. Pneumonia (Bacterial)- Lobar opacity= consolidation

b. Klebsiella pneumoniae- Bulging fissure sign

c. Pneumocystis pneumonia- B/l interstitial infiltrates

d. Staph aureus pneumonia- Cavity, pneumatocoele

e. TB- Upper lobe consolidation, cavity, millet dots

f. Aspiration pneumonia- Infiltrates in dependent lung segments

g. Meniscus sign- Pleural effusion (Bacterial, TB), Water bottle heart- Pericardial effusion

h. Deep sulcus (costophrenic angle) + mediastinal shift - Pneumothorax

i. Blunted costophrenic angle + layering of fluid- Hemothorax

j. Loculated/septate pleural fluid + thick pleura- Pleural empyema

k. Elevated hemidiaphragm + mediastinal shift + hyperinflated other lung- Lung atelectasis

h. Air fluid level in Chest Xray- Lung abscess

g. Air filled cavity in lung- TB, lung abscess

h. Gas under diaphragm- Hollow viscus perforation (H. pylori, Typhoid, Entamoeba)

i. Cannon ball metastasis in lung- Renal cell ca

Angiography done for Coronary Artery Disease/MI, aneurysm

Mammography done for breast ca/ lump screening, and follow up after breast ca Rx

HSG= Hysterosalpingography done for infertility evaluation (tubal block)

MCU done for neurogenic bladder/urethral function, vesicoureteric reflux, recurrent UTI

Barium swallow a. Bird beak- Achalasia cardia b. Cork screw- Esophageal spasm

Barium enema a. Saw tooth- Diverticulitis (MC lower GI bleed)

DEXA scan= Dual energy X ray absorptiometry (T/Z score)- Osteoporosis

2. CT = Computed Tomography

Uses high radiation X ray, C/I in pregnancy, safe in lactation, CT types

a. NCCT- Best for head injury, kidney stone, lung nodule, hemorrhage (SAH, EDH, SDH)

b. CECT (Iodinated contrast)- Infection, Tumour, vessels. Contraindicated in CKD

c. HRCT- Best for lung, temporal bone, inner ear

d. Spiral/Helical CT- Best for chest/abdomen

e. CT angiography- Best for blood vessel aneurysm, thromobosis, stenosis

CT appearance in infections

a. Complicated pneumonia- Cavity, necrosis, empyema

b. All abscess- Rim enhacing lesion c. Neurocysticercosis- Cyst with scolex

d. TB meningitis- Basal meningeal enhancement e. Aspergillus- Halo/ air crescent sign

f. Ground Glass Opacity (GGO)- CT chest→ Light hazy 1. Pneumocystis

2. Covid, TB, Atypical pneumonia, ARDS-N18, ILD, Cancer, Pulmonary edema

GGO Chest X-ray→ Neonatal resp distress syndrome = Hyaline membrane dis

Ground Glass Bone- Fibrous dysplasia Ground Glass Nuclei in liver cell- Hepatitis B

3. Ultrasonography=USG Uses high frequency non ionizing sound waves

USG probe receives Echoes (Isoechoic cant be imaged)- Safe in preg/lactation

a. Hyperechoic bright white- Bone, calcified tissues, fat

b. Hypoechoic dark- Abscess, muscle, tendon, lymph node, liver ca, kidney ca

c. Anechoic very dark- Fluid filled cyst, bladder, blood vessel

USG types a. 2D B mode USG- Obstetrics, abdomen, musculoskeletal imaging

b. 3D/4D USG- Obstetrics, fetus c. CEUS (Contrast enhanced USG)- Liver, kidney

d. FAST (Focused Assessment with sonography for trauma)- Abdomen/chest trauma

e. POCUS (Point of care USG)

f. Doppler USG (Color/Power/Spectral)- Blood flow, stenosis, aneurysm, varicose vein

g. Misc- Transvaginal USG, transrectal USG, endoscopic USG, elastography

USG appearance in infections T sign- Monochorionic twins

a. Hydatid cyst- Water Lily= Camalote sign due to detached membranes

b. Filaria- Filarial dance sign c. Acute Hepatitis- Starry sky pattern

d. Pyogenic/Amoebic liver abscess- Hypoechoic lesions

e. Cholelithiasis/Cholecystitis- Gallstones, thick wall, fluid collection

f. Appendicitis- Thick appendix >6mm, appendicolith, fluid collection

g. Pancreatitis- Thick pancreas, necrosis, fluid collection

h. Neural tube defect (↓Folic acid)- Lemon-Banana sign

i. Whirlpool- Midgut volvulus, ovarian torsion j. Snowstorm- Hydatidiform mole

k. String- Pyloric stenosis l. Bag of worm- Varicocoele m. Mickey Mouse- Portal triad

n. Ring of fire- Ectopic preg o. Starry sky USG- Acute hepatitis

p. TB- Golf hole ureter orifice

4. MRI = Magnetic Resonance Imaging Uses

Uses magnetic field + radiofrequency pulse to detect signal from hydrogen nuclei in body

Safe in pregnancy and lactation, C/I with pacemaker or metal implant, MRI types

a. T1 weighted MRI- Water dark, fat bright- For tumour, cyst

b. T2 weighted MRI- Water bright, fat less bright- For edema, inflammation, tumour

c. Others- FLAIR MRI- Brain, STIR MRI for bone marrow edema, DWI MRI, PWI MRI

d. CEMRI (Gadolinium contrast)- Tumour, inflammation, vascular imaging

e. MR angiography f. MR spectroscopy- Measures tissue metabolism

MRI appearance in infections

a. Neurocysticercosis- Del Brutto criteria + starry sky pattern + hole with dot sign

b. Herpes simplex- Temporal lobe encephalitis c. Toxoplasma- Ring enhancing lesions

d. CJD- Pulvinar = Hockey stick sign + Cortical ribboning e. Hanging drop- Potts spine

f. All abscess- Rim enhacing lesion g. Meningitis- Meningeal enhancement

5. FDG-PET = Positron Emission Tomography C/I in pregnancy and lactation

Uses radioactive tracers- TB, osteomyelitis, prosthetic joint infec, lymphadenitis, abscess

6. SPECT with Gamma camera C/I in pregnancy and lactation

Uses gamma ray + radioactive tracer

a. Thyroscan with Radioactive I-123/I-131 or Technetium-99

b. Myocardial perfusion with Technetium-99=Sestamibi scan or tetrofosmin

c. Renal function with DTPA/ DMSA d. Bone/ infection/ tumour scan

7. OCT = Optical coherence Tomography- Chorioretinitis, glaucoma, ARMD

8. CORNEAL TOPOGRAPHY- Astigmatism, befor LASIK, contact lens

9. OAE = Otoacoustic emission- Hearing loss in newborn, cochlear function

10. PAI = Photoacoustic imaging- Breast ca, vascular imaging

11. PCG = Phonocardiogram- Heart murmurs

12. BIRADS- Breast Imaging Reporting and Data System b. PIRADS- Prostate

c. LIRADS- Liver d. TIRADS- Thyroid e. Lung-RADS f. NI-RADS- Head neck ca

INCUBATION PERIOD=IP Time b/w pathogen entry and 1st clinical features

1. <7 days- Influenza, meningitis, cholera, gonorrhea, anthrax, plague, diphtheria
2. 7 day- Tetanus, chlamydia, dengue, chikungunya, scrub tyhpus, Leptospira
3. 7-14 days- MMRV, Typhoid, Malaria exc P. malariae, Pertussis, Polio, JE
4. ~30 days to many months- Hepatitis A,B,C,E, rabies, Kala azar, Filaria, HIV

WINDOW PERIOD=WP (Pathogen in in the body but not detectable)

1. General definition- Time b/w pathogen entry and 1st positive lab test. Every test has a different window period in all infections including HIV (HIV- HIV RNA RTPCR 7-10day, HIV p24 antigen 12day, IVgen ELISA 16day- Most sensitive, IIIgen ELISA 21day)
2. Hepatitis B has two window periods

(a) Early virological WP- HBV DNA PCR 7-14 days, HBsAg 1-2 month, HBeAg 1-3 mnth, Anti HBc IgM 1-3 mnth, Anti HBs 4-8 mnth, Anti HBc IgG >4 mnth

(b) Late serological WP (LSWP)= 1 month Diagnostic gap between disappearance of HBsAg and appearance of Anti HBs (3rd-4th month). Only IgM Anti HBc positive in LSWP

LATENT PERIOD Incubation period of non-communicable disease

GENERATION TIME Time interval b/w pathogen entry and max infectivity

SERIAL INTERVAL Time interval b/w 1^0 (1st case in population) and 2^0 case

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ICEBERG CONCEPT -N24- 20% infec diagnosed and Rx, 80% remain undiagnosed

1. Iceberg- TB, HIV, Viral Hepatitis, JE, Gonorrhea, Helminthiasis, Influenza, Covid, Cancer, CKD, Lifestyle dis, psychiatric dis, drug addiction
2. No iceberg (Disease always symptomatic, patients report)- Rabies, snake bite, measles, chickenpox, diphtheria, pertussis, tetanus, meningitis, CJD
3. Reverse Iceberg (If ↑Surveillance/ reporting)- SARS, Ebola, Bird flu

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JAUNDICE = HYPERBILIRUBINEMIA -INI25 Dark urine

1. Pre-hepatic/hemolytic/medical- Malaria, dengue, EHEC, sepsis, breastfeed

↑indirect=unconjugated bilirubinemia, pale stool, ↑spleen, ↑reticulocyte-young RBC

2. Intrahepatic/hepatocellular- Viral hepatitis (MC jaundice), alcohol, drugs

↑indirect=unconjugated bilirirubinemia, pale stool, hepatomeg, ascites, ↑liver enzyme

3. Post-hep/obstructive/surgical- Cholangitis, cholangioca, pancreatic ca -N18

↑direct=conjugated bilirubin, pale greasy clay stool, itching, Courvoisier sign -N22

KERNICTERUS- Brain damage after NJ Rx. Exchange transfusion

MECONIUM aspiration- Post term baby

CEPHALHEMATOMA crosses sutures, **CAPUT SUCCEDANEUM** doesn't

NEONATAL JAUNDICE=NJ a. Physiological- Starts 2nd-3rd day goes 1-2 wks after birth .Reason- Preterm baby, ↑Bilirubin, immature liver .Rx Phototherapy, ↑breastfeed
b. Early onset NJ (<24-48h)- Rh incompatibility, infection, ↓G6PD .Rx of cause
c. Late onset NJ (>72h)- ↓Breastfeed, Breast milk jaundice, liver dis, infec .Rx of cause
MC neonatal mortality- Prematurity

LYMPHO 1. Lymphoma 2. Lymphocytopenia 3. Lymphoproliferative 4. Lymphokine
5. Lymph node- Lymphadenopathy, Lymphadenitis, Lymphadenectomy
6. Lymphangitis (MC Strep pyogenes), Lymphoedema 7. Lymphangiosarcoma

LUPUS~Wolf 1. Lupus vulgaris- MC skin TB- Apple jelly nodule -N20/25
2. SLE (Lupus arthritis/nephritis)-Autoimmune 3. Lupus anticoagulant- ↑Blood clot

LEUKO = White 1. Leukocyte, leucocytosis, leukopenia 2. Leukoplakia-N22-EBV
3. Leukemia- EBV 4. Leukotriene- Mediator 5. Leukoderma- Skin patch
6. Leukorrhea-N21- Vaginal discharge 7. Leukocoria- Pupil 8. Leukotrichia- Hair
9. Leuconychia- Nail 10. Leucoblast, leucoblastosis 11. Leucopheresis
12. Leucomalacia, Leucomelanosis, Leucoencephalopathy, Leukemoid reaction

METALS (BMW- Blue bin) 1. Silver=Sv a. Sv sulfadiazine/nitrate (Burns)
b. Sv impregnation stain- Syphilis c. Grocotts Silver stain- Pneumocystis
2. Gold a. Gold pigment- Staph aureus b. Quantiferon TB gold test- Latent TB
3. Copper=Cu a. Cu T b. Cu penny body- Chromoblastomycosis c. ↑Cu- Wilson dis
4. Iron deficiency anemia- Hookworm 5. Mercury spill- Sulfur powder
6. Lead=Ld a. Ld aborbs radiocactivity b. Ld pipe pattern- Ulcerative colitis
7. Titanium- Implants 8. Stainless steel- Surgical instruments, implants
8. Zinc- Wound heal, powder ↓diaper rash 9. Cobalt- Pacemaker
10. Lithium- Psychiatric drug 11. Bismuth- Rx of H. pylori-F7/25

MIDDLE PATH RX 1. TB- Potts spine (MC paradiscal), TB meningitis, TB osteomyelitis
2. Ca- Burkitt lymphoma, NHL, ALL 3. Misc- Rheumatoid arthritis, sarcoidosis

MACROPHAGES 1. Connective tissue- Histiocyte 2. Peripheral bld- Monocyte
3. Bone- Osteoclast 4. Brain-N20,22- Microglia 5.Liver-F1/23,7/24- Kupffer cells
6. Kidney- Mesangial cells 7. Lung- Dust/alveolar macrophage -F 7/24
8. Lepra=Virchow lipid laden foamy cells
9. Lymphoid follicle- Tingible body macrophage
10. Spleen-Marginal Zone, red pulp, Metallophilic littoral macrophage
11. Placenta- Hofbauer cell 12. Joints- Synoviocyte= Synovial macrophage
13. Chr inflam- Langhans giant cell 14. Skin- Langerhans (Dendritic) cell-N19
15. Multiple myeloma- Hodgkins + Flame + Mott cell + Russell body -F 6/19/20,N24

MONOCLONAL ANTIBODIES Artificial antibody made in mouse
1. Anthrax- Obiltotaximab, Raxibacumab 2. Pseudomonas- Panobacumab
3. C. difficile- Bezlotoxumab, Actoxumab 4. Staphylococcus- Svratoxumab
5. Ebola- Cosfroviximab, Larcaviximab 6. RSV- Pavilizumab
7. Hepatitis B- Libivirumab, Tuvirumab 8. HIV- Ibalizumab, Leronlimab
9. Rabies- Foravirumab, Rafivirumab 10. Viral infec- Bavituximab, Suvizumab
11. Covid- Tixagevimab-Cilgavimab, Tocilizumab, Itolizumab
12. Candida- Enfungumab 13. TNF α inhibitor- Infliximab
14. Liver ca- Nivolumab, Camrelizumab 15. NHL, CLL- Rituximab, ibritumomab

NATIONAL HEALTH PROGRAMS
1. National TB Elimination Program 2. National Leprosy Eradication Program
3. National Rabies Control Program 2023 4. National Viral Hepatitis Control Program
5. National AIDS Control Program 6. National Vector Borne Disease Control Program
7. Anemia Mukt Bharat Prog-F1/26 (6 beneficiary x 6 intervention x 6 mechanism strategy)
8. Universal Immunization Program- Mission Indradhanush 2.0
9. 2025 National One Health Mission- Triple threat pathogens (Human+ animal+ wildlife)
(i) Nat Action Plan for Dog-mediated Rabies Elimination (NAPRE) 2030
(a) Bacteria- Anthrax, Plague, Brucella, Leptopspira
(b) Virus- Bird flu, Nipah, KFD, CCHF
(c) Vector Borne- JE, Dengue, Scrub typhus

NECROSIS 1. Caseous- MC TB, Histoplasma -F1/26 2. Fat- Calcium (Ac pancreatitis)
3. Liquefactive- Brain infarct/abscess 4. Coagulative- Most tissues except brain
5. Fibrinoid- RHD, SLE, PAN, vasculitis, malignant HTN

NECROTISING FASCIITIS 1. MC GABHS 2. Staph aureus 3. Gas gangrene

OVA 1. Cestode=Tapeworm- Polar knob + yolk granule + hexacanth embryo
2. Trematode= Fluke- Operculated ova except Schistosoma= Blood fluke
3. Shistosoma- Spinous ova a. S. hematobium- Terminal b. S. mansonii- Lateral
4. Ascaris= Roundworm a. Fertilized- Coat+ space+ float b. Unfert- None of 3
5. Trichuris= Whipworm- Barrel ova + mucus plugs *Barrel chest- Emphysema
6. Ancylostoma/Necator= Hookworm- Segmented ova
7. Enterobius= Pinworm- D shaped planoconvex ova

-OSIS 1. Bacteria- Tuberculosis, Donovanosis, Vaginosis, Mycoplasmosis, Brucellosis, Leptospirosis, Meliodosis, Rhinosporidiosis
2. Fungus- Aspergillosis, Mucormycosis = Zygomycosis, Maduramycosis, Histoplasmosis, Sporotrichosis, Chromoblastomycosis, Dermatophytosis,
3. Virus- Mononucleosis 4. Parasite- Cysticercosis, Hydatidosis-N23
5. Clinical- Cirrhosis, pustulosis, dermatosis, endometriosis, osteoporosis, apoptosis, siderosis, sarcoidosis, amyloidosis, madarosis, ketoacidosis-N23, ptosis, miosis, anhydrosis, granulomatosis, cholesterosis, thrombosis -N20 scoliosis, kyphosis, lordosis, coal worker pneumonconiosis=athracosis
6. Misc- Zoonosis, necrosis, fibrosis, stenosis, prognosis, osmosis, pyknosis

PARASITE 1. Hemoparasite- Malaria, Filaria 2. STI- Trichomonas-N19/20
3. Neuroparasite- Neurocysticercosis, Toxoplasma 4. Dog- Hydatid cyst
5. Orofecal- Entamoeba, Giardia, Cryptosporidium-N19 6. Fish- Fish tapeworm
7. TORCHES- Toxoplasma, Malaria 8. Vector- Malaria, Kala azar
9. Soil- Hookworm, Strongyloides 10. Pork- Taenia solium, Trichinella-F1/26
11. Hookworm- Ancylostoma, Necator 12. Filaria worm- Wuchereria, Brugia
13. Ca bladder- Schistosoma hematobium -N18 14. No ova- Strongyloides-F1/26
15. Cholangioca-N23- Clonorchis, Opisthorchis 16. No cyst-Trichomonas-N20

POWDER 1. Bleaching $CaOCl_2$- Chlorination 2. Sulfur- Mercury spill
3. Antifungal Miconazole- ↓Athlete foot, jock itch 4. Calamine- ↓Itch/inflam
6. Antiseptic- Wounds 7. Zinc oxide talc- ↓Diaper rash

PROCEDURES 1. -desis= Fusion- Arthrodesis, pleurodesis

2. -plasty=Reconstruction- Mammoplasty, rhinop, blepharo, cheiro, phallo, otoplasty

3. -ectomy=Excision=remove- Tonsillectomy (MC), mastoidectomy, stapedectomy, tympanectomy, epiglottidectomy, adenoidectomy, sialadenectomy, laryngectomy, mandibulectomy, parotidectomy, thymectomy, uvulo-palato-pharyngectomy, tubectomy, hysterectomy, endometrectomy, tracheelctomy (cervix), colpectomy (vagina), clitorectomy (clitoris), salpingo-oophorectomy (fallopian tube + ovary), vulvectomy, orchiectomy, vasectomy, epididymectomy, mastectomy, lumpectomy, pinealectomy, lobectomy (lung/liver/thyroid/temporal), pneumonectomy, segmentectomy (lung), aneurysmectomy, atherectomy, endarterectomy, thrombectomy, embolectomy, hemicraniectomy, cerebellectomy, thalamectomy, rhizectomy (nerve root), neurectomy, pallidectomy (globus pallidus), sympathectomy (symp ganglionectomy), lipectomy (fat), laminectomy (vertebra), diskectomy, meniscectomy, synovectomy, bursectomy, bunionectomy, myomectomy (muscle), osteectomy (bone), phalanectomy (phalanx), facetectomy (vertebra facet), costectomy (rib), apocrinectomy, thyroidectomy, pericardectomy, diaphragmectomy, lympadenectomy, glossectomy, pancreatectomy, appendicectomy, cholecystectomy, omentectomy, splenectomy, hepatectomy, oesophago-gastrectomy, procto-colectomy, Whipple pancreatio-duodenectomy, sigmoidectomy, cecectomy (cecum), anorectectomy, amygdalo-hippocapectomy, nephrectomy, adrenalectomy, ureterectomy, urethrectomy, cysto-prostatectomy (bladder), iridectomy, vitrectomy, sclerectomy, trabeculectomy (glaucoma), dacryocystadenectomy,

4. -otomy=Open only- Episiotomy, colpotomy, laparotomy, gastrotomy, enterotomy, phlebotomy, iridotomy, dermotomy, pyelotomy, ureterolithotomy, patellotomy, craniotomy, ventriculotomy, thoracotomy, sternotomy, cardiotomy, myringotomy, tracheotomy, laryngotomy, tympanotomy, fasciotomy, escharotomy, osteotomy, cordotomy (spinal cord), tendonotomy, cricothyroidotomy, Heller myotomy- Achalasia

5. -ostomy= Opening stays- Tracheostomy, gastro-duodeno-jejunostomy, ileostomy, colostomy, cystostomy, nephrostomy, vesicostomy (bladder), pyelostomy (kidney)

6. -pexy= Suspension- Mastopexy, nephro, hystero, orchio, recto, gastropexy

7. -rrhaphy= Repair- Herniorrhaphy, neuro, arthro, cysto, colpo, tenorrhaphy

8. -centesis= Fluid removal- Amniocentesis, thoraco, cardio, paracentesis

9. -scopy=See- Colposcopy (vagina), hystero, procto, cysto, arthro, laparo, fundo/retino, stroboscopy (vocal cord), broncho, nephro, oto, feto, embryo, microscopy, horoscope ☺

10. -graphy=Imaging- Mammography, ECG, EEG, EMG, ERCP, angiography, ultrasonography, computed tomography (CT scan), electroretino/cochleography, echocardiography, cardiotocography, plethysmography (blood flow test), polysomnography (sleep test), radiography, hysterosalpingography, scannography, lymphoscintigraphy, micturating cystourethrography (MCU), tympanography
11. -tripsy= Breaking- Lithotripsy (stone), cholelitho, nephrolithotripsy
12. -metry=measurement- Pulse oximetry, Bone densitometry, spirometry, pure tone audiometry (PTA), goniometry (glaucoma), optometry, cardiac telemetry

PSEUDO 1. Pseudohyphae+budding yeast- Candida -F1/26,7/25,INI25
2. Pseudomembrane a. Throat- Diphtheria b. Colitis- C. difficile- Volcano eruption -N22
3. Pseudopodia- Entamoeba 4. Burkholderia pseudomallei- Meliodosis 5. Pseudogout
6. Pseudofracture (Looser zone)- Osteomalacia 7. Pseudopolyp- Ulcerative colitis
8. Pseudobubo= Reactive lymph node- Trauma/ insect bite 9. Pseudomonas-N20,21 Bacteria
10. Pseudo-Nikolsky- Bullous pemphigoid 11. Pseudocyesis- False pregnancy

PERFORATION 1. Hollow viscus (gut- oesophagus, stomach, small intestine, colon, Appendix, diverticular) 2. Tympanic memb 3. Cornea 4. Uterus 5. Bladder 6. Lung

PLACES (Only Important) 1. Delhi boil/Oriental sore- Cutaneous Leishmania
2. Dum dum fever (Kolkata)- Kala azar 3. Madura foot (Madurai) -F1/26
4. Malta = Undulant fever- Brucellosis-F1/26 5. South East Asian liver fluke
6. China a.Letter- Diphtheria b.Liver fluke- Clonorchis c.Fried rice- B. cereus
7. Japanese Encephalitis 8. Spanish/Bombay flu= H1N1 (1918- 1920) -N20
9. Japan- Ogawa Inaba Hikojima Kanagawa Wagatsuma Nagayama Katayama
10. Kyasanur Forest disease (Shimoga, Karnataka) 11. Nipah- Malaysia
12. Meghalaya- 2024 VAPP, planned mandatory premarital HIV testing 2025
13. Australia antigen = HBsAg 14. Lyme disease (US)- Borrelia burgdorferi
15. African sleeping sickness- T. brucei- Tse tse fly 16. Ebola- Africa river
17. South American Chagas dis- T. cruzi- Kissing bug 18. West Nile virus
19. Rocky Mountain spotted fever (US)- Tick- Rickettsia 20. German measles
21. Congo red stain- Amyloidosis-F7/25 22. Singapore ear- Pseudomonas
23. Valley = California fever = Desert rheumatism (Coccidioides fungus) -N24
24. North/South American, European Blastomycosis (Fungus) -N24
25. Middle East Resp syndrome coronavirus 26. Hantavirus- South Korea

PIGMENT 1. Red-Serratia 2. Pteridine- Fungus 3. Hemozoin- Malaria
4. Pyocyanin green- Pseudomonas 5. Staphyloxanthin golden- Oil paint pattern

PENCIL 1. Pencil tip sign-N23- Croup 2. Pencil shape- Trichophyton
3. Pencil in cup deformity- Psoriatic arthritis

PANDEMIC LEVEL DISEASES Alcoholism, smoking, substance abuse, HTN, obesity, chronic pain, Diabetes, CKD, Asthma, Psychiatric/ CNS/ Oral disease

PLANTS 1. Flower/rosette pattern- Sporothrix -F23/24/25,N20
2. Tree twig/plant injury- Fungal keratitis 3. Scrub/wild gass- Scrub typhus
4. Bamboo/box car shape- Bacillus anthracis, Clostridium perfringens
5. Fruity smell- Pseudomonas, acetone breath in DKA 6. Lichenoid drug eruption
7. a. Cotton wool- Aspergilloma, pneumonia b. Lactophenol cotton blue fungal stain
b. Cotton fluffy colony- Black fungus Mucormycosis-F1/26

PALSY=py- Partial weakness of muscle 1. Bells py= MC facial py= Idiopathic facial py
2. Cerebral py (TORCHES/Low borth weight/prematurity) 3. Bulbar py- Polio
4. Ulnar nerve palsy- Claw hand, Froment sign 5. Radial nerve Saturday night palsy
6. Median nerve palsy- Ape thumb deformity 7. Long thoracic nerve palsy- Winged scapula
8. Erb py= Waiter/Policeman tip hand- Upper brachial plexus injury- C_5-C_6
9. Klumpke py= Claw hand + Horners syndrome - Lower brachial plexus injury- C_8-T_1
10. Anterior interosseous nerve palsy- Ok sign weakness

PARALYSIS=ps Complete loss of muscle function-N24-

Features	Lower motor neuron-Flaccid	Upper motor neuron-Spastic
1. Site	Anterior horn cell, nerve, NMJ	Brain, spinal cord
2. Muscle tone/power/reflex	↓↓↓	↑↑↑
3. Muscle activity	Fasciculations	Clonus
4. Babinski sign/Plantars	↓	↑
5. Examples	BPOGNP (Botulism, Polio, organophosphate poisoning, GB syndrome, Neurotoxic snake bite, peripheral neuritis)	TLCSSSM (Tetanus, Lathyrism, Cerebral palsy, Strychnine poisoning, stroke, spinal cord Injury, multiple sclerosis)

Ascending sym flaccid ps- OP poisoning, GBS
Descending sym flaccid ps- Botox, neurotoxic snake, MS
Systemic sym flaccid ps- Polio, hypokalemia, myasthenia

PINPOINT 1. Pupil- Opioid/ organophosphate toxicity, pontine hemorrhage, stroke
2. Rash (Petechiae)- Dengue, thrombocytopenia, meningococcemia, endocarditis, leukemia

QUALITY CONTROL Levey Jenning chart, Shewhart chart, Westgard rules
NABH- National Accreditation Board for Hospitals
NABL- National Accred Board for Calibration and Test Labs

REACTION(rxn) 1. Lepra- Leprosy-F1/26 2. Quellungs- Capsule 3. Arthus- Type3 HS
3. McFadyean- Anthrax-N25 4. Drug rxn 4. Transfusion rxn 5. Polymerase chain rxn

RECEPTORS 1. Sialic acid- H. pylori, Influenza 2. HACE2- Covid-19 spike protein
3. CD4, CXCR4, CCR5- HIV 4. CD46- Measles, Mumps 5. CD155- Polio
6. Acetylcholine- Rabies, botox-F1/26, Myasthenia-F7/24/25, organophosphate, cobra venom
7. GM1 Ganglioside-N22- Cholera, Tetanus, Botox, Guillain Barre syndrome-N19

RING 1. Ringworm= Tinea-F1/26 2. Signet ring- Plasmodium, gastric ca
3. KF ring- Wilson dis-N23,F7/25 4. Ring vacc- KFD, Ebola, MMR, Mpox
5. Waldeyer ring- Tonsil lymphoid tissue

SHAPE OF BACTERIA 1. Bamboo- Anthrax 2. Boxcar- Clostridium perfringens
3. Curved=comma- Vibrio cholera 4. Spiral- Treponema, Leptospira, Borrelia
5. Curved umbrella handle- Leptospira 6. Seagull/Gull wing- Campylobacter
7. Safety pin=Bipolar- Yersinia pestis (Wayson/Wright stain), Burkholderia
8. D- Neisseria meningitides 9. Grape- Staphylococcus 10. Chain- Streptococcus
11. Filamentous- Actinomyces, Nocardia 12. Beaded + curved- M. tuberculosis
13. Chinese letter=cuneiform, picket fence/club- Corynebacterium diphtheriae

STRING 1. Test- H. pylori, Giardia, Strongyloides, gastric bleed
2. String of pearl- Anthrax, PCOS, Gallstones, intestinal obstruction
3. String Sign of Kantor- Crohn dis
4. Egg on String/side- TGA 5. String sign- Hypertrophis pyloric stenosis
5. Figure of 8/ Snowman- TAPVC, Blastomyces 6. Boot shape heart- TOF
. MC congenital heart dis (CHD)- VSD . MC cyanotic CHD- TOF

SATELLITE 1. Virus- Hepatitis D 2. Satellitism- Hemophilus influenzae -F1/26,INI25

STEP LADDER 1. Fever- Typhoid 2. Deformity- Spondylolisthesis

SKIN INFECTION/ DISEASE 1. Hair follicle- Folliculitis- MC Staph aureus
2. Sweat gland- Hidradenitis suppurativa= Acne inversa - MC Staph aureus in axilla
3. Pilosebaceous unit- Acne- Cutibacterium (Propionibacterium) acnes
4. Skin TB- MC Lupus vulgaris> Scrofuloderma 5. Skin lesion in systemic TB- Tuberculid
6. Impetigo (Honey coloured crusts)- Non-bullous and bullous impetigo-
MC Staph aureus> Strep pyogenes
7. Ecthyma = Ulcerative form of impetigo (Strep pyogenes + Staps aureus)
8. Lichen planus- 5 Ps- Purple, Polygonal, Pruritic, Planar, Papule
9. Pityriaisis rosea- Herald patch + Christmas tree distribution
10. Fish scales + String of prearls on knuckles- Ichthyosis vulgaris
11. Café au lait macule + axillary freckle- Neurofibromatosis Type 1
12. Multiple angiofibroma + Shagreen patch- Tuberous sclerosis
13. Basal cell ca + jaw cyst + bifid ribs- Basal cell nevus- Gorlin syndrome
14. Colloidon/Harlequin baby- Ichthyosis congenita
15. Koebner phenomenon- Lesion at trauma site- Psoriasis, Lichen planus, Vitiligo
16. Wickhams striae- White lacy streaks- Lichen planus
17. MC eczema- Contact dermatitis 18. Chronic scratching in eczema→ Lichenification
19. Acanthosis nigricans (Dark velvety skin patches)- Seen in Insulin resistance,
Cushing syndrome, PCOD, gastric ca, steroid Rx, hormone Rx
20. Urticaria=Hives- Transient itchy red wheal+ edema+ Rx antihistaminics
21. Pemphigus vulgaris- Flaccid rupturing/crusting blister/bullae, Rx steroid/Rituximab
22. Fatty streak- Earliest sign of atherosclerosis 23. Calabar swelling- Loa loa
24. Tinea capitis- Inside hair- Endothrix, outside hair- Ectothrix, Kerion, favus
25. Xeroderma= Dry skin- Atopic dermatitis, ichthyosis vulgaris, hypothyroidism,
Renal failure, ↓Vit A, diabetes, malnutrition, dehydration, use of harsh soap/retinoids

SULPHUR/SULFUR 1. Granules- Actinomyces 2. Powder- for mercury spill
3. Sulfonamides- Antibiotic class 4. Disulfiram- Alcohol dependence

SUN/MOON/STAR 1. Sunray fungus- Actinomyces bacteria
2. Sunset sign- Hydrocephalus 3. Sunrise sign- Epiglottitis
4. Sunburst + Codman triangle- Osteosarcoma
5. Sunflower KF Ring cataract- Wilsons dis 6. Moon face- Cushings, Kwashiorkor
7. Starry sky- Burkitt lymphoma, acute hepatitis, neurocysticercosis

SELECTIVE MEDIA/ AGAR=ag -F7/24, 1/22/25- Allows growth of only one bacteria

1. Staph aureus- Salt milk/ Mannitol salt/ Ludlams agar
2. Strep pyogenes- Crystal violet blood agar 3. Pseudomonas- Cetrimide agar
4. Tuberculosis- Lowenstein Jensen ag 5. Neisseria- Thayer Martin/ New York ag
5. Coryne. diphtheriae- Potassium tellurite blood/ Tinsdale/ Loefflers serum slope
6. Salmonella + Shigella- DCA/ XLD/ SSA/ Hektoen enteric agar
7. Salmonella- Wilson Blair ag-F1/23 8. Leptospira- EMJH/ Fletcher/ Korthoff/ Cox ag
9. Anaerobe(Clostridium perfringens, tetani, botulinum)-Robertson cooked meat ag
10. Vibrio cholera- Alkaline bile salt/ Monsurs GTTA/ TCBS agar
11. Clostridioides diffcile- CCFA/ CCYA ag 12. Burkholderia- Ashdown agar
13. Mycoplasma- PPLO/ Eaton ag 14. Hemophilus influenzae- Levinthal/ Filde agar
15. Bacillus anthracis- PLET agar 16. Bacillus cereus- MYPA/ PEMBA agar
16. Bordetella- Regan Lowe medium/ Bordet Gengou glycerine potato blood agar
17. Listeria- PALCAM ag 18. Legionella- BCYE agar 19. Malassezia- Dixon agar
20. Cryptococcoccus- Bird/Niger seed agar 21. Nocardia- Paraffin bait -NEET 25
22. Acinetobacter isolation ag 23. P. falciparum- RPMI 1640 medium
24. Leishmania- NNN/ Grace insect/ Schneider drophila medium
25. Entamoeba- Polyxenic (NIH,Nelson,Balamuth,Robinson ag), Axenic diamond ag
26. Hookworm + Strongyloides- Harada Mori, Agar plate, Bayermann, Charcoal ag

TRANSPORT MEDIA/ AGAR 1. Salmonella/ Shigella- Cary Blair medium
2. Vibrio cholera- Cary Blair, Venkatraman Ramakrishnan medium

ENRICHMENT MEDIA/ AGAR 1. Vibrio cholera- Alkaline peptone water
2. Salmonella/ Shigella- Selenite F broth -N21, Tetrathionate broth -F6/19

PIGMENT MEDIA/ AGAR 1. Pseudomonas- Kings A/B 2. Strep agalactiae- Islam medium

COLONY TYPES 1. Rough, tough, buff on LJ media- Mycobacterium tuberculosis
2. β hemolytic- Staphylococcus, Strep pyogenes and agalactiae, Listeria
3. α hemolytic- Strep pneumoniae and viridans, Enterococcus
4. Carrom coin/ Draughtsman/ autolytic- Streptococcus pneumoniae
5. Daisy head/Frog egg/Poached egg- Corynebacterium diphtheria
6. Medusa head, string of pearls, inverted fir-Christmas tree- Bacillus anthracis
7. Spider molar teeth- Actinomyces 8. Orange wrinkled- Nocardia
9. Stalactite- Yersinia pestis 10. Satellitism- Hemophilus influenza-F1/26
11. Fried egg-F1/26- Mycoplasma, Malassezia 12. Turnip/napiform on gelatin- V. cholerae

SYNDROMES=sy 1. Red man sy- Vancomycin 2. Gray baby sy- Chloramphenicol
3. Reiters sy- Conjunctivitis/Uveitis-N24 + Urethritis + Arthritis-N19- MC Chlamydia
4. Scalded Skin sy/Toxic epidermal necrolysis/ Ritters sy- S.aureus- Nikolsky sign +ve-F7/25
5. Toxic Shock sy- Staph aureus, Strep pyogenes, MMR 4. Weils sy-Leptospira-F1/26
6. Guillain Barre sy=AIDP-N19 Ascend flaccid paralysis- MC- Campylobacter
7. Ramsay Hunt sy-N22- Facial palsy + ear/tongue rash (Shingles in geniculate ganglia)
8. Asherman sy-F7/25/21,N23 Endometrial adhesion in PID (Chlamydia, TB, gonorrhea)
9. Fitz Hugh Curtis sy- Perihepatitis+ Peritonitis in PID (Chlamydia, Gono, TB)
10. Hemolytic uremic sy (Renal failure)- EHEC O157:H7, Dengue, Malaria
11. IRIS= Immune Reconstitution Inflam sy- ↑Immunity after HIV Rx
12. Steven Johnson sy- Risky- Skin/mucosal lesion- HSV, HIV, NSAID
13. Horner sy-N19/24- Ptosis+ miosis+ anhidrosis+ Enopthalmos+ weak face
14. Gradenigo sy- Otorrhea + V cranial nerve pain + diplopia after infection
15. Reye sy- Measles, flu virus→ Fatty liver+ Encephalopathy (↑↑aspirin, paracetamol)
16. Compartment sy-N18- Gas gangrene, snake/insect bite, crush injury, burns, gunshots -N22, diabetic ketoacidosis, ↑↑bandage, ↑↑exercise 17. MEN sy
18. Floppy baby sy- Botulism 19. Dengue Shock sy 20. Hyper IgE sy-N19
21. Waterhouse Fridericksen sy- Neisseria 22. Empty Nose sy- Ozaena-F1/26
23. Nephritic sy-N22- Hematuria a. PSGN-RBC cast+ Lumpy Bumpy granular deposit
b. Pyelonephritis-WBC cast c. Spike and dome- Membranous nephropathy
d. Tram track- Membranoproliferative glomerulonephritis
e. Crescent- Rapidly progressing glomerulonephritis
24. Nephrotic sy- Proteinuria >3.5g/day- Fatty cast 25. Chronic renal failure- Waxy cast
26. Ac tubular necrosis- Muddy brown granular cast 27. Normal hyaline cast in urine
28. Loefflers sy- MC Ascaris 29. Sheehan sy-N21/24- PPH + pituitary necrosis
30. Bulls eye syn-Chloroquine 31. Systemic Inflammatory Response sy (Before sepsis)
32. Goodpasture sy (Hematuria + hemoptysis)- Anti GBM Ab- Type2 hsn + Type4 collagen
33. Felty sy- Rheumatoid arthritis+ neutropenia+ Splenomegaly
33. Jilted Lover sy= Erotomanic delusion 34. Burning feet sy- ↓Vitamin B_5
35. Idiopathic sy - Cause unknown (Idiopathic thrombocytopenic purpura)
36. Iatrogenic sy- Due to healthcare process/intervention/surgery
37. Barrett esophagus- Intestinal metaplasia (Squamous→ columnar)
38. Carpal tunnel- Median nerve 39. DRESS=DIHS (Drug-Induced Hypersens Syn)

40. Teratogenic sy= Fetal warfarin/ hydantoin/ alcohol sy

a. Teratogenic= Organ malformation in fetus, organogenesis- 1st trimester

b. Teratogens- TORCH, ↓folic acid, alcohol, smoking, lithium, warfarin, ↓VitA, diabetes, isotretinoin(acne), antiepileptic, anticancer, thalidomide, ACEi

c. Preg C/I -F19,23- CT, PET, ↑Xray, angiography, bone/thyroid scan

41. Conn sy= 1^0 aldosteronism- ↑aldosterone, ↑BP-Na, ↓K, metabolic alkalosis

42. Addison dis (MC in India-TB)-↓cortisol↓aldosterone↓BP-Na,↑K,metabolic acidosis

43. Trisomy=T sy a. Patau sy- T13 + cleft lip/palate+ small eye/head+ polydactyly

b. Edwards sy- T18 + clenched fist+ strawberry skull+ rocker botton feet+ small jaw

c. Downs sy- T21 (MC) -F7/25 + palmar crease+ small ear/neck+ epicanthic fold

44. Sex chromosome a. Turner sy- 45,X + webbed neck b. Klinefelter sy- 47,XXY-F7/25

45. Genetic deletion=del sy a. Cri du chat sy(cat cry)- 5p del b. Prader-Willi sy- 15q del

46. Acute retroviral sy- Fever + maculopapular rash 47. Stockholm sy- Kidnapper

48. Carcinogenic sy- Chemical + Radiation + Infection + Genetics

49. Cushing sy (Pituitary/adrenal tumour or glucocorticorticoid Rx)- ↑cortisol, ↑BP-glucose-Na, ↓K, metab alkalosis, moon face, ↑weight, buffalo hump, purple striae=stretch marks, hirsutism, cortical adrenal adenoma

.Marbling- Purple veins after livor mortis

50. Capgras sy- Stranger appears familiar 51. Fregoli sy- Familiar appears stranger

52. Resp sy- ARDS, SARS, MERS 53. Alice in Wonderland/Lilliput sy- EBV

54. Marfan- Tall stature, long limbs, arachnodactyly, aortic root dilation, lens dislocation

55. Ehler Danlos- Hypermobile joints, skin hyperextensibility, tissue fragility

56. Sjogren- Dry eyes, dry mouth, arthritis, autoantibodies

57. Antiphospholipid- Thrombosis, recurrent miscarriages, livedo reticularis, ↓platelets

58. Hemochromatosis- Iron overload, skin pigment, liver dis, diabetes, cardiomyopathy

59. Zollinger-Ellison- Gastrinoma, excessive acid production, peptic ulcers, diarrhea

60. Carcinoid- Flushing, diarrhea, wheeze, heart valve dis, carcinoid heart disease

61. Lambert Eaton myasthenic sy 62. Alport sy 63. Bartter sy

64. Beckwith-Wiedeman sy 65. CHARGE sy 66. DiGeorge sy 67. Gardner sy

68. Dubowitz sy 69. Holt Oram sy 70. Kartagener sy 71. McCune Albright sy

72. Noona sy 73. Prader Willi sy 74. Rett sy 75. Russel Silver sy 76. William sy

77. Dressler sy- Post MI 78. Fanconi sy 79. Serotonin sy- MAOI/SSRI

80. Tumour lysis sy- ↑uric acid, hyperkalemia, hyperphosphatemia, hypocalcemia, AKI

81. CREST- Calcinosis, Raynauds, Esophageal dysmotility, sclerodactyly, telangiectasia
82. Churg Strauss- Asthma + eosinophilia + vasculitis

1. Cold SORE-HSV-1 2. Delhi=Oriental Sore- Cutaneous Leishmaniasis

SPOTS 1. Rose- Typhoid 2. Koplik- Measles-F1/26 3. Forcheimmer- Rubella
4. Roths-N18- Infec endocard (Splinter hemorrhages) 5. Nagayama- HHV6 Roseola
6. Cherry Red- Tay Sachs dis, retinal artery occlusion, Macular degeneration
7. Spotted wing- Anopheles 8. Spotted fever- Rickettsia 9. Bitot- ↓Vit A
10. Café au lait- Neurofibromatosis 11. Ash leaf + Shagreen patch- Tuberous sclerosis

SURVIVAL TIME- Rule Of 3 (Death afterwards) 1. Survival for 3min- Injury/ hypoxia
2. 3 hrs- Physiologial decompensation 3. Survival for 3 days- Dehydration
4. Survival for 3 weeks- Starvation 5. Survival for 3 months- Loneliness/boredom
6. 3 years- Loss of purpose/meaning in life 7. Survival for 3 decades- Climate change

SPAULDING CLASSIFICATION 1. Non critical- Touch skin- Decontamination=Clean
2. Semi critical- Touch mucosa- High level disinfection/ Sterilization
3. Critical- Touch sterile tissue- High level disinfection/ Sterilization

SPLENECTOMY= Splec 1. Indications→ Leukemia, lymphoma, myelofibrosis, rupture, cyst, tumour, infarction, thrombosis, abscess, Kala azar, Hypersplenism, thalassemia, sickle cell, hemolytic anemia, immune thrombocytopenic purpura, Felty syndrome, Banti disease, wandering spleen
2. MC organ injured in blunt trauma abdomen→ Spleen
3. OPSI/PSS- Overwhelming post-splec infection or sepsis (MC Pneumococcus)
4. Prophylaxis after splenectomy- Amoxy/ Cefuroxime/Clarithromycin 250mg daily
5. NHS capsular vacc <2wk before or >1wk after splenectomy and ↓complement

SUSTAINABLE DEVELOPMENT GOALS 3- End epidemics by 2030
TB, HIV-AIDS, Malaria, Hepatitis, Neglected Tropical Disease, Water borne disease

STRIDOR (Upper airway obs) -N19 1. Diphtheria 2. Epiglottitis 3. Croup 4. Pertussis

SADDLE 1. Saddleback fever- Dengue 2. Saddle nose- Syphilis, Leprosy

SILENT 1. Silent killer snake- Common krait 2. Silent MI in elderly
3. Silent Chest- Severe asthma- Life threatening 4. Silent migraine 5. Silent reflux

SAFETY 1. Biosafety precautions for infectious pathogens 2. Patient safety

SIGN 1. Tripod- Polio, epiglottits, meningitis, COPD, asthma, emphysema, CHF
2. Thumb+ Sunrise- Epiglottitis 3. Steeple+ Pencil tip-N23- Croup 4. String- Crohn dis
5. Auspitz/candle wax- Psoriasis-N22 6. Darier (Urticaria)- Mastocytosis
7. Cartwheel- Ac otitis media(ASOM) *Cartwheel nucleus- Entamoeba
8. Nikolsky- SSSS/TEN/Pemphigus vulgaris 9. Pseudo-Nikolsky- Bullous pemphigoid
10. Kernig + Brudzinski-F1/26- Meningitis 11. Murphys + Boas- Cholecystitis
12. Double tongue- Ludwigs angina 13. Faget- Relative bradycardia
14. Sunset- Hydrocephalus -F1/22,N20 15. Water Lily=Camalote- Hydatid cyst
16. Grey Turner+ Cullen+ Fox- Pancreatitis 17. Courvoisier- Obstructive jaundice
18. Blumbergs+ Rebound- Peritonitis 19. Kehrs+ Balance- Splenic injury
20. McBurney+ Dance+ Rovsing+ Dunphy+ Obturator+ Psoas- Appendicitis
21. Lighthouse- ASOM -N22, Mastoiditis 22. Sherman- Palpable dorsalis pedis
23. Reservoir- Mastoiditis 24. Target/ Halo- CSF leak 25. Filarial dance- USG
26. Trousseau+ Chvostek+ carpopedal- Tetany=↓Ca 27. Dural tail- Meningioma
28. Sheesh Kebab- Candida oesophagitis 29. Popeye -N20- Biceps tendon rupture
30. Raccoon+ Battle- Skull base fracture 31. Spalding- Fetal death = IUFD
32. Homan- Thrombophlebitis 33. Hennebert- Congenital syphilis 34. Groove- LGV-F1/26
35. Winterbottom- Sleeping sickness 36. Romana- Chagas dis
37. Gower- Duchenne muscular dystrophy 38. Klein- Mesentric adenitis
39. Babinski- ↑Plantar 40. Lhermitte- Multiple sclerosis 41. Claw-Intussusception
42. Goodell/Osiander-Cervix soft 43. Chadwick-Cervix blue 44. Hoover- COPD
45. Hegar-Uterus isthmus soft -F7/25 46. Hamman crunch- Pneumomediastinum
47. Kussmaul + Pulsus paradoxus- Constrictive pericarditis 48. Martel- Gout
49. Ring of fire/ Bagel/ Tubal ring/ Interstitial line- Ectopic pregnancy
50. Muller- Pulsatile uvula 51. Morton-Mahon- Facial flushing/blanching
52. Bozzolo- Pulsatile nasal mucosa 53. Minervini- Pulsatile tongue
54. Rosenbach- Pulsatile big liver 55. Gerhardt- Pulsatile spleen
56. Corrigan- Dancing caortids 57. Hill- ↑Lower limb BP/ Aortic regurgitation
58. Quincke- Capillary pulsation in nail bed 59. Hollman Miller- Angiofibroma
60. Meniscus= Crescent- Pleural effusion, Meniscal tear 61. Levine- Angina
62. Dalrymple/ Lid lag/ Von Graefe- Hyperthyroidism 63. Trail- Aortic aneurysm
64. Queen Anne- Hypothyroidism 65. Silhouette- Lobar pneumonia
66. Froment- Ulnar nerve palsy 67. Kanavel- Infective flexor tenosynovitis

68. Chik- Chikungunya 69. Courvoisier law- Ca head of pancreas
70. Dagger + Scotty dog with collar- Spondylolysis 71. Pathergy- Behcets disease
72. Double density + Crescent- ANV hip 73. Fallen fragment- Simple bone cyst
74. Bone within bone- Osteopetrosis 75. Shawl +V- Dermatomyositis
76. Candle grease/ Nail pitting- Psoriasis 77. Oil drop (nail)- Psoriasis
78. Duroziez- Double intermitent murmur over femoral artery 79. Shoulder pad-Amyloidosis
80. Carpet tack- Discoid lupus erythematosus 81. Schwartz- Otosclerosis
82. Hampton hump + Westermark- Pulm embolism 83. Rigler- Pneumoperitoneum
84. Coffee bean- Sigmoid volvulus 85. Colon cutoff- Ac pancreatitis
86. Air bronchogram- Pneumonia 87. Codman triangle- Osteosarcoma
88. Landolfi- Pulsatile iris 89. Becker- Retinal artery pulsation

SMEAR 1. Peripheral blood-Malaria,Filaria-F24,N24 2. Starry sky-Burkitt lymphoma-EBV
3. PAP- Ca cervix-95% ethanol fixative-F7/25 4. Slit Skin- Leprosy 5. Tzanck-F24/25- HSV, VZV

STAIN 1. Cell wall- Gram 2. TB/Leprosy/Cryptosporidium- Acid fast ZN stain
a. TB/Non TB mycobacteria - ZN 5%H_2SO_4 b. Leprosy- Fite Faraco, ZN 5%H_2SO_4
c. Cryptosporidium/Cyclospora/Cystoisospora/Nocardia- Kinyoun cold ZN 1%H_2SO_4
3. TB screening- Fluorescent Auramine-Rhodamine
4. Capsule- India Ink= Nigrosin-N19 5. Syphilis- Silver impregnation stain
6. Diphtheria- Albert/ Neisser/ Ponder stain for Babes-Ernst granules
7. Malaria/Filaria- Giemsa, Leishman, JSB, fluorescent acridine orange
8. Fungal- PAS, LPCB, KOH, Grocotts, Fluorescent Calcufluor white
9. Malaria- JSB, Wright, Giemsa 10. Mucicarmine- Cryptococcus, adenoca
11. Mast cell- Toluidine blue 12. Amyloid- Congo Red 13. Glycogen- PAS
14. Collagen- Masson trichome 15. Melanin- Masson Fontana
16. Calcium- von Kossa, Alizarin 17. Copper- Rodonine, Rubeanic acid
18. Fat- Oil red O, Sudan Black B 19. Hemosiderin- Perl Prussian blue
20. Ceruloplasmin- Orcein 21. Corneal ulcer- Fluorescein
22. H. pylori, T. pallidum- Warthin starry silver
23. Mycoplasma- Diene stain 24. Proteus- Diene phenomenon
24. Fluorescent stains- FITC, DAPI, Hoechst, propidium iodide, GFP, alexa fluor
Auramine Rhodamine- TB, Acridine orange- Malaria, Calcofluor white- Fungi

SEPSIS= SIRS+infec 1. SOFA-6 points-ICU pt 4. qSOFA-3 pts- Non ICU pt -F20/21/23

STRATEGY=St 1. RIGHT St- Snake bite first aid (Not for Rx)
2. WASH- Water, sanitation, hygiene (Water washed dis- poor hygiene- Scabies, Trachoma)
3. HIV testing strategy- I blood donation, IIa epidem, IIb symptomatic, III asymptomatic
4. HIV 95-95-95 strategy- 95% diagnosed, 95% Rx, 95% virally suppressed
5. HIV prognostic strategy- U=U- Undetectable=Untransmissible 6. TB- 90-90-90
7. Cervical ca- 90-70-90- 90% vacc, 70% screen, 90% Rx 8. SAFE- Trachoma
9. Anemia Mukt Bharat-F1/26- 6 beneficiary x 6 intervention x 6 mechanism strategy

SURVEILLANCE 1. Active- Actively finding patients door to door
2. Passive- Patients coming to hospital 3. Sentinel- Finding missing patients

SOAP BUBBLE Giant cell tumour/Osteoclastoma
DOUBLE BUBBLE in X ray- Duodenal atresia **TRIPLE BUBBLE**- Jejunal atresia

SKULL-F25,N19 1. Salt & pepper- Hyperparathyroidism 2. Moth eaten- Multiple myeloma
3. Hair on end=Crew cut- Thalassemia, sickle cell, iron def anemia, neuroblastoma
4. Cotton wool- Paget dis

STOOL COLOUR 1. Pea soup green/yellow (2nd week)- Typhoid 2. Clay- Giardia
2. Red currant jelly-Rotavirus, Intussuception (MC ileocolic) 3. Rice water- Cholera-F1/26

SPUTUM 1. Rusty- Strep pneumoniae 2. Red curant jelly- Klebsiella
3. Bloody- Hemoptysis (MC TB) 4. Frothy- Pulmonary edema 5. Green- Pseudomonas
6. Yellow/ purulent- Bacterial infection 7. White- Viral infec
8. Anchovy sauce- Ameobiasis 9. Mucoid/sticky- Chr bronchitis

SMELL 1. Vagina a. Curdy- Candida b. Fishy- Bact vaginosis/Trichomoniasis
2. Nose- Foul- Atrophic rhinitis 3. Pus a. Fruity- Pseudomonas b. Foul- Anaerobe
4. Stool- Foul- Entamoeba 5. Rotten egg- H_2S poisoning (Green colour)
6. Mothball- Napthalene poisoning 7. Bitter almond- Cyanide

SURVIVAL TIME- Rule Of 3 (Death afterwards) 1. Survival for 3min- Injury/hypoxia
2. 3 hrs- Physiologial decompensation 3. 3 days- Dehydration
4. 3 weeks- Starvation 5. 3 months- Loneliness/boredom
6. 3 years- Loss of purpose/meaning in life 7. 3 decades- Climate change

SPINE 1. Potts- TB 2. Rugger Jersey- Hyperparathyroidism 3. Codfish- Osteoporosis
4. Bamboo- Ankylosing spondylitis HLA B27 5. Ivory- Prostate ca metastasis
6. H shape- Sickle cell dis

SCLERO = Hardening ± scarring ± fibrosis due to collagen/ plaque/ bone

1. Scleroderma= Systemic sclerosis- Hard skin+ lungs+ heart+ kidney+ Anti centromere Ab
2. Sclerodactyly- Hard skin of finger + toe, associated with systemic sclerosis
3. Scleredema- Hard thick skin, associated with diabetes
4. Osteosclerosis- Hard bone
5. Otosclerosis- Hard stapes + 1st conductive→ Mixed deafness+ carhart notch
6. Scleradenitis- Hard glands 7. Phlebosclerosis- Hard veins
8. Arteriosclerosis- Hard thick artery wall- Less blood flow
9. Atherosclerosis- Plaque (fat, cholesterol, calcium)- Hard narrow artery- MI/ Stroke
10. Rhinoscleroma-F1/26- Hard nasal tissue + congestion + crusting + deformity
11. Multiple sclerosis- Scarring and demyelination, poor nerve signals
12. Tuberous sclerosis- Bening tumour causing hard brain, heart, lung, kidney, skin Shagreen patch + Ash leaf macules/spots
13. Sclerotherapy for hemorrhoids- Dilated engorged veins are made hard
14. SSPE- Subacute Sclerosing Panencephalitis- Complications of measles
15. Copper penny/ Medlar/ Sclerotic body- Chromoblastomycosis
16. Sclera of eye- Scleritis (inflammation), scleromalacia (after inflammation), sclerectomy

TEST 1. Kirby Bauer- Antibiotic resistance -N19/20 2. Germ tube- Candida -N20,22,F1/26
3. Horrocks/ Orthotoluidine- Chlorine 4. Phosphatase-N18 Pasteurization
5. Rideal Walker -N18/20/21 New disinfectant 6. Lepromin- Leprosy
7. Eleks gel+ Schick- Diphtheria 8. Standard+ Milk ring- Brucella
9. CAMP- GBBHS 10. Reverse CAMP- Clostridium perfringens
11. Widal- Typhoid (Not Typhus) 12. Weil-Felix- Typhus (Rickettsia)
13. Coombs+ Tridot+ Western blot- HIV 14. Burrow Ink- Scabies
15. DEC Provocation- Filariasis -N20 16. Sabin Feldman dye -N18 Toxoplasma
17. Casoni- Hydatid cyst- Type 1 hypersensitivity -N23 18. Tourniquet- Dengue, Shock
19. Kleihauer Betke- Anti D dose 20. Monospot & Paul Bunnel- EBV-F1/26
21. CBNAAT/ UDST/ CDST/ Mantoux/ Tuberculin/ Quantiferon gold- TB
22. Whiffamine- Bacterial vaginosis, Trichomoniasis -F1/7/20/23/24,N19/20/23,INIC24
23. Hair perforation-Trichophyton mentagrophytes-N21/22, Microsporum canis
24. Rapid Immunochromatography-N20-Preg, HIV, HepB, HepC, Dengue, Malaria, Chikungunya, Leishmania, Leptospira, Syphilis, Typhoid, Cryptococcus
25. Urea breath (C_{13}>C_{14}), Rapid urease CLO biopsy, coproantigen test- Helicobacter

26. Stool antigen= Coproantigen test- H. pyloti, Giardia, Rotavirus, Noroviri]us, IBD
27. Scotch/cello tape- Pin worm -INI25 28. Naeglers α toxin- Gas gangrene
29. 20min whole blood clotting-Viper bite -N23 30. Hanging drop- Cholera-F1/26
31. Any infec- CRP, Procalcitonin -INI25 32. Patch=Allergy- Type1 hypersensitivity
33. G6PD test- Before starting chloroquine/primaquine
34. Syphilis standard test- VDRL/RPR . Syphilis confirm test- TPHA, TPPA, TPLA
. Syphilis test with live Treponemes- TPI . Syphilis most specific test- DFA/TPI
. Syphilis Tertiary/ congenital syphilis/ Most sensitive/ 1st +ve test- FTA Abs
35. Rinne/Weber- Deafness 36. Caloric- Vestibulitis 37. Shake- Vacc freeze
38. Baldwing, obturator- Appendicitis 39. Precipitin- Human vs animal blood
40. Bonney- Stress incontinence 41. Guaiac- Stool occult blood
42. Treadmill/Stress- Heart function 43. Schirmer- Dry eye
44. Schilling- Vit B12 defi 45. HLA B27- Ankylosing spondylitis
46. Snellen- Distance vision 47. Jegar- Near vision 48. Isihara- Colour vision
49. Confrontation- Central-peripheral vision 50. Roll over- Pre-eclampsia
51. Triple- Down syn 52. Trendelenburg- Varicose veins 53. Bangle- PEM
54. Romberg- Balance/gait 55. Knee kiss- Meningitis in child 56. Binnet- IQ
57. Water deprivation- Diabetes insipidius 58. Water loading- SIADH
59. VMA- Pheochromocytoma 60. ACTH stimulation/Cosyntropin- Addison dis
61. Glucose challenge- Acromegaly 62. Saline infusion- Conn syn
63. Dexamethasone suppression- ACTH/Pituitary tumour/ Cushings syn
64. Gettler/Diatom- Drowning 65. Creatine clearance- GFR 66. Patrick Faber- Hip
67. McMurray- Meniscus tear 68. Ortolani-Barlow- Congenital hip dislocation
69. Teichman= Hemin crystal- Hemoglobin 70. Takayama= Hemochoromogen crystal
71. Blood (Forensic)- Benzedine, o-toluidine, KastleMeyer, Teichman, Takayama tests
72. Sperm (Forensic)- Barberio, Florence, Barbital tests 73. Adam- Scoliosis
74. Allen- ABG analyis 75. Kveim- Sarcoidosis 76. Fern/Nitrizine- Amniotic fluid leak
77. Tensilon- Myasthenia gravis 78. Sweat chloride- Cystic fibrosis -F7/25
79. D-Dimer- DVT/ Pulm embolism 80. Holter- 24 hr ECG monitoring
81. Dix Hallpike- BPPV 82. Methylene blue- Milk contamination (E. coli)
84. Presumptive coliform- Water contamination (MC E.coli)
85. Differential coliform- Recent water contamination (Thermotolerant E.coli)
86. Slit air sampler/ Settle plate- Air contamination

THERAPY=TREATMENT=Rx 1. Medical Rx (Research/Evidence based- Standardized- Regulated- Approved- Can be prescribed)- Pharmaco Rx (Empirical Rx= Presumptive Rx, adjunct Rx= Additional, Definitive= Radical Rx=Full cure, Clinical Rx=Symptomatic cure), Hormone Rx, Oxygen Rx, Immuno Rx, Antibiotic Rx (Monodrug, multidrug), Chemo Rx (Induction, Neoadjuvant, Adjuvant, Targeted, Consolidation), Radiation Rx (Brachy Rx), Photo Rx (UVA/UVB- Psoriasis Rx, Blue light- Neontal jaundice Rx, Red light, Laser Rx, Infrared Rx, Photocoagulation Rx, Woods lamp- Fungus diagnosis), Cryo Rx, Nano Rx, Hyperthermia Rx, Hypothermia Rx, Ultrasound Rx, Chelation Rx, Biofeedback Rx, Phlebotomy Rx, Sclero Rx, Electro Rx (Electroconvulsive Rx, Interferential Rx, Deep Brain Stmulation Rx, Pacemaker, Defibrillator, TENS), Rehabilitation Rx (Physio Rx, Speech Rx, Wax Rx, Infrared Rx, Hydro/Aqua Rx), Apheresis/Plasmapheresis Rx, Dietary/food Rx, Palliative=Hospice=End of Life Rx

2. Psychological Rx (Supportive- Approved- Can be prescribed)- Emotional Rx, Psychodynamic Rx, Counseling Rx, Cognitive/Dialectical Behaviour Rx, Tele Rx, Deaddiction Rx, Mirror Rx, Recreational Rx, Humanistic Rx, Interpersonal Rx, Exposure Rx, EMDR Rx, Marriage/Couple Rx, Family Rx, Group Rx, Hypnotic Rx, Self hypnotic Rx= Autogenic Rx, Psychodrama Rx, Attachment-Detachment Rx, Placebo Rx, Acceptance & Commitment Rx, shadow Rx, Imagination Rx, Therapeutic dialogue Rx, Sleep deprivation Rx (Depression)

3. Lifestyle and Wellness Rx (Supportive- Approved- Can be prescribed)- Occupational Rx, Exercise Rx, Dietary Rx, Massage=Manual Rx, Lifestyle Rx

4. Alternative/complementary Rx (Limited research/evidence- Safety/Potency/ Efficacy unknown- Can be beneficial/ dangerous- Not approved)- Gene Rx, Herbal Rx, Acupressure Rx, Acupuncture Rx, Leech Rx, Fish Rx, Yoga Rx, Meditation Rx, Breathwrok Rx, Art Rx, Music Rx, Dance Rx, Cinema Rx, Color/Chromo Rx, Heat/Cold exposure Rx, Laughter Rx, Light Rx, Sweat lodge Rx, Aroma Rx, Bach flower/Flower essence Rx, Sea/Thalasso Rx, Cannabis Rx, Animal assisted Rx, Api Rx = Bee Rx, Magnetic field/Electromagnetic Rx, Biblio Rx, Anthroposophical =Spiritual=Religious=Faith based Rx, activaed charcoal Rx, Cupping Rx, Reiki Rx, Relexology Rx, Chiropractor Rx, Enema Rx, Trigger point Rx Salt/Speleo Rx, Bath/Balneo/Ice bath Rx, Osteopathic Rx, Weighted Blanket Rx,

Traditional/Cultural Rx (Ayurveda, Homeopathy, Unani, Siddha, Naturopathy, Tibetan Sowa Rigpa, Chinese Tai Chi/ Qi Gong/ Tui na/ Moxibustion, Korea Oriental, Japan Kampo/ Shiatsu, Europe botanical/ Kneipping/Alexander, Scandinavian Napra, German Heilpraktiker/ Feldenkrais, Shamanic Rx, Amazonian Ayahuasca/ San Pedro)

TERMS 1. Hyper→ Hyperthyroidism, hypercarbia 2. Hypo→Hypoxia, hypokalemia
3. EU=Normal→ Eukaryote, Euglycemia, Euvolemia, Eupnea, Eumycetoma, Euthyroid, Eumenorrhea, Euploidy (Normal chromosomes), Eutocia (Normal labour), Eustress (Good stress), Eupepsia, Eurhythmia, Eythymia (mood), Euphoria/Eudaimonia (well being)
4. PRO=Old/before→ Prokaryote, Prophylaxis, Prognosis, Proximal, Prolapse, Probiotic, Promastigote (Infective stage of Kala azar), Prodrug, Progesterone, Progestasert, Procaine, Prolactin, Prostaglandin, Propranolol, Probenecid, Prochlorperazine, Promethazine, Procainamide, Propofol, Propoxur, Proguanil
5. emia=In blood→ Bacteremia (typhoid, meningococcus, sepsis), Viremia (HIV, Hepatitis B/C), fungemia (Candida), parasitemia (Malaria, filarial, Leishmania), toxaemia (Tetanus, botulism, gas gangrene), septicaemia, pyemia, hyperemia (↑blood flow), ischemia (↓blood flow), uremia, azotemia (urea+creat), anemia, leukemia, dyslipidemia, thalassemia, galactosemia, polycythemia, hypo/hyper- voluemia/glycemia/natremia/kalemia/calcemia
Ca colon- Bacteremia by Clostridium septicum and Streptococcus gallolyticus
6. uria=In urine→ Pyuria, bacteriuria, hematuria, hemoglobinuria, ketonuria, chyluria (filaria), proteinuria, glycosuria, nocturia, homocystinuria, oxaluria, bilirubinuria, microalbuminuria- Earliest sign of diabetic nephropathy, polyuria (>3litre), oliguria (<400ml), anuria (<100ml)
7. Dys=Difficult→ Dysphagia, Dyspnea 8. Odyno=Painful→ Odynophagia
9. Pyo=Pus→ Pyoderma, pyometra, pyosalpinx, pyothorax, pyelonephritis, pyemia, empyema (pleural cavity), hypopyon (pus in eye-anterior chamber)
10. Penia=Less→ Leuco/neutro/thromobytopenia, pancytopenia, osteopenia, Myo/Neuroglycopenia 11. A=absence→ Anemia, anuria
12. Poly→ Polyuria, polydipsia, polyarthritis, polyphagia, polymyalgia, polyneuropathy, polymorphonuclear, polydactyly, polycythemia, polydactyly, polymorphism, polymerase, polysaccharide
13. -ismus=Spasm a. Strabismus-N19/20 Cross eye/squint b. Trismus= Lockjaw c. Meningismus d. Vaginismus e. Tenesmus- Feeling of ↓bowel evacuation

14. -ache/algia/dynia= Pain a. Headache, backache, heartache, toothache
b. Myalgia, fibromyalgia, arthralgia, neuralgia, cervicalgia, otalgia (ear)
c. Pleurodynia, coccydynia, osteodynia, gastrodynia, mastodynia, acrodynia
15. -axis a. Anaphylaxis b. Tachyphylaxis- Rapid development of drug tolerance
c. Epistaxis= Nose bleed- MC nose picking (young)/ HTN (old), atrophic rhinitis
MC Little area- Kiesselbach plexus- Rx- Epistaxis balloon catheter
Posterior epistaxis- Woodruff plexus
d. Chemotaxis- Cell movement towards/away from stimuli due to cytokines
16. -coele= swelling/herniation a. Hydrocoele- Scrotal fluid- MC filariasis
b. Cystocoele- Bladder herniates in vagina c. Rectocoele d. Enterocoele
e. Meningocoele f. Meningomyelocoele g. Syringomyelocoele
h. Cerebrocoele i. Gastrocoele j. Hepatocoele k. Omentocoele
l. Pneumocoele= Pneumatocoele- MC Staph aureus *Fecolith- Hard feces
17. -lithiasis= Stones a. Cholelithiasis b. Choledocholithiasis- Bile duct stones
c. Enterolithiasis d. Sialolithiasis- Salivary gland stone e. Pancreaticolithiasis
f. Hepatolithiaisis g. Broncholithiasis h. Urolithiasis/Cystolithiasis- Urolith
i. Nephrolithiasis- Staghorn calculus- Takes shape of pelvicalyceal system
k. MC stone- Calcium > struvite (magnesium phosphate) in UTI by Proteus
18. -megaly= ↑Size- Hepatosplenomegaly, cardiom, acromegaly (↑hand+ feet)
19. -nychia= Nails a. Paronychia- MC Staph aureus b. Leuconychia
c. Koilonychia= Spoon shape nail (Iron deficiency anemia, poor nutrition)
20. -angi = Vessel/duct a. Angina, angiography, angioplasty, angiostenosis
b. Cholangitis-N19 Bile duct c. Lymphangitis- Lymph duct d. Angiotensin
e. Angioinvasive- Angiosarcoma, Angiofibroma, Angiolipoma, Angiomatosis
f. Angioedema g. Angiodysplasia h. Microangiopathy i. Telangiectasia
21. OID= Like something→ Human ~ Humanoid, Nucleus ~ Nucleoid, Chancre ~ Chancroid, Typhus ~Typhoid, Tuberculosis ~ Tuberculoid, Rheumatic ~ Rheumatoid, Toxin ~ Toxoid, Thyroid, Mastoid, Ethmoid, Styloid, Deltoid, Corticoid, Steroid, Adenoid, Keloid, Arachnoid, Crystalloid, Colloid, Epitheloid, Histoid, Scrombroid, Asteroid, Sarcoid, Amyloid, Fibrinoid, Fibroid, Coccoid, Pemphigoid, Discoid, Mucoid, Cystoid, Alkaloid, Discoid, Opioid, Android

22. -karyo=nucleus a. Karyolysis- Nucleus necrosis b. Karyokinase- Enzyme
c. Karyorrhexis- Nucleus fragmented in apoptosis d. Karyogamy- Fusion of two nuclei
e. Karyotyping- Chromosomal analysis in Trisomy (Down 21, Edwards 18, Patau 13), Monosomy (Turner 45XO), Klinefelter (47XXY), chromosomal deletion (Cri du chat syn, Prader Willi syn), duplication, mosaicism, chimerism, translocation, aneuploidy,
f. Karyoplasm- Protoplasm of nucleus g. Karyophage- Cell eating nucleus
h. Orphan anne eye nuclei- Papillary thyroid ca
23. Epi= Above/over/upon a. Epiglottis, Epidermis, Epidural, Epiphysis, Epithelium, Epididymis, Epicondyle, Epigastric, Epicanthus, Epiretinal, Episcleral, Epithalamus
b. Epilepsy, Epistaxis, Epiphora, Epimenorrhea, Episiotomy, Epiploic,
c. Epitope, Epidermophyton, Epinephrine, Epidemic, Epidemiology

TARGET=T 1. T lesion- Steven Johnson syn, Lyme dis, erythema multiforme
2. T cell- Thalassemia, liver dis 3. T hemolysis- Clostridium perfringens
4. T sign- CSF leak
5. Elimination T- India TB 2025, Malaria 2027, Kala azar 2023, Filaria 2030
World 2030- Malaria + TB + Ca cervix + Hepatitis + Filaria + Kala azar + Trachoma + Blood fluke + Chagas dis + mother to child HIV
6. Eradication T- Polio 2026, Leprosy 2027, Yaws + Guinea worm 2030-F1/26
7. Malaria Annual Parasite Index Target <1

TONGUE=T 1. Double tongue sign- Ludwigs angina 2. Magenta red T- ↓Vit B_2
3. Strawberry (Beefy red large T)- Scarlet fever, Kawasaki disease, ↓Vit B_{12}
4. Rash on T- Ramsay Hunt syn 5. Smooth T- Atrophic glossitis (↓Vit B_{12}, ↓iron)
6. Fissured+ protruding T- Down syn 7. Geographic T- Psoriasis 8. Black hairy T- ↑Tobacco

TROPICAL=Tr 1. Tr pyomyositis- Staph aureus 2. Tr ulcer- Polymicrobial
3. Tr pulm eosinophilia-N21-Filaria 4. Tr sprue 5. Tr spastic paraparesis-HTLV

THYMUS 1. 1^0 lymphoid 2-lobed organ in mediastinum producing thymosin
2. Develops/matures T cells 3. Grows till puberty→ Shrinks afterwards
4. Thymus prevents autoimmune disease→ Thymoma/ thymic hyperplasia causes myasthenia gravis-F7/24/25 due to antibodies against acetylcholine receptor→ Rx- Neostigmine, steroids, monoclonal antibody, ivig, thymectomy
5. Absence/aplasia= DiGeorge syndrome→ Rx- Thymus transplant in thigh

TREE 1. Inverted fir tree colony-Anthrax 2. Tree in bud (CT)-TB,bronchiolitis,cystic fibrosis

TRIAD/ TETRAD/ PENTAD 1. FSH- Meningitis-Fever, headache, neck stiffness-F7/25

2. Charcot/ Reynolds- Cholecystitis- Pain + fever + jaundice + confusion + ↓BP

3. Charcot- Multiple sclerosis- Scanning speech + Intentional Tremor + Nystagmus

4. Beck= BMJ- Cardiac tamponade- ↓BP + Muffled heart sound + ↑JVP +

Pulsus paradoxus, waterbottle/moneybag pattern, ECG-Low voltage, Rx pericardiocentesis

5. Cushings= HRP- ↑ICP- ↓Heart rate + ↓↑ Resp (Cheyne Stokes) + ↑Pulse pressure

6. Adam= Hakim- Normal pressure hydrocephalus- DAC- Dementia, ataxia, urin incontinenc

7. Virchows=HBE- Hypercoagulability + Blood stasis + Endothelial injury

Virchow node- Gastric ca Virchow foamy cell- Leprosy

8. PHD- Pheochromocytoma- Palpitation+ Headache+ Diaphoresis

9. Hutchison- C. syphilis- KDH- Keratitis+ Deafness+ Hutchison teeth/ Mulberry molar

Hutchison sign in Herpes zoster ophthalmicus (vesicles on nose tip)

Hutchison/Chauffeur fracture- Radial styloid

Harrison sulcus/Rachitic Rosary/Infant craniotabes/X ray cupping-fraying- Rickets

10. RRR- Leptospirosis= Weils disease- Rain+ Rice field+ Rodent/rat urine

11. RRSET- Scrub typhus- Rain+ Rodent+ Scrub grass+ Eschar+ Trombiculid chigger mite

12. HIT- Toxoplasma- Hydrocephalus+ Intracerebral calcif+ Torch/headlight chorioretinitis

13. DCH- Congenital rubella- Deafness VIII nerve+ Cataract + Heart- MC PDA

14. PJH- Congenital CMV- Petechiae + Jaundice + Hepatosplenomegaly

15. SSP- Mycetoma= Madura foot-F1/26- Swelling + Sinus + Pus

16. Asherman-F7/25/21,N23 Endometrial adhesions+ amenorrhea+ recurrent preg loss

17. Trotter- Nasopharygeal ca- EBV 18.

19. Terrible- Knee (ACL + MCL + Medial mensicus tear) 20. Sandblom- Hemobilia

21. Unhappy- Knee (ACL + MCL + Lateral meniscus tear) 22. Bezold- Otosclerosis

23. Meniere dis- HTV- Hearing loss + Tinnitus + Vertigo 24. Dieulafoy- Appendicitis

25. Hakim- Normal pressure hydrocephalus- Ataxia + Dementia + Incontinence

26. HUS- Hemolytic anemia + Thrombocytopenia + Renal failure

27. Wernicke encephalopathy- GOA- Global confusion, Opthalmoplegia, Ataxia

28. Miller Fischer type GBS- AOA- Arreflexia, Opthalmoplegia, Ataxia

29. Oslers/ Austrian triad- Pneumonia + Endocarditis + Meningitis

30. Whipple-Insulinoma 31. Vogt- Tuberous sclerosis 32. Rigler- Gallstone ileus

33. Marshall- 2^0 blast injury 34. Macklers- Boerhaave syndrome

35. O'donoghues- ACL injury 36. Saint- Gall stone+Hiatus hernia+Diverticulosis

37. Samter=AERD- Aspirin sensitivity + Asthma + Nasal polyp -N19

38. Fat embolism- Resp distree+ neurological signs+ petechiae

TRAVEL 1. Traveler diarrhea- MC- ETEC 2. Hajj- Neisseria meningitides

3. Africa- Ebola, Yellow fever, Neisseria meningitides, sleeping sickness

4. Pre-travel vacc- YF, Typhoid, Hepatitis A/E/B, Polio, TdaP, cholera, rabies, meningococcal for Hajj, JE, chickenpox

5. Motion sickness- Hyoscine/ Scopolamine/ Dimenhydrinate

6. Sea sickness- Meclizine 7. Mountain sickness- Acetazolamide

8. Morning sickness- Doxylamine, Vit B6

URINE COLOUR 1. Orange/Red- Rifampin, Warfarin, Heparin, PSGN, hemorrhagic UTI

2. Tea-Leptospira 3. Green-Pseudomonas 4. Brown- Nitrofurantoin, Metronidazole

5. White/Milky- UTI, pyuria, chyluria-Filaria, bacteriuria, phosphaturia

6. Blue- Cimetidine 7. Blue-green- Propofol, Amitryptiline

8. Bright yellow- Multivitamins 9. Dark yellow- Chloroquine

10. Cola color urine- Hemolytic anemia, hemo/myoglobinuria, alkaptonuria-N21, phenylketonuria-N24, pheochromocytoma, bilirubinuria, porphyria, Dubin-Johnson sy

ULCERS 1. **Ileocaecal ulcer** a. Transverse constricting non bleeding- TB

b. Longitudinal perforating bleeding- Typhoid c. Flask shape perforating bleeding- Amoebiasis -N19 d. Serpiginous rose thorn skip- Crohn disease

2. **Peptic (Gastric + Duodenal) ulcer** a. Infection- Helicobacter pylori-F7/25

b. Curling stress- Burns c. Cushing- Brain injury d. NSAIDs e. Stress

3. **Skin ulcer** a. Pressure/Decubitus ulcer= Bed sore b. Meleney ulcer

c. Burns→Marjolin ulcer→Squamous cell ca d. Black necrotic- Eschar-F7/25

d. Snail track- Syphilis e. Trophic (Nerve damage)- Diabetic foot, neuropathy

4. **STI ulcer** a. Painless-LGV, Syphilis,Donovanosis b. Painful-Chancroid,HSV-F1/25/26

5. **Eye ulcer** a. Dendritic=epithelial + geographical- Herpes simplex -F1/26/25,N19

b. Contact lens- Pseudomonas>Acanthamoeba-F7/25 c. Mooren d. Shield-VKC e. Mason

6. **Miscellaneous** a. Aphthous- Mouth b. Venous- Stasis-F7/25 c. Buruli- M. ulcerans

d. Superficial continuous ulcer + polyp + rectal disease- Ulcerative colitis

e. Kissing ulcer- 2 ulcers touch . STI- 2 Inguinal ulcers . TB- 2 larynx ulcers

f. Indurated= hard ulcer- TB, Leprosy, Syphilis, Lupus vulgaris, diabetic ulcer, basal cell ca, squamous cell ca, vasculitis ulcers, pyoderma gangrenosum

HOT POTATO VOICE (Muffled dysphonic voice) 1. Epiglottitis 2. Quinsy=Peritonsillitis -F21/23,N22 3. Supraglottic ca 4. Retropharyngeal abscess 5. Ludwing angina 6. Angioedema

HOARSE VOICE 1. Laryngitis 2. Croup 3. TB larynx 4. Common cold 5. Influenza 6. Diphtheria 7. Mononucleosis 8. Epiglottis 9. Bronchitis 10. Bronchiolitis

VACCINES 1. **Acellular/Subunit**- Pertussis DTaP/DaPT/TdaP, Rotavirus, Covid-19 Corbevax/Novovax, Influenza, Shingles Shingrix, Hepatitis B, HPV

2. **Toxoid**- Diptheria Park William8, Pertussis, Tetanus, Botulinum, Anthrax

Tetanus prophylaxis (TT/Td) in previously immunized (valid for 5 yrs)

>5yr <10yr- 1dose TT, >10yr- Clean wound- 1dose TT, unclean wound- 1dose TT+ Ig

3. **Killed**- Cholera-N24, Anthrax, Pertussis, Plague, HepA, Polio Salk, MeningitisB vacc protects 30% gonorrhea- UK 2025, Influenza H3N2+H1N1, Covid19 Covaxin/Incovacc, JE JEEV/JENVAC Kolar, Rabies egg/cell culture vacc Rabipur/Verorab/Vaxirab/Indirab, Leptovac, KFD

4. **Live**-N18- BCG Danish1331, Cholera-N24, Typhoid Ty21a, Brucella Rev1, Hep A, Leptospira, Measles Edmonston Zagreb-F7/25,N18, Mumps Jeryll Lynn, Polio Sabin, Rubella RA 27/3, YF 17D, Varicella Chickenpox/Shingles Oka, Smallpox/Mpox Jynneos/Imvamune/Imvanex, JE Imojev SA 14-14-2, Dengue Dengvaxia CYD-TDV, Ebola rVSV-Zebov, Influenza H3N2+H1N1 Nasovac 2023 Indian, Plague

5. **Capsular**/Capsular conjugate- Neisseria meningitides ACYW135, Strepto pneumoniae PPSV23/PCV13, Hemophilus influenza b,Typhoid Typhim Vi/TCV

6. **Recombinant**- Hepatitis B,E, HPV-N23 Cervarix/Gardasil/Cervavac, Shingrix, Malaria Mosquirix/Matrix M, RSV Arexvy/Abrysvo 7. DNA- Covid Zy-Cov-D

8. **mRNA**- Covid-19 Pfizer/Moderna/Gemcovac, RSV mResvia

9. **Viral vector**- Covid-19 Covishield/Sputnik, Chikungunya Ixchiq

10. **Diluents** a. BCG- Normal saline b. Measles- Distilled water c. JE- Phosphate buffer

11. **Delayed vacc** a. Give till 1yr- BCG, Rota, Penta, IPV, PCV

b. Give till 5ys- OPV, Measles, MR, Vit-A c. Give till 7yr- DPT

12. **Adverse effects** a. Rota- Intussusceptionb. Measles/MR/MMR- Toxic shock syn

13. Most important in cold chain- ILR= Ice Lined Refrigerator

VULGARIS (Vulg)=COMMON 1. Lupus vulgaris- Skin TB -F1/20/24, F6/22, N20/25

2. Verruca vulgaris-N18- HPV 3. Acne vulgaris- Cutibacterium acnes -F1/25N23

4. Sycosis vulgaris- Staph aureus boil of beard 5. Impetigo vulgaris- Strep pyogenes

6. Non infectious- Pemphigus vulgaris, Psoriasis vulgaris, Ichthyosis vulgaris

WORLD HEALTH DAY 7Apr theme 2025- Healthy Beginnings, hopeful futures

.MC World Health Day theme- Universal Health- 2018/19/23

.WHO Pandemic Agreement 2025- World Health Assembly

WHO 'ESSENTIAL DIAGNOSTICS' 2018- TB, Syphilis, HIV, Hep B-C, HPV, Malaria

1. Priority 1- Critical- Carbapenem resistance (Acinetobacter, Pseudomonas),

ESBL= Extended Spectrum beta lactamase in Gram negative bacteria

2. Priority 2- High- VRSA, VRE, MRSA **WHO 'PRIORITY PATHOGENS' 2017**

3. Priority 3- Medium- Strep pneumoniae, Hemophilus influenza, Shigella

WATER 1. Watery diarrhea- Cholera-F1/26, ETEC, Rotavirus 2. Warts- Molluscum-F7/24/25

3. Water-Lily sign- Hydatid cyst 4. Blackwater fever- Malaria hemoglobinuria

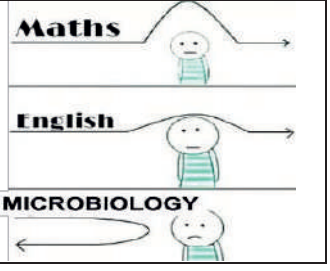
5. Fresh water- Anopheles, Aedes 6. Dirty water- Culex 7. Rice water- Cholera-F1/26

8. Waterhouse Friderichsen Syn=Adrenal insufficiency in Neisseria meningitides

9. Water birds- Japanese encephalitis 10. Water flea=Cyclops 11. WASH strategy

FMGE 1/26, 7/25, NEET PG 8/25 PYQS ANALYSIS			
H-YIELD TOPIC	**FMGE 1/26 ~51 MCQs**	**FMGE 7/25 ~55 MCQs**	**NEET 8/25 ~24 MCQs**
GEN MICRO- 10	BMW- Expired med + TPN	BMW-Cotton needle syring	PCR TB
BMW- Urobag Red	Canned food vehicle trans	BMW recycling logo	Multiplex PCR meningitis
	Airbrone infec pic	BMW Non chlorinate	
	Bacteriophag transduction	Abscess- USG I and D	
DNA fingerprint VNTR	Sec prev- Screenin+ match	Secondary prevention	
	Autoclave bio indicator		
BACTERIO- 24	VSD- Pansystolic murmur	RHD+ coronary aneurysm	Meningococc prophylaxis
TB meningitis	MS-Mid diastolic murmur	RHD-MS-Middiastolic mur	MS-Mid diastolic murmur
Clue cell- Bac vagi	Pneumonia RUL- X ray	Severe pneumonia	HIV + TB Rx
Syphilis- Painless Chancroid- Painful LGV- LN painful	Meningitis-Ceftriax+ Vanco	Bacterial meningitis	BPaLM regimen
	Cholera- Rice water	Trachoma-Herbert pit	Skin TB-Lupus vulgaris
	Leptospiremic phase	Serious otitis media-Fluid	Weils dis- Leptospira
Klebsiella ozaena- Foul smell nose		STSS- Tampon + SSSS	Anthrax-McFadyean rxn
Klebsiella rhinoscleromatis- Antibiotic + Surgery		Staph food poison 3hrs	Brucella farmer cattle
Mycoplasma fried egg colony		Ulnar nerve monitoring	Nocardia- Paraffin bait
Liver abscess CT	Brucella Malta fev- Sheep	Brucella- Cattle sheep	
Typhoid- Ceftriaxo	H. influenz- Chocolat agar	Scrub typhus-Eschar-Doxy	
UTI- No fosfomycin	Botox inh acetylcholine	Asherman Synd + Stye	
Tonsillitis pic	Lepra rxn- BT leprosy	H.pylori Rx + Vaccutainer	
Larynomalacia	Anthrax- Wool+ Eschar	GBS Demyelination	
VIROLOGY- 8	Viral meningitis	Influenza ssRNA	HIV-Nevirapine+Zidovudi
	HBsAg-Rx Tenofovir	HBsAg-Rx Lamivudine	
	Chickenpox- Centripetal	HPV- Oropharyngeal ca	HPV- Oropharyngeal ca
	HSV-Dendritic ulcer	HSV-Dendritic ulcer	Oral herpes
	Pap smear	Ca cervix- HPV+PAP (2)	Pap smear
	Measles- Koplik spots	Measles live v EZ	
	EBV- Kiss- Paul Bunnel	HSV1 whitlow/felon dentist	
	EBV Burkitts lymphoma	Hand foot mo-Coxsackie A	
		Thick vagina mucosa	
		OCP safe in HIV	
		Co test HPV	
		Tzanck smear-HSV	
		Rabies samp nape of neck	
		Molluscum contagiosum	
PARASITO- 5	Anemia Mukt Bharat	Megaloblastic anemia	Anopheles identify
MALARIA	P. falci Rx artesunate	P. falci Rx artesunate	
	Manipur AL+Priamaq day2	Cyclospora chr diarrhea	Toxoplasma pregnancy
Guineaworm	Strongyloides, Trichinela	Acanthamoeba contactlens	Ova- Pinworm
MYCOLOGY- 5	Madura foot Tamil farmer	Aspergilloma Voriconazole	Aspergillu septate hypha
Pseudohyphae	Vagina Candidiasis	Sportothrix lymphatic	Aspergillus flavus
	Mucormycosis-Diabetics	Mucormycosis-Diabetics	Histoplasma
	Ringworm hand dorsum	Sabouraud dextrose agar	
	Histoplasma	Pneumocystis pneumo HIV	Dapsone
		Ergotism-Bajra	
IMMUNOLOGY- 1	Rheumatoid arth- Ty3 HSN	IgM- 1st Ab after infection	Neisseria-C5C9complem
Myasthenia gravis		Open/closed vial policy	Flow cytometry (2)
Kaposi sarcoma		Multiple myeloma-M spike	
Yellow fev 17D Nig		BCG vac-Reassure mother	

FMGE 7/24, 1/24, 7/23 PYQS ANALYSIS			
H-YIELD TOPIC	**FMGE 7/24 ~40 PYQs**	**FMGE 1/24 ~40 PYQs**	**F 7/23 ~40 PYQs**
GEN MICRO	Relative bradycardia- Typhoid, Brucella	Capsule stain, Kinyoun stain Conjugation-Sex pili	Staph epidermidis Modified ZN stain Rickettsia Multinuc giant cell MMR, TT vacc
Sterilization	Breast inf-Staph	Ethylene oxide- B. globigii	
Biomedical Waste	Yellow Bag- Incineration	Needle with fix syringe-White Blue bag- Metal Implant Red-Plastic, rubber, vacutainer	
BACTERIOLOGY	Diphtheria toxin EF2	Gas gangrene-Crepitus	Gas gangrene
TUBERCULOSIS 7/24-7-8 MCQs	Miliary TB Chest X ray Ethamb-Optic neuritis Nikshay app, TB hip	Skin TB India- Lupus vulgaris MC genital TB-Salpingitis First TB test-CBNAAT	1st TB test-CBNAAT Strep agalactiae- CAMP test positive
Cholera/Vibrio Darting motility	Selective media-TCBS Secretory diarrhea	Image of Fourniers gangrene Charcots triad- Cholangitis	V. parahemolyticus Seafood poisoning
Scrub typhus	Orientia tsutsugamushi	Eschar=Black necrotic ulcer	TB in child, PSGN
Staphylococcus	Mastitis-Rx Cloxacillin Staph epid-Conjunctiva Nikolsky sign-SSSS	Klebsiella- Ozaena=A. rhinitis Endocarditis- Dukes criteria Leprosy	Strep pyogenes bacitracin sensitive Anthrax-Cat A
Streptococcus/ Rheumatic fever	↑ASO- Rheumatic fever Guttate psoriasis Typhoid blood culture	Image of mitral stenosis S. pneumoniae- Gram +ve capsulated diplococci	Glomerulonephritis Jones criteria-RHD Sore throat-GABHS
Other	CSF=VP shunt	Epiglottitis-Tripod/steeple sign	Argyll Robertson p
VIROLOGY	Mpox diagnosis- PCR	Nipah-Fruit Bats-Kerala	Leprosy, Acne
Herpes virus gp Incl EBV, CMV	Tzanck smear- Multinucleate giantcell	-	HHV8-Kaposi sarc Endocarditis
HPV/ Other	Molluscum contagiosum- Water wart	Condyloma acuminata Rx Pap smear- Cervical cancer	Condy acuminata Condy lata-Syphilis
HIV	Kaposi sarcoma-CD4 200 CMV- Owl eye appear	-	Diarrhea in HIV- Cryptosporidium
Miscellaneous	Tenofovir-Rx- HIV&Hep B	Mumps orchitis	Measles, Mpox
PARASITOLOGY	Itch mite-Scabies	Gambusia eats mosq larva	Aedes-Freshwater
MALARIA 7/24-3 MCQs	Rx of P. falciparum Banana gametocyte P. falci- Multiple ring pic	Rx in pregnancy P. vivax Cerebral malaria	-
Leishm=Kalaazar	Paris green kill Anophele	India reservoir-Only human	Falling leaf-Giardia
Entamoeba Trichomonas	Flask shape ulcer Quadrinucleate cyst	CT/MRI Image of liver abscess	Hydatid cyst-Dog Bladder ca Rx
Filaria	Successful parasitism	Filaria- Lymphangitis	Elephantiasis
Toxoplasmosis	Bence Jones-Myeloma	Toxoplasma from cat feces	Toxoplasm-TORCH
MYCOLOGY	Aspergilloma image	Cryptococcus meningitis	Fungal keratitis
Dermatophyte	Epidermophyton-Club	-	Trichophyton, Tinea
Candida	Oral thrush=Candidiasis	Vaginal Candidiasis	Exotoxin. BMWaste
Miscellaneous	Pneumocystis in HIV	Sporotrichosis-Rose Gardener	Chr inflam-TNF α
IMMUNOLOGY	1^0 lymphoid org-Thymus Albumin↓in inflammation	ABO/Rh incompatibility- IgG Organ Transplant	Bence Jones pro- Multiple myeloma
CLINICAL INFEC	Myasthenia gravis	TB meningitis-Cobweb coagulum	Newborn of HIV +
Meningitis	CSF bac meningitis	CSF bacterial meningitis	CSF bac meningitis
Hemoph ducreyi- Chancroid	Painful ulcer + painful lymph node = bubo pic	Painful inguinal bubo pic- Seen in Chancroid and LGV	Penicillin anemia- Type 2 HSR
Bioterrorism	Ebola-Cat A, biosafety 4	Bact vaginosis- Amsel criteria	Bact vaginosis

FMGE 1/25, 1/23, NEET 8/24 PYQS ANALYSIS			
H-YIELD!	**FMGE 1/25 ~40 PYQs**	**NEET 8/24 ~50 PYQs**	**FMGE 1/23 ~40 PYQs**
GEN MICRO	Condom catheter-Boy 2y	Subclinical case- Influenza	McConkey agar
	Lactation mastitis-Staph	Multiplex PCR- Meningitis	Wilson Blair medium
	Psoriasis- Auspitz sign SSSS-Nikolsky sign	Strongyloides in HIV Gono-Thayer Martin, stain	Plasma sterilization India Ink-Cryptococcus
BACTERIO	Bacillus cereus friedrice	Pili=fimbriae for adherence	Guillain Barre syndrome
TB 1/25-5MCQ	Isoniazid-katG mutation Potts spine, LJ medium	TB pleural effusion-Lights Lepromatous Leprosy	TB Rx, Leprosy Rx Bladder ca Rx- BCG
Klebsiella ozaena-Mercif anosmia	Meningitis Rx- Ceftriaxone+Vancomycin	Meningitis+ rash=Neisseria V. parahemolytic-Seafood	Secondary syphilis Chancre in syphilis
	Shigella bloody diarrhea	EHEC O157:H7, pneumonia	Mycobacterium avium
Botox mech H. pylori	Prosthetic valve endocard <12m-S. epider, >12m	Staph epidermidis catheter Generation time infectivity	TB meningitis. miliary
Leprosy Rx- Rifampicin Ulnar nerve	RHD- Pulm stenosis- Graham Steel murmur	Rheumatic fever- Murmur Syphilis, Rx of diphtheria	Fournier gangrene Gas gangrene, anthrax
	Leptospira- Rat urine	Leptospirosis- Rice farmer	PSGN, Autoimmunity
VIROLOGY	HAART+ATT- IRIS	Fruit bat- Nipah, Ringworm	Multiple myeloma
Congenital Rubella	Tzanck smear-HSV, VZV Rabies3-Wash+Vacc+Ig	Tzanck smear-HSV, VZV Genital herpes pic	Tzanck smear-HSV, VZV Owl eye- CMV
	Ca cervix Rx- Cisplatin	Ca cervix-Pap smear	Ramsay Hunt syndrom
	Measles-Vit A+Home isol	Rotavirus, cholera vaccine	Measles
	Dengue hemorrhagic fever	Dengue thrombocytopenia	SAFE strategy-Trachom
PARASITO	Toxo-Chorioretinitis	Toxo-Chorioretinitis	Trichomoniasis
MALARIA 7/24-3 MCQs	Acne Rx- Isotretinoin Acu Hep-High infectivity	Acne- Rx-Isotretinoin Chr Hep-High infectivity	ACT- AL, ACT-ASP Cerebral malaria
	IgM Anti HBc- Ac infec	Delhi sore, Sandfly pic	QSOFA- Sepsis
	H. ducreyi- Both painful LGV- Lymph node pain	Twitching motility Reed Sternberg cell	Mast cell-Type 1 hyper. Reed Sternberg cell
	-	Neurocysticercosis	Schistosoma
MYCOLOGY	Mucor- Amphotericin B	Histoplasmosis, Blastomyc	Mucor- Amphotericin B
	Cryptococcus- India Ink Amphotericin+ Flucytosine	Aspergillosis- Voriconazol	Sporothrix-Rose
		Candida budding-Catheter	Fungal sepsis
	Sporothrix-Gardener	Desert rheumatism	Ulnar nerve, RNA
IMMUNOLOGY	Heterophile Ag- Typhus	Endothrix-Trichophyton	Hepatitis A
	No live vacc in pregnancy	Acanthamoeba keratitis	
	Widal test- Typhoid	C difficile colonoscopy pic	Kupffer cells- Liver, IgG
fIPV-6w & 14w	Bruton agammaglobuline- Recurrent S. aureus infec	Mott cell-Multiple myeloma	Antibiotic action
		Plasma cell, SCID	Lumbar puncture
CLINICAL INF	Pseudomonas-Green colony-Pyocyanin	Gonorrhea- Rx-Cefixime Trachoma- Herberts pits	Lobar pneumonia Cystic fibrosis
Pneumonia-2y Fev+cough	Hepatocell ca- High AFP	Catheter with biofilm, OPV	
	Covid-Real time RTPCR	Lepromin test, Nocardia	
Steeple sign Croup=ALTB	Burkitt lymphoma-cmyc	Wuchereria pointed tail	
	HFMD-Tomato-hox gene	Bacterial vag-Nugents	
HUS-E.coli	Rx malaria-G6PD screen	Active Herpes- LSCS	
	Culex Anopheles larva	Quarantine period	
	Kill mosq larva-Prim prev	Dengue hematocrit	
	Deworm Day-10Feb10Aug	Sapna-Leprosy mascot	

FMGE Myths 1. I can focus on score 150 (Always target score 200/300, else no success)

2. I can focus on individual subject (Always focus on integrated concept to solve clinical MCQs)

FMGE Nutrition Plan 1. Balanced home style diet + snacks/comfort foods/fast foods at ease

2. Multiple light meals. Walk for 5min after meals. This will act as natural antidepressant

3. Fresh fruits/juices, dry fruits, veggies, proteins. They boost energy and memory

4. Comfort foods (Chocolate, icecream). They are mood boosters, they reduce loneliness and stress

5. Mental health is more important than physical health. You can afford to lose weight anytime, but you can never ever afford to lose your mind. More weight + More marks in FMGE exam

6. Drink water 25-50ml per kg body weight. It boosts energy and improves focus

7. Supplements (Multivitamins, calcium, creatine, caffeinated energy drinks, modafinil) not necessarily required if taking a good diet. Minimize tea/ coffee/ alcohol/ nicotine/ stuff

FMGE Repeaters Checklist 1. Workbooks remain same- All updates are posted online

2. Are your concepts clear? If you score >60-70% in GTs/Mocks, concepts are clear

3. If concepts/contents not clear- More classes/videos required (Self study not sufficient)

4. If concepts clear- Read workbook/ notes/ rapid revision and invest time in revision + MCQs

FMGE Pre-Exam Checklist (One day before Exam) Aadhaar update takes 1-10 days, do it earlier

1. Revise high yield notes + images + PYQs. No new topic

2. Colour printed admit card both pages back to back on single sheet

3. Update Aadhaar SMS phone number + fingerprint, unlock biometrics on mAadhaar app/website

4. Light non spicy dinner, hydrate suitably avoid tea/ coffee /alcohol/ nicotine/ stuff

5. Think positive, pray for success, take deep breaths/ watch a movie/ scroll to destress

6. Sleep by 8pm, ensure 7-8 hrs of refreshing sleep

FMGE Exam Day Checklist (Exam Day) First of all, arrange post exam party resources ☺

1. Get up @ 5am, refreshing breakfast. No over excitement, no overthinking, just smart confidence

2. If feeling tensed, take deep breaths, listen meditation/happy/upbeat youtube music for 10-20min

3. If getting very tensed (Palpitations/sweating/tremors), if pulse >100, you may take tab Propranolol 10 mg at 6am (avoid after 7am). It reduces performance anxiety. Avoid unproven drugs (Modafinil)

4. Simple clothes (Sweater, multiple layers, cloth shoes). No jacket/ hoodie/ boots/ watch/ jewellery/ belt/ cap/ wallet/ handbag/ goggles. Keep snacks/chocolates to eat enroute

5. Take any 2 ID proof hard copy (Aadhaar/ PAN/ Voter Card/ Passport/ Driving License) + notes

6. Reach exam centre between 7-7:30 am, get formalities done, visit washroom before 8:15am

7. If you feel tensed, take slow deep breaths before and during exam

8. After paper 1 @11:30 am, eat light, drink water, revise notes (no MCQs), don't discuss paper 1

9. After paper2 @ 4:30pm, eat/drink well, talk to friends/gfs/bfs, relax, sleep like there's no tomorrow

10. Carpe diem! Seize the Day! See you on the ladder of success beyond 200/300 soon!

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